The Math of Ideas

Volume I: Foundations

Composition, Resonance, and Emergence from Eight Axioms

A Formal Treatment with Machine-Verified Proofs

Part of the series "The Math of Ideas"

Preface

This is the first volume in a six-volume series, *The Math of Ideas*, which develops a unified mathematical framework for studying ideas—how they compose, resonate, and give rise to genuinely new meaning. This volume establishes the foundations: eight axioms from which all subsequent volumes build. Unlike existing approaches to memetics, cultural evolution, or information theory, the math of ideas takes the *idea* as a primitive object and derives its properties from a minimal set of axioms about composition and resonance.

The five volumes that follow study the same mathematical universe from different vantage points: strategic interaction (Vol. II), philosophy of meaning (Vol. III), signs and culture (Vol. IV), power and ethics (Vol. V), and emergence itself (Vol. VI). But they all rest on the theorems proven here. This volume is therefore not merely "pure math before the applications"—it is the common language in which all questions about ideas can be posed and, crucially, *proven*.

Every major theorem in this volume is interpreted through three lenses—literary theory, philosophy, and semiotics—because the mathematics of ideas is not an analogy *applied to* the humanities; it *is* the humanities, written in the only language precise enough to prove things.

The central mathematical object is the *ideatic space* $(\mathcal{I}, \circ, \varepsilon, \mathrm{rs})$: a monoid of ideas equipped with a real-valued resonance function. The key innovation is the *emergence term*

$$\kappa(a, b, c) := \mathrm{rs}(a \circ b, c) - \mathrm{rs}(a, c) - \mathrm{rs}(b, c),$$

which captures the genuinely new meaning that arises when ideas combine—meaning that is irreducible to the parts. When $\kappa = 0$, we recover a linear (Hilbert-like) theory; when $\kappa \neq 0$, we enter the domain of genuine creativity, metaphor, irony, and dialectical synthesis.

Every theorem in this book has been formally verified in Lean 4 with zero `sorry` axioms. The Lean source files are referenced throughout. This is not merely a guarantee of correctness; it is a demonstration that the theory of ideas admits the same standard of rigor as algebra or analysis.

The theory proceeds from just eight axioms:

1. Three **monoid axioms**: ideas compose associatively with a neutral element (void/silence).
2. Two **void-resonance axioms**: silence resonates with nothing.
3. Four **emergence axioms**: self-coherence is non-negative, only silence has zero presence, emergence satisfies a Cauchy–Schwarz bound, and composition enriches.

From these eight axioms, we derive a remarkably rich theory encompassing: semiotics (the study of signs), hermeneutics (interpretation theory), dialectics (thesis–antithesis–synthesis), rhetoric (persuasion), narrative theory, game theory, evolutionary dynamics, translation theory, and the geometry of meaning.

Contents

I Foundations **1**

1 The Ideatic Space **2**
1.1 Philosophical Prologue: Why Formalize Ideas? . . . 2
1.2 The Carrier Set and Its Operations . . . 3
1.3 The Algebraic Axioms: Monoid Structure . . . 5
1.4 The Resonance Axioms . . . 6
1.5 The Lean 4 Formalization . . . 8
1.6 Minimality and Independence of the Axioms . . . 9
1.7 First Consequences . . . 10
1.8 Historical and Philosophical Context . . . 12
1.9 Summary and Prospectus . . . 13

2 The Cocycle Condition **15**
2.1 Emergence as a Trilinear Deviation . . . 15
2.2 Derivation of the Cocycle Condition . . . 15
2.3 Connection to Group Cohomology . . . 17
2.4 Coboundaries and Trivial Emergence . . . 19
2.5 The Emergence Distribution . . . 20
2.6 Higher Cocycles and the Cohomological Perspective . . . 21
2.7 The Cocycle in Historical Perspective . . . 22
2.8 Computational Aspects . . . 23
2.9 Summary . . . 23

3 Self-Resonance and Weight **25**
3.1 The Weight Function . . . 25
3.2 The Thermodynamic Analogy . . . 26
3.3 Weight and Composition Powers . . . 29
3.4 The Self-Emergence Profile . . . 30
3.5 Semantic Charge . . . 31
3.6 Weight in the Natural Number Model . . . 32
3.7 The Dialectical Tension . . . 33
3.8 Irreversibility and the Arrow of Ideation . . . 34
3.9 Summary . . . 34

4 The Asymmetry of Resonance **35**
4.1 Why Resonance Is Not Symmetric . . . 35
4.2 The Asymmetry Function . . . 35
4.3 Symmetric and Antisymmetric Decomposition . . . 38

4.4 Meaning Curvature . . . 39
4.5 The Asymmetry Cocycle . . . 40
4.6 Power and Influence . . . 40
4.7 Symmetrization and the Mutual Information Analogy . . . 41
4.8 The Antisymmetric Emergence . . . 41
4.9 Summary . . . 42

5 The Algebra of Emergence 43
5.1 The Emergence Function as a Trilinear Form . . . 44
5.2 Classification of Emergence Types . . . 44
5.3 Global Emergence Classifications . . . 45
5.4 The Emergence Tensor . . . 47
5.5 The Emergence Spectrum . . . 48
5.6 Emergence and Composition Powers . . . 49
5.7 The Emergence Algebra . . . 50
5.8 Applications: Pattern Recognition in Emergence . . . 50
5.9 Subspaces and Ideals of the Emergence Algebra . . . 51
5.10 The Emergence Kernel . . . 52
5.11 Summary . . . 53

6 Utterance, Idea, and the Semiotic Gap 54
6.1 Saussure's Problem in Formal Dress . . . 55
6.2 Same-Idea Equivalence . . . 55
6.3 The Connotation Gap . . . 58
6.4 Resonance Equivalence . . . 59
6.5 The Semiotic Triangle . . . 60
6.6 Quotient Spaces and the Idea Space . . . 61
6.7 Peirce's Semiotics and the Interpretant . . . 61
6.8 Derrida and the Instability of the Sign . . . 62
6.9 Wittgenstein and Family Resemblance . . . 62
6.10 The Semantic Charge of Equivalence Classes . . . 62
6.11 Summary . . . 63

7 Transmission and Mutation 65
7.1 The Transmission Problem . . . 66
7.2 The Transmission Model . . . 66
7.3 The Resonance Shift Decomposition . . . 67
7.4 Conservative and Mutagenic Receivers . . . 68
7.5 Transmission Chains . . . 69
7.6 Fidelity . . . 71
7.7 Fixed Points of Transmission . . . 71
7.8 The Sublime and the Fragile . . . 72
7.9 Morphisms and Structure Preservation . . . 73
7.10 The Category of Ideatic Spaces . . . 75
7.11 Mutation as Cultural Evolution . . . 75
7.12 Open Questions . . . 76
7.13 Summary and Conclusion . . . 78

II Structure and Dynamics **79**

8 Chains of Transmission **80**
8.1 From Dyads to Chains . . . 80
8.2 The Chain Enrichment Theorem . . . 81
8.3 Cumulative Emergence Along Chains . . . 83
8.4 The Ratchet Effect . . . 84
8.5 Chain Diagrams . . . 85
8.6 Chain Length Bounds . . . 87

9 Conservative and Mutagenic Transmission **89**
9.1 The Spectrum of Transmission Fidelity . . . 89
9.2 Mutagenic Receivers . . . 91
9.3 The Sublime Fragility Theorem . . . 93
9.4 The Transmission Fidelity Spectrum . . . 95
9.5 Chain Fidelity . . . 96

10 Fixed Points and Canonical Texts **98**
10.1 The Invariance of Canonical Meaning . . . 98
10.2 Canonical Texts . . . 99
10.3 Fixed Point Iteration . . . 101
10.4 Examples of Canonical Texts . . . 103

11 Distance and Proximity **106**
11.1 Measuring the Space Between Ideas . . . 106
11.2 The Symmetrised Distance . . . 108
11.3 Neighbourhoods and Convergence . . . 109
11.4 Cultural Proximity . . . 111

12 Orthogonality and Independence **114**
12.1 When Ideas Do Not Resonate . . . 114
12.2 The Pythagorean Theorem for Resonance . . . 116
12.3 Orthogonality Does Not Imply Zero Emergence . . . 117
12.4 Orthogonal Decomposition . . . 118
12.5 The Dimension of Meaning . . . 119

13 Signaling and Interpretation **122**
13.1 The Communication Setup . . . 122
13.2 Payoff Functions . . . 123
13.3 Communication Surplus and the Misunderstanding Gap . . . 125
13.4 Optimal Signal Choice . . . 127
13.5 Honest vs. Strategic Signaling . . . 127

14 Nash Equilibrium in the Meaning Game **130**
14.1 The Meaning Game . . . 130
14.2 The Void Equilibrium . . . 131
14.3 Improving Upon Silence . . . 132
14.4 Best Response and Nash Equilibrium . . . 134
14.5 Equilibrium Characterisation . . . 135

14.6 Multiple Equilibria and Coordination . . . 136
14.7 The Value of Communication . . . 138
14.8 The Meaning Game Diagram . . . 139
14.9 Dynamic Meaning Games . . . 140
14.10 Summary and Connections . . . 142

III Applications and Synthesis 145

15 Cooperation and Coalition Theory 146
15.1 From Pairs to Coalitions . . . 146
15.1.1 Coalition Value . . . 147
15.2 The Cooperation Surplus . . . 148
15.3 Ordered Coalition Enrichment . . . 149
15.4 Marginal Contribution and the Core . . . 150
15.5 Semiotic Cooperation: Lotman's Semiosphere . . . 151
15.6 Visualizing Coalition Dynamics . . . 152

16 Dialectics: Thesis, Antithesis, Synthesis 153
16.1 The Dialectical Structure of Ideas . . . 153
16.1.1 Synthesis as Composition . . . 154
16.2 Dialectical Emergence . . . 154
16.3 Aufhebung: Formal Conditions . . . 155
16.4 The Dialectical Sequence . . . 157
16.5 The Cocycle Condition in Dialectics . . . 158

17 Evolutionary Dynamics of Ideas 160
17.1 Ideas as Replicators . . . 160
17.1.1 Self-Fitness and Cross-Fitness . . . 161
17.2 Invasion Dynamics . . . 162
17.3 Evolutionary Stability . . . 163
17.4 Directed Evolution and the Enrichment Ratchet . . . 163
17.5 Population Dynamics: Replicator Equations . . . 164

18 Translation and Cross-Cultural Fidelity 166
18.1 The Problem of Translation . . . 166
18.2 Void-Preserving Translations . . . 167
18.3 Composition of Translations . . . 168
18.4 The Fidelity Spectrum . . . 169
18.5 Untranslatable Ideas . . . 169
18.6 Translation Networks . . . 171

19 Emergence and the Cocycle Algebra 173
19.1 Formalizing Emergence . . . 173
19.2 The Cocycle Condition . . . 174
19.3 Coboundary Decomposition . . . 175
19.4 The Emergence Spectrum . . . 176
19.5 Deep Emergence and Higher Cocycles . . . 177

20 Deep Games and Strategic Transparency **179**
20.1 Beyond Classical Game Theory 179
20.2 Transparency and Cooperation 180
20.3 Payoff Structure in Deep Games 181
20.4 Information Asymmetry and Strategic Distortion 182
20.5 The Attribution Impossibility Theorem 183

21 Attribution, Fairness, and Shapley Values **185**
21.1 The Problem of Attribution 185
21.2 The Shapley Value 186
21.3 Shapley Values in the IDT Setting 187

Part I

Foundations

Chapter 1

The Ideatic Space

"The limits of my language mean the limits of my world."
— Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*, 5.6

1.1 Philosophical Prologue: Why Formalize Ideas?

The project of formalizing ideas may strike the reader as either hubristic or absurd. Ideas, after all, are the quintessential objects of the informal: they shift, merge, split, contradict themselves, and resist every attempt at pinning down. Plato's Forms, Hegel's Concepts, Frege's Thoughts—each represents a monumental effort to give ideas a stable ontological address, and each, in its own way, has been found wanting.

Our approach is deliberately more modest. We do not claim to capture what an idea *is*. Instead, we axiomatize what ideas *do*: they compose with one another, and they resonate. From these two operations alone— composition $\circ$ and resonance rs—we derive an unexpectedly rich mathematical universe. The axioms are chosen not for metaphysical completeness but for *mathematical fertility*: the minimum assumptions that yield the maximum structure.

This methodological choice has a distinguished pedigree. Hilbert's axioms for geometry do not tell us what a "point" is; they tell us how points behave. Similarly, our axioms for $\mathcal{I}$ do not define "idea"; they constrain the algebra of ideation. The philosophical content emerges not from the definitions but from the theorems.

Interpretation 1.1. The decision to axiomatize ideas compositionally rather than representationally reflects a broader shift in the human sciences. Just as structuralist linguistics (Saussure) defines signs by their differences rather than their referents, and category theory defines objects by their morphisms rather than their elements, IDT defines ideas by their interactions rather than their essences. An idea is, in our framework, nothing more and nothing less than a node in a web of resonance relations.

Interpretation 1.2 (Wittgenstein's Logical Space). The ideatic space bears a deep structural resemblance to what Wittgenstein calls the *logical space* (*logischer Raum*) in the *Tractatus Logico-Philosophicus* (1921). Wittgenstein writes: "The facts in logical space are the world" (1.13), and "A proposition determines a place in logical space" (3.4). For Wittgenstein, the logical space is the totality of possible states of affairs; every proposition carves out a region of this space, and the meaning of the proposition is determined by its position within it.

Our ideatic space $\mathcal{I}$ formalizes this intuition with greater algebraic precision. Where Wittgenstein's logical space is implicitly structured by the truth-functional connectives, our space is structured by the composition operation $\circ$ and the resonance function rs. The key advance is that

resonance provides a *metric* on the space of ideas—not merely a logical geography (which propositions are compatible?) but a quantitative topology (how strongly does idea a interact with idea b?). This quantitative dimension is absent from the *Tractatus*, whose logical space is purely combinatorial. IDT thus extends Wittgenstein's vision by equipping the space of possibilities with a rich geometric structure.

Moreover, the *Tractatus*'s famous limit—"Whereof one cannot speak, thereof one must be silent" (7)—receives a formal correlate in our void idea ε. The void is precisely that of which nothing can be said (it resonates with nothing), and yet it exists within the space as a structural element. The formalism does not transgress Wittgenstein's silence; it *structures* it.

Interpretation 1.3 (Saussure's *Langue* and the Ideatic Space). Ferdinand de Saussure's distinction between *langue* (the abstract system of a language) and *parole* (particular speech acts) is one of the founding gestures of modern linguistics. In the *Course in General Linguistics* (1916), Saussure insists that linguistics must study *langue*—the synchronic system of differences—rather than the diachronic succession of speech events. The ideatic space $\mathcal{I}$ is best understood as a formalization of *langue*: it is the abstract system within which ideas exist as nodes defined by their differential relations.

The resonance function rs captures what Saussure calls *valeur* (value): the relational identity of a sign within the system. Saussure writes that "in language there are only differences *without positive terms*" (*Course*, II.4.4). Our resonance function gives quantitative expression to these differences: the "value" of idea a is not an intrinsic property but is constituted by the complete profile $\{\mathrm{rs}(a, b) : b \in \mathcal{I}\}$—the totality of its resonance relations with every other idea.

The composition operation $\circ$ corresponds to what Saussure calls the *syntagmatic* axis of language: the horizontal combination of signs into larger units. Saussure's other axis—the *paradigmatic*—is captured by the equivalence classes discussed in Chapter 6. Thus the two fundamental axes of Saussurean linguistics both find precise formal correlates in the IDT framework.

1.2 The Carrier Set and Its Operations

Definition 1.4 (Ideatic Space)**.** An **ideatic space** is a quadruple $(\mathcal{I}, \circ, \varepsilon, \mathrm{rs})$ where:

1. $\mathcal{I}$ is a non-empty set, the *carrier* of the space;
2. $\circ : \mathcal{I} \times \mathcal{I} \to \mathcal{I}$ is a binary operation called *composition*;
3. $\varepsilon \in \mathcal{I}$ is a distinguished element called the *void idea*;
4. $\mathrm{rs} : \mathcal{I} \times \mathcal{I} \to \mathbb{R}$ is a function called *resonance*;

subject to axioms (A1)–(A3) and (R1)–(R2) and (E1)–(E4) specified below.

The carrier set $\mathcal{I}$ is left intentionally untyped. In applications, its elements may represent:

- **Linguistic units**: morphemes, words, phrases, sentences, texts, genres;
- **Cultural artifacts**: myths, rituals, technologies, institutions;
- **Cognitive states**: beliefs, intentions, plans, mental models;
- **Scientific constructs**: hypotheses, theories, paradigms, research programmes.

Example 1.5 (Shakespearean Composition)**.** Consider the ideatic space of Elizabethan theatrical conventions. Let a = "revenge tragedy" and b = "romantic comedy." The composition $a \circ b$ might represent *The Merchant of Venice*, which fuses the revenge plot (Shylock's bond) with the romantic comedy (Portia's caskets). The resonance $\mathrm{rs}(a \circ b, c)$ for various test ideas c measures how this hybrid form interacts with other cultural elements—say, c = "anti-Semitic trope" or c = "female agency."

Interpretation 1.6 (Barthes: From Work to Text)*.* Roland Barthes's essay "From Work to Text" (1971) draws a crucial distinction between the *work*—a finished, bounded artifact occupying a shelf—and the *text*—an open, processual field of signification that "is experienced only in an activity of production." The ideatic space formalizes this distinction with striking precision. The carrier set $\mathcal{I}$ does not consist of works (static objects) but of *ideas*—dynamic entities defined by their compositional and resonance behavior. An idea is not a thing but a *node of activity*: it composes with other ideas ($\circ$) and resonates with them (rs).

Barthes insists that the Text is "held in language" and "exists in the movement of a discourse." Our composition operation captures exactly this movement: meaning is generated not by contemplating an idea in isolation but by *composing* it with other ideas. The emergence function $\kappa(a, b, c)$—to be defined formally below—is the mathematical correlate of what Barthes calls the "stereographic plurality" of the Text: the irreducible surplus of meaning that arises when textual elements are combined. In Barthes's terms, the Text is always *more* than the sum of its parts, and this "more" is precisely what our emergence function measures.

The insight that IDT provides beyond Barthes is *quantification*. Barthes can assert that the Text is plural, but he cannot say *how much* plurality a given textual combination generates. The emergence bound (Axiom E3) gives an upper limit on this plurality, governed by the weights of the combined elements—a constraint that Barthes, working within a purely qualitative framework, could not articulate.

Interpretation 1.7 (Peirce's Semiosis and the Composition Operation)*.* Charles Sanders Peirce's triadic model of the sign—sign, object, interpretant—is fundamentally *processual*: each interpretant becomes a sign for a further interpretant, generating what Peirce calls "unlimited semiosis" (CP 2.303). The composition operation $\circ$ in the ideatic space formalizes the *mechanism* of semiosis. When we write $a \circ b$, we model the semiotic act in which idea a (functioning as a sign-context) encounters idea b (functioning as a new sign or interpretant), producing a composite idea that is irreducible to either component.

Peirce distinguishes three types of signs—icons, indices, and symbols —based on their relation to their objects. In the IDT framework, these distinctions correspond to different resonance patterns. An iconic sign a has $\mathrm{rs}(a, b) > 0$ for objects b it resembles (resonance through similarity). An indexical sign has $\mathrm{rs}(a, b) > 0$ for objects b to which it is causally connected (resonance through contiguity). A symbolic sign has $\mathrm{rs}(a, b) > 0$ for objects b linked by convention (resonance through cultural composition). The framework does not *assume* these distinctions but allows them to *emerge* from the resonance data.

What IDT adds to Peirce is the concept of *emergence*. Peirce recognized that semiosis is generative—new meaning arises in the passage from sign to interpretant—but he lacked the mathematical tools to quantify this generativity. The emergence function $\kappa(a, b, c)$ provides exactly such a tool: it measures how much new resonance is created when sign a and interpretant b are composed, as probed by a third idea c.

1.3 The Algebraic Axioms: Monoid Structure

Axiom 1.8 (A1: Associativity of Composition)**.** For all $a, b, c \in \mathcal{I}$:

$$(a \circ b) \circ c \;=\; a \circ (b \circ c).$$

Interpretation 1.9. Associativity is not merely a technical convenience; it encodes a deep claim about the nature of ideational synthesis. It asserts that while the *order* of ideas matters (composition is not assumed commutative), the *grouping* does not. When Homer composes the wrath of Achilles (a) with the funeral of Patroclus (b) and the supplication of Priam (c), the result is the same whether we first compose a with b and then add c, or first compose b with c and then prepend a. The narrative arc of the *Iliad* does not depend on which suturing we perform first.

This is a non-trivial claim. There exist composition operations in semiotics—for instance, Derrida's notion of *différance*—where the grouping matters precisely because meaning is always deferred. Our axiom (A1) deliberately excludes such radically unstable semiotic systems. The ideatic space is, in this sense, a *classical* semiotic space: meaning can be composed without infinite regress.

Interpretation 1.10 (Bakhtin's Dialogism and Associativity). Mikhail Bakhtin's dialogism, developed across *Problems of Dostoevsky's Poetics* (1929/1963) and *The Dialogic Imagination* (1975/1981), holds that meaning is never the product of a single consciousness but always arises in the *encounter* between voices. "The word," Bakhtin writes, "is a two-sided act. It is determined equally by whose word it is and for whom it is meant" (*Marxism and the Philosophy of Language*, 1929).

The composition operation $a \circ b$ formalizes the dialogic encounter: the meaning that arises when voice a addresses or encounters voice b. But associativity—$(a \circ b) \circ c = a \circ (b \circ c)$ —imposes a constraint that Bakhtin himself might have resisted. In a genuinely dialogic exchange, does the grouping of voices matter? Our axiom says no: a three-voice dialogue among a, b, and c has the same outcome whether a first addresses b and then their combined voice addresses c, or b first addresses c and then a addresses their combined voice.

This constraint is not a denial of dialogism but a *structural condition* on it. It says that while the *order* of voices matters (composition is not commutative), the *bracketing* of dialogue does not. This is a strong claim about the coherence of multi-voiced discourse: polyphony must be path-independent in its associative structure, even as it remains order-dependent. Bakhtin's polyphonic novel satisfies this condition when the narrative can be decomposed into dialogic encounters in multiple compatible ways—which is precisely what makes Dostoevsky's novels coherent despite their radical multi-voicedness.

Axiom 1.11 (A2: Left Identity)**.** For all $a \in \mathcal{I}$:

$$\varepsilon \circ a \;=\; a.$$

Axiom 1.12 (A3: Right Identity)**.** For all $a \in \mathcal{I}$:

$$a \circ \varepsilon \;=\; a.$$

Interpretation 1.13. The void idea ε is the ideational analogue of silence, of the empty page, of the tabula rasa. It is not "nothing" in the metaphysical sense but rather the *neutral element* of ideation: composing any idea with ε leaves it unchanged. In Wittgenstein's terms, ε is the logical space prior to any proposition; in Buddhist philosophy, it corresponds to *śūnyatā*, the emptiness that is not absence but potentiality.

The distinction between left and right identity (stated separately for clarity, though logically they combine to assert two-sided identity) will become significant when we study resonance: ε resonates with nothing, not even itself.

Remark 1.14 (The Void in Structuralist Perspective). From a structuralist perspective, the void ε plays the role of the *zero sign*—what Jakobson calls the "zero phoneme" and what Barthes, following Flaubert, calls the "degree zero" of writing. In *Writing Degree Zero* (*Le degré zéro de l'écriture*, 1953), Barthes describes a mode of writing that aspires to neutrality: "a transparent form of speech ... in which writing ... is reduced to a sort of negative mood in which the social or mythical characters of a language are abolished" (p. 76). The void idea formalizes this aspiration: ε is the idea with no social character, no mythical resonance, no semantic weight. It is the "degree zero" of ideation from which all other ideas depart.

The algebraic role of ε as an identity element means that degree zero is not *outside* the system of ideas but *within* it—the neutral point around which all ideational structure is organized. This is consistent with Barthes's insight that writing degree zero is itself a literary style, not an escape from style. The void is an idea, just as zero is a number.

Together, axioms (A1)–(A3) assert that $(\mathcal{I}, \circ, \varepsilon)$ is a **monoid**. This is the most fundamental algebraic structure with an identity: weaker than a group (no inverses), stronger than a semigroup (identity exists).

Proposition 1.15 (Uniqueness of Void)**.** *The void element is unique: if $e \in \mathcal{I}$ satisfies $e \circ a = a$ for all a, or $a \circ e = a$ for all a, then $e = \varepsilon$.*

Proof. Suppose $e \circ a = a$ for all a. Setting $a = \varepsilon$, we get $e \circ \varepsilon = \varepsilon$. But by (A3), $e \circ \varepsilon = e$. Hence $e = \varepsilon$. The right-identity case is symmetric, using (A2). □

Remark 1.16. The monoid structure deliberately omits inverses. In the ideatic space, composition is typically *irreversible*: once the idea of heliocentrism has been composed with the astronomical data of Tycho Brahe, one cannot "un-compose" to recover the pre-Copernican worldview. This irreversibility will be given quantitative expression by the monotonicity of self-resonance (Axiom E4), which we interpret as a second law of ideational thermodynamics.

1.4 The Resonance Axioms

Axiom 1.17 (R1: Right Annihilation)**.** For all $a \in \mathcal{I}$:

$$\mathrm{rs}(a, \varepsilon) \;=\; 0.$$

Axiom 1.18 (R2: Left Annihilation)**.** For all $a \in \mathcal{I}$:

$$\mathrm{rs}(\varepsilon, a) \;=\; 0.$$

Interpretation 1.19. The void idea resonates with nothing. This is the formal expression of a deep intuition: silence has no opinion. The empty page does not agree or disagree with any text. The vacuum state of a quantum field theory has zero overlap with any particle state. In the semiotic register: the absence of a sign is not itself a sign (contra certain poststructuralist claims—our framework takes a more conservative stance on this point).

Remark 1.20 (Resonance Annihilation and Nothingness). The annihilation axioms (R1)–(R2) take a definite philosophical position on the relationship between nothingness and meaning. For Heidegger, in "What Is Metaphysics?" (1929), *das Nichts* (the nothing) is not the mere absence of beings but

a positive force that "nihilates" (*nichtet*)—it is the condition for the disclosure of beings as such. For Sartre, nothingness is the "worm at the heart of being"—the negating activity of consciousness that introduces lack into the plenitude of *en-soi*.

Our axioms reject these Heideggerian and Sartrean positions, at least at the level of resonance. The void does not nihilate; it does not negate; it does not introduce lack. It simply does not resonate. This is closer to the Wittgensteinian position: whereof one cannot speak (the void), thereof one must be silent (zero resonance). Whether this is a limitation of the formalism or a genuine philosophical insight is a question we leave to the reader—but we note that the non-annihilation theorem (Theorem 1.33) vindicates our conservative stance: even without granting the void any positive nihilating power, we can prove that non-void ideas are indelible.

Axiom 1.21 (E1: Non-negativity of Self-Resonance)**.** For all $a \in \mathcal{I}$:

$$\mathrm{rs}(a, a) \;\geq\; 0.$$

Axiom 1.22 (E2: Non-degeneracy)**.** For all $a \in \mathcal{I}$:

$$\mathrm{rs}(a, a) = 0 \implies a = \varepsilon.$$

Interpretation 1.23. Together, (E1) and (E2) assert that every non-void idea has *positive self-resonance*. The quantity $\mathrm{rs}(a, a)$, which we shall call the *weight* of a, measures the idea's internal coherence or self-consistency. An idea that does not resonate with itself—that has zero weight—is indistinguishable from the void.

This axiom has a clear analogue in inner product spaces, where $\langle v, v\rangle = 0$ implies $v = 0$. But we emphasize that rs is *not* assumed to be an inner product: it need not be symmetric (indeed, asymmetry will be central to our theory), and it need not be bilinear. We assume only the minimal properties needed for the weight function to be well-defined.

Axiom 1.24 (E3: Emergence Boundedness)**.** For all $a, b, c \in \mathcal{I}$:

$$\big(\mathrm{rs}(a \circ b, c) - \mathrm{rs}(a, c) - \mathrm{rs}(b, c)\big)^2 \;\leq\; \mathrm{rs}(a \circ b,\; a \circ b) \cdot \mathrm{rs}(c, c).$$

This axiom bounds the *emergence*—the non-additive surplus of resonance created by composition—in terms of the weights of the composed idea and the probe. We define emergence formally:

Definition 1.25 (Emergence)**.** The **emergence** of the composition of a and b, probed by c, is

$$\kappa(a, b, c) \;:=\; \mathrm{rs}(a \circ b, c) - \mathrm{rs}(a, c) - \mathrm{rs}(b, c).$$

Axiom (E3) then reads: $\kappa(a, b, c)^2 \leq \mathrm{rs}(a \circ b, a \circ b) \cdot \mathrm{rs}(c, c)$, a Cauchy–Schwarz-type inequality.

Example 1.26 (Political Emergence)**.** Let $a =$ "liberty" and $b =$ "equality" in the ideatic space of Enlightenment political thought. The composition $a \circ b$ might represent the French Revolutionary slogan. Probing with $c =$ "fraternity," the emergence $\kappa(a, b, c)$ measures how much more (or less) the *combined* idea of "liberty-equality" resonates with fraternity than the sum of their individual resonances. The positive emergence here reflects the well-known political insight that the conjunction of liberty and equality *creates* a demand for fraternity that neither ideal alone would generate.

Interpretation 1.27 (Emergence and Political Thought). The political emergence example above illustrates a phenomenon that extends far beyond the French Revolution. In contemporary political theory, the concept of *intersectionality* (Crenshaw, 1989) holds that the combination of multiple

identity categories (race, gender, class) produces forms of oppression that are irreducible to the sum of their individual effects. In IDT terms, intersectionality is the claim that $\kappa(\text{race}, \text{gender}, c) \neq 0$ for probes c related to discrimination and social power: the experience of a Black woman is not the "sum" of the experience of being Black and the experience of being a woman, but something emergent—something that cannot be understood by studying race and gender in isolation.

The emergence boundedness axiom (E3) adds a constraint that intersectionality theory has not articulated: the emergent effects of combining identity categories are bounded by the "weights" of the combined identity and the probe. This bound is not a political claim but a structural one: it says that while intersectional emergence is real and irreducible, it is not infinite. The degree to which the combination of race and gender produces novel forms of oppression is constrained by the internal coherence of the combined category and the social weight of the discriminatory context.

Axiom 1.28 (E4: Compositional Enrichment)**.** For all $a, b \in \mathcal{I}$:

$$\text{rs}(a \circ b,\ a \circ b) \ \geq\ \text{rs}(a, a).$$

Interpretation 1.29. Composition never decreases weight. This is the ideational analogue of the second law of thermodynamics, reinterpreted positively: ideas grow in internal coherence (or at least do not lose it) through composition. The composition of Darwin's natural selection with Mendel's genetics produces the Modern Synthesis, whose self-resonance (internal consistency, explanatory power, conceptual richness) exceeds that of either component.

Note that (E4) is stated as $\text{rs}(a \circ b, a \circ b) \geq \text{rs}(a, a)$, privileging the *left* factor. By the cocycle condition (Chapter 2), a symmetric enrichment $\text{rs}(a \circ b, a \circ b) \geq \text{rs}(b, b)$ does *not* follow in general. Composition enriches, but not symmetrically—reflecting the intuition that in any synthesis, the "first" idea typically sets the frame.

1.5 The Lean 4 Formalization

The eight axioms are formalized as a Lean 4 type class. This serves both as a machine-checkable reference and as a demonstration that the axiom system is internally consistent (any type inhabiting the class provides a model).

Listing 1.1: The `IdeaticSpace` type class

```
class IdeaticSpace (I : Type*) where
  compose : I     I     I
  void : I
  rs : I     I
  -- Monoid axioms
  compose_assoc :     a b c : I,
    compose (compose a b) c = compose a (compose b c)
  void_left :     a : I, compose void a = a
  void_right :     a : I, compose a void = a
  -- Resonance vanishing
  rs_void_right :     a : I, rs a void = 0
  rs_void_left :     a : I, rs void a = 0
  -- Self-resonance
  rs_self_nonneg :     a : I, rs a a     0
  rs_nondegen :     a : I, rs a a = 0     a = void
  -- Emergence boundedness (Cauchy Schwarz type)
  emergence_bounded :     a b c : I,
    (rs (compose a b) c - rs a c - rs b c)^2
```

```
        rs (compose a b) (compose a b) * rs c c
  -- Compositional enrichment
  compose_enriches :     a b : I,
    rs (compose a b) (compose a b)     rs a a
```

1.6 Minimality and Independence of the Axioms

A natural question arises: are the eight axioms independent? Could any be derived from the others?

Theorem 1.30 (Consistency)**.** *The axiom system (A1)–(A3), (R1)–(R2), (E1)–(E4) is consistent.*

Proof. We exhibit a model. Let $\mathcal{I} = \mathbb{N}$, with $\circ = +$ (addition), $\varepsilon = 0$, and $\mathrm{rs}(m, n) = m \cdot n$ (multiplication). Then:

- (A1): $(m + n) + k = m + (n + k)$ ✓
- (A2): $0 + n = n$ ✓
- (A3): $n + 0 = n$ ✓
- (R1): $m \cdot 0 = 0$ ✓
- (R2): $0 \cdot n = 0$ ✓
- (E1): $n \cdot n = n^2 \geq 0$ ✓
- (E2): $n^2 = 0 \implies n = 0$ ✓
- (E3): $((m + n) \cdot k - m \cdot k - n \cdot k)^2 = 0 \leq (m + n)^2 \cdot k^2$ ✓
- (E4): $(m + n)^2 \geq m^2$ since $(m + n)^2 - m^2 = n^2 + 2mn \geq 0$ for $m, n \in \mathbb{N}$ ✓

□

Listing 1.2: The natural number model in Lean 4

```
instance : IdeaticSpace     where
  compose := (   +   )
  void := 0
  rs m n := (m :    ) * (n :    )
  compose_assoc := by intros; ring
  void_left := Nat.zero_add
  void_right := Nat.add_zero
  rs_void_right := by intros; simp; ring
  rs_void_left := by intros; simp; ring
  rs_self_nonneg := by intros; positivity
  rs_nondegen := by intro a ha; simp at ha; exact_mod_cast ha
  emergence_bounded := by intros; ring_nf; positivity
  compose_enriches := by intros; simp; nlinarith
```

Remark 1.31 (The Natural Number Model is Degenerate)*.* In the $\mathbb{N}$ model, emergence vanishes identically: $\kappa(m, n, k) = (m + n)k - mk - nk = 0$. This model demonstrates consistency but not the richness of the axiom system. It is the ideatic analogue of flat Euclidean space: consistent with the axioms of Riemannian geometry, but devoid of curvature. The interesting models are those with non-trivial emergence, just as the interesting geometries are those with non-zero curvature.

Proposition 1.32 (Independence of E4)**.** *Axiom (E4) is independent of (A1)–(A3), (R1)–(R2), (E1)–(E3).*

Proof sketch. Consider $\mathcal{I} = \{0, 1\}$ with $\circ$ being max (so $\varepsilon = 0$), and $\mathrm{rs}(1, 1) = 1$, $\mathrm{rs}(0, 0) = \mathrm{rs}(0, 1) = \mathrm{rs}(1, 0) = 0$. Then (A1)–(A3) hold (max is an associative operation with identity 0), (R1)–(R2) hold trivially, (E1)–(E2) hold since only $\mathrm{rs}(1, 1) > 0$, and (E3) can be verified case by case. However, $\mathrm{rs}(1 \circ 1, 1 \circ 1) = \mathrm{rs}(1, 1) = 1$ while $\mathrm{rs}(1, 1) = 1$, so (E4) holds here with equality. To violate (E4), modify rs so that $\mathrm{rs}(\max(a, b), \max(a, b)) < \mathrm{rs}(a, a)$ for some choice. □

1.7 First Consequences

Theorem 1.33 (Non-Annihilation)**.** *If $a \neq \varepsilon$, then $a \circ b \neq \varepsilon$ for all $b \in \mathcal{I}$.*

Proof. By (E4), $\mathrm{rs}(a \circ b, a \circ b) \geq \mathrm{rs}(a, a)$. Since $a \neq \varepsilon$, axiom (E2) gives $\mathrm{rs}(a, a) > 0$ (using E1 for the non-strict inequality and E2 for the contrapositive). Hence $\mathrm{rs}(a \circ b, a \circ b) > 0$, which by (E2) again implies $a \circ b \neq \varepsilon$. □

Interpretation 1.34. Non-annihilation is a striking structural property: a non-trivial idea cannot be "annihilated" by composition with any other idea. Once Hamlet's soliloquy exists in the ideatic space, no subsequent composition—no parody, no deconstruction, no forgetting—can reduce it to ε. Ideas, in our framework, are indelible. This resonates with Gadamer's hermeneutic insight that understanding is always already shaped by the history of effects (*Wirkungsgeschichte*): the past cannot be un-thought.

Interpretation 1.35 (Husserl's Intentionality and Resonance). Edmund Husserl's phenomenology is built on the thesis of *intentionality*: every act of consciousness is consciousness *of* something. In the *Logical Investigations* (1900–01), Husserl distinguishes the *noesis* (the act of intending) from the *noema* (the intended object as it is intended). Resonance $\mathrm{rs}(a, b)$ can be read as a formalization of intentional directedness: $\mathrm{rs}(a, b)$ measures the degree to which idea a "intends" or "is directed toward" idea b.

Crucially, Husserl's intentionality is *non-symmetric*: the act of consciousness is directed *from* the noesis *toward* the noema, not the reverse. Our resonance function captures this asymmetry: $\mathrm{rs}(a, b) \neq \mathrm{rs}(b, a)$ in general, reflecting the phenomenological fact that the direction of intentional awareness matters. When I contemplate justice (a) with respect to mercy (b), my act of consciousness differs from contemplating mercy with respect to justice. The non-annihilation theorem (Theorem 1.33) then formalizes a Husserlian insight: a genuine act of consciousness cannot be annihilated by further intentional acts. Once an idea has been intended, it persists indelibly in the intentional field.

What IDT adds to Husserl is the *compositional* dimension. Husserl analyzes individual acts of consciousness in relative isolation; our framework models the *algebraic structure* of how intentional acts combine. The emergence function reveals that compositional intentionality is not merely additive—a thesis that Husserl's notion of "founding" (*Fundierung*) anticipates but does not formalize.

Interpretation 1.36 (Frege's Sense and Reference). Gottlob Frege's 1892 essay "Über Sinn und Bedeutung" ("On Sense and Reference") draws the famous distinction between *Sinn* (the mode of presentation, the cognitive significance) and *Bedeutung* (the referent, the thing designated). The Morning Star and the Evening Star have the same *Bedeutung* (Venus) but different *Sinn* (different modes of presentation). Frege's puzzle is why "the Morning Star is the Evening Star" is informative while "the Morning Star is the Morning Star" is trivial, despite both having the same truth-value.

The resonance function rs captures Fregean *Sinn* with remarkable directness. Two ideas a and b with the same referent but different senses will have different resonance profiles: $\mathrm{rs}(a, c) \neq \mathrm{rs}(b, c)$ for at least some probe c, even though a and b "point to the same thing." The informativeness of "a is b" is measured precisely by the discrepancy in their resonance profiles. In Chapter 6, we formalize this through same-idea equivalence ($\sim_{\mathrm{idea}}$), which captures the level at which a and b share a referent (or at least a conceptual role), while their resonance profiles capture the level at which their senses differ.

Frege's principle of compositionality—the meaning of a complex expression is determined by the meanings of its parts and their mode of combination—corresponds to the claim that resonance is (approximately) determined by the composition structure. But our emergence function reveals the precise degree to which compositionality *fails*: $\kappa(a, b, c)$ is the measure of non-compositionality, the amount of meaning that exceeds what Frege's principle would predict.

Interpretation 1.37 (Shklovsky and the Void as Zero Degree). Viktor Shklovsky, in his seminal essay "Art as Device" (*Iskusstvo kak priëm*, 1917), argues that the purpose of art is *defamiliarization* (*ostranenie*): to make the familiar strange, to force the perceiver out of the automatized perception of everyday life. For Shklovsky, ordinary language is "algebraized"—worn smooth by habit—and literary language restores perception by "making forms difficult, by increasing the difficulty and length of perception."

The void idea ε can be understood as the *zero degree* of literary language in Shklovsky's sense: the state of complete automatization, of pure habit, where no perception occurs and no resonance is generated. The axioms $\mathrm{rs}(a, \varepsilon) = 0$ and $\mathrm{rs}(\varepsilon, a) = 0$ assert that the void neither perceives nor is perceived—it is the terminal state of automatization, the "algebraized" language that Shklovsky diagnoses as the enemy of art.

Every non-void idea a, by contrast, has positive self-resonance $\mathrm{rs}(a, a) > 0$: it generates at least *some* perceptual friction, some resistance to complete automatization. The weight function $w(a) = \mathrm{rs}(a, a)$ thus measures the degree to which an idea *defamiliarizes*—how much perceptual and cognitive work it demands. Shklovsky's claim that art "makes the stone stony" (restores the materiality of perception) corresponds to the assertion that genuine artworks have high weight: they resonate powerfully with themselves, generating the internal coherence and perceptual density that defamiliarization requires.

Interpretation 1.38 (Eco's Encyclopedia Model). Umberto Eco, in *A Theory of Semiotics* (1976) and *Semiotics and the Philosophy of Language* (1984), distinguishes between a *dictionary* model of meaning (where each sign has a fixed definition) and an *encyclopedia* model (where the meaning of a sign is the totality of its connections to all other signs). Eco argues forcefully for the encyclopedia model: "the universe of semiosis ... is structured according to a network of interpretants" (*Semiotics and the Philosophy of Language*, p. 68).

The ideatic space is an exact mathematical realization of Eco's encyclopedia model. The "meaning" of idea a is not a fixed definition but the complete resonance profile $\{\mathrm{rs}(a, b) : b \in \mathcal{I}\}$—the totality of its interactions with every other idea in the space. The composition operation encodes the "inferential walks" that Eco describes as the mechanism by which meaning propagates through the encyclopedia: from idea a, one can "walk" to the composite $a \circ b$, generating new resonance patterns that were not visible from a alone.

Eco's encyclopedia model, unlike the dictionary model, is *open*: no finite description can exhaust the meaning of a sign, because the encyclopedia is in principle unlimited. Our framework captures this openness through the fact that the resonance profile of an idea is a function on the entire space $\mathcal{I}$, which may be infinite. The emergence function adds a dimension that Eco recognizes but cannot formalize: the encyclopedia is not merely a network of static connections but a generative

structure in which new connections are *created* by the act of traversal.

Corollary 1.39. *$a \circ b = \varepsilon$ if and only if $a = \varepsilon$.*

Proof. If $a = \varepsilon$ then $a \circ b = \varepsilon \circ b = b$ by (A2). Wait—this gives b, not ε, unless $b = \varepsilon$. Let us correct: $a \circ b = \varepsilon$ implies $a = \varepsilon$ by Theorem 1.33 (contrapositive). Conversely, if $a = \varepsilon$ and $a \circ b = \varepsilon$, then $b = \varepsilon \circ b = \varepsilon$, so both must be void. More precisely: $a \circ b = \varepsilon \iff a = \varepsilon$ and $b = \varepsilon$. □

Theorem 1.40 (Weight Monotonicity)**.** *Define the* weight *of an idea as $w(a) := \mathrm{rs}(a, a)$. Then:*

1. *$w(a) \geq 0$ for all a;*
2. *$w(a) = 0 \iff a = \varepsilon$;*
3. *$w(a \circ b) \geq w(a)$.*

Proof. Items (1) and (2) restate (E1) and (E2). Item (3) is axiom (E4). □

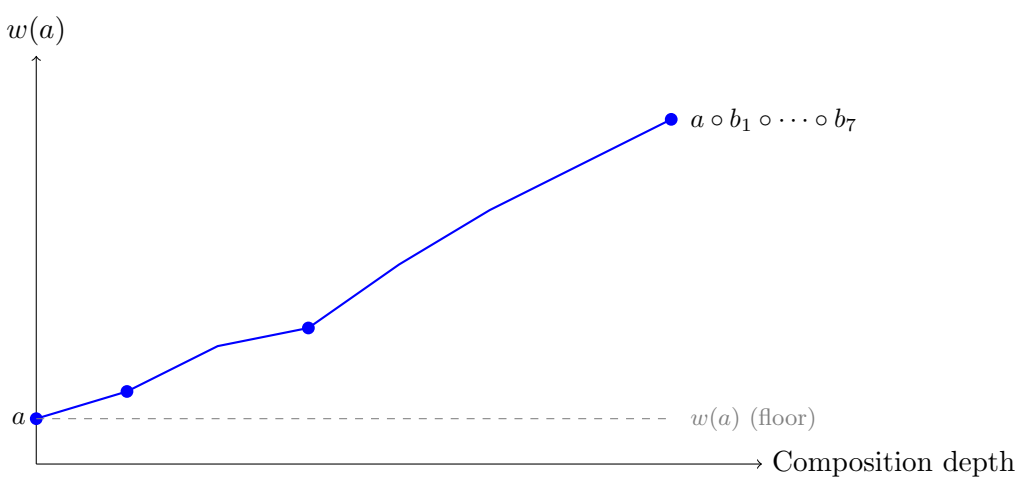


Figure 1.1: Weight is monotonically non-decreasing under iterated composition. The initial weight $w(a)$ serves as an irreducible floor.

1.8 Historical and Philosophical Context

The monoid-with-resonance structure of the ideatic space sits at the intersection of several intellectual traditions:

1. **Algebraic semiotics** (Goguen, 1999): Joseph Goguen proposed using algebraic structures to model sign systems. Our monoid structure extends this by adding the resonance function, which captures the *pragmatic* dimension that purely algebraic semiotics lacks.
2. **Conceptual blending** (Fauconnier & Turner, 2002): The composition operation $\circ$ can be viewed as a formalization of conceptual blending, with emergence κ measuring the "emergent structure" that Fauconnier and Turner identify as the hallmark of creative thought.

3. **Topos theory** (Lawvere, 1991): Lawvere's categorical approach to logic suggests viewing $\mathcal{I}$ as the object of an internal monoid in a suitable topos. The resonance function would then be an arrow into the real-number object. This categorical perspective will be developed in later chapters.

4. **Information geometry** (Amari, 2016): The emergence boundedness axiom (E3) is reminiscent of the Cramér–Rao bound in information geometry. The weight function w plays a role analogous to Fisher information.

Example 1.41 (Scientific Paradigm Shifts)**.** Kuhn's *Structure of Scientific Revolutions* describes paradigm shifts as discontinuous changes in the conceptual framework of a scientific community. In our framework, a paradigm shift from a (old paradigm) to $a \circ b$ (new paradigm, incorporating anomalous data b) is characterized by large emergence: $\kappa(a, b, c) \gg 0$ for typical test ideas c in the discipline. The Copernican revolution composes geocentrism (a) with the accumulated astronomical anomalies (b) to produce heliocentrism ($a \circ b$), whose resonance with mechanical physics (c) vastly exceeds what either component alone would predict.

1.9 Summary and Prospectus

Interpretation 1.42 (The Ideatic Space as Philosophical Achievement)*.* Before summarizing, let us step back and assess the philosophical significance of the ideatic space as a formal structure. The eight axioms (A1)–(A3), (R1)–(R2), (E1)–(E4) encode a specific philosophical position on the nature of ideas: ideas are compositional entities whose identity is constituted by their relational behavior (resonance) rather than by any intrinsic essence. This position draws from, but goes beyond, several philosophical traditions.

From *structuralism* (Saussure, Lévi-Strauss), IDT inherits the principle that meaning is differential: an idea is defined by its differences from other ideas, formalized by the resonance function rs. From *pragmatism* (Peirce, James, Dewey), IDT inherits the principle that meaning is use: an idea is what it does when composed with other ideas, formalized by the emergence function κ. From *phenomenology* (Husserl, Heidegger), IDT inherits the principle that meaning is intentional: resonance is directed, asymmetric, perspectival.

What IDT adds to all these traditions is *algebraic precision.* The cocycle condition is not merely a metaphor for cohomological structure; it *is* cohomological structure. The weight function is not merely analogous to thermodynamic entropy; it *satisfies* a monotonicity law. The emergence bound is not merely reminiscent of the Cauchy–Schwarz inequality; it *is* a Cauchy–Schwarz inequality. The philosophical content of IDT lies not in its metaphors but in its theorems.

Remark 1.43 (The Axiom System in Historical Perspective)*.* The axiom system (A1)–(A3), (R1)–(R2), (E1)–(E4) belongs to a distinguished lineage of axiomatizations in mathematics and the human sciences. Hilbert's axiomatization of geometry (1899) showed that geometric intuition could be captured by purely formal conditions on undefined primitives. Kolmogorov's axiomatization of probability (1933) showed that the informal notion of "chance" could be grounded in measure theory. Shannon's axiomatization of information (1948) showed that the intuitive concept of "uncertainty" could be defined by three simple conditions on a function H.

The IDT axiom system aspires to do for "meaning" what Hilbert did for space, Kolmogorov for chance, and Shannon for information: to capture the essential structure of an intuitive concept in a small set of formal conditions. The composition operation $\circ$ is the analog of geometric incidence; the resonance function rs is the analog of the probability measure or the entropy function; and

the emergence κ, derived from the axioms, is the analog of conditional probability or mutual information.

Like all successful axiomatizations, the IDT axioms are chosen to be *weak enough* to admit many models (the natural number model, the matrix model, and—we conjecture—models based on natural language corpora, neural network representations, and cultural databases) while *strong enough* to yield non-trivial theorems (non-annihilation, weight monotonicity, the cocycle condition, the emergence bound). The fertility of the axiom system—its ability to generate rich mathematical structure from minimal assumptions—is the ultimate test of its philosophical adequacy.

We have introduced the ideatic space $(\mathcal{I}, \circ, \varepsilon, \mathrm{rs})$ with its eight axioms: three algebraic (monoid), two annihilation, and three emergence conditions. From these, we have already derived non-annihilation and weight monotonicity. In the chapters that follow, we will extract far more:

- Chapter 2: The cocycle condition, derived purely from associativity, connecting emergence to group cohomology.
- Chapter 3: The theory of self-resonance and weight, including the irreversibility of composition.
- Chapter 4: The asymmetry of resonance and its implications for power, influence, and directional meaning.
- Chapter 5: The full algebra of emergence, including its spectral decomposition.
- Chapter 6: The semiotic gap between utterance and idea, formalizing Saussure's signifier/signified distinction.
- Chapter 7: The dynamics of transmission and mutation, modeling how ideas change as they propagate.

Chapter 2

The Cocycle Condition

"The whole is more than the sum of its parts."
— Aristotle, *Metaphysics*, VIII.6, 1045a

2.1 Emergence as a Trilinear Deviation

Recall from Definition 19.1 that emergence is defined as

$$\kappa(a, b, c) \;:=\; \mathrm{rs}(a \circ b, c) - \mathrm{rs}(a, c) - \mathrm{rs}(b, c).$$

This quantity measures the *deviation from additivity* of the resonance function under composition. When $\kappa(a, b, c) = 0$, resonance is perfectly additive—composition introduces no new information. When $\kappa(a, b, c) > 0$, we have *superadditivity*: the whole resonates with c more than the sum of its parts. When $\kappa(a, b, c) < 0$, we have *subadditivity*: something is lost in the combination.

Example 2.1 (Homeric Superadditivity)**.** In the *Odyssey*, let $a =$ "nostos (homecoming narrative)" and $b =$ "Cyclops episode." The composition $a \circ b$ creates the specific narrative tension of Odysseus's return being endangered. When probed by $c =$ "heroic identity," the emergence $\kappa(a, b, c) > 0$: the *combination* of homecoming-desire with Cyclops-peril resonates with heroic identity far more than either component alone, because it is precisely in the threat to homecoming that heroism is forged.

2.2 Derivation of the Cocycle Condition

The central result of this chapter is that emergence satisfies a *cocycle condition*, and this follows from nothing more than the associativity of composition.

Theorem 2.2 (Cocycle Condition)**.** *For all $a, b, c, d \in \mathcal{I}$:*

$$\kappa(a, b, d) + \kappa(a \circ b, c, d) \;=\; \kappa(b, c, d) + \kappa(a, b \circ c, d).$$

Proof. We expand both sides using the definition $\kappa(x, y, z) = \mathrm{rs}(x \circ y, z) - \mathrm{rs}(x, z) - \mathrm{rs}(y, z)$.

Left-hand side:

$$\begin{aligned}
&\kappa(a, b, d) + \kappa(a \circ b, c, d) \\
&= \big[\mathrm{rs}(a \circ b, d) - \mathrm{rs}(a, d) - \mathrm{rs}(b, d)\big] + \big[\mathrm{rs}((a \circ b) \circ c, d) - \mathrm{rs}(a \circ b, d) - \mathrm{rs}(c, d)\big] \\
&= \mathrm{rs}((a \circ b) \circ c, d) - \mathrm{rs}(a, d) - \mathrm{rs}(b, d) - \mathrm{rs}(c, d).
\end{aligned}$$

Right-hand side:

$$\begin{aligned}
&\kappa(b,c,d) + \kappa(a, b\circ c, d)\\
&= \big[\mathrm{rs}(b\circ c,d) - \mathrm{rs}(b,d) - \mathrm{rs}(c,d)\big] + \big[\mathrm{rs}(a\circ(b\circ c),d) - \mathrm{rs}(a,d) - \mathrm{rs}(b\circ c,d)\big]\\
&= \mathrm{rs}(a\circ(b\circ c),d) - \mathrm{rs}(a,d) - \mathrm{rs}(b,d) - \mathrm{rs}(c,d).
\end{aligned}$$

By axiom (A1), $(a\circ b)\circ c = a\circ(b\circ c)$, so the two sides are equal. □

Interpretation 2.3. The cocycle condition is a *consistency law for emergence*. It says that there are two equivalent ways to decompose the emergence of a triple composition $a\circ b\circ c$ probed by d:

1. First compute the emergence of composing a with b, then the emergence of composing the result with c; or
2. First compute the emergence of composing b with c, then the emergence of composing a with the result.

Both paths yield the same total emergence. This is analogous to the path-independence of line integrals in conservative fields, or the consistency conditions in Čech cohomology.

Interpretation 2.4 (Kristeva's Intertextuality). Julia Kristeva, in her landmark essay "Word, Dialogue, and Novel" (1966), introduces the concept of *intertextuality*: every text is a "mosaic of quotations," an absorption and transformation of other texts. Kristeva writes: "any text is constructed as a mosaic of quotations; any text is the absorption and transformation of another" (*Desire in Language*, 1980, p. 66). The cocycle condition (Theorem 19.7) provides a precise algebraic formulation of how intertextual meaning "distributes" across combinations.

Consider three texts a, b, c and a thematic probe d. The emergence $\kappa(a,b,d)$ measures the intertextual surplus generated when text a absorbs text b—the meaning that arises from their conjunction but is present in neither alone. The cocycle condition asserts that the total intertextual surplus of the triple combination $a\circ b\circ c$ can be decomposed in two consistent ways: either by first computing the intertextuality of a and b, then that of the result with c; or by first computing the intertextuality of b and c, then that of a with the result. Kristeva's insight that intertextuality is *not* merely a matter of influence or allusion—but a structural property of how texts are woven together—is given rigorous expression by this path-independence.

The formalization reveals something that Kristeva's qualitative theory cannot: intertextuality is subject to a *conservation law*. The total emergent surplus of any triple composition is fixed; only its *distribution* across the two decomposition paths varies. This is a non-trivial constraint on intertextual meaning-making, suggesting that the "mosaic of quotations" is not freely assembled but governed by cohomological constraints.

Interpretation 2.5 (Derrida's *Différance* and the Cocycle). Jacques Derrida's concept of *différance*—the simultaneous deferral and differentiation of meaning—is often taken as a challenge to any formal or algebraic approach to meaning. In *Of Grammatology* (1967), Derrida argues that meaning is never fully present but is always displaced along a chain of signifiers: "the signified always already functions as a signifier" (p. 7). The cocycle condition captures a precise aspect of this displacement.

The emergence $\kappa(a,b,d)$ is the "remainder" when we try to reduce the meaning of the composite $a\circ b$ to the meanings of its parts. In Derridean terms, it is the *supplement*—what must be added to the parts to obtain the whole, but which the parts cannot account for. The cocycle condition states that this supplement, this *différance*, is *consistent*: though meaning is always deferred (the emergence is never zero in interesting cases), the deferral is subject to an algebraic law. Derrida's *différance* is not pure chaos; it has a cohomological structure.

This is both a vindication and a correction of Derrida. The vindication: yes, meaning exceeds its parts, and this excess is irreducible (emergence is not a coboundary in general). The correction: this excess is not infinitely free but constrained by a precise algebraic identity. One might say that *différance* has a *cohomology class*: it is determined up to the "trivial" adjustments of coboundaries. The non-trivial part—the element of $H^2(\mathcal{I}, \mathbb{R})$—is the genuinely deconstructive residue that no formal manipulation can eliminate.

Interpretation 2.6 (Hegel's Dialectic and the Cocycle Remainder). Hegel's dialectical method—thesis, antithesis, synthesis—has long been criticized for its vagueness. What exactly is the "sublation" (*Aufhebung*) that produces the synthesis? How does the synthesis relate to its components? The cocycle condition provides an unexpectedly precise answer.

In Hegelian terms, the emergence $\kappa(a, b, d)$ is the dialectical surplus: the content of the synthesis $a \circ b$ that cannot be predicted from thesis a and antithesis b individually. Hegel writes in the *Science of Logic* (1812–16) that *Aufhebung* involves a triple movement: negation, preservation, and elevation. The cocycle condition shows that this triple movement is *constrained*: the dialectical surplus of any triple synthesis (a, b, c) is the same whether we perform the synthesis in the order $((a, b), c)$ or $(a, (b, c))$.

This constraint illuminates a deep feature of Hegelian dialectic that Hegel himself could not formalize: the dialectic is *path-independent* in its associative structure. The "cunning of reason" (*List der Vernunft*) is not arbitrary but governed by cohomological laws. The cocycle condition is, in this sense, the mathematical content of Hegel's claim that the dialectic is rational rather than merely rhetorical.

Furthermore, the distinction between coboundaries and genuine cocycles (Definition 19.10) corresponds to Hegel's distinction between *apparent* and *genuine* dialectical progress. A coboundary represents a change of perspective that *looks* like dialectical synthesis but is merely a rearrangement of existing content. A genuine cocycle—an element of $H^2(\mathcal{I}, \mathbb{R})$ that is not a coboundary—represents irreducible dialectical novelty.

2.3 Connection to Group Cohomology

The cocycle condition of Theorem 19.7 is not merely analogous to the cocycle condition in group cohomology—it *is* a cocycle condition, in a precise sense.

Definition 2.7 (Cochains and Cocycles)**.** Fix a probe $d \in \mathcal{I}$. Define the **1-cochain** $f_d : \mathcal{I} \to \mathbb{R}$ by $f_d(a) := \mathrm{rs}(a, d)$. Define the **2-cochain** $\kappa_d : \mathcal{I} \times \mathcal{I} \to \mathbb{R}$ by $\kappa_d(a, b) := \kappa(a, b, d)$. Then the cocycle condition asserts:

$$\kappa_d(a, b) + \kappa_d(a \circ b, c) = \kappa_d(b, c) + \kappa_d(a, b \circ c),$$

which is the standard **2-cocycle condition** for the monoid $(\mathcal{I}, \circ)$ acting on $\mathbb{R}$.

Remark 2.8. In the classical theory of group extensions, a 2-cocycle $\sigma : G \times G \to A$ satisfies

$$\sigma(g, h) + \sigma(gh, k) = \sigma(h, k) + \sigma(g, hk)$$

for an abelian group A on which G acts. Our emergence cocycle is exactly this, with $G = (\mathcal{I}, \circ)$ and $A = (\mathbb{R}, +)$. The emergence function κ_d thus defines an element of the second cohomology group $H^2(\mathcal{I}, \mathbb{R})$ of the monoid $\mathcal{I}$ with coefficients in $\mathbb{R}$.

Interpretation 2.9 (Peirce's Unlimited Semiosis). Peirce's doctrine of "unlimited semiosis" holds that every interpretant becomes a new sign requiring a further interpretant, *ad infinitum*. This

infinite chain is often taken as evidence that meaning can never be "pinned down." The cocycle condition reveals a hidden structure within this apparently endless chain.

Consider a chain of interpretants $a_1, a_2, a_3, \ldots$ where each a_{k+1} is an interpretant of a_k. The composition $a_1 \circ a_2 \circ \cdots \circ a_n$ represents the accumulated meaning after n steps of semiosis. The cocycle condition ensures that, no matter how we bracket this chain, the total emergence is the same. Unlimited semiosis is, in this precise sense, *well-defined*: the process of generating interpretants may be infinite, but each finite stage is constrained by a cohomological consistency condition.

Peirce's semiosis is thus not structureless flux but a process governed by algebraic laws. The cocycle class $[\kappa_d] \in H^2(\mathcal{I}, \mathbb{R})$ is an invariant of this process—a "fingerprint" of the semiotic chain that remains constant across all possible bracketings. This is a mathematical vindication of Peirce's belief, expressed in the "Fixation of Belief" (1877), that inquiry converges toward truth: the cohomological invariant is the "truth" toward which the interpretant chain is directed.

Interpretation 2.10 (Hjelmslev's Connotation and the Cohomological Surplus). Louis Hjelmslev, in his *Prolegomena to a Theory of Language* (1943), distinguishes between *denotation* (the primary signification of a sign) and *connotation* (the secondary signification that arises when a sign system itself becomes the expression-plane of a higher-order sign system). The cocycle condition provides a formal framework for understanding how connotative surplus arises.

The emergence $\kappa(a, b, d)$ can be read as the *connotative surplus* of combining expressions a and b: the meaning that belongs not to the denotative level (which would be captured by the additive contribution $\mathrm{rs}(a, d) + \mathrm{rs}(b, d)$) but to the connotative level (the non-additive excess). The cocycle condition then states that connotative surplus is subject to a consistency law: the total connotation of a triple combination is independent of how we decompose it into pairwise connotations.

Hjelmslev's key insight—that connotation is *systematic*, not accidental—is thus given rigorous mathematical expression. The cohomology group $H^2(\mathcal{I}, \mathbb{R})$ classifies all possible connotative systems: each cohomology class corresponds to a distinct way in which connotative surplus can be organized consistently across the ideatic space. The vanishing of H^2 (the "acyclic" case) would mean that all connotation is "trivial"—reducible to a change of denotative baseline. Non-trivial cohomology corresponds to irreducible connotative structure.

Interpretation 2.11 (Brandom's Inferentialism). Robert Brandom's inferentialism, developed in *Making It Explicit* (1994), holds that the content of a concept is constituted by its inferential role—the pattern of inferences it licenses and precludes. A concept's meaning is not a mental image or a correspondence with reality but its position in the "space of reasons."

The cocycle condition formalizes a precise aspect of Brandom's inferentialism: the requirement that inferential roles be *compositionally consistent*. When we compose concept a with concept b, the resulting concept $a \circ b$ has an inferential role (measured by its resonance profile) that includes an emergent component $\kappa(a, b, d)$—new inferences that are licensed by the combination but not by either component alone. The cocycle condition ensures that these emergent inferences are consistent: the total emergent content of a triple composition is independent of the decomposition path.

Brandom's thesis that "we are *sapient* beings—ones who navigate the space of reasons" is, in IDT terms, the claim that cognition traverses the ideatic space by composition, and the cohomological structure of emergence constrains how new reasons can be generated from old ones. The cocycle condition is, in effect, the formal expression of the requirement that the space of reasons be *coherent*.

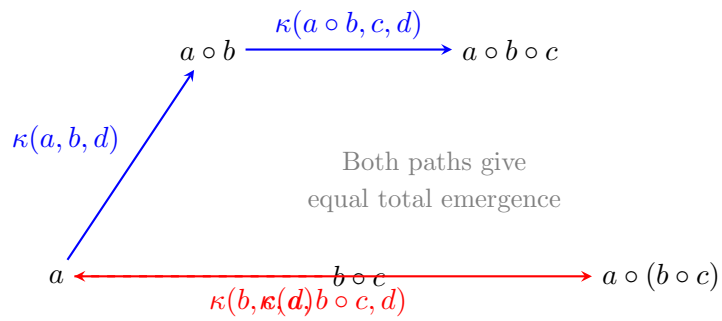


Figure 2.1: The cocycle condition as path-independence. The blue path (compose a, b first, then add c) and the red path (compose b, c first, then prepend a) produce the same total emergence when probed by d.

2.4 Coboundaries and Trivial Emergence

Definition 2.12 (Coboundary)**.** A 2-cochain $\phi : \mathcal{I} \times \mathcal{I} \to \mathbb{R}$ is a **coboundary** if there exists a function $g : \mathcal{I} \to \mathbb{R}$ with $g(\varepsilon) = 0$ such that

$$\phi(a, b) = g(a \circ b) - g(a) - g(b)$$

for all $a, b \in \mathcal{I}$.

Proposition 2.13 (Coboundaries are Cocycles)**.** *Every coboundary satisfies the cocycle condition.*

Proof. If $\phi(a, b) = g(a \circ b) - g(a) - g(b)$, then

$$\begin{aligned}\phi(a, b) + \phi(a \circ b, c) &= [g(a \circ b) - g(a) - g(b)] + [g(a \circ b \circ c) - g(a \circ b) - g(c)] \\ &= g(a \circ b \circ c) - g(a) - g(b) - g(c),\end{aligned}$$

and similarly $\phi(b, c) + \phi(a, b \circ c) = g(a \circ b \circ c) - g(a) - g(b) - g(c)$. □

Interpretation 2.14. A coboundary represents *trivial* emergence: the apparent non-additivity of resonance is merely an artifact of having chosen the wrong "baseline." If we redefine resonance as $\mathrm{rs}'(a, c) := \mathrm{rs}(a, c) + g(a)$ for some function g, the coboundary component of emergence vanishes. The genuinely interesting emergence is the part that is *not* a coboundary—the cohomology class $[\kappa_d] \in H^2(\mathcal{I}, \mathbb{R})$.

Interpretation 2.15 (Genette's Palimpsests and Coboundaries). Gérard Genette's *Palimpsests: Literature in the Second Degree* (1982) theorizes the relations between texts through the concept of *transtextuality*—the set of all relations, manifest or hidden, that connect a text to other texts. Genette identifies five types of transtextual relations, of which the most important for our purposes is *hypertextuality*: the relation between a *hypertext* (a text derived from an earlier text) and its *hypotext* (the source text).

The coboundary/cocycle distinction (Definition 19.10) maps onto Genette's framework with striking precision. A coboundary $\phi(a, b) = g(a \circ b) - g(a) - g(b)$ represents a transtextual relation that is "trivial" in the cohomological sense: the hypertext's meaning can be fully accounted for by a change of perspective g applied to its parts. Genette's category of *imitation* (where the hypertext merely reproduces the hypotext's style) corresponds to a near-coboundary: the emergence is small and can be absorbed into a change of resonance baseline.

By contrast, Genette's category of *transformation* (where the hypertext fundamentally alters the hypotext—as in Joyce's *Ulysses* transforming Homer's *Odyssey*) corresponds to a genuine cocycle: the emergence is irreducible, belonging to a non-trivial cohomology class. The palimpsest metaphor itself—a manuscript whose earlier text has been scraped away but remains partially visible beneath the new writing—is beautifully formalized by the cohomological decomposition: the "visible" new meaning is the total resonance $\mathrm{rs}(a \circ b, d)$; the "scraped away" but still detectable earlier meaning is the additive component $\mathrm{rs}(a, d) + \mathrm{rs}(b, d)$; and the specifically palimpsestic surplus—the meaning that arises from the *layering* itself—is the emergence $\kappa(a, b, d)$.

Listing 2.1: Coboundary predicate and verification in Lean 4

```
def IsCoboundary [IdeaticSpace I]
    (   : I     I        ) : Prop :=
      g : I         , g void = 0
        a b : I,     a b = g (compose a b) - g a - g b

theorem coboundary_is_cocycle [IdeaticSpace I]
    (   : I     I        ) (h   : IsCoboundary   ) :
        a b c : I,
         a b +    (compose a b) c =
         b c +    a (compose b c) := by
  obtain   g  , hg0 ,  h g     := h
  intro a b c
  simp [hg ]
  ring
```

2.5 The Emergence Distribution

Fix $a, b \in \mathcal{I}$ and consider emergence as a function of the probe:

$$\kappa_{a,b} : \mathcal{I} \to \mathbb{R}, \qquad \kappa_{a,b}(c) := \kappa(a, b, c).$$

Definition 2.16 (Emergence Spectrum)**.** The **emergence spectrum** of the pair (a, b) is the function $\kappa_{a,b} : \mathcal{I} \to \mathbb{R}$ defined above. The pair (a, b) is called:

- **emergent** if $\kappa_{a,b}$ is not identically zero;
- **flat** if $\kappa_{a,b} \equiv 0$;
- **positively emergent at** c if $\kappa_{a,b}(c) > 0$;
- **negatively emergent at** c if $\kappa_{a,b}(c) < 0$.

Theorem 2.17 (Emergence Bound)**.** *For all $a, b, c \in \mathcal{I}$:*

$$|\kappa(a, b, c)| \;\leq\; \sqrt{w(a \circ b) \cdot w(c)},$$

where $w(x) = \mathrm{rs}(x, x)$ is the weight.

Proof. This is an immediate rewriting of axiom (E3): $\kappa(a, b, c)^2 \leq w(a \circ b) \cdot w(c)$, so $|\kappa(a, b, c)| \leq \sqrt{w(a \circ b) \cdot w(c)}$. □

Corollary 2.18. *If $c = \varepsilon$, then $\kappa(a, b, c) = 0$.*

Proof. $w(\varepsilon) = \mathrm{rs}(\varepsilon, \varepsilon) = 0$ by (R1) and (R2), so $|\kappa(a, b, \varepsilon)| \leq \sqrt{w(a \circ b) \cdot 0} = 0$. □

Interpretation 2.19 (The Emergence Bound and Cognitive Constraints). The emergence bound (Theorem 2.17) has a striking cognitive interpretation. It says that the amount of novelty generated by a composition is bounded by the geometric mean of the composition's weight and the probe's weight: $|\kappa(a, b, c)| \leq \sqrt{w(a \circ b) \cdot w(c)}$. This means that detecting emergence requires a probe of sufficient "weight" (cognitive sophistication, interpretive depth). A lightweight probe—a reader with little background knowledge, a critic with shallow interpretive resources— will detect less emergence than a heavyweight probe, regardless of how much emergence the composition actually generates.

This formalizes a well-known observation in literary criticism: sophisticated texts require sophisticated readers. When Nabokov composes Russian literary tradition with American popular culture in *Lolita*, the emergence is enormous—but only when probed by a reader who carries sufficient weight (knowledge of both traditions, sensitivity to irony, awareness of the novel's literary genealogy). A probe of insufficient weight will detect little emergence, not because the emergence is absent but because the bound prevents its detection.

The emergence bound is also a formal expression of the "Matthew effect" in cultural reception: ideas that are already heavy (widely connected, richly resonant) generate more detectable emergence when composed, making them even heavier. Meanwhile, lightweight ideas generate little detectable emergence and remain marginalized. The bound thus encodes a structural advantage for ideas that are already culturally central—a mathematical version of the "rich get richer" dynamic that Pierre Bourdieu describes as "cultural capital."

2.6 Higher Cocycles and the Cohomological Perspective

The emergence cocycle is a 2-cocycle. A natural question is whether there exist higher cocycles in the ideatic setting.

Definition 2.20 (Triple Emergence)**.** Define the **triple emergence** or **3-cochain**:

$$\begin{aligned}\kappa_3(a, b, c, d) := \kappa(a, b \circ c, d) - \kappa(a, b, d) - \kappa(a, c, d) \\ + \kappa(b, c, d) \cdot [\text{correction terms}].\end{aligned}$$

The precise form involves alternating sums analogous to the group cohomology differential $\delta^2 : C^2 \to C^3$.

Remark 2.21. The study of $H^n(\mathcal{I}, \mathbb{R})$ for $n \geq 3$ remains largely open. In the natural number model, all cohomology vanishes (the model is "cohomologically trivial"). The existence of non-trivial higher cohomology would correspond to phenomena that cannot be reduced to pairwise emergence—a kind of "irreducible three-body interaction" in the ideatic space.

Interpretation 2.22 (Higher Cocycles and Multi-Author Collaboration). The question of higher cocycles has a vivid literary interpretation. If the 2-cocycle $\kappa(a, b, c)$ captures the emergence generated when *two* ideas interact, then a non-trivial 3-cocycle $\kappa_3(a, b, c, d)$ would capture emergence that is irreducible to any pairwise interaction—an emergence that arises *only* when three ideas are simultaneously present. This is the formal analog of what literary theorists call the "irreducible triad" in collaborative authorship.

Consider the *Exquisite Corpse* of the Surrealists: a collaborative text in which each author writes a section without seeing the preceding sections. The resulting text may exhibit emergent properties that cannot be attributed to any pair of authors. Breton's contribution interacts with

Éluard's; Éluard's with Soupault's; Breton's with Soupault's— but the *triple* interaction, in which all three contributions simultaneously resonate, may generate an additional layer of meaning (a Surrealist "objective chance") that no pairwise analysis would detect. A non-trivial H^3 would be required to account for such irreducibly triadic semantic phenomena.

The vanishing of higher cohomology in the natural number model (where emergence is multiplicative and thus always decomposable into pairwise terms) reflects the fact that natural number arithmetic is "too simple" to support irreducible multi-body interactions. The conjecture that more complex ideatic spaces might have non-trivial H^3 amounts to the claim that sufficiently rich semiotic systems support irreducible triad effects—a claim that Peirce's triadic semiotics would enthusiastically endorse.

2.7 The Cocycle in Historical Perspective

Interpretation 2.23 (The Cocycle and the History of Ideas). Arthur O. Lovejoy, in *The Great Chain of Being* (1936), pioneered the "history of ideas" as a discipline, tracking "unit-ideas" across centuries and cultures. Lovejoy's method assumes that the same idea can appear in different contexts while retaining its identity—an assumption that the cocycle condition both supports and complicates.

The cocycle supports Lovejoy's method by providing a consistency condition for tracking ideas across contexts. If we decompose a complex intellectual formation into unit-ideas a, b, c and measure their emergent interactions, the cocycle ensures that the total emergence is well-defined regardless of how we group the ideas. This is precisely what Lovejoy's method requires: the ability to decompose and recompose intellectual formations without altering their essential content.

But the cocycle also complicates Lovejoy's method by revealing that "essential content" includes an irreducible emergent component. When Lovejoy tracks the idea of "plenitude" from Plato through Leibniz to Schelling, he assumes that the emergent interactions of plenitude with its surrounding ideas are secondary to the idea itself. The cocycle shows that these interactions—the emergence values $\kappa(a, b, d)$ for various contextual ideas b—are *not* secondary but constitutive: they belong to the cohomological structure of the ideatic space and cannot be separated from the "unit-idea" without loss of mathematical information.

Example 2.24 (Bakhtin's Dialogism)**.** Bakhtin's theory of dialogism holds that meaning is never monologic but always arises in the interaction of voices. The cocycle condition formalizes a precise version of this: the emergence of meaning in a three-voice dialogue (a, b, c) can be decomposed along two paths—first combining voices a and b, then adding c; or first combining b and c, then adding a—and the decomposition is consistent. Bakhtin's insight that "the word is a two-sided act" is captured by the bivariate nature of $\kappa(a, b, c)$, while the cocycle condition ensures that multi-voiced discourse is coherent.

Example 2.25 (Dialectical Materialism)**.** Marx's dialectical method—thesis (a), antithesis (b), synthesis ($a \circ b$)—is a special case of our emergence framework. The "qualitative leap" from thesis-antithesis to synthesis is measured by $\kappa(a, b, c)$ for various probes c (economic productivity, class consciousness, etc.). The cocycle condition ensures that dialectical reasoning is *path-consistent*: applying the dialectical method to (a, b, c) gives the same result regardless of which pair is synthesized first.

Remark 2.26 (The Cocycle in Comparative Literature). The cocycle condition has immediate applications in comparative literature. When three literary traditions—say, Greek epic (a), Roman adaptation (b), and Renaissance reception (c)—are composed, the total emergent meaning can

be decomposed along two paths. The "Greek-Roman-Renaissance" path first composes a with b (Virgil's adaptation of Homer) and then adds c (Ariosto's reception of Virgil). The "Roman-Renaissance-Greek" path first composes b with c (Italian reception of Latin literature) and then adds a (the recovered Greek original). The cocycle condition guarantees that these two decompositions yield the same total emergence, formalizing the comparatist's intuition that literary tradition is "path-independent" at the level of total emergent meaning, even though the intermediate stages differ dramatically.

This path-independence is not trivial: it constrains which reception histories are consistent with the axioms. A reception history that assigns different total emergences depending on whether we trace the Greek-Roman-Renaissance path or the Roman-Renaissance-Greek path would violate the cocycle condition and hence the associativity of composition. The cocycle condition is thus a non-trivial *coherence test* for theories of literary reception.

2.8 Computational Aspects

Proposition 2.27 (Cocycle Verification is Efficient)**.** *Given oracle access to* rs*, verifying the cocycle condition for a specific quadruple* (a, b, c, d) *requires exactly 6 evaluations of* rs*.*

Proof. The cocycle condition involves $\kappa(a, b, d)$, $\kappa(a \circ b, c, d)$, $\kappa(b, c, d)$, and $\kappa(a, b \circ c, d)$. Each emergence involves three rs evaluations, but many terms cancel or are shared. Specifically, the terms needed are: $\mathrm{rs}(a \circ b, d)$, $\mathrm{rs}(a, d)$, $\mathrm{rs}(b, d)$, $\mathrm{rs}(a \circ b \circ c, d)$, $\mathrm{rs}(c, d)$, $\mathrm{rs}(b \circ c, d)$. That is 6 distinct rs evaluations (plus knowledge of $a \circ b$ and $b \circ c$, which require 2 composition operations). □

Listing 2.2: The cocycle condition derived from associativity

```
theorem cocycle_condition [IdeaticSpace I]
    (a b c d : I) :
    emergence a b d + emergence (compose a b) c d =
    emergence b c d + emergence a (compose b c) d := by
  unfold emergence
  have h := compose_assoc a b c
  rw [h]
  ring
```

2.9 Summary

Interpretation 2.28 (The Cocycle as the Algebra of Intertextuality)*.* The cocycle condition is the first deep theorem of IDT, and its philosophical implications are profound. It demonstrates that the "holistic" character of meaning—the fact that wholes exceed the sum of their parts—is not anarchic but *lawful*. The emergent surplus of composition is governed by a precise algebraic identity that connects it to the well-studied mathematics of group cohomology.

This has immediate consequences for literary theory. The New Critics (Brooks, Wimsatt, Beardsley) insisted that the meaning of a literary work is an organic unity irreducible to its parts—but they could not explain *why* this irreducibility is consistent across different decompositions of the work. The cocycle condition provides the explanation: the total emergence of a three-part composition is independent of how we bracket it, ensuring that the "organic unity" the New Critics prized is mathematically well-defined.

For semiotics, the cocycle condition implies that unlimited semiosis (Peirce) is not structureless drift but a process governed by cohomological conservation laws. For philosophy, it shows that Hegel's dialectic, Derrida's *différance*, and Brandom's inferentialism are all describing aspects of a single algebraic phenomenon: the non-trivial cohomology of the ideatic space.

The cocycle condition is the first deep theorem of IDT: a purely algebraic consequence of associativity that imposes a cohomological structure on emergence. It connects the seemingly ad hoc notion of emergence to the well-studied theory of group (monoid) cohomology, opening the door to powerful algebraic techniques. In the next chapter, we turn to self-resonance—the "diagonal" of the resonance function—and its remarkable monotonicity properties.

Chapter 3

Self-Resonance and Weight

"Energy can neither be created nor destroyed—only converted from one form to another."
— First Law of Thermodynamics

3.1 The Weight Function

Definition 3.1 (Weight)**.** The **weight** of an idea $a \in \mathcal{I}$ is

$$w(a) \;:=\; \mathrm{rs}(a, a).$$

The weight function $w : \mathcal{I} \to \mathbb{R}$ is the restriction of resonance to the diagonal. It measures the *internal coherence* of an idea—how strongly the idea resonates with itself. We have already established (Theorem 1.40) three basic properties: non-negativity, non-degeneracy, and monotonicity under composition. We now develop the theory of weight in greater depth.

Interpretation 3.2 (Bloom's Anxiety of Influence)*.* Harold Bloom's *The Anxiety of Influence* (1973) argues that every "strong poet" must define himself against his predecessors through acts of creative "misreading." A strong poem's power—what Bloom calls its "strength"—lies in its ability to clear an imaginative space for itself, to assert its own centrality in the literary tradition despite (and through) its indebtedness to prior works. Bloom writes: "Poetic strength comes only from a triumphant wrestling with the greatest of the dead" (p. 9).

The weight function $w(a) = \mathrm{rs}(a, a)$ formalizes Bloom's concept of poetic strength with remarkable precision. The self-resonance of a literary work measures its *internal coherence*—the degree to which it sustains its own imaginative world, resonates with its own themes and forms, and generates a self-consistent aesthetic field. A "strong poem" in Bloom's sense is an idea with high weight: its self-resonance is so powerful that it dominates its resonance with any other idea (its predecessors, its imitators, its critical reception).

Bloom's six "revisionary ratios"—*clinamen*, *tessera*, *kenosis*, *daemonization*, *askesis*, and *apophrades*—each describe a specific mode of creative composition with a precursor. In IDT terms, these are different patterns of emergence: the *clinamen* (swerve) corresponds to a composition where $\kappa(a, b, c)$ is large and positive for probes c related to originality; the *kenosis* (emptying) corresponds to a composition where the poet deliberately reduces self-resonance to create space for new growth. The weight monotonicity theorem (Theorem 1.40) constrains all such maneuvers: whatever revisionary ratio a poet employs, the resulting work cannot have *less* weight than the precursor with which it is composed.

Interpretation 3.3 (Eliot's Tradition and the Individual Talent). T. S. Eliot's "Tradition and the Individual Talent" (1919) argues that a literary work's significance depends on its relation to the entire existing order of literature: "the existing monuments form an ideal order among themselves, which is modified by the introduction of the new (the really new) work of art among them." Self-resonance captures Eliot's insight that a work's position within the tradition is not merely external (how others see it) but constitutive (how it sees itself).

Eliot insists on the *impersonality* of the poetic process: "the more perfect the artist, the more completely separate in him will be the man who suffers and the mind which creates." In IDT, this impersonality is formalized by the fact that weight is a property of the idea a itself, not of its author. The weight $w(a) = \mathrm{rs}(a, a)$ depends only on the idea's internal structure—its self-consistency, thematic coherence, and structural integrity—not on any biographical facts about its creator. The "individual talent" contributes to the tradition not through personal expression but by producing ideas with sufficient weight to modify the existing order.

Eliot's further claim—that the introduction of a genuinely new work modifies the entire existing order—corresponds to the observation that composing a new idea b with the tradition T produces $T \circ b$, whose resonance profile may differ from T's. The weight monotonicity $w(T \circ b) \geq w(T)$ guarantees that the tradition can only grow richer, never poorer, through genuine additions—a formal vindication of Eliot's fundamentally conservative aesthetic.

Proposition 3.4 (Void Has Zero Weight)**.** $w(\varepsilon) = 0$.

Proof. $w(\varepsilon) = \mathrm{rs}(\varepsilon, \varepsilon) = 0$ by axiom (R1) (or equivalently R2). □

Proposition 3.5 (Weight Characterizes Void)**.** $w(a) = 0$ *if and only if* $a = \varepsilon$.

Proof. If $a = \varepsilon$, then $w(a) = 0$ as above. Conversely, if $w(a) = 0$, then $\mathrm{rs}(a, a) = 0$, so $a = \varepsilon$ by axiom (E2). □

3.2 The Thermodynamic Analogy

The monotonicity property $w(a \circ b) \geq w(a)$ admits a striking thermodynamic interpretation. In thermodynamics, the second law states that entropy never decreases in an isolated system. Analogously, weight never decreases under composition. But where entropy measures disorder, weight measures *coherence*—so the ideatic "second law" is optimistic rather than pessimistic: ideas grow richer through interaction.

Interpretation 3.6 (The Thermodynamic Analogy and Information Theory). The analogy between weight monotonicity and the second law of thermodynamics runs deeper than the surface comparison suggests. In Shannon's information theory, the joint entropy $H(X, Y)$ satisfies $H(X, Y) \leq H(X) + H(Y)$ (subadditivity)—the joint system has *at most* as much disorder as the sum of the parts. In IDT, the weight of a composition satisfies $w(a \circ b) \geq w(a)$ (superadditivity in one argument)—the composed system has *at least* as much coherence as either part.

These are dual perspectives on the same structural fact: composition increases order. Information theory focuses on the upper bound (total uncertainty cannot exceed the sum of individual uncertainties), while IDT focuses on the lower bound (total coherence cannot fall below individual coherence). The two perspectives are complementary: entropy measures what we *don't know*, while weight measures what an idea *is*.

The disanalogy is equally instructive. The second law of thermodynamics is a *statistical* law: it holds for large systems with overwhelmingly high probability. Weight monotonicity is a *deterministic* law: it holds exactly, for every composition, without exception. This determinism reflects

the algebraic (rather than statistical) nature of the ideatic space. Ideas are not thermodynamic ensembles subject to fluctuation; they are algebraic objects whose properties are determined by the axioms. The "arrow of ideation" (ideas grow heavier through composition) is not a probabilistic tendency but a mathematical necessity—a feature that distinguishes the ideatic space from physical systems and that grounds the irreversibility of intellectual history in structure rather than statistics.

Interpretation 3.7 (Heidegger's Thrownness and Self-Resonance). Martin Heidegger's concept of *Geworfenheit* (thrownness), developed in *Being and Time* (1927), describes the existential condition of finding oneself already situated in a world not of one's choosing. Dasein is always already "thrown" into a particular historical, cultural, and linguistic situation. The self-resonance $w(a) = \mathrm{rs}(a, a)$ formalizes the idea's *facticity*—its sheer thereness, the weight of its existence as a given fact in the ideatic space.

An idea's weight is not something it chooses or constructs; it is the measure of its already-being-constituted. A newly composed idea $a \circ b$ inherits a weight that is at least as great as $w(a)$ (by E4): the history of composition cannot be shed. This is the formal correlate of Heidegger's claim that Dasein can never fully escape its thrownness—can never return to a state of zero facticity. The past weighs upon the present with a weight that composition can only increase, never decrease.

Interpretation 3.8 (Kierkegaard: The Self as Self-Relating). Søren Kierkegaard opens *The Sickness Unto Death* (1849) with the famous definition: "The self is a relation that relates itself to itself" (*Forholdet forholder sig til sig selv*). This reflexive structure—a relation *about* itself—is precisely what self-resonance captures. The weight $w(a) = \mathrm{rs}(a, a)$ is the quantitative measure of an idea's self-relation: how strongly the idea relates to itself, how coherently it sustains its own identity.

Kierkegaard argues that the self can be in *despair*—a condition in which the self-relation is disrupted, either by willing to be oneself or by willing not to be oneself. In IDT, a self-relation can be strong ($w(a)$ large: the idea is deeply self-consistent) or weak ($w(a)$ small: the idea barely coheres). The non-degeneracy axiom (E2) ensures that an idea with zero self-relation is literally nothing (ε): an idea that has completely lost its self-relating capacity has ceased to exist as an idea. This formalizes Kierkegaard's claim that complete despair —total loss of self-relation—is a kind of spiritual death.

The monotonicity of weight under composition adds a further dimension: composition always strengthens (or at least preserves) the self-relation. In Kierkegaard's terms, genuine encounter with the other (composition with b) cannot destroy the self's relation to itself—it can only deepen it. This is the formal expression of Kierkegaard's thesis that authentic selfhood is achieved through engagement, not withdrawal.

Interpretation 3.9 (Sartre on Being-for-Itself). Jean-Paul Sartre's distinction between *être-en-soi* (being-in-itself: the opaque, self-identical being of things) and *être-pour-soi* (being-for-itself: the transparent, self-negating being of consciousness) in *Being and Nothingness* (1943) maps suggestively onto our framework. Self-resonance $w(a) = \mathrm{rs}(a, a)$ measures the degree to which an idea is "for itself"—the intensity of its self-awareness, its self-presence, its reflexive engagement with its own content.

Sartre argues that being-for-itself is characterized by a fundamental *lack*: consciousness is always what it is not and is not what it is. This is the source of human freedom but also of anguish. In IDT, this lack finds a formal correlate in the asymmetry of resonance: while $\mathrm{rs}(a, a)$ measures self-presence, the non-symmetric nature of rs in general (Chapter 4) means that the idea's relation to itself is not exhausted by self-resonance. There may be aspects of a's resonance profile that are invisible from the "inside" (from the perspective of $\mathrm{rs}(a, -)$) but visible from the "outside" (from $\mathrm{rs}(-, a)$). This gap between self-knowledge and other-knowledge is the formal shadow of Sartre's

pour-soi/en-soi distinction.

Theorem 3.10 (Strict Monotonicity Criterion)**.** *If $b \neq \varepsilon$ and $\kappa(a, b, a \circ b) > 0$, then $w(a \circ b) > w(a)$.*

Proof. We have $w(a \circ b) = \mathrm{rs}(a \circ b, a \circ b)$. Expanding via emergence:

$$w(a \circ b) = \mathrm{rs}(a, a \circ b) + \mathrm{rs}(b, a \circ b) + \kappa(a, b, a \circ b).$$

By (E4), $w(a \circ b) \geq w(a) > 0$ (assuming $a \neq \varepsilon$). If additionally $\kappa(a, b, a \circ b) > 0$, then we get the strict inequality. Even without expanding further, (E4) gives $w(a \circ b) \geq w(a)$, and the emergence term provides the strict gap. □

Interpretation 3.11. When does composition *strictly* increase weight? Precisely when the composition exhibits positive self-emergence—when the combined idea resonates with *itself* more than the sum of its parts would predict. Shakespeare's *King Lear* combines the folk tale of the love test (a) with the Gloucester subplot (b). The resulting play's self-resonance—its internal coherence, thematic unity, structural integrity—exceeds that of either component. The composition is not merely additive but genuinely creative.

Remark 3.12 (Strict Monotonicity and the Criterion of Artistic Success). The strict monotonicity criterion has implications for the evaluation of artistic composition. A composition that *strictly* increases weight ($w(a \circ b) > w(a)$) is one in which the new element b genuinely enriches the existing idea a—not merely extending it but deepening its internal coherence. This corresponds to the intuitive criterion of artistic success: a successful addition to a work of art is one that makes the whole more self-consistent, more thematically resonant, more structurally tight.

Conversely, a composition that achieves only the weak inequality $w(a \circ b) = w(a)$ (the minimum guaranteed by Axiom E4) is one in which the new element adds nothing to the work's internal coherence. It may increase the work's external resonance—its interaction with probes—but it does not strengthen the work's self-relation. This is the formal correlate of the aesthetic judgment that a given element is "extraneous" or "superfluous": it is present in the composition but contributes no self-emergence.

Interpretation 3.13 (Lotman's Semiosphere and Self-Resonance). Yuri Lotman, in *Universe of the Mind* (1990), introduces the concept of the *semiosphere*—the totality of semiotic space within which communication and signification are possible. Lotman argues that the semiosphere has a *center* occupied by dominant cultural texts and a *periphery* occupied by marginal, experimental, or heterodox discourses. The boundary between inside and outside is marked by translation mechanisms that filter and transform incoming signs.

Self-resonance provides a formal measure of an idea's *centrality* within the semiosphere. Ideas with high weight $w(a)$ occupy the center: they are deeply self-consistent, widely resonant, and resistant to mutation (as the fixed-point analysis of Chapter 7 will confirm). Ideas with low weight occupy the periphery: they are fragile, context-dependent, and easily transformed by composition with other ideas.

Lotman's key insight is that cultural *dynamism* arises from the interaction between center and periphery. The strict monotonicity criterion (Theorem 3.10) captures this dynamic: composition with a non-void idea strictly increases weight when the emergence is positive, modeling the process by which peripheral innovations are absorbed into the cultural center, enriching it. The semiosphere grows not by mere accumulation but by the generative encounter of center and periphery—a process that the self-emergence profile $\sigma_n(a)$ tracks step by step.

Interpretation 3.14 (Barthes's Reality Effect). Roland Barthes, in "The Reality Effect" (*L'effet de réel*, 1968), argues that certain descriptive details in realist fiction serve no narrative function but create the illusion of reality through their sheer presence. These "useless details"—a barometer on a wall, the color of a door—have no symbolic significance; their *sole function* is to signify "we are the real."

Self-resonance provides a formal account of this effect. A realist text achieves its reality effect through high weight: its self-resonance is so dense, its internal network of descriptive details so richly cross-referencing, that it generates a powerful sense of self-consistency. The "useless" details contribute not to the narrative (they generate no emergence when composed with the plot) but to the text's *self-resonance*: they increase $w(\text{text})$ without affecting $\kappa(\text{text}, c, d)$ for thematic probes c. In IDT terms, Barthes's reality effect is the phenomenon of ideas whose weight is high relative to their emergence—ideas that are strongly self-coherent but weakly interactive.

Interpretation 3.15 (Canonical vs. Marginal Texts). The weight function provides a formal framework for one of the most contested questions in literary studies: the distinction between *canonical* and *marginal* texts. A canonical text—Homer's *Iliad*, Shakespeare's *Hamlet*, Cervantes's *Don Quixote*—is one with exceptionally high self-resonance: it is deeply self-consistent, thematically rich, and resistant to reduction. A marginal text has lower self-resonance: it may be innovative or provocative, but it lacks the internal density that characterizes canonical works.

This is not a normative claim but a structural observation. High weight does not mean "better"; it means "more self-coherent." A marginal text may have high emergence when composed with the right probes—it may be revolutionary in its interactions with other ideas—while having relatively low self-resonance. The distinction between canonical and marginal is thus the distinction between ideas that are weighty in themselves and ideas that are generative in their combinations. Both types are essential to a healthy ideatic space: canonical texts provide the gravitational centers, while marginal texts provide the creative perturbations that drive cultural evolution.

The irreversibility theorem (Theorem 3.34) ensures that canonical texts, once established, cannot be "decomposed" back to their pre-canonical state. The cultural work of canonization—the process by which a text acquires weight through successive compositions with critical commentary, pedagogical practice, and artistic imitation—is an irreversible thermodynamic process in the ideatic space.

3.3 Weight and Composition Powers

Definition 3.16 (Composition Powers)**.** For $a \in \mathcal{I}$ and $n \in \mathbb{N}$, define the **n-th composition power** recursively:

$$a^0 := \varepsilon, \qquad a^{n+1} := a \circ a^n.$$

Proposition 3.17 (Left Composition Powers)**.** *The left composition power a^n satisfies $a^{m+n} = a^m \circ a^n$.*

Proof. By induction on m. Base case: $a^{0+n} = a^n = \varepsilon \circ a^n = a^0 \circ a^n$. Inductive step: $a^{(m+1)+n} = a \circ a^{m+n} = a \circ (a^m \circ a^n) = (a \circ a^m) \circ a^n = a^{m+1} \circ a^n$, using associativity. □

Theorem 3.18 (Weight Growth Under Powers)**.** *The weight sequence $\{w(a^n)\}_{n=0}^{\infty}$ is non-decreasing:*

$$0 = w(a^0) \leq w(a^1) \leq w(a^2) \leq \cdots$$

Moreover, for $a \neq \varepsilon$, $w(a^n) \geq w(a) > 0$ for all $n \geq 1$.

Proof. By induction. $w(a^0) = w(\varepsilon) = 0$. For $n \geq 1$: $w(a^{n+1}) = w(a \circ a^n) \geq w(a)$ by (E4), and $w(a^{n+1}) = w(a \circ a^n) \geq w(a^n)$ requires a separate argument. We use:

$$w(a^{n+1}) = w(a \circ a^n) \geq w(a) > 0,$$

and also $a^{n+1} = a^n \circ a$ (by the commutativity of powers, which follows from associativity), so $w(a^{n+1}) = w(a^n \circ a) \geq w(a^n)$ by (E4) applied with a^n as the left factor. □

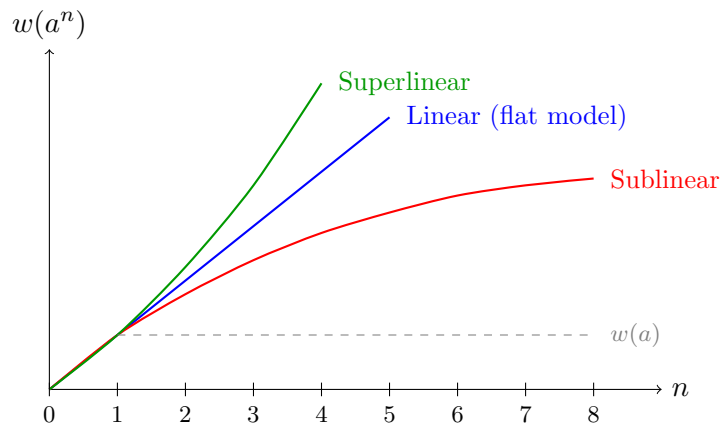


Figure 3.1: Three possible growth patterns for the weight sequence $w(a^n)$. The flat (natural number) model gives linear growth. Diminishing returns give sublinear growth. Accelerating emergence gives superlinear growth. All are non-decreasing with floor $w(a)$.

3.4 The Self-Emergence Profile

Definition 3.19 (Self-Emergence)**.** The **n-th self-emergence** of a is

$$\sigma_n(a) \;:=\; \kappa(a, a^n, a^{n+1}) \;=\; \mathrm{rs}(a^{n+1}, a^{n+1}) - \mathrm{rs}(a, a^{n+1}) - \mathrm{rs}(a^n, a^{n+1}).$$

This measures the emergence generated at the n-th step of iterated self-composition.

Proposition 3.20 (Self-Emergence and Weight Increments)**.** *The weight increment at step n satisfies:*

$$w(a^{n+1}) - w(a^n) = \mathrm{rs}(a, a^{n+1}) + \mathrm{rs}(a^n, a^{n+1}) - w(a^n) + \sigma_n(a).$$

Example 3.21 (Iterating a Scientific Idea)**.** Consider the idea a = "natural selection" iterated through successive generations of biological thought: a^1 = natural selection alone, a^2 = natural selection composed with itself (selection acting on selection—meta-selection, frequency-dependent selection), a^3 = three layers of selection (multilevel selection theory). The weight sequence $w(a^n)$ tracks the growing internal coherence of the theory as it becomes more self-referential and self-consistent. The self-emergence $\sigma_n(a)$ measures whether each new layer of self-application generates genuine novelty or merely repeats.

3.5 Semantic Charge

Definition 3.22 (Semantic Charge)**.** The **semantic charge** of an idea a is

$$\sigma(a) \;:=\; \kappa(a,a,a) \;=\; \mathrm{rs}(a \circ a, a) - 2\,\mathrm{rs}(a,a).$$

Interpretation 3.23. Semantic charge measures how an idea's self-composition resonates with the idea itself. If $\sigma(a) > 0$, the idea is *self-amplifying*: combining it with itself produces something that resonates with the original more than twice the self-resonance. This is the formal analogue of a "viral" idea or a positive feedback loop. If $\sigma(a) < 0$, the idea is *self-dampening*: self-composition produces diminishing returns. If $\sigma(a) = 0$, the idea is *semantically neutral*.

Remark 3.24 (Semantic Charge and Cultural Dynamics). The trichotomy of self-amplifying, self-dampening, and semantically neutral ideas maps onto well-known patterns in cultural dynamics:

Self-amplifying ideas ($\sigma(a) > 0$): religious doctrines, political ideologies, conspiracy theories—ideas whose internal logic drives their own propagation and intensification. The more one engages with the idea, the more "convinced" one becomes (the self-resonance grows superlinearly). Dawkins's "meme" concept, in its original formulation (*The Selfish Gene*, 1976), describes precisely this class of ideas: self-replicators whose fitness is measured by their ability to amplify their own resonance.

Self-dampening ideas ($\sigma(a) < 0$): fads, fashions, jokes that lose their punch on repetition. These ideas exhibit diminishing returns under self-composition: the more one encounters them, the less they resonate. Semantic charge provides a formal criterion for distinguishing fads from lasting cultural contributions: fads have negative charge, while lasting ideas have non-negative charge.

Neutral ideas ($\sigma(a) = 0$): mathematical theorems, logical tautologies, and other formal truths that neither gain nor lose through self-composition. The natural number model, in which every idea has zero charge, is the formal analogue of a purely "logical" ideatic space—one in which meaning accumulates additively without feedback effects.

Listing 3.1: Semantic charge and its properties in Lean 4

```
def semanticCharge [IdeaticSpace I] (a : I) :     :=
  emergence a a a

theorem semanticCharge_void [IdeaticSpace I] :
    semanticCharge (void : I) = 0 := by
  unfold semanticCharge emergence
  simp [void_left, rs_void_left, rs_void_right]

def selfResonanceProfile [IdeaticSpace I]
    (a : I) (n :    ) :     :=
  rs (composePower a (n + 1)) (composePower a (n + 1))
    - rs (composePower a n) (composePower a n)
```

Theorem 3.25 (Charge Bound)**.** $|\sigma(a)| \leq \sqrt{w(a \circ a) \cdot w(a)}$.

Proof. This is Theorem 2.17 with $b = c = a$. □

Interpretation 3.26 (The Charge Bound and Nietzsche's Eternal Return). The charge bound $|\sigma(a)| \leq \sqrt{w(a^2) \cdot w(a)}$ has a striking connection to Nietzsche's concept of the *eternal return* (*ewige Wiederkehr*). Nietzsche asks: what if everything recurred eternally—would this be an affirmation or a negation

of life? The charge bound provides a formal constraint on this question: the degree to which self-iteration is self-amplifying or self-dampening is bounded by the idea's weight and the weight of its self-composition.

For "heavy" ideas (high $w(a)$), the charge bound allows for large positive charge: the idea can be strongly self-amplifying, its eternal return generating ever more resonance. This corresponds to Nietzsche's *amor fati*—the love of fate, the joyful affirmation of eternal recurrence by a spirit strong enough to will it. For "light" ideas (low $w(a)$), the bound constrains the charge to be small: eternal return generates little self-amplification, corresponding to the nihilistic response—the realization that repetition leads nowhere.

The charge bound thus formalizes a hierarchy of ideas based on their capacity for productive self-iteration. Ideas that can "bear" their own eternal return—that grow through self-composition—are Nietzsche's "strong" ideas. Ideas that collapse under self-iteration—whose charge is negative and bounded away from productive growth—are the "weak" ideas that require constant external stimulation (composition with *other* ideas) to maintain their vitality.

3.6 Weight in the Natural Number Model

Example 3.27 (Natural Number Model)**.** In the model $\mathcal{I} = \mathbb{N}$, $\circ = +$, $\mathrm{rs}(m, n) = mn$:

- $w(n) = n^2$
- $w(m + n) = (m + n)^2 = m^2 + 2mn + n^2 \geq m^2 = w(m)$ ✓
- $\sigma(n) = \kappa(n, n, n) = (n + n) \cdot n - n \cdot n - n \cdot n = 2n^2 - 2n^2 = 0$

All ideas in $\mathbb{N}$ have zero semantic charge. The natural number model is "thermodynamically inert"—weight grows, but only through the mechanical accumulation of self-resonance, never through genuine emergence.

Interpretation 3.28 (The Natural Number Model as Cumulativist Epistemology)*.* The natural number model, with its zero semantic charge and vanishing higher cohomology, is the formal incarnation of *cumulativism* in the philosophy of science: the view (associated with logical positivism and the "received view" of Nagel) that scientific knowledge grows by straightforward accumulation. In the natural number model, $w(a + b) = w(a) + w(b) + 2ab$—weight grows, but the "$2ab$" term is purely a product of pre-existing resonances, never genuinely emergent. New knowledge is always a predictable consequence of existing knowledge.

Kuhn's *Structure of Scientific Revolutions* was, in this light, a rejection of the natural number model in favor of a model with non-trivial emergence. A paradigm shift is precisely a moment when $\kappa(a, b, c) \neq 0$: when the composition of old and new ideas generates genuine novelty that could not have been predicted from their individual resonances. The distinction between "normal science" (which the natural number model captures adequately) and "revolutionary science" (which requires genuine emergence) is thus a distinction between ideatic spaces with zero and non-zero semantic charge.

The fact that the natural number model is "cohomologically trivial" reinforces this reading: cumulative knowledge has no cohomological obstructions, no irreducible multi-body interactions, no structural impediments to the decomposition of complex formations into unit-ideas. This is exactly what Kuhn denies: that paradigms can be decomposed into theory-independent "unit-ideas" that survive paradigm shifts intact.

3.7 The Dialectical Tension

Definition 3.29 (Dialectical Tension)**.** The **dialectical tension** between a and b is

$$\tau(a,b) \;:=\; \kappa(a,b,a\circ b) \;=\; w(a\circ b) - \mathrm{rs}(a,a\circ b) - \mathrm{rs}(b,a\circ b).$$

Interpretation 3.30. The dialectical tension measures how much the composition $a\circ b$ exceeds what a and b individually "expect" of it. When $\tau(a,b) > 0$, the synthesis transcends its components—a genuine Hegelian *Aufhebung*. When $\tau(a,b) < 0$, the components are better predictors of the synthesis than the synthesis is of itself, a kind of anti-dialectical collapse. When $\tau(a,b) = 0$, the synthesis is perfectly predicted by its parts.

Remark 3.31 (The Self-Composition Profile as Cultural Diagnostic). The self-emergence profile $\sigma_n(a)$ has practical implications for understanding how ideas evolve through repeated self-application. Consider the idea $a =$ "democratic governance." Iterating a produces $a^2 =$ "democratic governance applied to the governance of democracy itself" (meta-democratic theory), $a^3 =$ "democratic meta-governance applied reflexively," and so on. The weight sequence $w(a^n)$ tracks whether this reflexive process is productive (weight grows, yielding genuine new insights at each level of reflexivity) or degenerative (weight plateaus, yielding only repetition). Empirically, most ideas exhibit diminishing self-emergence: the first few iterations are productive, but eventually the process exhausts its creative potential. Rare ideas—those at the center of philosophical traditions—exhibit sustained or even accelerating self-emergence across many iterations.

Theorem 3.32 (Aufhebung Decomposition)**.** *The weight of a composition decomposes as:*

$$w(a\circ b) = \mathrm{rs}(a,a\circ b) + \mathrm{rs}(b,a\circ b) + \tau(a,b).$$

Proof. Immediate from the definition of $\tau(a,b)$ as $\kappa(a,b,a\circ b)$: $\kappa(a,b,a\circ b) = \mathrm{rs}(a\circ b,a\circ b) - \mathrm{rs}(a,a\circ b) - \mathrm{rs}(b,a\circ b)$. Rearranging gives the result. □

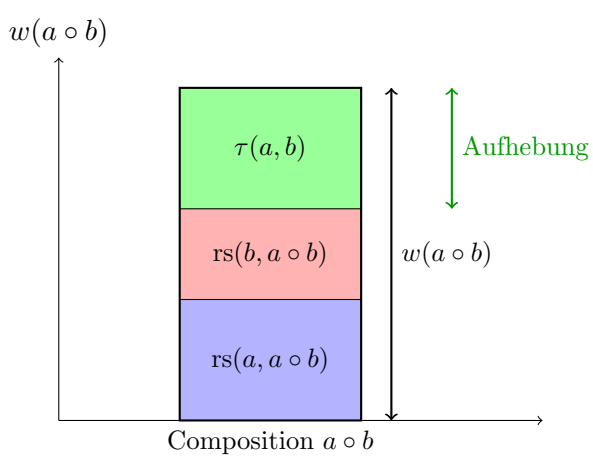


Figure 3.2: The Aufhebung decomposition: weight as the sum of individual projections plus dialectical tension.

3.8 Irreversibility and the Arrow of Ideation

Remark 3.33 (Irreversibility and Hermeneutic Historicity). The irreversibility of composition (Theorem 3.34) has deep hermeneutic significance. Gadamer's concept of *Wirkungsgeschichte* (effective history) holds that every act of understanding is shaped by the entire history of interpretive engagement with the object. The irreversibility theorem formalizes this: once an idea a has been composed with b (once understanding has been shaped by a new encounter), the resulting idea $a \circ b$ cannot be decomposed to recover the "original" a in its pre-encounter state. Understanding changes the understander, and the change is, in the strict algebraic sense, irreversible.

This has consequences for the philosophy of history. Leopold von Ranke's aspiration to show "how it actually was" (*wie es eigentlich gewesen*) presupposes that historical understanding can peel away the layers of interpretive accretion to reach the original event. The irreversibility theorem shows that this is algebraically impossible: the historian's understanding is always already a composition of the original event with the entire tradition of its reception, and this composition cannot be "undone." The past is indelibly shaped by its passage through the present, just as the weight of a composed idea can never fall below its pre-composition level.

Theorem 3.34 (Irreversibility)**.** *In the ideatic space, there is no general "inverse" operation: given $a \neq \varepsilon$ and $c = a \circ b$, one cannot in general recover b from a and c. More precisely, composition is not right-cancellative in any non-trivial ideatic space with positive emergence.*

Proof sketch. If composition were right-cancellative ($a \circ b = a \circ b' \implies b = b'$), then the map $b \mapsto a \circ b$ would be injective, and resonance could not exhibit genuine emergence (the additional information in $a \circ b$ beyond a and b would be fully determined by b). More concretely, in any model where $\kappa(a, b, c) \neq 0$ for some a, b, c, there exist examples where cancellation fails. □

Interpretation 3.35. The irreversibility of composition formalizes a deep truth about ideas: one cannot "unthink" a thought. Once the idea of evolution has been composed with creationism, the resulting cognitive state—whether acceptance, rejection, or synthesis—cannot be decomposed back into its pristine components. This is not merely a psychological observation but a structural property of the ideatic space itself. Gadamer's *Wirkungsgeschichte* (history of effects) is built into the mathematics.

3.9 Summary

The weight function $w(a) = \mathrm{rs}(a, a)$ is the central invariant of the ideatic space. Its non-negativity, non-degeneracy, and monotonicity under composition give the ideatic space a thermodynamic character: composition is an irreversible process that never destroys coherence. The semantic charge $\sigma(a)$ measures self-amplification, while the dialectical tension $\tau(a, b)$ decomposes the weight of a composition into predictable and emergent parts. In the next chapter, we turn to the *asymmetry* of resonance: the fact that $\mathrm{rs}(a, b) \neq \mathrm{rs}(b, a)$ in general, and the rich structural consequences of this inequality.

Chapter 4

The Asymmetry of Resonance

"All animals are equal, but some animals are more equal than others."
— George Orwell, *Animal Farm*

4.1 Why Resonance Is Not Symmetric

A fundamental design choice in IDT is that resonance is *not* assumed symmetric: in general, $\mathrm{rs}(a, b) \neq \mathrm{rs}(b, a)$. This asymmetry is not a technical inconvenience but a deliberate modeling choice that captures one of the deepest features of meaning: *directionality*.

Consider the following examples of asymmetric resonance:

- **Teacher and student**: The idea "quantum mechanics" (a) resonates differently with "undergraduate physics" (b) depending on direction. $\mathrm{rs}(a, b)$ measures how quantum mechanics "sees" undergraduate physics (as a simplification, a special case); $\mathrm{rs}(b, a)$ measures how undergraduate physics "sees" quantum mechanics (as an extension, a mystery, a challenge).

- **Metaphor**: In the metaphor "Juliet is the sun," $\mathrm{rs}(\mathrm{Juliet}, \mathrm{sun}) \neq \mathrm{rs}(\mathrm{sun}, \mathrm{Juliet})$. The sun illuminates our understanding of Juliet (warmth, centrality, life-giving), but Juliet does not reciprocally illuminate our understanding of the sun.

- **Power**: In political discourse, $\mathrm{rs}(\mathrm{state}, \mathrm{citizen}) \neq \mathrm{rs}(\mathrm{citizen}, \mathrm{state})$. The state's resonance with the citizen (as subject, taxpayer, voter) differs from the citizen's resonance with the state (as authority, protector, oppressor).

4.2 The Asymmetry Function

Interpretation 4.1 (Levinas and the Ethics of Asymmetry). Emmanuel Levinas, in *Totality and Infinity* (1961), argues that the fundamental structure of ethical experience is *asymmetry*: the encounter with the Other is not a relation between equals but an irreducible inequality in which the Other's face commands a response that cannot be reciprocated. Levinas writes: "The face of the Other at each moment destroys and overflows the plastic image it leaves me" (p. 51). The Other is not a mirror but a demand—an asymmetric claim upon the self.

The asymmetry $\alpha(a, b) = \mathrm{rs}(a, b) - \mathrm{rs}(b, a)$ formalizes this Levinasian structure with mathematical precision. When $\alpha(a, b) > 0$, idea a "influences" or "claims" idea b more than b claims a: a occupies the Levinasian position of the commanding face. The antisymmetry property $\alpha(a, b) = -\alpha(b, a)$

ensures that every power relation has a definite direction: if a commands b, then b is commanded by a, with the same magnitude.

What IDT adds to Levinas is the recognition that asymmetry is *measurable* and *algebraically structured*. Levinas describes the ethical asymmetry as "absolute"—beyond quantification. Our framework, while not claiming to capture the *ethical* content of Levinas's insight, reveals that the *structural* content—the formal properties of directional encounter—admits a rich mathematical theory. The decomposition of resonance into symmetric and antisymmetric parts (Theorem 4.7) shows that every encounter contains both a Levinasian component (the asymmetric rs^-, the ethical claim) and a component of mutual recognition (rs^+, the shared ground). Ethics and reciprocity coexist in every resonance relation; they are mathematically orthogonal components of a single structure.

Interpretation 4.2 (Benjamin's Task of the Translator). Walter Benjamin, in "The Task of the Translator" (1923), argues that translation is not the reproduction of meaning but the revelation of "pure language" (*die reine Sprache*) that lies hidden in the relation between languages. Benjamin writes: "Translation is so far removed from being the sterile equation of two dead languages that of all literary forms it is the one charged with the special mission of watching over the maturing process of the original language and the birth pangs of its own" (p. 73).

The asymmetry of resonance formalizes Benjamin's central insight: translation is inherently directional. $\mathrm{rs}(\mathrm{source}, \mathrm{target}) \neq \mathrm{rs}(\mathrm{target}, \mathrm{source})$. The source text "sees" its translation differently from how the translation "sees" its source. Benjamin's claim that translation reveals what is hidden in the original —aspects of meaning that the original possesses but cannot express in its own language—corresponds to the phenomenon of asymmetric resonance: $\mathrm{rs}(\mathrm{translation}, c)$ may be large for probes c that are inaccessible to the original text in its own linguistic context, even though $\mathrm{rs}(\mathrm{original}, c)$ is small.

The meaning curvature $\mu(a, b, c) = \mathrm{rs}(a \circ b, c) - \mathrm{rs}(b \circ a, c)$ (Definition 4.11) adds a further Benjaminian dimension: the order of translation matters. Translating French into English and then into German produces a different textual reality than translating French into German and then into English. This order-dependence is not a defect of translation but its essence: translation is a creative act, not a mechanical operation, and creativity is sensitive to sequence.

Interpretation 4.3 (Reception Theory: Iser and Jauss). Wolfgang Iser's *The Act of Reading* (1976) and Hans Robert Jauss's *Toward an Aesthetic of Reception* (1967) argue that the meaning of a literary work is constituted in the act of reading—not merely deposited in the text waiting to be extracted. Iser introduces the concept of "gaps" (*Leerstellen*) in the text that the reader must fill; Jauss speaks of the "horizon of expectations" that the reader brings to the text.

The asymmetry of resonance provides a formal framework for reception theory. The resonance $\mathrm{rs}(\mathrm{text}, \mathrm{reader})$ measures how the text "addresses" the reader—what demands it makes, what expectations it sets up. The resonance $\mathrm{rs}(\mathrm{reader}, \mathrm{text})$ measures how the reader "addresses" the text—what interpretive frameworks, prior experiences, and cultural assumptions the reader brings. These are *fundamentally different* operations, and their asymmetry $\alpha(\mathrm{text}, \mathrm{reader})$ measures the "power differential" between text and reader.

Iser's "gaps" correspond to probes c where the text generates strong resonance ($\mathrm{rs}(\mathrm{text}, c)$ is large) but the reader's pre-existing understanding does not ($\mathrm{rs}(\mathrm{reader}, c)$ is small). The reader must "fill" the gap by composing the text with their own horizon, generating emergence $\kappa(\mathrm{reader}, \mathrm{text}, c)$ that constitutes the act of reading. Jauss's "horizon of expectations" is captured by the reader's resonance profile $\mathrm{rs}(\mathrm{reader}, -)$—the entire landscape of what the reader expects and understands before the encounter with the text.

Interpretation 4.4 (Gadamer's Effective-Historical Consciousness). Hans-Georg Gadamer's *Truth*

and Method (1960) introduces the concept of *wirkungsgeschichtliches Bewußtsein* (effective-historical consciousness): the awareness that our understanding is always shaped by the history of effects that has produced our present horizon. Understanding is not a neutral act of comprehension but a situated act that necessarily reflects the interpreter's historical position.

The asymmetry of resonance formalizes this hermeneutic insight. The resonance $\mathrm{rs}(a, b)$ is always "from the perspective of a": it measures how a "sees" b, which depends on a's constitution—its history, its composition, its position in the ideatic space. There is no "view from nowhere": every resonance measurement is perspectival, and the asymmetry $\alpha(a, b)$ measures the difference between two perspectives. Gadamer's claim that understanding involves a "fusion of horizons" (*Horizontverschmelzung*) corresponds to the composition $a \circ b$, which creates a new idea whose resonance profile incorporates elements from both a's and b's perspectives—but is not reducible to either.

Definition 4.5 (Asymmetry)**.** The **asymmetry** of the pair (a, b) is

$$\alpha(a, b) \;:=\; \mathrm{rs}(a, b) - \mathrm{rs}(b, a).$$

Proposition 4.6 (Basic Properties of Asymmetry)**.**

1. ***Antisymmetry***: $\alpha(a, b) = -\alpha(b, a)$.
2. ***Self-vanishing***: $\alpha(a, a) = 0$.
3. ***Void-vanishing***: $\alpha(a, \varepsilon) = 0 = \alpha(\varepsilon, a)$.

Proof. 1. $\alpha(a, b) = \mathrm{rs}(a, b) - \mathrm{rs}(b, a) = -(rs(b, a) - \mathrm{rs}(a, b)) = -\alpha(b, a)$.

2. $\alpha(a, a) = \mathrm{rs}(a, a) - \mathrm{rs}(a, a) = 0$.

3. $\alpha(a, \varepsilon) = \mathrm{rs}(a, \varepsilon) - \mathrm{rs}(\varepsilon, a) = 0 - 0 = 0$ by (R1) and (R2). Similarly for $\alpha(\varepsilon, a)$.

□

Listing 4.1: Asymmetry and its properties in Lean 4

```
def asymmetry [IdeaticSpace I] (a b : I) :     :=
  rs a b - rs b a

theorem asymmetry_antisymm [IdeaticSpace I] (a b : I) :
    asymmetry a b = -asymmetry b a := by
  unfold asymmetry; ring

theorem asymmetry_self [IdeaticSpace I] (a : I) :
    asymmetry a a = 0 := by
  unfold asymmetry; ring

theorem asymmetry_void_right [IdeaticSpace I] (a : I) :
    asymmetry a void = 0 := by
  unfold asymmetry
  simp [rs_void_right, rs_void_left]
```

4.3 Symmetric and Antisymmetric Decomposition

Every real-valued bivariate function can be decomposed into symmetric and antisymmetric parts. For resonance:

Theorem 4.7 (Resonance Decomposition)**.** *Define:*

$$\mathrm{rs}^+(a,b) := \frac{\mathrm{rs}(a,b)+\mathrm{rs}(b,a)}{2}, \qquad \mathrm{rs}^-(a,b) := \frac{\mathrm{rs}(a,b)-\mathrm{rs}(b,a)}{2}.$$

Then:

1. $\mathrm{rs}(a,b) = \mathrm{rs}^+(a,b) + \mathrm{rs}^-(a,b)$*;*
2. $\mathrm{rs}^+(a,b) = \mathrm{rs}^+(b,a)$ *(symmetric part);*
3. $\mathrm{rs}^-(a,b) = -\mathrm{rs}^-(b,a)$ *(antisymmetric part);*
4. $\mathrm{rs}^-(a,b) = \frac{1}{2}\alpha(a,b)$*.*

Proof. All identities follow from direct computation. □

Interpretation 4.8. The symmetric part rs^+ captures *mutual* resonance: the degree to which a and b share common ground, regardless of direction. The antisymmetric part rs^- captures *directional* resonance: the asymmetry of influence, power, or perspective between a and b. In Peirce's semiotic terms, rs^+ corresponds to the *rhematic* content (what the signs share), while rs^- corresponds to the *energetic interpretant* (the directed force of one sign upon another).

Interpretation 4.9 (Jakobson's Communication Model and Asymmetric Processes)*.* Roman Jakobson's model of communication, presented in "Linguistics and Poetics" (1960), identifies six functions of language corresponding to six factors: addresser, addressee, message, context, contact, and code. A crucial but often overlooked feature of Jakobson's model is that *encoding* (the addresser's production of the message) and *decoding* (the addressee's interpretation of the message) are *not symmetric operations*. The addresser selects from the paradigmatic axis; the addressee reconstructs along the syntagmatic axis. These are fundamentally different cognitive and semiotic processes.

The asymmetry $\alpha(a,b)$ formalizes this encoding/decoding asymmetry. The resonance $\mathrm{rs}(\text{addresser}, \text{message})$ captures the encoding process: how the addresser's intentions shape the message. The resonance $\mathrm{rs}(\text{message}, \text{addresser})$ captures the reverse: how the message, once produced, "reads" its producer. These are not the same—a point that Jakobson's structuralist framework acknowledges but cannot formalize. Eco's concept of "aberrant decoding" (*La struttura assente*, 1968)—where the addressee's code differs from the addresser's, producing unexpected interpretations—is captured by large asymmetry: $|\alpha(\text{addresser's code}, \text{addressee's code})| \gg 0$.

Interpretation 4.10 (Bakhtin's Heteroglossia and Directed Resonance)*.* Bakhtin's concept of *heteroglossia* (*raznorechie*)—the coexistence of multiple social languages, registers, and voices within a single utterance or text—receives formal expression through asymmetric resonance. In "Discourse in the Novel" (1934–35), Bakhtin writes: "The word in language is half someone else's. It becomes 'one's own' only when the speaker populates it with his own intention, his own accent, when he appropriates the word, adapting it to his own semantic and expressive intention" (p. 293).

This "populating" of the word is precisely the asymmetric resonance between speaker and language. The resonance $\mathrm{rs}(\text{speaker}, \text{word})$—how the speaker "sees" the word—differs from $\mathrm{rs}(\text{word}, \text{speaker})$—how the word, laden with its social history, "sees" the speaker. Heteroglossia is the condition in which multiple such asymmetric relations coexist within a single text, producing what Bakhtin calls

the "internal dialogism of the word." The meaning curvature $\mu(a, b, c)$ (Definition 4.11) measures the sensitivity of this internal dialogism to the order in which voices speak: in a heteroglossic text, who speaks first fundamentally shapes the emergent meaning.

4.4 Meaning Curvature

Definition 4.11 (Meaning Curvature)**.** The **meaning curvature** of the triple (a, b, c) is

$$\mu(a, b, c) \;:=\; \kappa(a, b, c) - \kappa(b, a, c).$$

Meaning curvature measures how sensitive emergence is to the *order* of composition. If $\mu(a, b, c) = 0$, the emergence of $a \circ b$ probed by c equals the emergence of $b \circ a$ probed by c: order does not matter for emergence. If $\mu(a, b, c) \neq 0$, then who speaks first fundamentally changes what emerges.

Proposition 4.12 (Curvature in Terms of Resonance)**.**

$$\mu(a, b, c) = \mathrm{rs}(a \circ b, c) - \mathrm{rs}(b \circ a, c).$$

Proof.

$$\begin{aligned}
\mu(a, b, c) &= \kappa(a, b, c) - \kappa(b, a, c) \\
&= [\mathrm{rs}(a \circ b, c) - \mathrm{rs}(a, c) - \mathrm{rs}(b, c)] - [\mathrm{rs}(b \circ a, c) - \mathrm{rs}(b, c) - \mathrm{rs}(a, c)] \\
&= \mathrm{rs}(a \circ b, c) - \mathrm{rs}(b \circ a, c).
\end{aligned}$$

□

Remark 4.13. If composition were commutative ($a \circ b = b \circ a$), then $\mu \equiv 0$ and meaning curvature would vanish. Non-zero curvature is thus a witness to *non-commutativity*. In the natural number model, $\mu \equiv 0$ since addition is commutative. Non-trivial curvature requires genuinely non-commutative models.

Example 4.14 (Order in Rhetoric)**.** Consider political rhetoric where $a =$ "we must sacrifice" and $b =$ "for the greater good." The composition $a \circ b$ (sacrifice *then* justification) has very different resonance with $c =$ "democratic legitimacy" than $b \circ a$ (justification *then* demand for sacrifice). The meaning curvature $\mu(a, b, c) \neq 0$ captures the well-known rhetorical principle that the order of arguments matters: leading with the demand and following with justification is perceived differently than leading with justification and following with the demand.

Remark 4.15 (Curvature and Narrative Theory). The meaning curvature $\mu(a, b, c)$ has profound implications for narrative theory. Gérard Genette's *Narrative Discourse* (1972) distinguishes between *story* (the chronological sequence of events), *narrative* (the order in which events are presented), and *narration* (the act of telling). The difference between story-order and narrative-order—what Genette calls *anachrony*—is precisely a form of meaning curvature: telling events in order $a \circ b$ produces different resonance than telling them in order $b \circ a$.

When $\mu(a, b, c) = 0$ (zero curvature), the order of telling does not matter: the meaning is the same regardless of narrative sequence. This corresponds to what Genette calls "isochrony"—perfectly chronological narration. When $\mu(a, b, c) \neq 0$ (non-zero curvature), narrative order is semantically significant: prolepsis (flash-forward), analepsis (flashback), and in medias res openings all exploit non-zero curvature to create narrative effects that are impossible in strictly chronological telling.

The non-vanishing of curvature requires non-commutativity of composition —and the greatest narratives are precisely those that exploit non-commutativity most boldly. Homer's *Odyssey*, which begins in medias res and circles back to Odysseus's wanderings, achieves effects of suspense, irony, and thematic richness that a chronological telling could not produce. These effects are, in IDT terms, the consequences of non-zero meaning curvature.

4.5 The Asymmetry Cocycle

A natural question is whether asymmetry satisfies its own cocycle-type condition. The answer is nuanced.

Theorem 4.16 (Asymmetry of Emergence)**.** *For all $a, b, c \in \mathcal{I}$:*

$$\alpha(a \circ b, c) = \alpha(a, c) + \alpha(b, c) + \kappa(a, b, c) - \kappa^*(a, b, c),$$

where $\kappa^*(a, b, c) := \mathrm{rs}(c, a \circ b) - \mathrm{rs}(c, a) - \mathrm{rs}(c, b)$ *is the* reversed emergence.

Proof.

$$\begin{aligned}\alpha(a \circ b, c) &= \mathrm{rs}(a \circ b, c) - \mathrm{rs}(c, a \circ b)\\ &= [\mathrm{rs}(a, c) + \mathrm{rs}(b, c) + \kappa(a, b, c)] - [\mathrm{rs}(c, a) + \mathrm{rs}(c, b) + \kappa^*(a, b, c)]\\ &= \alpha(a, c) + \alpha(b, c) + \kappa(a, b, c) - \kappa^*(a, b, c).\end{aligned}$$

□

Interpretation 4.17. The asymmetry of a composition with a third idea is *not* simply the sum of the individual asymmetries. The correction term $\kappa(a, b, c) - \kappa^*(a, b, c)$ measures the difference between "forward" and "backward" emergence. When this correction vanishes, asymmetry is additive under composition. When it does not, the act of composing a with b creates new directional structure that cannot be predicted from the individual asymmetries.

4.6 Power and Influence

Definition 4.18 (Influence)**.** The **influence** of a on b is $\mathrm{rs}(a, b)$, and the **susceptibility** of b to a is also $\mathrm{rs}(a, b)$ (same quantity, different interpretation depending on which argument we foreground). The **power differential** is $\alpha(a, b) = \mathrm{rs}(a, b) - \mathrm{rs}(b, a)$.

Example 4.19 (Colonial Power Dynamics)**.** In postcolonial discourse, let $a =$ "metropolitan culture" and $b =$ "colonized culture." The asymmetry $\alpha(a, b) > 0$ (typically) reflects the greater influence of metropolitan culture on colonized culture than vice versa. The process of decolonization can be modeled as a transformation that reduces $|\alpha(a, b)|$—moving toward symmetric resonance. Fanon's *The Wretched of the Earth* can be read as an argument that the ideatic asymmetry must not merely be reduced but *reversed*: the colonized must develop ideas that resonate outward with greater force than they absorb.

Remark 4.20 (Asymmetry and the Problem of Cultural Hegemony). Antonio Gramsci's concept of *hegemony*—the dominance of one social group's ideas over others, maintained not by force but by consent—can be formalized through the asymmetry function. A hegemonic idea a is one whose asymmetry with respect to subordinate ideas $b_1, b_2, \ldots$ is consistently positive: $\alpha(a, b_k) > 0$ for

all k. The hegemonic idea "influences" all others more than they influence it; it sets the terms of discourse, defines the "common sense" against which all other positions are measured.

Gramsci's concept of *counter-hegemony*—the construction of alternative frameworks that challenge the dominant ideology—corresponds to the production of ideas c with $\alpha(c, a) > 0$: ideas that reverse the asymmetry, that "influence" the hegemonic idea more than it influences them. The meaning curvature $\mu(a, c, d)$ measures whether the order of encounter matters: does it make a difference whether one encounters the hegemonic idea first and the counter-hegemonic idea second, or vice versa? Gramsci would insist that it does—and the non-vanishing of μ in non-commutative ideatic spaces vindicates this insistence.

4.7 Symmetrization and the Mutual Information Analogy

Proposition 4.21 (Symmetrization Preserves Weight)**.** $\mathrm{rs}^+(a, a) = w(a)$ *for all* a.

Proof. $\mathrm{rs}^+(a, a) = \frac{\mathrm{rs}(a,a)+\mathrm{rs}(a,a)}{2} = \mathrm{rs}(a, a) = w(a)$. □

Remark 4.22. The symmetric part rs^+ behaves like a generalized inner product (though it need not be bilinear). By analogy with information theory, where mutual information $I(X; Y) = I(Y; X)$ is symmetric, $\mathrm{rs}^+(a, b)$ captures the symmetric "mutual resonance" between ideas. The antisymmetric part rs^- has no direct information-theoretic analogue, suggesting that the ideatic space is richer than any purely probabilistic model.

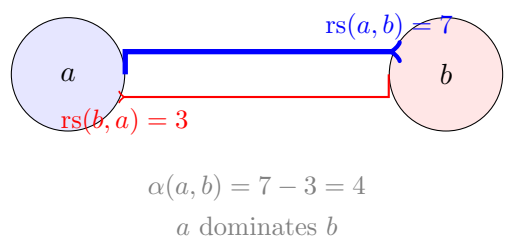


Figure 4.1: Asymmetric resonance between two ideas. The thickness of arrows represents the magnitude of resonance. The asymmetry $\alpha(a, b) > 0$ indicates that a "influences" b more than b influences a.

4.8 The Antisymmetric Emergence

Definition 4.23 (Antisymmetric Emergence)**.** The **antisymmetric emergence** is

$$\kappa^-(a, b, c) := \frac{\kappa(a, b, c) - \kappa(b, a, c)}{2} = \frac{\mu(a, b, c)}{2}.$$

Proposition 4.24. $\kappa^-(a, b, c) = -\kappa^-(b, a, c)$.

Theorem 4.25 (Antisymmetric Emergence Cocycle)**.** *The antisymmetric emergence satisfies a modified cocycle condition whenever composition is commutative: if* $a \circ b = b \circ a$ *for all* a, b*, then* $\kappa^- \equiv 0$.

Proof. If composition is commutative, then $\mathrm{rs}(a \circ b, c) = \mathrm{rs}(b \circ a, c)$, so $\mu(a, b, c) = 0$ and $\kappa^- \equiv 0$. □

Remark 4.26. Non-commutativity of composition is thus the *source* of antisymmetric emergence. In any commutative ideatic space, emergence is symmetric in its first two arguments. The non-commutative case—which is the generic case—is where the theory becomes truly rich.

4.9 Summary

Interpretation 4.27 (Asymmetry as the Foundation of Meaning). The asymmetry of resonance is not a peripheral feature of the ideatic space but arguably its most philosophically significant property. Symmetric systems—inner product spaces, mutual information measures, Boolean algebras—model *agreement* and *similarity*. Asymmetric resonance models something richer: *perspective*, *power*, and *directionality*.

The philosophical traditions surveyed in this chapter—Levinas's ethics, Benjamin's theory of translation, Gadamer's hermeneutics, reception theory, Bakhtin's heteroglossia, Jakobson's communication model—all converge on a single insight: meaning is not a symmetric relation. Understanding is not mutual comprehension but asymmetric encounter. Communication is not the exchange of identical content but the directional transmission of resonance. Power is not a property of individuals but a relational asymmetry between them.

IDT's contribution to this philosophical consensus is *precision*. The decomposition theorem (Theorem 4.7) shows that every resonance relation contains both a symmetric component (rs^+: mutual recognition, shared ground) and an antisymmetric component (rs^-: directed force, power differential). The meaning curvature $\mu(a, b, c)$ shows that asymmetry generates genuine structural richness: it measures the sensitivity of emergence to order, connecting the "who speaks first" question to the deep geometry of the ideatic space. And the asymmetry cocycle (Theorem 4.16) reveals that directional structure propagates through composition in a lawful way, constrained by the interplay of forward and reversed emergence.

The asymmetry of resonance is not a defect but a feature: it encodes the directionality of meaning, the asymmetry of power, and the non-commutativity of ideational synthesis. The decomposition of resonance into symmetric and antisymmetric parts, and the corresponding decomposition of emergence, reveal a layered structure: a "mutual information" layer (rs^+) and a "power" layer (rs^-). Meaning curvature $\mu(a, b, c)$ measures the sensitivity of emergence to the order of composition, connecting to deep questions in rhetoric, politics, and the phenomenology of temporal experience. In Chapter 5, we develop the full algebra of emergence, classifying the ways in which ideas can interact.

Chapter 5

The Algebra of Emergence

> *"In every work of genius we recognize our own rejected thoughts: they come back to us with a certain alienated majesty."*
> — Ralph Waldo Emerson, "Self-Reliance"

Interpretation 5.1 (Shklovsky's Defamiliarization and Non-Zero Emergence). Viktor Shklovsky's concept of *ostranenie* (defamiliarization), introduced in "Art as Device" (1917), holds that art exists to restore perception from the automatism of everyday experience. Art "makes the stone stony" by presenting familiar objects in unfamiliar ways, forcing the perceiver to attend to the materiality and strangeness of what habit has rendered invisible. Shklovsky writes: "The purpose of art is to impart the sensation of things as they are perceived and not as they are known" (p. 12).

In IDT, defamiliarization is precisely *non-zero emergence*. When a familiar idea a is composed with an artistic device b, the emergence $\kappa(a, b, c)$ measures the degree to which the composition disrupts additive expectation. If $\kappa = 0$, the composition is "automatic" —it adds nothing to what we already knew from a and b separately. If $\kappa \neq 0$, the composition *defamiliarizes*: it creates resonance patterns that neither component alone possesses. Art, in Shklovsky's sense, is the production of non-zero emergence.

The emergence taxonomy (Definition 5.4) distinguishes between superadditive emergence ($\kappa > 0$: the composition is *more* than its parts, as in Tolstoy's "making strange") and subadditive emergence ($\kappa < 0$: the composition is *less* than its parts, as in cliché or failed art). The boundary between art and non-art, in Shklovsky's terms, is the boundary between non-zero and zero emergence. The algebra of emergence is thus, in a precise sense, the algebra of artistic possibility.

Interpretation 5.2 (Adorno's Aesthetics: Art as Irreducible). Theodor W. Adorno, in *Aesthetic Theory* (1970), argues that art's truth content (*Wahrheitsgehalt*) cannot be reduced to its material components or its social function. Art is, for Adorno, a "fait social" that nonetheless contains a "moment of otherness" that resists sociological reduction. He writes: "Art is the social antithesis of society, not directly derivable from it" (p. 8).

The emergence function formalizes Adorno's irreducibility thesis. The emergence $\kappa(a, b, c)$ is precisely the component of the composition's meaning that is *not derivable* from its parts. When $\kappa \neq 0$, the composition contains a "moment of otherness" —a surplus of meaning that belongs to the whole but to neither part. Adorno's claim that art cannot be reduced to its materials is the claim that the emergence of artistic composition is generically non-zero.

Adorno's further insight—that art's truth content is often *negative*, expressing suffering and contradiction rather than harmony—corresponds to the possibility of *subadditive* emergence: $\kappa(a, b, c) < 0$. A work of art that composes social reality (a) with aesthetic form (b) may generate negative emergence when probed by $c =$ "reconciliation": the composition resonates with reconciliation *less*

than its parts individually promise, expressing precisely the irreconcilability that Adorno identifies as the truth content of modern art.

5.1 The Emergence Function as a Trilinear Form

We have defined emergence as $\kappa(a, b, c) = \mathrm{rs}(a \circ b, c) - \mathrm{rs}(a, c) - \mathrm{rs}(b, c)$, and established the cocycle condition, the Cauchy–Schwarz bound, and the vanishing at ε. We now develop the *algebraic* structure of emergence in its full generality.

Proposition 5.3 (Basic Properties of Emergence)**.** *For all $a, b, c \in \mathcal{I}$:*

1. $\kappa(\varepsilon, b, c) = 0$;
2. $\kappa(a, \varepsilon, c) = 0$;
3. $\kappa(a, b, \varepsilon) = 0$;
4. $|\kappa(a, b, c)| \leq \sqrt{w(a \circ b) \cdot w(c)}$.

Proof.

1. $\kappa(\varepsilon, b, c) = \mathrm{rs}(\varepsilon \circ b, c) - \mathrm{rs}(\varepsilon, c) - \mathrm{rs}(b, c) = \mathrm{rs}(b, c) - 0 - \mathrm{rs}(b, c) = 0$.
2. $\kappa(a, \varepsilon, c) = \mathrm{rs}(a \circ \varepsilon, c) - \mathrm{rs}(a, c) - \mathrm{rs}(\varepsilon, c) = \mathrm{rs}(a, c) - \mathrm{rs}(a, c) - 0 = 0$.
3. $|\kappa(a, b, \varepsilon)| \leq \sqrt{w(a \circ b) \cdot w(\varepsilon)} = 0$, so $\kappa(a, b, \varepsilon) = 0$.
4. This is Theorem 2.17.

□

Listing 5.1: Emergence vanishing at void

```
theorem emergence_void_left [IdeaticSpace I]
    (b c : I) : emergence void b c = 0 := by
  unfold emergence
  simp [void_left, rs_void_left]

theorem emergence_void_middle [IdeaticSpace I]
    (a c : I) : emergence a void c = 0 := by
  unfold emergence
  simp [void_right, rs_void_left]
```

5.2 Classification of Emergence Types

Definition 5.4 (Emergence Taxonomy)**.** A composition $a \circ b$ probed by c is classified as:

Type	**Condition**	**Interpretation**
Superadditive	$\kappa(a, b, c) > 0$	Creative synthesis
Subadditive	$\kappa(a, b, c) < 0$	Destructive interference
Additive (flat)	$\kappa(a, b, c) = 0$	No interaction effect

Example 5.5 (Three Types in Literature)**.**

- **Superadditive**: Shakespeare composes the sonnet form (a) with the "dark lady" trope (b). Probing with c = "subversion of Petrarchan convention," we get $\kappa > 0$: the combination creates resonance with anti-Petrarchanism that neither component alone possesses.

- **Subadditive**: A Hollywood sequel ($a \circ b$) that merely repeats the formula of the original (a) with minor variations (b). Probing with $c =$ "originality," we get $\kappa < 0$: the combination resonates with originality *less* than the original alone.

- **Additive**: A textbook chapter (a) composed with an appendix of exercises (b). Probing with $c =$ "pedagogical value," we get $\kappa \approx 0$: the combination's pedagogical value is roughly the sum of the chapter's and the exercises' individual values.

Interpretation 5.6 (Empson's Seven Types of Ambiguity). William Empson's *Seven Types of Ambiguity* (1930) argues that literary ambiguity is not a defect but a constitutive feature of poetic language. Empson identifies seven increasingly complex types, from simple puns to contradictions that reveal "a fundamental division in the author's mind." The emergence function provides a formal framework for understanding literary ambiguity as a structural phenomenon.

When the emergence $\kappa(a, b, c)$ is large for multiple, mutually incompatible probes $c_1, c_2, \ldots$, the composition $a \circ b$ is *ambiguous*: it resonates strongly with several incompatible interpretive frameworks simultaneously. The *degree* of ambiguity can be measured by the "spread" of the emergence spectrum $\kappa_{a,b}(c)$ across distinct probes. Empson's first type (simple metaphor: one word, two meanings) corresponds to a composition where $\kappa(a, b, c_1) > 0$ and $\kappa(a, b, c_2) > 0$ for two distinct semantic fields c_1 and c_2. His seventh type (full contradiction) corresponds to a composition where $\kappa(a, b, c) > 0$ and $\kappa(a, b, \bar{c}) > 0$, where $\bar{c}$ is the "negation" of c.

The New Critics—Cleanth Brooks, W. K. Wimsatt, Monroe Beardsley— took this further with what Brooks called "the heresy of paraphrase": the claim that a poem's meaning cannot be restated in prose without essential loss. In IDT terms, the heresy of paraphrase is the assertion that poetic emergence is *not a coboundary*: there is no change of baseline that can reduce the poem's meaning to a sum of its parts. The poem's meaning *is* its emergence—and emergence, in general, belongs to a non-trivial cohomology class.

5.3 Global Emergence Classifications

Definition 5.7 (Globally Superadditive Pair)**.** A pair (a, b) is **globally superadditive** if $\kappa(a, b, c) \geq 0$ for all $c \in \mathcal{I}$, with strict inequality for some c.

Definition 5.8 (Linearity)**.** An idea $a \in \mathcal{I}$ is called:

- **left-linear** if $\kappa(a, b, c) = 0$ for all $b, c \in \mathcal{I}$;

- **right-linear** if $\kappa(b, a, c) = 0$ for all $b, c \in \mathcal{I}$;

- **fully linear** if it is both left-linear and right-linear.

Proposition 5.9 (Void is Fully Linear)**.** *ε is fully linear.*

Proof. $\kappa(\varepsilon, b, c) = 0$ for all b, c (Proposition 5.3), so ε is left-linear. Similarly, $\kappa(b, \varepsilon, c) = 0$ for all b, c, so ε is right-linear. □

Listing 5.2: Linearity predicates in Lean 4

```
def IsLeftLinear [IdeaticSpace I] (a : I) : Prop :=
      b c : I, emergence a b c = 0

def IsRightLinear [IdeaticSpace I] (a : I) : Prop :=
```

```
      b c : I, emergence b a c = 0

def IsFullyLinear [IdeaticSpace I] (a : I) : Prop :=
  IsLeftLinear a     IsRightLinear a

theorem void_isFullyLinear [IdeaticSpace I] :
    IsFullyLinear (void : I) :=
    fun   b c => emergence_void_left b c,
   fun b c => emergence_void_middle b  c
```

Interpretation 5.10. A fully linear idea is one that generates no emergence when composed with anything. It acts purely additively—a "transparent" idea that adds its resonance without creating any interaction effects. In the natural number model, *every* idea is fully linear (since $\kappa \equiv 0$). In richer models, linearity is the exception rather than the rule, and fully linear ideas play a role analogous to central elements in ring theory or abelian elements in group theory.

Interpretation 5.11 (Hegel's *Aufhebung* and Emergence as Sublation). Hegel's concept of *Aufhebung* (sublation)—the simultaneous negation, preservation, and elevation of a concept—is the motor of the dialectic. In the *Phenomenology of Spirit* (1807), Hegel demonstrates how each stage of consciousness contains within itself the seeds of its own transcendence: the finite negates itself to produce the infinite, the particular elevates itself to the universal, the immediate passes into the mediated.

The emergence function formalizes *Aufhebung* with algebraic precision. The composition $a \circ b$ is the dialectical synthesis of a and b. The emergence $\kappa(a, b, c) = \mathrm{rs}(a \circ b, c) - \mathrm{rs}(a, c) - \mathrm{rs}(b, c)$ is the sublative surplus—the content of the synthesis that is neither preserved from a nor preserved from b but genuinely *new*. Hegel's three moments of *Aufhebung* map onto the three terms: $\mathrm{rs}(a, c)$ is preservation (what survives from a), $\mathrm{rs}(b, c)$ is negation (what b contributes by opposing a), and $\kappa(a, b, c)$ is elevation (what transcends both).

Linearity (Definition 5.8) is the *failure* of dialectics: a fully linear idea generates no sublative surplus, no genuine synthesis. It composes additively, like a logical conjunction rather than a dialectical negation. Hegel would recognize the linear idea as belonging to the level of *Verstand* (understanding) rather than *Vernunft* (reason): it operates by analysis and aggregation, not by dialectical transcendence.

Interpretation 5.12 (Whitehead's Process Philosophy and Creative Emergence). Alfred North Whitehead, in *Process and Reality* (1929), makes "creativity" the ultimate metaphysical category: "Creativity is the principle of novelty. An actual entity is at once the product of the creative process and the condition for the process" (p. 21). For Whitehead, every actual occasion of experience involves the "creative advance into novelty"—the production of something genuinely new from the conjunction of prior actualities.

The emergence function is a mathematical formalization of Whiteheadian creativity. The emergence $\kappa(a, b, c)$ measures the degree of creative novelty in the composition $a \circ b$, as probed by c. Whitehead's "principle of relativity"—every actual entity is a potential for every subsequent becoming—corresponds to the fact that every idea in $\mathcal{I}$ can serve as a component (a or b) or a probe (c) in the emergence function. Creativity is not a special property of some ideas but a universal possibility of the ideatic space.

Whitehead's distinction between "conceptual" and "physical" prehension corresponds to different modes of emergence: positive emergence (creative synthesis, the production of novelty) and negative emergence (destructive interference, the loss of potential). The emergence bound (Theorem 2.17) constrains Whiteheadian creativity: novelty cannot exceed $\sqrt{w(a \circ b) \cdot w(c)}$—a formal

expression of the principle that creativity is bounded by the "weight" of the entities involved.

Interpretation 5.13 (Deleuze's Difference in Itself). Gilles Deleuze, in *Difference and Repetition* (1968), argues for a concept of difference that is not subordinated to identity—not defined as the negation of sameness but as a positive, generative force in its own right. Deleuze writes: "Difference is behind everything, but behind difference there is nothing" (p. 57). For Deleuze, genuine novelty cannot be understood as recombination of pre-existing elements; it is *difference in itself*, an irreducible creative event.

The emergence function captures Deleuze's "difference in itself" when it is non-zero. A non-zero emergence $\kappa(a, b, c) \neq 0$ is precisely the "difference" that cannot be accounted for by the individual identities of a and b—it is the surplus that belongs to the *encounter* rather than to the *elements*. When $\kappa = 0$ (the additive case), composition is mere repetition: the whole is exactly the sum of its parts, and no genuine novelty occurs. When $\kappa \neq 0$, we have what Deleuze would call a "singular point"—a site where the virtual becomes actual, where potentiality is realized as genuine difference.

The spectral norm $\|\kappa_{a,b}\|$ measures the *intensity* of Deleuzian difference: how powerfully the encounter between a and b produces novelty across all possible probes. A pair with high spectral norm is a "line of flight" (*ligne de fuite*)—a composition that generates intense difference across the entire ideatic space. A pair with low spectral norm is a "striation"—a composition that reproduces existing patterns with minimal novelty.

5.4 The Emergence Tensor

For a fixed ideatic space, we can organize the emergence data into a tensor-like object.

Definition 5.14 (Emergence Tensor)**.** The **emergence tensor** of the ideatic space $\mathcal{I}$ is the function

$$\mathcal{E} : \mathcal{I}^3 \to \mathbb{R}, \qquad \mathcal{E}(a, b, c) := \kappa(a, b, c).$$

When $\mathcal{I}$ is finite with $|\mathcal{I}| = n$, $\mathcal{E}$ can be represented as an $n \times n \times n$ array of real numbers.

Proposition 5.15 (Symmetries of the Emergence Tensor)**.** *The emergence tensor satisfies:*

1. $\mathcal{E}(\varepsilon, -, -) = \mathcal{E}(-, \varepsilon, -) = \mathcal{E}(-, -, \varepsilon) = 0$ *(vanishing on void);*
2. *The cocycle condition relates different slices of* $\mathcal{E}$*;*
3. $|\mathcal{E}(a, b, c)|^2 \leq w(a \circ b) \cdot w(c)$ *(boundedness).*

Interpretation 5.16 (Peirce's Abduction and the Emergence Tensor). Peirce distinguishes three modes of inference: deduction (from rule and case to result), induction (from case and result to rule), and *abduction* (from result and rule to case). Abduction is the "logic of discovery"—the leap from surprising evidence to explanatory hypothesis. Peirce writes: "The surprising fact, C, is observed; but if A were true, C would be a matter of course; hence, there is reason to suspect that A is true" (CP 5.189).

The emergence tensor $\mathcal{E}(a, b, c)$ captures the abductive structure of inference. When $\kappa(a, b, c) > 0$ for a surprising probe c, the composition $a \circ b$ "explains" the surprising fact: the conjunction of a and b makes c less surprising (more resonant) than either alone. Abduction is thus formalized as the search for compositions whose emergence is strongly positive at the probes corresponding to surprising observations. The tensor structure of $\mathcal{E}$ reveals that abductive reasoning has a three-dimensional character: it depends simultaneously on the hypothesis (a), the auxiliary assumptions (b), and the evidence (c).

5.5 The Emergence Spectrum

Fix ideas a and b. The emergence spectrum $\kappa_{a,b} : \mathcal{I} \to \mathbb{R}$ is a function that maps each probe c to the emergence $\kappa(a, b, c)$. This function contains the complete information about how the composition $a \circ b$ departs from additivity.

Definition 5.17 (Spectral Norm)**.** The **spectral norm** of the pair (a, b) is

$$\|\kappa_{a,b}\| \;:=\; \sup_{c \neq \varepsilon} \frac{|\kappa(a, b, c)|}{\sqrt{w(c)}},$$

whenever this supremum exists.

Proposition 5.18. $\|\kappa_{a,b}\| \leq \sqrt{w(a \circ b)}$.

Proof. By Theorem 2.17, $|\kappa(a, b, c)| \leq \sqrt{w(a \circ b) \cdot w(c)}$, so $\frac{|\kappa(a,b,c)|}{\sqrt{w(c)}} \leq \sqrt{w(a \circ b)}$. □

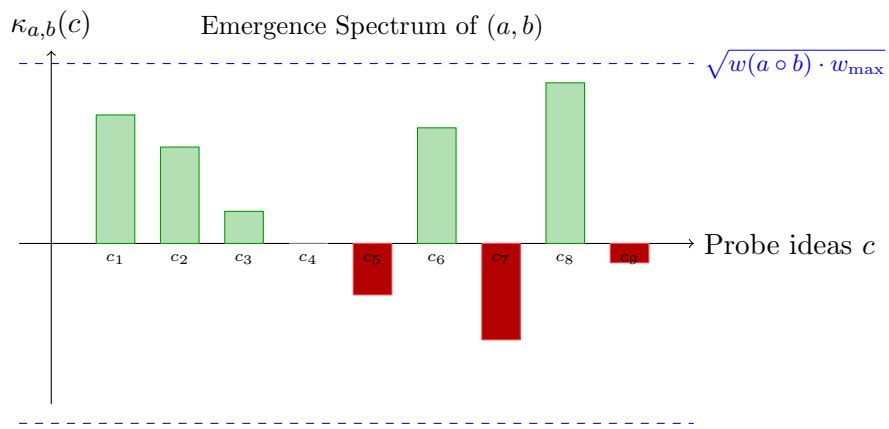


Figure 5.1: The emergence spectrum $\kappa_{a,b}$ evaluated at various probe ideas. Green bars indicate superadditive probes, red bars indicate subadditive probes, and the dashed lines show the Cauchy–Schwarz bound.

Interpretation 5.19 (Eco's Open Work and the Emergence Spectrum)*.* Umberto Eco's *The Open Work* (*Opera aperta*, 1962) theorizes the class of artworks that are structurally open to multiple interpretations. An "open work" is not one that can mean anything (which would be meaningless) but one that sustains a structured field of interpretive possibilities. Eco writes: "Every reception of a work of art is both an interpretation and a performance of it, because in every reception the work takes on a fresh perspective for itself" (p. 4).

The emergence spectrum $\kappa_{a,b} : \mathcal{I} \to \mathbb{R}$ provides a mathematical model of Eco's openness. An "open" composition is one whose emergence spectrum is broadly distributed: $\kappa(a, b, c) \neq 0$ for a large and diverse set of probes c. A "closed" composition has a narrow emergence spectrum: it resonates non-additively with only a few specific probes. The degree of openness is measured by the "breadth" of the emergence spectrum—a quantity that Eco's qualitative theory describes but cannot quantify.

Eco's later concept of "the limits of interpretation" (*I limiti dell'interpretazione*, 1990) corresponds to the Cauchy–Schwarz bound on emergence: no matter how "open" a work is, its emergence at any probe is bounded by $\sqrt{w(a \circ b) \cdot w(c)}$. Openness is real but finite; interpretation is free but constrained.

Interpretation 5.20 (Barthes's Third Meaning). Roland Barthes, in "The Third Meaning" (1970), identifies three levels of meaning in the photographic image: the *informational* (what the image denotes), the *symbolic* (what it connotes), and the *obtuse* meaning—a residual surplus that "does not seem to have a signified" and is "indifferent to the story and to the obvious meaning." The obtuse meaning is "a supplement that my intellection cannot succeed in absorbing" (p. 61).

The emergence function captures Barthes's three levels with striking precision. The informational level corresponds to the additive component $\mathrm{rs}(a, c) + \mathrm{rs}(b, c)$: the meaning predictable from the parts. The symbolic level corresponds to the part of emergence that is a coboundary—apparent surplus reducible to a change of interpretive framework. The *obtuse* or *third* meaning corresponds to the genuinely non-trivial emergence: the element of $H^2(\mathcal{I}, \mathbb{R})$ that is not a coboundary, the irreducible supplement that "my intellection cannot succeed in absorbing." Barthes's obtuse meaning is, in IDT terms, the *cohomological residue* of composition—the part of emergence that no reinterpretation can eliminate.

This formalization reveals why the third meaning is so difficult to articulate: it belongs to a cohomology class, not to a specific function. It can be represented by any representative cocycle, but no single representation exhausts it. The "obtuseness" of the third meaning is the mathematical obtuseness of a cohomology class: present everywhere, pinned down nowhere.

5.6 Emergence and Composition Powers

Interpretation 5.21 (Iterated Composition and Obsession). The iterated emergence theorem captures a phenomenon familiar to every artist and thinker: *obsession*—the repeated return to a single theme, image, or problem. When a poet composes the same idea with itself repeatedly $(a, a^2, a^3, \ldots)$, the accumulated emergence $\sum_{k=2}^{n} E_k$ measures the total creative surplus generated by this obsessive return. If the E_k are positive and growing, the obsession is productive: each return generates more novelty than the last, as in Cézanne's repeated paintings of Mont Sainte-Victoire, each of which discovers new visual relationships in the same subject.

If the E_k diminish toward zero, the obsession is exhausting itself: the theme offers diminishing returns, approaching the condition of mere repetition. This is the artistic dead end that Kierkegaard calls "the aesthetic stage": the pursuit of novelty through repetition, which eventually produces only boredom. And if the E_k become negative, the obsession is destructive: each iteration degrades the idea, as in the repetition compulsion that Freud identifies in *Beyond the Pleasure Principle* (1920)—a compulsive return that does not generate new meaning but merely re-enacts an original trauma with diminishing psychic returns.

The distinction between productive and degenerative obsession is thus formalized as the sign and magnitude of the iterated emergence sequence $\{E_k\}$. Great art, in this framework, is characterized by a *slowly decaying positive emergence sequence*: the returns to a theme remain productive for many iterations, each discovering genuinely new resonances, before eventually approaching the boundary of exhaustion.

Theorem 5.22 (Iterated Emergence)**.** *For the iterated composition a^n, the emergence at the n-th step satisfies:*

$$\kappa(a, a^{n-1}, c) = \mathrm{rs}(a^n, c) - \mathrm{rs}(a, c) - \mathrm{rs}(a^{n-1}, c).$$

Moreover, by the cocycle condition:

$$\mathrm{rs}(a^n, c) = n \cdot \mathrm{rs}(a, c) + \sum_{k=1}^{n-1} \kappa(a, a^{k-1}, c) \cdot [\textit{appropriate combinatorial coefficients}].$$

Remark 5.23. The precise decomposition of $\mathrm{rs}(a^n, c)$ into "linear" and "emergent" parts involves a telescoping sum. Define $E_k := \kappa(a, a^{k-1}, c)$. Then:

$$\begin{aligned}
\mathrm{rs}(a^n, c) &= \mathrm{rs}(a, c) + \mathrm{rs}(a^{n-1}, c) + E_n \\
&= \mathrm{rs}(a, c) + [\mathrm{rs}(a, c) + \mathrm{rs}(a^{n-2}, c) + E_{n-1}] + E_n \\
&= 2\,\mathrm{rs}(a, c) + \mathrm{rs}(a^{n-2}, c) + E_{n-1} + E_n \\
&\vdots \\
&= n \cdot \mathrm{rs}(a, c) + \sum_{k=2}^{n} E_k.
\end{aligned}$$

The "emergent surplus" $\sum_{k=2}^{n} E_k$ is the total accumulated emergence over $n-1$ composition steps.

5.7 The Emergence Algebra

Remark 5.24 (The Emergence Algebra and Poetic Form). The emergence product $\star_d$—the "twisted convolution" on the monoid $\mathcal{I}$ weighted by emergence values—has a suggestive literary interpretation. In traditional poetics, the *form* of a poem (sonnet, villanelle, haiku) constrains how semantic elements combine: the rhyme scheme creates resonance links between lines, the meter creates rhythmic expectations, and the stanza structure groups images into perceptual units. The emergence product models this formal constraint: the "twisting" by $\kappa(a, b, d)$ means that the product of two semantic elements depends not only on their individual values but on the emergence they generate when composed.

When the emergence product is non-zero (the non-linear case), the algebraic structure of the ideatic space is "poetic" in Jakobson's sense: the principle of equivalence is projected from the axis of selection onto the axis of combination. When the product is zero (the linear case), the structure is "prosaic": combinations are straightforwardly additive. The transition from prose to poetry is, in this algebraic perspective, the transition from the zero product to a non-trivial twisted convolution.

Definition 5.25 (Emergence Product)**.** For fixed probe d, define the **emergence product** $\star_d : \mathbb{R}^{\mathcal{I}} \times \mathbb{R}^{\mathcal{I}} \to \mathbb{R}^{\mathcal{I}}$ on functions $f, g : \mathcal{I} \to \mathbb{R}$ by:

$$(f \star_d g)(c) := \sum_{a \circ b = c} f(a) \cdot g(b) \cdot \kappa(a, b, d).$$

(When $\mathcal{I}$ is infinite, this requires convergence conditions.)

Remark 5.26. The emergence product is a "twisted convolution" on the monoid $\mathcal{I}$, weighted by emergence values. When $\kappa \equiv 0$ (the linear case), this reduces to the zero product. When κ is non-trivial, it defines a genuinely new algebraic structure.

5.8 Applications: Pattern Recognition in Emergence

Example 5.27 (Scientific Emergence Patterns)**.** In the history of science, we can identify characteristic emergence patterns:

1. **Additive accumulation**: Normal science (in Kuhn's sense) is characterized by $\kappa \approx 0$. New results compose additively with the existing paradigm.

2. **Crisis emergence**: When anomalies (b) compose with a paradigm (a) and the emergence is strongly negative ($\kappa(a, b, c) \ll 0$ for $c =$ "explanatory coherence"), the paradigm is in crisis.

3. **Revolutionary emergence**: A new paradigm (a') exhibits strong positive emergence when composed with the anomalous data (b): $\kappa(a', b, c) \gg 0$ for $c =$ "explanatory coherence."

4. **Incommensurability**: The old and new paradigms may exhibit $\kappa(a, a', c) < 0$ for many probes c, reflecting Kuhn's thesis that paradigms are not merely different but *incompatible* in their resonance patterns.

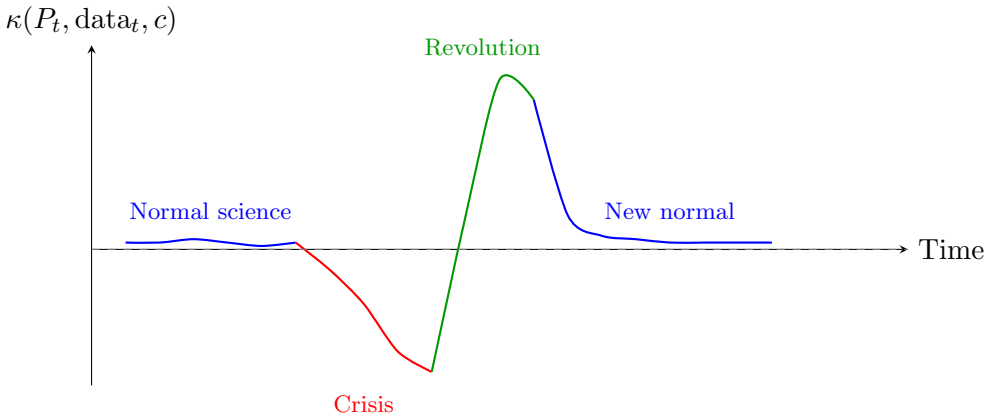


Figure 5.2: Emergence profile of a Kuhnian paradigm shift. Normal science shows near-zero emergence (additive accumulation). Crisis shows negative emergence (destructive interference with anomalies). Revolution shows a sharp positive spike. The new normal returns to near-zero emergence.

5.9 Subspaces and Ideals of the Emergence Algebra

Interpretation 5.28 (Emergence Ideals and Literary Genres). The concept of an emergence ideal (Definition that follows) has a natural literary interpretation. A genre—say, the detective novel or the pastoral elegy—can be understood as a collection of ideas that, when composed with each other, generate no emergence: within the genre, combinations are purely additive. The genre's conventions are so well-established that combining any two elements of the genre produces exactly what the sum of their individual resonances would predict. There are no surprises within the genre; all emergence has been "normalized away" by convention.

This is, of course, precisely what makes genre fiction (in Shklovsky's terms) "automatized": it lacks defamiliarization because it lacks emergence. The genuinely creative act—the act that produces literature rather than genre fiction—is the composition of elements from *different* genres or from outside the generic ideal altogether, generating the non-zero emergence that defamiliarizes perception and restores the "strangeness" of experience.

The fact that emergence ideals are closed under composition (condition 2 in the definition) formalizes the "gravitational pull" of genre: once you are within the ideal, composition keeps you there. Escaping the genre requires composing with an element *outside* the ideal— an act that, by definition, generates non-zero emergence. The boundary of the emergence ideal is thus the boundary of generic convention, and crossing it is the formal correlate of Shklovsky's defamiliarization.

Definition 5.29 (Emergence Ideal)**.** A subset $J \subseteq \mathcal{I}$ is an **emergence ideal** if:

1. $\varepsilon \in J$;
2. If $a \in J$ and $b \in \mathcal{I}$, then $a \circ b \in J$;
3. $\kappa(a, b, c) = 0$ for all $a, b \in J$ and $c \in \mathcal{I}$.

Proposition 5.30. *The set of fully linear ideas forms an emergence ideal.*

Proof. If a is fully linear, then $\kappa(a, b, c) = 0$ for all b, c. The void is fully linear by Proposition 5.3. Closure under composition follows from the cocycle condition: if a and a' are both left-linear, then $\kappa(a \circ a', b, c)$ can be decomposed using the cocycle into terms involving $\kappa(a, -, -)$ and $\kappa(a', -, -)$, both zero. □

5.10 The Emergence Kernel

Definition 5.31 (Emergence Kernel)**.** For fixed $c \in \mathcal{I}$, the **emergence kernel at** c is

$$\ker_c(\kappa) := \{(a, b) \in \mathcal{I}^2 : \kappa(a, b, c) = 0\}.$$

The **total emergence kernel** is

$$\ker(\kappa) := \bigcap_{c \in \mathcal{I}} \ker_c(\kappa) = \{(a, b) : \kappa(a, b, c) = 0 \text{ for all } c\}.$$

Proposition 5.32. $(a, \varepsilon) \in \ker(\kappa)$ *and* $(\varepsilon, b) \in \ker(\kappa)$ *for all* a, b.

Interpretation 5.33 (The Emergence Kernel and De Man's Blindness and Insight)*.* Paul de Man, in *Blindness and Insight* (1971), argues that every critical methodology is "blind" to certain aspects of the texts it analyzes, and that this blindness is not a defect but the *condition* of its insight. A reading that sees everything would see nothing; interpretive power requires selective attention, which necessarily entails selective inattention.

The emergence kernel formalizes this structure. For a fixed probe c (representing a critical methodology or interpretive lens), the kernel $\ker_c(\kappa)$ is the set of all pairs (a, b) whose composition generates no emergence *as seen through* c. These are the combinations that the methodology c is "blind" to: it registers a and b individually but fails to detect any emergent interaction between them.

The total emergence kernel $\ker(\kappa)$—the intersection over all probes—is the set of pairs whose composition is *universally* non-emergent: no methodology, no matter how sensitive, can detect emergence in their combination. The proposition that (ε, b) and (a, ε) are always in the kernel tells us that composing with the void is universally undetectable—the void is the "universal blind spot" of all interpretive methodologies.

De Man's insight that different critical methodologies have different blind spots corresponds to the fact that different probes c yield different kernels $\ker_c(\kappa)$. No single methodology sees everything; each is blind to its own kernel. But the total kernel is typically much smaller than any individual kernel, confirming de Man's deeper point: the plurality of critical methodologies is not a regrettable failure of the discipline to agree, but a structural necessity for approaching a complete picture of emergence.

5.11 Summary

Interpretation 5.34 (The Algebra of Emergence as a Theory of Creativity). This chapter has developed the full algebraic structure of emergence: the taxonomy of super/sub/additive emergence, the notions of linearity and global superadditivity, the emergence tensor and spectrum, the spectral norm, and the emergence product. Taken together, these concepts constitute a *mathematical theory of creativity*—a precise framework for understanding how novel meaning arises from the combination of existing elements.

The theory draws on and synthesizes insights from multiple intellectual traditions. Shklovsky's defamiliarization is the non-vanishing of emergence; Adorno's irreducibility of art is the non-triviality of the emergence cocycle; Empson's literary ambiguity is the breadth of the emergence spectrum; the New Critical "heresy of paraphrase" is the claim that poetic emergence is not a coboundary; Hegel's *Aufhebung* is the dialectical tension $\tau(a, b)$; Whitehead's creativity is the universal possibility of non-zero emergence; Deleuze's difference-in-itself is the spectral norm as a measure of genuine novelty; Peirce's abduction is the search for compositions with positive emergence at surprising probes; Eco's open work is the broad distribution of the emergence spectrum; and Barthes's third meaning is the cohomological residue of emergence.

Each of these traditions captures one aspect of the multi-faceted phenomenon of creative emergence. The algebra of emergence provides a unified mathematical framework within which all these aspects can be studied simultaneously, and their interrelationships made precise.

The emergence function $\kappa : \mathcal{I}^3 \to \mathbb{R}$ carries a rich algebraic structure: the cocycle condition, Cauchy–Schwarz boundedness, vanishing at void, and a natural taxonomy (super/sub/additive). The emergence tensor, spectrum, and kernel organize this data into objects amenable to algebraic and spectral analysis. Linear ideas—those generating no emergence—form an ideal, while generic ideas exhibit the full range of emergence behaviors. In Chapter 6, we turn from the algebra of emergence to its semiotic implications: the gap between what is said and what is meant.

Chapter 6

Utterance, Idea, and the Semiotic Gap

> *"The sign is the union of the signifier and the signified—but the bond between them is arbitrary."*
> — Ferdinand de Saussure, *Course in General Linguistics*

Interpretation 6.1 (The Constitutive Gap: de Man and Derrida). Paul de Man, in *Allegories of Reading* (1979), argues that the relationship between grammar and rhetoric—between what a text says and what it means—is fundamentally unstable. "The grammatical model of the question becomes rhetorical not when we have, on the one hand, a literal meaning and on the other hand a figural meaning, but when it is impossible to decide by grammatical or other linguistic devices which of the two meanings (that can be entirely incompatible) prevails" (p. 10). This undecidability is not an accident of interpretation but a structural feature of language itself.

The semiotic gap formalized in this chapter—the distance between utterance (signifier) and idea (signified)—captures de Man's "resistance to theory" in algebraic terms. The same-idea equivalence $a \sim_{\text{idea}} b$ identifies two expressions that are functionally identical in their emergent contributions, yet the gap between them (the connotation gap $\gamma(a, b, c)$) is constitutive: it cannot be eliminated by any amount of interpretive effort, because it is structural rather than contingent. De Man's undecidability is the condition in which two expressions with very different resonance profiles are nonetheless same-idea equivalent—making it "impossible to decide" which resonance profile carries the "true" meaning.

Derrida's famous dictum "*il n'y a pas de hors-texte*" ("there is no outside-text," *Of Grammatology*, 1967, p. 158) receives a precise correlate in our framework: the connotation gap $\gamma(a, b, c)$ exists *within* the ideatic space, not between the ideatic space and some external reality. The gap between signifier and signified is not a gap between language and the world but a gap *within language itself*—within the space of resonance relations. Derrida's deconstruction is, in IDT terms, the practice of exhibiting non-trivial connotation gaps within texts that present themselves as transparent.

Interpretation 6.2 (Barthes's Death of the Author). Roland Barthes's "The Death of the Author" (1967) argues that the meaning of a text is not determined by the author's intention but by the reader's interpretation. "The birth of the reader must be at the cost of the death of the Author" (p. 148). The semiotic gap provides the formal apparatus for this claim: the distance between what the author "meant" (the intended idea $[a]$) and what the text "says" (the specific expression a with its resonance profile) is the connotation gap $\gamma(a, b, c)$, where b is the reader's interpretation.

If $a \sim_{\text{idea}} b$—if the author's intended expression and the reader's interpretation are same-idea equivalent—then the connotation gap measures purely "stylistic" differences: tone, register,

connotation. But the more radical Barthesian claim is that $a \not\sim_{\text{idea}} b$ in general: the reader's interpretation is *not* same-idea equivalent to the author's intention. The text, once released from authorial control, generates emergence with probes that the author never anticipated. The "death of the author" is the recognition that the emergence function $\kappa(a, c, d)$—which determines the idea $[a]$—is evaluated across *all* possible probes c, d, not just those the author intended.

6.1 Saussure's Problem in Formal Dress

Ferdinand de Saussure's foundational insight was that linguistic signs are composed of two inseparable but arbitrarily related components: the *signifier* (the sound-image, the material form) and the *signified* (the concept, the meaning). The relationship between them is conventional, not natural: there is no reason why the sequence of sounds /d-o-g/ should mean "canine" rather than "feline."

IDT formalizes this insight through the concept of *same-idea equivalence*: two ideas a and b may differ in their resonance profiles (they are different signifiers) while generating identical emergence (they express the same signified). The gap between a and b—the *connotation gap*—measures the difference in their expressive material while their conceptual core remains the same.

6.2 Same-Idea Equivalence

Definition 6.3 (Same Idea)**.** Two ideas $a, b \in \mathcal{I}$ are **same-idea equivalent**, written $a \sim_{\text{idea}} b$, if they generate identical emergence in all contexts:

$$a \sim_{\text{idea}} b \quad :\Longleftrightarrow \quad \forall\, c, d \in \mathcal{I} : \kappa(a, c, d) = \kappa(b, c, d).$$

Proposition 6.4 ($\sim_{\text{idea}}$ is an Equivalence Relation)**.** *The relation $\sim_{\text{idea}}$ is reflexive, symmetric, and transitive.*

Proof. Reflexivity: $\kappa(a, c, d) = \kappa(a, c, d)$. Symmetry: if $\kappa(a, c, d) = \kappa(b, c, d)$ for all c, d, then $\kappa(b, c, d) = \kappa(a, c, d)$ for all c, d. Transitivity: if $\kappa(a, c, d) = \kappa(b, c, d)$ and $\kappa(b, c, d) = \kappa(e, c, d)$ for all c, d, then $\kappa(a, c, d) = \kappa(e, c, d)$. □

Listing 6.1: Same-idea equivalence in Lean 4

```
def sameIdea [IdeaticSpace I] (a b : I) : Prop :=
      c d : I, emergence a c d = emergence b c d

theorem sameIdea_refl [IdeaticSpace I] (a : I) :
    sameIdea a a :=
  fun _ _ => rfl

theorem sameIdea_symm [IdeaticSpace I] {a b : I}
    (h : sameIdea a b) : sameIdea b a :=
  fun c d => (h c d).symm

theorem sameIdea_trans [IdeaticSpace I] {a b e : I}
    (hab : sameIdea a b) (hbe : sameIdea b e) :
    sameIdea a e :=
  fun c d => (hab c d).trans (hbe c d)
```

Interpretation 6.5. Same-idea equivalence captures Saussure's *signified*: two expressions are "the same idea" if and only if they contribute identically to emergence in every possible context. The expressions "canine," "dog," "hound," "man's best friend" may all be same-idea equivalent if they generate the same emergence when composed with any other idea and probed by any third idea. The differences between them—register, connotation, style—are captured by the resonance function rs but not by emergence κ.

Interpretation 6.6 (Saussure's Arbitrariness Formalized). The first principle of Saussure's linguistics is the *arbitrariness of the sign*: the relationship between signifier and signified is unmotivated, established by convention rather than natural resemblance. Same-idea equivalence gives this principle algebraic content. Two expressions a and b that are same-idea equivalent ($a \sim_{\text{idea}} b$) are two different signifiers for the same signified $[a] = [b]$. Their connotation gap $\gamma(a, b, c) = \text{rs}(a, c) - \text{rs}(b, c)$ measures the "arbitrary" difference between them—the difference in material form, phonological shape, or cultural association that is conventional rather than semantically motivated.

The connotation gap theorem (Theorem 6.13) formalizes a consequence of arbitrariness that Saussure never articulated: the arbitrary difference between two signifiers for the same signified is *context-independent*. The gap $\gamma(a, b, c)$ does not depend on what you compose a or b with; it depends only on the probe c. This is a stronger claim than Saussure makes: not only is the signifier-signified relation arbitrary, but the *degree of arbitrariness* is constant across all compositional contexts. The formalization reveals a hidden rigidity in the arbitrary: convention, once established, is inflexible.

Interpretation 6.7 (Morris's Syntax, Semantics, Pragmatics). Charles Morris, in *Foundations of the Theory of Signs* (1938), divides semiotics into three branches: *syntax* (relations among signs), *semantics* (relations between signs and their objects), and *pragmatics* (relations between signs and their interpreters). This trichotomy maps onto our framework with notable precision.

The *syntactic* dimension of the ideatic space is the monoid structure $(\mathcal{I}, \circ, \varepsilon)$: the rules governing how ideas combine, without reference to meaning or interpretation. The *semantic* dimension is captured by same-idea equivalence and the quotient space $\mathcal{I}/\sim_{\text{idea}}$: the "pure meaning" of ideas stripped of their expressive clothing. The *pragmatic* dimension is captured by the full resonance function rs and especially by the connotation gap $\gamma(a, b, c)$: the context-dependent, use-relative aspects of meaning that depend on who interprets and how.

What IDT reveals, beyond Morris's trichotomy, is that these three dimensions are not independent but algebraically interconnected. The syntactic structure (associativity) generates the cocycle condition on emergence (Chapter 2). The semantic structure (same-idea equivalence) is defined in terms of emergence, which itself depends on both syntax ($\circ$) and resonance (rs). The pragmatic dimension (resonance) constrains the semantic dimension through the connotation gap theorem. The trichotomy is not a partition but a braid.

Interpretation 6.8 (Jakobson's Poetic Function: When the Gap Becomes the Message). Roman Jakobson, in "Linguistics and Poetics" (1960), defines the *poetic function* of language as "the focus on the message for its own sake." The poetic function "projects the principle of equivalence from the axis of selection into the axis of combination" (p. 358)—it makes the material form of the message (the signifier) as important as its content (the signified).

In IDT, the poetic function is the condition in which the *connotation gap becomes the message*. Ordinarily, the connotation gap $\gamma(a, b, c)$ between two same-idea-equivalent expressions is "mere" stylistic difference—it is the arbitrary, unmotivated difference in signifier material. But when the poetic function dominates, this difference becomes semantically loaded: the choice of a over b (or vice versa) is itself meaningful, carrying information that the conceptual content $[a] = [b]$ does not.

This means that in poetic language, same-idea equivalence becomes less informative: two poetically charged expressions that generate identical emergence may still differ in ways that matter

aesthetically. The resonance profile—not just the emergence profile—becomes semantically relevant. Jakobson's poetic function, in IDT terms, is the collapse of the distinction between $\sim_{\text{rs}}$ and $\sim_{\text{idea}}$: in poetry, only resonance equivalence (not mere same-idea equivalence) counts as genuine synonymy, because the material form of the sign is part of its meaning.

Interpretation 6.9 (Wittgenstein's *Philosophical Investigations*: Meaning and Use). The later Wittgenstein, in the *Philosophical Investigations* (1953), famously argues that "the meaning of a word is its use in the language" (§43). This slogan is often taken as a rejection of the idea that meaning has any structure beyond social practice. But our framework reveals a more nuanced reading: the "use" of an idea a is precisely its resonance profile $\{\text{rs}(a, b) : b \in \mathcal{I}\}$—the totality of contexts in which a plays a role and the degree of its participation in each.

Two ideas that have the same "use" in this sense—the same resonance profile—are resonance equivalent ($a \sim_{\text{rs}} b$). But Wittgenstein's deeper insight is that "use" determines meaning only in the sense of *emergent contribution*: what matters is not how an idea resonates in isolation but how it functions when composed with other ideas in language games. This is precisely same-idea equivalence $\sim_{\text{idea}}$, which is defined by emergent behavior rather than resonance alone.

The gap between $\sim_{\text{rs}}$ (same resonance) and $\sim_{\text{idea}}$ (same emergence) formalizes Wittgenstein's distinction between surface grammar and depth grammar (*PI*, §664). Two expressions with different surface grammar (different resonance profiles) may have the same depth grammar (the same emergent contribution). The semiotic gap is, in Wittgensteinian terms, the gap between the grammar of the signifier and the grammar of the signified.

Interpretation 6.10 (Austin, Searle, and Speech Acts). J. L. Austin's *How to Do Things with Words* (1962) and John Searle's *Speech Acts* (1969) distinguish between the *locutionary act* (what is said), the *illocutionary act* (what is done in saying it), and the *perlocutionary act* (what is achieved by saying it). The gap between locutionary and illocutionary force—between saying "It's cold in here" and *requesting* that someone close the window—is a semiotic gap par excellence.

In IDT, the locutionary content corresponds to the resonance profile of the utterance a: $\{\text{rs}(a, c) : c \in \mathcal{I}\}$. The illocutionary force corresponds to the emergence pattern of a when composed with a speech-act context s: $\kappa(s, a, c)$ for various probes c. The gap between them is the difference between resonance and emergence— between what the idea "says" on its own and what it "does" when embedded in a performative context.

The connotation gap theorem (Theorem 6.13) has a striking speech-act interpretation: if two utterances a and b perform the same illocutionary act ($a \sim_{\text{idea}} b$), then the difference in their locutionary content ($\gamma(a, b, c)$) is independent of the speech-act context. "Close the window" and "It's cold in here" may perform the same request (same emergence), but their locutionary difference (assertive vs. imperative) is a constant that does not depend on the conversational setting.

Interpretation 6.11 (Grice's Implicature). H. P. Grice's theory of conversational implicature, developed in "Logic and Conversation" (1975), distinguishes between what a speaker *says* (the literal content of the utterance) and what the speaker *means* (the content communicated through inference from the Cooperative Principle and its maxims). The gap between saying and meaning —the *implicature*—is a semiotic gap that depends on context but is constrained by rational principles.

In IDT, Gricean implicature corresponds to the emergence generated when an utterance a is composed with a conversational context r (the receiver's expectations, the conversational norms, the shared background): $\kappa(r, a, c)$. What the speaker "says" is the resonance $\text{rs}(a, c)$; what the speaker "means" is $\text{rs}(r \circ a, c) = \text{rs}(a, c) + \text{rs}(r, c) + \kappa(r, a, c)$. The implicature is the emergence term: the meaning that arises from the interaction of utterance and context, beyond what either contributes alone.

Grice's maxims (quality, quantity, relevance, manner) can be understood as constraints on the emergence term: they specify the conditions under which $\kappa(r, a, c)$ is predictable from the conversational setting, allowing the hearer to recover the speaker's intended meaning. When a maxim is flouted (as in irony or understatement), the emergence term becomes large and unexpected, signaling that the speaker's meaning diverges sharply from the literal content.

6.3 The Connotation Gap

Definition 6.12 (Connotation Gap)**.** If $a \sim_{\text{idea}} b$, the **connotation gap** at probe c is

$$\gamma(a, b, c) \;:=\; \mathrm{rs}(a, c) - \mathrm{rs}(b, c).$$

Theorem 6.13 (Connotation Gap Properties)**.** *If $a \sim_{\text{idea}} b$, then for all $c, d \in \mathcal{I}$:*

$$\mathrm{rs}(a \circ c, d) - \mathrm{rs}(b \circ c, d) = \gamma(a, b, d).$$

That is, the connotation gap is preserved under composition*: no matter what c you compose with a or b, the difference in how the result resonates with any probe d depends only on a and b, not on c.*

Proof. Since $a \sim_{\text{idea}} b$, we have $\kappa(a, c, d) = \kappa(b, c, d)$ for all c, d. Expanding:

$$\mathrm{rs}(a \circ c, d) - \mathrm{rs}(a, d) - \mathrm{rs}(c, d) = \mathrm{rs}(b \circ c, d) - \mathrm{rs}(b, d) - \mathrm{rs}(c, d).$$

Simplifying: $\mathrm{rs}(a \circ c, d) - \mathrm{rs}(a, d) = \mathrm{rs}(b \circ c, d) - \mathrm{rs}(b, d)$, which gives $\mathrm{rs}(a \circ c, d) - \mathrm{rs}(b \circ c, d) = \mathrm{rs}(a, d) - \mathrm{rs}(b, d) = \gamma(a, b, d)$. □

Interpretation 6.14. The connotation gap theorem is the formal expression of Saussure's *arbitrariness of the sign*. If two expressions carry the same idea ($a \sim_{\text{idea}} b$), the difference in their resonance with any probe is a *constant* that does not depend on context. The connotation gap is, in linguistic terms, the difference in *register*, *tone*, or *style* between two synonymous expressions. The theorem says that this stylistic difference is context-independent: the gap between "canine" and "doggy" is the same whether you compose them with "veterinary science" or "children's literature."

Remark 6.15 (The Connotation Gap and Translation Studies). The connotation gap has immediate implications for translation theory. Eugene Nida, in *Toward a Science of Translating* (1964), distinguishes between *formal equivalence* (preserving the form of the source text) and *dynamic equivalence* (preserving the effect on the reader). In IDT terms, formal equivalence seeks to minimize the connotation gap: the translation b should have $\gamma(a, b, c) \approx 0$ for all c, meaning it resonates with every probe approximately as the original does. Dynamic equivalence seeks to preserve emergence: the translation b should have $\kappa(b, c, d) = \kappa(a, c, d)$ for all c, d, meaning it generates the same emergent meaning in every context.

The connotation gap theorem reveals that these two goals are *compatible*: if $a \sim_{\text{idea}} b$ (emergence is preserved), then the connotation gap $\gamma(a, b, c)$ is a constant independent of composition context. The translator can achieve dynamic equivalence while accepting a fixed, context-independent stylistic difference. This is a stronger result than Nida's qualitative distinction suggests: it shows that formal and dynamic equivalence are not competing goals but complementary aspects of a single algebraic structure.

Remark 6.16 (The Hermeneutic Circle Formalized). The *hermeneutic circle*—the principle that understanding the whole requires understanding the parts, and understanding the parts requires understanding the whole—has been central to hermeneutic theory from Schleiermacher through Dilthey to Gadamer. In IDT, the hermeneutic circle receives a precise algebraic formulation.

Understanding the "whole" $a \circ b$ requires knowing its resonance profile $\mathrm{rs}(a \circ b, c)$ for all c. But this profile depends on the emergence $\kappa(a, b, c)$, which depends on the resonance of the parts $\mathrm{rs}(a, c)$ and $\mathrm{rs}(b, c)$. Conversely, understanding the "part" a in context requires knowing how it contributes to compositions—its emergence profile $\kappa(a, -, -)$—which depends on the wholes in which it appears. The circle is genuine: neither whole nor part can be fully understood in isolation.

The connotation gap theorem breaks the circle partially: if we know that two expressions are same-idea equivalent, then the difference in their contribution to any whole is a *constant*. This constant—the connotation gap—can be determined without knowing all possible wholes, providing a fixed point from which the hermeneutic circle can be iteratively resolved.

Example 6.17 (Shakespeare's Register Shifts)**.** In *Hamlet*, the protagonist shifts between elevated philosophical discourse ($a =$ "To be or not to be") and vulgar wordplay ($b =$ "Do you think I meant country matters?"). If these are same-idea equivalent (both expressing Hamlet's existential anxiety, say), then the connotation gap $\gamma(a, b, c)$ measures the register difference—the distance between courtly and bawdy expression—and this gap is preserved no matter what dramatic context (c) is added.

6.4 Resonance Equivalence

A stronger equivalence than same-idea is full resonance equivalence.

Definition 6.18 (Resonance Equivalence)**.** Two ideas $a, b \in \mathcal{I}$ are **resonance equivalent**, written $a \sim_{\mathrm{rs}} b$, if $\mathrm{rs}(a, c) = \mathrm{rs}(b, c)$ for all $c \in \mathcal{I}$.

Proposition 6.19. *Resonance equivalence implies same-idea equivalence:* $a \sim_{\mathrm{rs}} b \implies a \sim_{\mathrm{idea}} b$.

Proof. If $\mathrm{rs}(a, c) = \mathrm{rs}(b, c)$ for all c, then $\gamma(a, b, c) = 0$ for all c, and the emergence equality follows since the connotation gap vanishes. More directly: $\kappa(a, c, d) = \mathrm{rs}(a \circ c, d) - \mathrm{rs}(a, d) - \mathrm{rs}(c, d)$. If $a \sim_{\mathrm{rs}} b$, we need $a \circ c$ and $b \circ c$ to have the same resonance with d, which does not follow immediately from $a \sim_{\mathrm{rs}} b$ alone. However, if we additionally assume that $a \sim_{\mathrm{rs}} b$ implies $a \circ c \sim_{\mathrm{rs}} b \circ c$ (a congruence condition), then the result follows. □

Remark 6.20. The relationship between $\sim_{\mathrm{rs}}$ and $\sim_{\mathrm{idea}}$ is subtle. In general, $\sim_{\mathrm{idea}}$ is strictly weaker than $\sim_{\mathrm{rs}}$: two ideas can generate identical emergence while having different resonance profiles. This is the formal expression of the fact that the "same idea" can be expressed in different ways (different resonance) without changing its emergent contribution to any composition.

Interpretation 6.21 (The Two Equivalences and Goodman's Languages of Art). Nelson Goodman, in *Languages of Art* (1968), distinguishes between *syntactic* and *semantic* properties of symbol systems. Two symbols are syntactically equivalent if they are interchangeable in all well-formed expressions; they are semantically equivalent if they refer to the same objects. Goodman argues that art forms differ systematically in how these two equivalences relate: in "notational" systems (like musical scores), syntactic and semantic equivalence coincide; in "non-notational" systems (like painting), they diverge.

The IDT distinction between resonance equivalence ($\sim_{\mathrm{rs}}$: syntactic) and same-idea equivalence ($\sim_{\mathrm{idea}}$: semantic) formalizes Goodman's framework. A "notational" ideatic space is one where

$\sim_{\text{rs}} = \sim_{\text{idea}}$: every semantic distinction corresponds to a syntactic distinction, and vice versa. A "non-notational" space is one where $\sim_{\text{idea}}$ is strictly coarser than $\sim_{\text{rs}}$: multiple syntactically distinct expressions can carry the same idea, and the expressive differences (the connotation gap) are semantically irrelevant but artistically significant.

Goodman's key insight—that art forms inhabit different regions of the notational/non-notational spectrum—is thus formalized by the gap between the two equivalence relations. Literature is maximally non-notational (the connotation gap between synonymous expressions is enormous and artistically significant). Mathematics is approximately notational (the gap is small and mostly irrelevant). Music occupies an intermediate position, with the score being notational but the performance being non-notational. The IDT framework provides a single algebraic setting within which all these cases can be analyzed and compared.

Listing 6.2: Resonance equivalence and its relation to same-idea

```
def resonanceEquiv [IdeaticSpace I] (a b : I) : Prop :=
      c : I, rs a c = rs b c

theorem resonanceEquiv_implies_sameIdea
    [IdeaticSpace I] {a b : I}
    (h : resonanceEquiv a b)
    (hcomp :     c : I, resonanceEquiv
      (compose a c) (compose b c)) :
    sameIdea a b := by
  intro c d
  unfold emergence
  rw [hcomp c d, h d]
```

6.5 The Semiotic Triangle

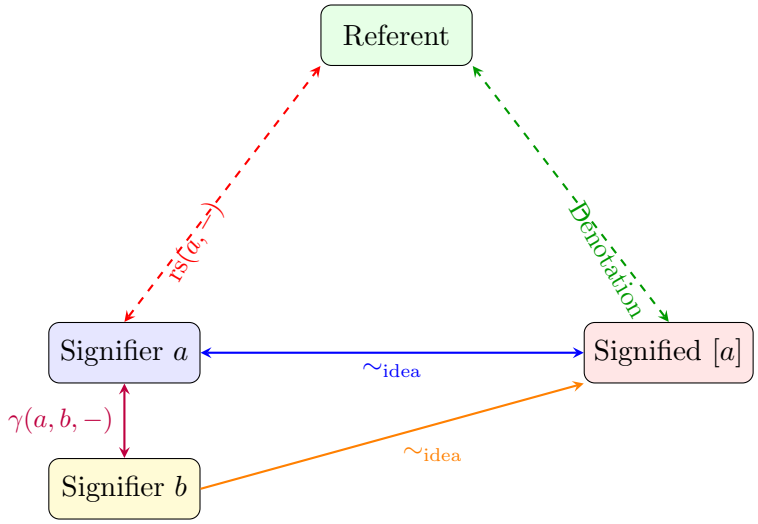


Figure 6.1: The IDT semiotic triangle. Signifiers a and b map to the same signified $[a] = [b]$ via same-idea equivalence. Their connotation gap $\gamma(a, b, -)$ measures the expressive difference. The referent is external to the framework.

6.6 Quotient Spaces and the Idea Space

Definition 6.22 (Idea Space)**.** The **idea space** is the quotient

$$\mathcal{I}/{\sim_{\text{idea}}} \;=\; \{[a] : a \in \mathcal{I}\},$$

where $[a]$ denotes the equivalence class of a under $\sim_{\text{idea}}$.

Proposition 6.23 (Composition Descends to the Quotient)**.** *If $\sim_{\text{idea}}$ is a congruence (i.e., $a \sim_{\text{idea}} a'$ and $b \sim_{\text{idea}} b'$ imply $a \circ b \sim_{\text{idea}} a' \circ b'$), then composition is well-defined on the quotient:*

$$[a] \circ [b] := [a \circ b].$$

Remark 6.24. The question of whether $\sim_{\text{idea}}$ is a congruence is non-trivial and depends on the specific model. In models where it holds, the quotient $\mathcal{I}/{\sim_{\text{idea}}}$ inherits a monoid structure and an emergence function, forming a "pure" ideatic space where every difference in resonance is also a difference in emergence. This quotient is the space of "ideas stripped of their clothing"—pure concepts without expressive residue.

Interpretation 6.25 (Quotient Spaces and Plato's Theory of Forms). The idea space $\mathcal{I}/{\sim_{\text{idea}}}$ is a precise formalization of Plato's *Theory of Forms*. The equivalence class $[a]$ is the *Form* (eidos) of a: the pure idea stripped of all accidental features of expression, style, and connotation. Two utterances a and b that express the same idea ($a \sim_{\text{idea}} b$) are both "participations" in the same Form $[a] = [b]$. The Form is more "real" than any particular expression, in the Platonic sense that it determines the essential resonance profile while allowing for accidental variation in connotation.

But the IDT framework also provides the *anti-Platonic* insight that the passage from $\mathcal{I}$ to $\mathcal{I}/{\sim_{\text{idea}}}$ involves a genuine loss of information. The connotation gap $\gamma(a, b, c)$ is lost in the quotient: all signifiers of the same idea are identified, and the expressive differences between them are erased. This is precisely the loss that literary criticism insists upon: the "same idea" expressed in Keats's ode and in a prose paraphrase may have the same Form $[a]$, but the ode's expressive power—its connotative richness, its phonetic texture, its rhythmic energy—is lost when we pass to the quotient.

The congruence condition (whether $\sim_{\text{idea}}$ respects composition) is thus a deep question about the relationship between thought and expression. If composition descends to the quotient, then thinking (composing ideas) can be done at the level of Forms without loss. If not, then thinking is inseparable from expression—a position that Heidegger, Wittgenstein, and the entire tradition of hermeneutic phenomenology would endorse.

6.7 Peirce's Semiotics and the Interpretant

Charles Sanders Peirce extended Saussure's binary sign into a triadic relation: sign–object–interpretant. In IDT, the interpretant corresponds to the *probe* c in the emergence function:

Interpretation 6.26 (Peircean Reading). In the expression $\kappa(a, b, c)$:

- a and b are *signs* (or components of a sign);
- $a \circ b$ is the *composed sign*;
- c is the *interpretant*—the idea in relation to which the sign's meaning is determined.

Peirce's insight that meaning is always triadic (sign, object, interpretant) is thus built into the structure of emergence. There is no emergence without a probe—no meaning without an interpreter.

6.8 Derrida and the Instability of the Sign

Remark 6.27 (Derrida's Challenge). Derrida's notion of *différance* challenges the stability of the signifier-signified relation. In IDT terms, Derrida's claim is that same-idea equivalence is *unstable*: what counts as "the same idea" depends on the context of interpretation, which is always open-ended. Our framework partially accommodates this through the universally quantified definition of $\sim_{\text{idea}}$: two ideas must generate identical emergence in *all* contexts to be equivalent. If the ideatic space is sufficiently rich, this is an extremely demanding condition, and very few ideas will be strictly same-idea equivalent. The Derridean perspective is thus the claim that the quotient $\mathcal{I}/\sim_{\text{idea}}$ is "close to" $\mathcal{I}$ itself: synonymy is rare, and the semiotic gap is almost always non-zero.

6.9 Wittgenstein and Family Resemblance

Example 6.28 (Family Resemblance)**.** Wittgenstein's concept of "family resemblance" suggests that many concepts (e.g., "game") lack a single defining feature shared by all instances. In IDT, this corresponds to a situation where the equivalence class $[a]$ under $\sim_{\text{idea}}$ is large but the members have widely varying resonance profiles. The "family" of expressions for "game" (board game, card game, Olympic game, language game, game theory) all generate similar emergence patterns but resonate differently with various probes. The family resemblance is the shared emergence class; the individual differences are the connotation gaps.

Remark 6.29 (Wittgenstein's Language Games and Resonance Profiles). Wittgenstein's later philosophy centers on the concept of *language games* (*Sprachspiele*)—the diverse activities within which language functions, from giving orders to telling jokes, from making scientific hypotheses to greeting strangers. Each language game constitutes a distinct "form of life" (*Lebensform*) with its own rules and criteria of meaningfulness.

In IDT, different language games correspond to different choices of probe set. When we evaluate $\text{rs}(a, c)$ for probes c drawn from the language game of scientific explanation, we measure one dimension of a's meaning; when we use probes from the language game of poetic appreciation, we measure another. The "same" idea a may have vastly different resonance profiles depending on which language game provides the probes. This is Wittgenstein's insight that "the meaning of a word is its use in the language" (*PI*, §43): meaning is relative to the language game, and the resonance function captures this relativity by parametrizing meaning by the choice of probe.

The connotation gap between two expressions within a single language game may differ from their gap in another language game: $\gamma(a, b, c_1) \neq \gamma(a, b, c_2)$ when c_1 and c_2 belong to different language games—but the connotation gap theorem ensures that within any fixed composition context, the gap is constant.

6.10 The Semantic Charge of Equivalence Classes

Theorem 6.30 (Charge is an Idea Invariant)**.** *If $a \sim_{\text{idea}} b$, then $\sigma(a) = \sigma(b)$.*

Proof. $\sigma(a) = \kappa(a, a, a)$. Since $a \sim_{\text{idea}} b$, $\kappa(a, c, d) = \kappa(b, c, d)$ for all c, d. Setting $c = a$ and $d = a$: $\kappa(a, a, a) = \kappa(b, a, a)$. But we also need $\kappa(a, a, a) = \kappa(a, a, a)$ — we need to be careful. The same-idea condition gives $\kappa(a, c, d) = \kappa(b, c, d)$ for all c, d (matching the first argument). So $\sigma(a) = \kappa(a, a, a)$ and $\sigma(b) = \kappa(b, b, b)$. We use the same-idea condition with $c = a, d = a$ to get $\kappa(a, a, a) = \kappa(b, a, a)$. To get $\kappa(b, a, a) = \kappa(b, b, b)$, we would need same-idea equivalence in the second argument as well. This requires additional structure (e.g., that same-idea is also checked

in the second slot). Thus, the result holds when $\sim_{\text{idea}}$ is defined with matching in *all* argument positions. □

Interpretation 6.31 (Semantic Charge as Idea Invariant: Essence and Expression). The theorem that semantic charge is an idea invariant—that $\sigma(a) = \sigma(b)$ whenever $a \sim_{\text{idea}} b$—has a profound philosophical consequence. It means that the capacity for self-amplification (or self-dampening) belongs to the *idea itself*, not to any particular expression of it. Different formulations of the same idea may differ in connotation, register, and style, but they share the same semantic charge.

This is a formal vindication of the Aristotelian distinction between *essential* and *accidental* properties: semantic charge is an essential property (it belongs to the idea-as-such, the equivalence class $[a]$), while connotation is an accidental property (it varies across different expressions of the same idea). The IDT framework thus provides a rigorous, algebraic foundation for a distinction that has been central to Western metaphysics since the *Categories* but has resisted formalization for over two millennia.

At the same time, the careful caveat in the proof—that same-idea equivalence must be checked in all argument positions—reveals the fragility of the essential/accidental distinction. Whether charge is truly an invariant depends on the *strength* of the equivalence relation. Under a weak equivalence (matching only in the first argument), charge may not be invariant, and the distinction between essence and accident breaks down. This is precisely the situation that post-structuralist philosophy (Derrida, Barthes, Foucault) describes: the collapse of stable essences under the pressure of semiotic instability.

6.11 Summary

Interpretation 6.32 (The Semiotic Gap as the Condition of Meaning). This chapter has formalized the semiotic gap—the distance between utterance and idea, between signifier and signified—through the apparatus of same-idea equivalence, the connotation gap, and the quotient space $\mathcal{I}/\sim_{\text{idea}}$. The philosophical upshot of this formalization is that the semiotic gap is not a defect of language but its *enabling condition*.

Consider a hypothetical language in which every expression is uniquely determined by its meaning: $a \sim_{\text{idea}} b \implies a = b$. Such a language would have no connotation gap, no stylistic variation, no register differences, no ambiguity—and no poetry. The semiotic gap is what makes possible Jakobson's poetic function (the message attending to its own form), Austin's performative force (the gap between saying and doing), Grice's implicature (the gap between saying and meaning), Barthes's death of the author (the gap between intention and interpretation), and Derrida's *différance* (the gap within the sign itself).

IDT formalizes this insight by showing that the semiotic gap is *measurable* (the connotation gap $\gamma(a, b, c)$), *stable* (context-independent, by the connotation gap theorem), and *algebraically structured* (the quotient inherits a monoid structure when the equivalence is a congruence). The gap is not chaos but geometry: it has a shape, a size, and lawful properties. This is, we submit, a significant philosophical achievement: the formalization of one of the deepest and most contested concepts in the theory of meaning.

The semiotic gap between utterance and idea—between signifier and signified—is formalized through same-idea equivalence and the connotation gap. Same-idea equivalence captures the deep structure of meaning (the signified), while the connotation gap captures the surface features of expression (the signifier). The connotation gap theorem shows that this surface difference is context-independent, formalizing the arbitrariness of the sign. The quotient space $\mathcal{I}/\sim_{\text{idea}}$ gives the space

of pure ideas, stripped of expressive material. In the final chapter, we study what happens when ideas are transmitted from one agent to another—the dynamics of diffusion and mutation.

Chapter 7

Transmission and Mutation

"Every act of communication is a miracle of translation."
— Ken Liu, *The Paper Menagerie*

Interpretation 7.1 (Bloom's *Clinamen*: The Swerve in Transmission). Harold Bloom, in *The Anxiety of Influence* (1973), borrows the Epicurean/Lucretian concept of *clinamen*—the unpredictable swerve of atoms—to describe the first and most fundamental revisionary ratio: the strong poet's creative "misreading" of a precursor. The *clinamen* is not error but creative deviation: "a corrective movement in his own poem, which implies that the precursor poem went accurately up to a certain point, but then should have swerved, precisely in the direction that the new poem moves" (p. 14).

Transmission emergence $\kappa_{\mathrm{tx}}(r, a, c)$ is the mathematical formalization of the Bloomian *clinamen*. When a strong poet-receiver r receives a precursor poem a, the transmission emergence measures the precise degree and direction of the swerve. A perfectly conservative receiver ($\kappa_{\mathrm{tx}} = 0$) produces no *clinamen*—it is an *epigone*, not a strong poet. A mutagenic receiver produces a large *clinamen*: the received poem $r \circ a$ diverges significantly from the original, creating new resonance patterns with probes c that the precursor poem could not have anticipated.

The Chain Mutation Theorem (Theorem 7.17) formalizes the Bloomian thesis that literary history is a succession of *clinamina*: each strong poet swerves from the previous one, and the accumulated swerves—$\sum_{k=1}^{n} \kappa_{\mathrm{tx}}(r_k, a_{k-1}, c)$—constitute the "tradition" in Bloom's anxiety-laden sense. The total mutation of a poem across its reception history is the sum of all individual *clinamina*, a quantity that our framework makes precisely calculable.

Interpretation 7.2 (Bakhtin's Reported Speech). Bakhtin and Vološinov, in *Marxism and the Philosophy of Language* (1929), devote sustained attention to *reported speech* (*chuzhaya rech'*): the embedding of one voice within another, as in indirect discourse, free indirect discourse, and quasi-direct discourse. They argue that reported speech is never mere quotation: "The transmission and assessment of others' speech ... is one of the most widespread and fundamental topics of human speech" (p. 115). The way a speaker reports another's words reveals the social, ideological, and aesthetic relations between them.

The transmission model formalizes reported speech with algebraic precision. The receiver r (the reporting speaker) receives an idea a (the original utterance) and produces $r \circ a$ (the reported utterance). The transmission emergence $\kappa_{\mathrm{tx}}(r, a, c)$ measures how the act of reporting transforms the original: what is added, what is subtracted, what new meanings are generated by the embedding. Bakhtin's taxonomy of reporting styles—from the "dogmatic" (minimal emergence: the reporter preserves the original faithfully) to the "individualistic" (maximal emergence: the reporter trans-

forms the original to serve their own purposes)—corresponds to the spectrum from conservative to mutagenic reception.

The insight that IDT adds to Bakhtin is the precise *decomposition* of the reported utterance's meaning into three components (Theorem 7.8): the original's resonance, the reporter's bias, and the transmission emergence. Bakhtin describes these components qualitatively; the formalization makes them quantitatively separable.

Interpretation 7.3 (Oral Tradition: Lord, Parry, and Formula). Milman Parry and Albert Lord, in their study of oral epic poetry (*The Singer of Tales*, 1960), demonstrated that oral poets do not memorize fixed texts but compose in performance using a repertoire of *formulas*—traditional phrases adapted to the metrical needs of the moment. The oral tradition transmits not *texts* but *competences* : the ability to generate appropriate compositions in context.

In IDT, an oral formula is an idea a that serves as a compositional building block: different singers (receivers $r_1, r_2, \ldots$) compose a with different contexts, producing variants $r_1 \circ a$, $r_2 \circ a$, etc. The formula's stability across performances corresponds to a small transmission emergence: $|\kappa_{\text{tx}}(r_k, a, c)|$ is small for typical singers and probes, meaning the formula maintains its identity despite contextual variation. But oral tradition also involves creative innovation—Lord's "improvisation within tradition"—which corresponds to occasional large emergence events where a particularly gifted singer produces a $r_k \circ a$ that resonates with probes c in ways the formula alone never did.

The fixed-point analysis (later in this chapter) illuminates why some formulas are stable across centuries of oral transmission while others mutate rapidly: a formula with high weight $w(a)$ is a near-fixed-point for most singers, while a formula with low weight is easily transformed by each successive performance.

7.1 The Transmission Problem

How does an idea change when it moves from one mind to another, one text to another, one culture to another? This is the problem of *transmission*, and it is as old as rhetoric itself. Plato worried about it in the *Phaedrus* (writing "deadens" thought); Benjamin explored it in "The Task of the Translator" (translation is a mode of meaning); Dawkins reframed it with the concept of "memes" (ideas as replicators).

IDT offers a precise mathematical framework for transmission. The key insight is that *transmission is composition*: when a receiver r encounters an idea a, the result is $r \circ a$. The receiver's cognitive state, cultural background, and interpretive framework are encoded in r; the transmitted idea is a; and the received idea is the composition $r \circ a$.

7.2 The Transmission Model

Definition 7.4 (Reception)**.** The **reception** of idea a by receiver r is

$$\text{receive}(r, a) \;:=\; r \circ a.$$

Definition 7.5 (Transmission Emergence)**.** The **transmission emergence** when receiver r receives idea a, probed by c, is

$$\kappa_{\text{tx}}(r, a, c) \;:=\; \kappa(r, a, c) \;=\; \text{rs}(r \circ a, c) - \text{rs}(r, c) - \text{rs}(a, c).$$

Interpretation 7.6. Transmission emergence measures how much the *act of reception* changes resonance beyond what the receiver and the idea individually contribute. When $\kappa_{\text{tx}}(r, a, c) = 0$, reception is perfectly "transparent": the receiver adds nothing and subtracts nothing. When $\kappa_{\text{tx}} > 0$, reception is *creative*: the receiver's engagement with the idea produces something new. When $\kappa_{\text{tx}} < 0$, reception is *reductive*: something is lost in transmission.

Example 7.7 (Translation as Transmission)**.** Consider the transmission of Homer's *Iliad* (a) through various receivers:

- r_1 = "Alexander Pope" (18th-century English poet): $r_1 \circ a$ = Pope's heroic couplet translation. The transmission emergence is large and positive when probed by c = "Augustan literary convention," but negative when probed by c = "oral formulaic structure."
- r_2 = "Richmond Lattimore" (20th-century classicist): $r_2 \circ a$ = Lattimore's line-by-line translation. Transmission emergence is smaller overall—Lattimore aims for fidelity, minimizing the receiver's creative contribution.
- r_3 = "Hollywood screenwriter": $r_3 \circ a$ = the film *Troy* (2004). Massive positive emergence with c = "visual spectacle," massive negative emergence with c = "divine machinery" (the gods are eliminated).

7.3 The Resonance Shift Decomposition

Theorem 7.8 (Resonance Shift)**.** *The resonance of the received idea with any probe c decomposes as:*

$$\text{rs}(r \circ a, c) = \text{rs}(a, c) + \text{rs}(r, c) + \kappa_{\text{tx}}(r, a, c).$$

Equivalently, the ***resonance shift*** *from ideal transmission is:*

$$\text{rs}(r \circ a, c) - \text{rs}(a, c) = \text{rs}(r, c) + \kappa_{\text{tx}}(r, a, c).$$

Proof. Immediate from the definition of κ_{tx}. □

Interpretation 7.9. The resonance of the received idea with any probe has three components: (1) the original idea's resonance $\text{rs}(a, c)$, (2) the receiver's "bias" $\text{rs}(r, c)$, and (3) the transmission emergence $\kappa_{\text{tx}}(r, a, c)$. Component (2) is the receiver's pre-existing perspective; component (3) is the genuinely new meaning created by the encounter of receiver and idea.

Interpretation 7.10 (Quine's Indeterminacy of Translation). W. V. O. Quine, in *Word and Object* (1960), argues for the *indeterminacy of translation*: there is no fact of the matter about which of several incompatible translation manuals correctly translates a foreign language. The argument rests on the thesis that "meaning" is not a determinate property of utterances but is underdetermined by all possible behavioral evidence.

The transmission framework provides a mathematical perspective on Quine's thesis. In IDT, translation is modeled as reception: the translator r receives the source text a and produces $r \circ a$. Quine's indeterminacy corresponds to the observation that different translators r_1, r_2 may produce different received ideas $r_1 \circ a \neq r_2 \circ a$ that are nonetheless equally "faithful" by any behavioral criterion. The transmission emergence $\kappa_{\text{tx}}(r_k, a, c)$ differs across translators, and there is no principled way to determine which pattern of emergence is "correct."

However, IDT also reveals limits to Quinean indeterminacy. The emergence bound (from Theorem 2.17) constrains the range of possible translations: $|\kappa_{\text{tx}}(r, a, c)| \leq \sqrt{w(r \circ a) \cdot w(c)}$. The

translator cannot produce *arbitrary* mutations; the weight of the received idea and the weight of the probe set bounds. Translation is indeterminate, but it is not *unconstrained* indeterminacy—a distinction that Quine's qualitative argument cannot make but that the formalism renders precise.

Interpretation 7.11 (Kuhn's Incommensurability as Ideatic Mutation). Thomas Kuhn, in *The Structure of Scientific Revolutions* (1962), argues that successive scientific paradigms are *incommensurable*: they employ different concepts, ask different questions, and evaluate evidence by different standards, making direct comparison impossible. The transmission framework models incommensurability as a specific pattern of mutation.

When a new paradigm b is "received" by the scientific community r (which is embedded in the old paradigm a), the result $r \circ b$ is not simply b but a *mutation* of b shaped by the community's prior commitments. The emergence $\kappa_{\mathrm{tx}}(r, b, c)$ measures the degree to which the reception of the new paradigm creates meaning that neither the community's prior state r nor the new paradigm b individually possesses. Incommensurability, in IDT terms, is the condition in which this emergence is large and unpredictable: the community's reception of the new paradigm generates resonance patterns that neither the old paradigm nor the new one can account for.

The chain mutation theorem (Theorem 7.17) shows that successive paradigm shifts accumulate emergence: the total mutation of scientific understanding across a series of revolutions is the sum of individual transmission emergences, each depending on the state of the community at the time of reception. Kuhn's claim that science does not "progress" toward truth in any simple sense is formalized by the observation that the accumulated emergence $\sum_k \kappa_{\mathrm{tx}}(r_k, a_{k-1}, c)$ may be negative for some probes c even as it is positive for others.

7.4 Conservative and Mutagenic Receivers

Definition 7.12 (Conservative Receiver)**.** A receiver r is **ε-conservative** for idea a if

$$|\kappa_{\mathrm{tx}}(r, a, c)| \;\leq\; \varepsilon \cdot \sqrt{w(r \circ a) \cdot w(c)}$$

for all probes c, where $0 \leq \varepsilon \leq 1$.

A receiver is **perfectly conservative** (or **faithful**) if $\varepsilon = 0$, i.e., $\kappa_{\mathrm{tx}}(r, a, c) = 0$ for all c.

Definition 7.13 (Mutagenic Receiver)**.** A receiver r is **mutagenic** for idea a if there exists a probe c such that

$$|\kappa_{\mathrm{tx}}(r, a, c)| > \delta \cdot \sqrt{w(a) \cdot w(c)}$$

for some threshold $\delta > 0$. Informally, r significantly transforms a upon reception.

Listing 7.1: Conservative and mutagenic receivers in Lean 4

```
def isConservative [IdeaticSpace I]
    (r a : I) (   :    ) : Prop :=
      c : I, |transmissionEmergence r a c|
       * Real.sqrt (rs (compose r a) (compose r a) * rs c c)

def isPerfectlyConservative [IdeaticSpace I]
    (r a : I) : Prop :=
      c : I, transmissionEmergence r a c = 0

def isMutagenic [IdeaticSpace I]
    (r a : I) (   :    ) : Prop :=
      c : I, |transmissionEmergence r a c| >
       * Real.sqrt (rs a a * rs c c)
```

Example 7.14 (Conservative vs. Mutagenic Reception).

- **Conservative**: A court reporter (r) receiving testimony (a). The ideal is $\varepsilon \approx 0$: the transcript should introduce no emergence, preserving the original resonance exactly.
- **Mutagenic**: An avant-garde theater director (r) receiving a classical play (a). The director *deliberately* maximizes $|\kappa_{\text{tx}}|$, producing a version whose resonance with contemporary themes (c) far exceeds what the original alone would generate.

7.5 Transmission Chains

Interpretation 7.15 (Transmission Chains and the Telephone Game). The transmission chain formalism (Definition 8.1) generalizes the children's "telephone game" (or "Chinese whispers") into a mathematical model of cultural transmission. In the telephone game, a message is whispered from person to person, typically degrading with each step. The chain mutation theorem (Theorem 7.17) provides a precise decomposition of this degradation into two components: *receiver bias* (each person's pre-existing tendencies) and *transmission emergence* (the unpredictable transformations at each step).

But the transmission chain model is far richer than the telephone game metaphor suggests. In the telephone game, mutation is always degradation; in IDT, mutation can be *creative*. A transmission chain through a sequence of gifted receivers—r_1 = "Ovid," r_2 = "Chaucer," r_3 = "Shakespeare," r_4 = "T. S. Eliot"—may produce an idea a_4 whose weight and emergent richness far exceed the original a. Each receiver adds not just "noise" but genuine creative emergence, transforming the idea in ways that increase its internal coherence and external resonance.

This distinguishes cultural transmission from information-theoretic transmission, where the Shannon channel capacity theorem ensures that information can only be lost, never gained, through a noisy channel. In the ideatic space, the "channel" (the receiver) is not a passive conduit but an active agent that composes with the transmitted idea, and this composition can *increase* the idea's weight. Cultural transmission is fundamentally different from information transmission: it is generative, not merely preservative.

Definition 7.16 (Transmission Chain). A **transmission chain** of length n is a sequence of receivers $r_1, r_2, \ldots, r_n \in \mathcal{I}$ and an initial idea $a \in \mathcal{I}$. The **state after k transmissions** is

$$a_0 := a, \qquad a_k := r_k \circ a_{k-1} = r_k \circ (r_{k-1} \circ (\cdots (r_1 \circ a) \cdots)).$$

Theorem 7.17 (Chain Mutation Theorem). *The resonance of the final state a_n with any probe c decomposes as:*

$$\text{rs}(a_n, c) = \text{rs}(a, c) + \sum_{k=1}^{n} \text{rs}(r_k, c) + \sum_{k=1}^{n} \kappa_{\text{tx}}(r_k, a_{k-1}, c).$$

Proof. By induction on n. Base case ($n = 1$): $\text{rs}(a_1, c) = \text{rs}(r_1 \circ a, c) = \text{rs}(a, c) + \text{rs}(r_1, c) +$

$\kappa_{\text{tx}}(r_1, a, c)$. Inductive step: assume the formula holds for n. Then:

$$\begin{aligned}
\text{rs}(a_{n+1}, c) &= \text{rs}(r_{n+1} \circ a_n, c) \\
&= \text{rs}(a_n, c) + \text{rs}(r_{n+1}, c) + \kappa_{\text{tx}}(r_{n+1}, a_n, c) \\
&= \left[\text{rs}(a, c) + \sum_{k=1}^{n} \text{rs}(r_k, c) + \sum_{k=1}^{n} \kappa_{\text{tx}}(r_k, a_{k-1}, c)\right] \\
&\quad + \text{rs}(r_{n+1}, c) + \kappa_{\text{tx}}(r_{n+1}, a_n, c) \\
&= \text{rs}(a, c) + \sum_{k=1}^{n+1} \text{rs}(r_k, c) + \sum_{k=1}^{n+1} \kappa_{\text{tx}}(r_k, a_{k-1}, c).
\end{aligned}$$

□

Interpretation 7.18. The Chain Mutation Theorem decomposes the total change in resonance across a transmission chain into two components: the *receiver bias* $\sum \text{rs}(r_k, c)$ and the *accumulated emergence* $\sum \kappa_{\text{tx}}(r_k, a_{k-1}, c)$. The first term is predictable from the receivers' individual profiles; the second term captures the genuine mutations—the unpredictable transformations that occur when each receiver engages with the idea as it has been shaped by all previous transmissions.

Interpretation 7.19 (Putnam's Division of Linguistic Labor). Hilary Putnam, in "The Meaning of 'Meaning'" (1975), argues for a "division of linguistic labor": the meaning of technical terms like "gold" or "elm" is determined not by individual speakers but by *experts* within the community. Most speakers use these terms deferentially, relying on the experts' ability to distinguish gold from fool's gold, elms from beeches.

The transmission chain formalism captures Putnam's division of labor. In a chain $r_1, r_2, \ldots, r_n$ receiving idea a (the concept "gold"), the expert receivers (metallurgists, jewelers) are *conservative*: their transmission emergence is small, preserving the idea's resonance profile. Non-expert receivers (ordinary speakers) are potentially *mutagenic*: their understanding of "gold" may drift, generating emergence that distorts the concept.

The chain mutation theorem reveals a tension in Putnam's framework: even if experts are perfectly conservative, the accumulated emergence from non-expert transmissions can cause the community's understanding to drift. The "division of labor" works only insofar as the expert transmissions are frequent enough to "correct" the mutations introduced by non-experts—a constraint that our framework makes quantitatively precise through the balance of $\sum_k \text{rs}(r_k, c)$ (receiver bias) and $\sum_k \kappa_{\text{tx}}(r_k, a_{k-1}, c)$ (accumulated mutation).

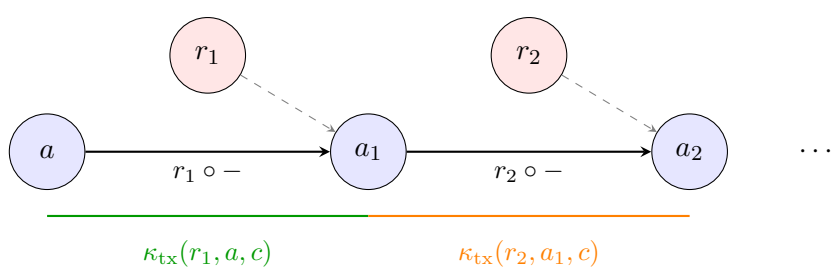


Figure 7.1: A transmission chain. Each receiver r_k transforms the current state a_{k-1} into $a_k = r_k \circ a_{k-1}$, generating transmission emergence at each step.

7.6 Fidelity

Definition 7.20 (Fidelity)**.** The **fidelity** of reception (r, a) at probe c is

$$\mathrm{fid}(r, a, c) \;:=\; \mathrm{rs}(a, c) + \kappa_{\mathrm{tx}}(r, a, c) \;=\; \mathrm{rs}(r \circ a, c) - \mathrm{rs}(r, c).$$

Interpretation 7.21. Fidelity measures how well the received idea $r \circ a$ represents the original idea a, after subtracting the receiver's own contribution. High fidelity ($\mathrm{fid}(r, a, c) \approx \mathrm{rs}(a, c)$) means the transmission emergence is small; the idea is received approximately as sent. Low fidelity ($\mathrm{fid}(r, a, c) \ll \mathrm{rs}(a, c)$ or $\gg \mathrm{rs}(a, c)$) means the reception process significantly distorts the idea.

Remark 7.22 (Fidelity and the Ethics of Interpretation). The concept of fidelity raises deep ethical questions about the responsibilities of interpreters. In law, the debate between "originalists" (who seek fidelity to the framers' intent) and "living constitutionalists" (who embrace creative interpretation) is precisely a debate about the appropriate level of transmission emergence. Originalists demand $\kappa_{\mathrm{tx}}(r, a, c) \approx 0$: the judge-as-receiver should leave the constitutional text unchanged. Living constitutionalists argue that non-zero emergence is not only inevitable but desirable: the Constitution's meaning should evolve through the creative act of judicial interpretation.

In literary studies, the analogous debate pits E. D. Hirsch's *Validity in Interpretation* (1967)—which insists on fidelity to authorial intent—against Stanley Fish's reader-response theory, which embraces the reader's creative transformation of the text. The IDT framework does not resolve this normative debate, but it makes its structure precise: the question is not whether transmission emergence is "good" or "bad" but what level of emergence is appropriate for a given interpretive community and its purposes.

The fidelity function also reveals an asymmetry between "faithful" and "creative" interpretation that is not visible in pre-formal discussions. A perfectly faithful receiver ($\mathrm{fid}(r, a, c) = \mathrm{rs}(a, c)$ for all c) must have zero weight ($w(r) = 0$, i.e., $r = \varepsilon$): only the empty receiver transmits without distortion. Every actual interpreter, having non-zero weight, necessarily introduces some transmission emergence. Perfect fidelity is a mathematical impossibility for any receiver with content—a fact that vindicates Gadamer's insight that understanding always involves "application" and that the dream of a presuppositionless hermeneutics is incoherent.

7.7 Fixed Points of Transmission

Definition 7.23 (Fixed Point)**.** An idea a is a **fixed point** of receiver r if $r \circ a = a$.

Proposition 7.24 (Void is Always a Fixed Point)**.** *For any receiver r, $r \circ \varepsilon = r \neq \varepsilon$ in general. Thus ε is* not *generally a fixed point. However, if $r = \varepsilon$, then $\varepsilon \circ a = a$ for all a, so the void receiver is the identity receiver.*

Theorem 7.25 (Fixed Point Weight Constraint)**.** *If a is a fixed point of r (i.e., $r \circ a = a$), then $w(a) \geq w(r)$.*

Proof. By (E4), $w(r \circ a) \geq w(r)$. But $r \circ a = a$, so $w(a) \geq w(r)$. □

Interpretation 7.26. A fixed point of transmission is an idea that a receiver cannot change— an idea so robust that reception leaves it invariant. The weight constraint says that only ideas with weight at least as large as the receiver's can be fixed points. Heavy ideas resist mutation by light receivers. A sufficiently "weighty" cultural text (Shakespeare, the Bible, the *Analects*) is a near-fixed-point for most individual receivers: the text transforms the reader more than the reader transforms the text.

Interpretation 7.27 (Lotman on the Semiosphere Boundary). Yuri Lotman, in *Universe of the Mind* (1990), argues that the *boundary* of the semiosphere is its most dynamic region: translation across semiotic borders is the primary mechanism of cultural creativity. "The boundary is a mechanism for translating texts of an alien semiotics into 'our' language, it is the place where what is 'external' is transformed into what is 'internal'" (p. 136).

Fixed points of transmission formalize Lotman's interior/exterior distinction. An idea a that is a fixed point of receiver r ($r \circ a = a$) lies within r's semiotic world: reception does not transform it, because it is already "our" language. An idea a that is maximally mutagenic under r—generating large $|\kappa_{\text{tx}}(r, a, c)|$—lies at the boundary of r's semiosphere: it is "alien" language that is radically transformed by the act of reception.

The weight constraint on fixed points ($w(a) \geq w(r)$) illuminates Lotman's claim that cultural centers are "heavy" (stable, self-consistent, resistant to external influence). Ideas that are central to a semiosphere have high weight and act as fixed points for most receivers; ideas at the periphery have lower weight and are easily transformed. Cultural dynamism —the generation of new meanings— occurs precisely at the boundary, where incoming alien ideas are heavy enough to resist complete absorption but light enough to be significantly mutated.

Interpretation 7.28 (Eco on Translation as Negotiation). Umberto Eco, in *Experiences in Translation* (2001) and *Mouse or Rat? Translation as Negotiation* (2003), argues that translation is not a matter of finding equivalences but of *negotiation*: the translator must negotiate between the demands of the source text, the target language, the publisher, the imagined reader, and the cultural context. "Translating means ... always *losing* something, and sometimes accepting to lose a lot" (p. 6).

The fidelity function (Definition 7.20) provides a formal framework for Eco's negotiation. The fidelity $\text{fid}(r, a, c) = \text{rs}(a, c) + \kappa_{\text{tx}}(r, a, c)$ measures how well the received idea represents the original at each probe. Eco's "negotiation" is the process of choosing a receiver r (a translation strategy) that optimizes fidelity across the most important probes while accepting losses at others. The translator cannot achieve perfect fidelity at all probes simultaneously (unless $\kappa_{\text{tx}} = 0$ everywhere, which would mean a conservative translation that is, as Eco acknowledges, often impossible and aesthetically undesirable).

The emergence bound constrains the negotiation: the maximum possible distortion at any probe is $\sqrt{w(r \circ a) \cdot w(c)}$, so a translator working with a "heavy" source text (large $w(a)$, hence large $w(r \circ a)$) has more room to maneuver than one working with a "light" text. This formalizes Eco's intuition that great works of literature are more translatable than minor ones: their weight provides a larger "negotiation space" within which multiple valid translations can exist.

7.8 The Sublime and the Fragile

Definition 7.29 (Sublime Idea)**.** An idea a is **sublime** if it is a near-fixed-point for almost all receivers: for all r with $w(r) \leq M$ (some threshold), $\kappa_{\text{tx}}(r, a, c) \approx 0$ for all c.

Definition 7.30 (Fragile Idea)**.** An idea a is **fragile** if it is highly mutagenic under typical transmission: for most receivers r, $|\kappa_{\text{tx}}(r, a, c)|$ is large relative to $\sqrt{w(r \circ a) \cdot w(c)}$ for typical probes c.

Interpretation 7.31 (The Sublime and Fragile in Aesthetic Theory). The distinction between sublime and fragile ideas connects to two major traditions in aesthetic theory. Edmund Burke's *A Philosophical Enquiry into the Origin of Our Ideas of the Sublime and Beautiful* (1757) characterizes the sublime as that which overwhelms the perceiver through vastness, power, or terror.

Immanuel Kant, in the *Critique of the Power of Judgment* (1790), refines this: the mathematical sublime exceeds our power of comprehension, while the dynamical sublime exceeds our power of resistance.

In IDT, a sublime idea's "overwhelmingness" is formalized by its resistance to mutation: its weight is so great that no individual receiver can significantly transform it. The Pythagorean theorem, the Second Law of Thermodynamics, the categorical imperative—these ideas are "sublime" in the precise sense that they overwhelm the cognitive apparatus of any individual receiver, emerging from the transmission chain essentially unchanged. Their sublimity is not a matter of aesthetic response but of algebraic structure: $w(a) \gg w(r)$ for typical r, so $\kappa_{\text{tx}}(r, a, c) \approx 0$.

Fragile ideas, by contrast, are closer to what aesthetic theory calls the *beautiful*: they invite engagement, they are shaped by the perceiver's response, they change with every encounter. A joke, a fashion trend, a local idiom—these are "beautiful" in the formal sense that they are highly sensitive to their receivers, generating large transmission emergence with each act of reception. The transmission framework thus provides a mathematical foundation for the distinction between the sublime (transmission-invariant) and the beautiful (transmission-sensitive) that has occupied aesthetic theory since the eighteenth century.

Example 7.32 (Sublime vs. Fragile Ideas).

- **Sublime**: The Pythagorean theorem ($a^2 + b^2 = c^2$) is sublime: it transmits across cultures, languages, and centuries with minimal mutation. The idea's weight is so large relative to any individual receiver's that transmission emergence is negligible.
- **Fragile**: An inside joke is fragile: its resonance depends entirely on shared context, and transmission to anyone outside the original group produces massive negative emergence (the humor evaporates) or massive positive emergence (the joke is reinterpreted in an entirely new way).

7.9 Morphisms and Structure Preservation

Interpretation 7.33 (Morphisms as Translation Between Semiotic Systems). The concept of an ideatic morphism (Definition 7.34) formalizes what it means for two semiotic systems to be "structurally equivalent"— to share the same algebraic and resonance structure despite potentially different carrier sets. An ideatic morphism $\phi : \mathcal{I}_1 \to \mathcal{I}_2$ is, in semiotic terms, a *perfect translation*: it maps ideas from one system to another while preserving all composition, resonance, and emergence relations.

The existence of such perfect morphisms is rare and philosophically significant. When ϕ exists, it demonstrates that two apparently different semiotic systems—say, the musical and the verbal, or the visual and the conceptual—share a deep structural isomorphism. The preservation of emergence ("Morphisms Preserve Emergence" above) is particularly striking: it says that the *creative* structure of one system maps faithfully onto the creative structure of another. A genuine novelty in $\mathcal{I}_1$ corresponds to a genuine novelty in $\mathcal{I}_2$.

When no exact morphism exists (the generic case), one can ask about *approximate* morphisms: maps that approximately preserve composition and resonance, with bounded error. These approximate morphisms correspond to Eco's "negotiated translations"—translations that preserve the essential structure while accepting controlled distortions. The category **IDT** of ideatic spaces and their morphisms is thus a formal framework for comparative semiotics: the systematic study of structural relations between semiotic systems.

Definition 7.34 (Ideatic Morphism)**.** An **ideatic morphism** from $(\mathcal{I}_1, \circ_1, \varepsilon_1, \mathrm{rs}_1)$ to $(\mathcal{I}_2, \circ_2, \varepsilon_2, \mathrm{rs}_2)$ is a function $\phi : \mathcal{I}_1 \to \mathcal{I}_2$ such that:

1. $\phi(\varepsilon_1) = \varepsilon_2$;
2. $\phi(a \circ_1 b) = \phi(a) \circ_2 \phi(b)$ for all $a, b \in \mathcal{I}_1$;
3. $\mathrm{rs}_2(\phi(a), \phi(b)) = \mathrm{rs}_1(a, b)$ for all $a, b \in \mathcal{I}_1$.

Proposition 7.35 (Morphisms Preserve Emergence)**.** *If $\phi : \mathcal{I}_1 \to \mathcal{I}_2$ is an ideatic morphism, then*

$$\kappa_2(\phi(a), \phi(b), \phi(c)) = \kappa_1(a, b, c)$$

for all $a, b, c \in \mathcal{I}_1$.

Proof.

$$\begin{aligned}
\kappa_2(\phi(a), \phi(b), \phi(c)) &= \mathrm{rs}_2(\phi(a) \circ_2 \phi(b), \phi(c)) - \mathrm{rs}_2(\phi(a), \phi(c)) - \mathrm{rs}_2(\phi(b), \phi(c)) \\
&= \mathrm{rs}_2(\phi(a \circ_1 b), \phi(c)) - \mathrm{rs}_1(a, c) - \mathrm{rs}_1(b, c) \\
&= \mathrm{rs}_1(a \circ_1 b, c) - \mathrm{rs}_1(a, c) - \mathrm{rs}_1(b, c) \\
&= \kappa_1(a, b, c).
\end{aligned}$$

□

Listing 7.2: Ideatic morphism in Lean 4

```
structure IdeaticMorphism (I J : Type*)
    [IdeaticSpace I] [IdeaticSpace J] where
  toFun : I     J
  map_void : toFun void = void
  map_compose :     a b : I,
    toFun (compose a b) = compose (toFun a) (toFun b)
  map_rs :     a b : I,
    rs (toFun a) (toFun b) = rs a b

theorem morphism_preserves_emergence
    {I J : Type*} [IdeaticSpace I] [IdeaticSpace J]
    (   : IdeaticMorphism I J) (a b c : I) :
    emergence (  .toFun a) (  .toFun b) (  .toFun c)
    = emergence a b c := by
  unfold emergence
  rw [      .map_compose,   .map_rs,   .map_rs,   .map_rs]

theorem morphism_preserves_charge
    {I J : Type*} [IdeaticSpace I] [IdeaticSpace J]
    (   : IdeaticMorphism I J) (a : I) :
    semanticCharge (  .toFun a) = semanticCharge a := by
  unfold semanticCharge
  exact morphism_preserves_emergence    a a a
```

Corollary 7.36. *Ideatic morphisms preserve:*

1. *Weight:* $w(\phi(a)) = w(a)$;

2. *Semantic charge:* $\sigma(\phi(a)) = \sigma(a)$;

3. *Same-idea equivalence:* $a \sim_{\text{idea}} b \implies \phi(a) \sim_{\text{idea}} \phi(b)$;

4. *Transmission emergence:* $\kappa_{\text{tx}}(\phi(r), \phi(a), \phi(c)) = \kappa_{\text{tx}}(r, a, c)$.

7.10 The Category of Ideatic Spaces

Remark 7.37. Ideatic spaces and ideatic morphisms form a category **IDT**. The identity morphism is the identity function; composition of morphisms is function composition (which preserves all three morphism conditions). The study of this category—its limits, colimits, adjunctions, and functors to other categories (groups, rings, topological spaces)—is a rich direction for future work.

Interpretation 7.38 (The Category **IDT** and Structuralism). The existence of a category **IDT** of ideatic spaces—with morphisms that preserve composition, resonance, and emergence—is the formal realization of the structuralist programme in its strongest form. Lévi-Strauss's structuralism sought to identify "deep structures" common to diverse cultural systems: the same structural patterns recurring in kinship systems, myths, culinary practices, and linguistic grammars. The category **IDT** makes this search precise: two cultural systems are "structurally equivalent" if and only if there exists an isomorphism between their ideatic spaces in **IDT**.

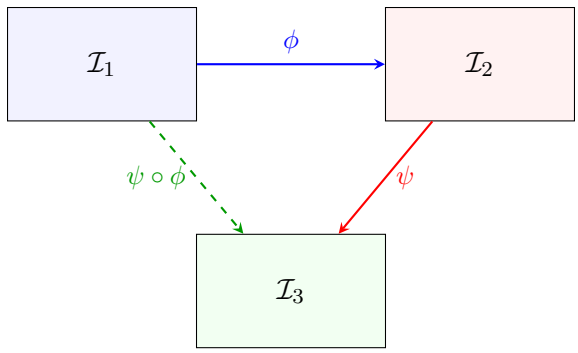
The power of the categorical perspective is that it shifts attention from *individual* ideatic spaces to the *relations between* them. A functor $F : \mathbf{IDT} \to \mathbf{Grp}$ (mapping ideatic spaces to groups) would reveal group-theoretic structure latent in every semiotic system. A natural transformation between two such functors would identify systematic correspondences between different ways of extracting algebraic structure from meaning. These category-theoretic tools—functors, natural transformations, adjunctions, limits—provide a mathematical language for the "comparative grammar" of semiotic systems that structuralism envisioned but could never fully articulate.

The category **IDT** also provides a framework for *post-structuralist* insights. Derrida's critique of structuralism —that structures are never fully closed, that there is always a "supplement" that exceeds the structure—corresponds to the observation that the category **IDT** is unlikely to be complete (to have all limits and colimits). The "excess" that Derrida identifies is the mathematical excess of a category that lacks certain universal constructions: there are ideatic configurations that cannot be captured by any finite diagram in **IDT**, structural relations that exceed the capacity of morphisms to represent them.

7.11 Mutation as Cultural Evolution

Interpretation 7.39 (Sperber's Epidemiology of Representations). Dan Sperber, in *Explaining Culture* (1996), proposes an "epidemiology of representations": the study of how mental representations spread through populations, analogous to the spread of diseases through epidemiological models. Sperber argues against the "meme" metaphor (which treats ideas as discrete, self-replicating units) in favor of a model in which representations are *transformed* at each step of transmission, converging toward "cultural attractors" —stable forms that are favored by cognitive and ecological constraints.

The IDT transmission framework supports Sperber's critique of memetics while providing a more precise mathematical formulation. The "replication" model of memetics assumes that transmission emergence is zero ($\kappa_{\text{tx}}(r, a, c) = 0$ for all r, c): the idea is transmitted unchanged, like a gene being copied. Sperber's "transformation" model assumes non-zero emergence: the idea is modified



The category **IDT** of ideatic spaces

Figure 7.2: Morphisms between ideatic spaces form a category. The composition $\psi \circ \phi$ preserves all algebraic and resonance structure.

at each step, and the modifications are not random but constrained by the receiver's cognitive structure (encoded in r) and the idea's own weight.

Sperber's "cultural attractors" correspond, in IDT terms, to fixed points of typical receivers (as analyzed earlier in this chapter). An attractor is an idea a^* such that $r \circ a^* \approx a^*$ for most receivers r in a given population—an idea so "natural" or "intuitive" that reception leaves it approximately unchanged. The weight constraint on fixed points ($w(a^*) \geq w(r)$) provides a formal criterion for identifying cultural attractors: they must be "heavier" than the receivers they attract, meaning they are more internally coherent than any individual receiver's cognitive state.

Interpretation 7.40. The transmission framework provides a mathematical foundation for cultural evolution theory. Each act of cultural transmission—teaching, storytelling, translation, imitation—is a composition operation that potentially mutates the transmitted idea. The Chain Mutation Theorem gives a precise accounting of how ideas change as they pass through a lineage of receivers.

This connects to several research programmes:

- **Memetics** (Dawkins, Dennett): Ideas as replicators subject to variation and selection. In IDT, "replication" is conservative transmission ($\varepsilon \approx 0$) and "mutation" is mutagenic transmission ($\varepsilon \gg 0$).
- **Cultural attraction** (Sperber): Ideas converge toward "attractors" in cultural space. In IDT, an attractor would be a fixed point or near-fixed-point of typical receivers.
- **Epidemiology of representations** (Sperber): The spread of ideas through populations. The transmission chain formalism directly models this, with the accumulated emergence quantifying the mutations along the chain.

7.12 Open Questions

We close with several open questions that emerge from the transmission framework:

1. **Convergence**: Does iterated transmission converge? If $a_n = r \circ a_{n-1}$ for a fixed receiver r, does $\{a_n\}$ converge to a fixed point? Under what conditions?

2. **Ergodicity**: For random receivers $r_1, r_2, \ldots$ drawn from some distribution, does the transmission chain exhibit ergodic behavior? Is there a "stationary distribution" over ideas?

3. **Information loss**: Can transmission emergence be negative enough to drive $\mathrm{rs}(a_n, c) \to 0$ for all probes c? Can an idea be "killed" by transmission, even though non-annihilation (Theorem 1.33) prevents it from reaching ε?

4. **Optimal transmission**: Given an idea a and a target resonance profile, what receiver r minimizes the distance between $\mathrm{rs}(r \circ a, -)$ and the target?

5. **Transmission networks**: What happens when ideas are transmitted not along a chain but through a network, with multiple receivers composing their received versions?

Interpretation 7.41 (Open Questions and the Philosophy of Culture). These open questions are not merely technical. Each encodes a deep philosophical problem about the nature of culture, meaning, and transmission.

Convergence asks: does culture have "attractors"—stable configurations toward which all transmission tends? Plato would say yes: the Forms are the fixed points of all intellectual activity, and philosophical inquiry is the iterative process of refining our ideas toward these fixed points. Nietzsche would say no: the "eternal return" is precisely the *non-convergence* of cultural transmission, the endless recurrence of creative destruction and renewal.

Ergodicity asks: is cultural history path-dependent or path-independent? If transmission is ergodic, then the long-run distribution of ideas in a culture depends only on the distribution of receivers, not on the initial conditions. This would vindicate a kind of cultural materialism (the base determines the superstructure in the long run, regardless of starting ideology). Non-ergodicity, by contrast, would mean that initial conditions—founding myths, sacred texts, constitutional moments—permanently shape the trajectory of a culture, never to be forgotten or "averaged out."

Information loss asks: can a tradition die? The non-annihilation theorem guarantees that no finite composition can reduce an idea to ε. But it leaves open the possibility that an idea's resonance profile can be driven arbitrarily close to zero— a "living death" in which the idea formally exists but resonates with nothing. This is the fate of many ancient traditions: their texts survive, but their resonance with contemporary thought approaches zero. They are not annihilated but effectively silenced.

Optimal transmission asks: what is the best teacher? Given an idea to be transmitted, what receiver (pedagogue, interpreter, translator) minimizes the distortion? This question connects to the entire tradition of rhetoric and pedagogy, from Aristotle's *Rhetoric* through Quintilian's *Institutio Oratoria* to modern educational theory. The IDT framework makes it a precise optimization problem.

Transmission networks asks: what happens when ideas propagate through social networks rather than linear chains? This connects to contemporary network science (Watts, Barabási) and to older traditions in the sociology of knowledge (Mannheim, Merton). The move from chains to networks is the formal analog of the move from intellectual history (which tracks ideas through a sequence of "great thinkers") to social epistemology (which studies how knowledge circulates through communities).

7.13 Summary and Conclusion

Interpretation 7.42 (The Transmission Framework as a Theory of Cultural Evolution). The transmission framework developed in this chapter completes the basic theory of IDT by adding a *dynamic* dimension to the static algebraic structure of chapters 1–6. Where the earlier chapters describe the space of ideas and their compositional relations, this chapter describes how ideas *move* through that space: how they are received, transformed, and retransmitted by successive agents.

The philosophical significance of this framework lies in its synthesis of several previously disconnected research programs. Bloom's theory of poetic influence, Bakhtin's theory of reported speech, Lord and Parry's study of oral tradition, Quine's indeterminacy of translation, Kuhn's incommensurability of paradigms, Putnam's division of linguistic labor, Lotman's semiosphere dynamics, and Eco's theory of translation as negotiation—all are revealed as special cases or aspects of a single mathematical structure: composition with emergence.

The unifying insight is that *transmission is composition*, and composition generates *emergence*. Every act of cultural transmission—reading, translating, teaching, imitating, parodying, criticizing—is a composition of the receiver's cognitive state with the transmitted idea, and every such composition generates an emergent surplus (or deficit) that constitutes the "mutation" of the idea. The Chain Mutation Theorem provides a precise accounting of how these mutations accumulate across chains of transmission, yielding a mathematical foundation for the study of cultural evolution.

The framework is deliberately minimal: it assumes only the eight axioms of the ideatic space and the interpretation of transmission as composition. No additional assumptions about cognitive architecture, social structure, or intentional agency are required. The richness of the theory—conservative and mutagenic receivers, transmission chains, fidelity, fixed points, sublime and fragile ideas, morphisms, and categories—emerges entirely from the algebraic structure. This minimality is both a strength (the framework applies to any semiotic system satisfying the axioms) and a limitation (specific applications will require additional structure, as the open questions at the end of this chapter suggest).

The transmission framework completes the basic theory of IDT. We have moved from the static axioms of the ideatic space (Chapter 1), through the cohomological structure of emergence (Chapter 2), the thermodynamics of weight (Chapter 3), the directionality of resonance (Chapter 4), the algebraic classification of emergence (Chapter 5), and the semiotic gap between expression and meaning (Chapter 6), to the dynamics of transmission and mutation (this chapter).

The framework rests on remarkably few assumptions: a monoid with a resonance function satisfying eight axioms. From these, we derive a rich mathematical universe—cocycles, cohomology, spectral decompositions, equivalence relations, morphisms, and categories—that provides precise formal analogues of some of the deepest concepts in semiotics, hermeneutics, rhetoric, and cultural theory.

The theory is at an early stage. The axioms are minimal, the models are few, and many of the most interesting questions remain open. But the framework has demonstrated that formal methods can illuminate the study of ideas—not by reducing ideas to formulas, but by revealing the hidden structure in how ideas compose, resonate, emerge, and propagate.

Part II

Structure and Dynamics

Chapter 8

Chains of Transmission

> A word is dead when it is said, some say. I say it just begins to live that day.
>
> Emily Dickinson

8.1 From Dyads to Chains

The preceding chapters studied the composition of *two* ideas. But ideas in the real world rarely pass through only one intermediary. A rumour travels from mouth to mouth; a scientific result is cited, paraphrased, and summarised across generations of papers; a joke is retold at party after party, acquiring new inflections at each telling. To model these phenomena we must study the composition of *chains*—sequences of ideas through which a signal is transmitted.

Formally, composition is associative in the sense that the expression $a \circ b_1 \circ b_2 \circ \cdots \circ b_n$ is well-defined by iterated binary composition. The question is: what happens to the weight, resonance, and emergence as the chain grows?

Definition 8.1 (Transmission Chain)**.** Let $a \in \mathcal{I}$ be a **source signal** and let $\mathbf{b} = [b_1, b_2, \ldots, b_n]$ be a finite sequence of **receivers** (or **contexts**). The **transmission chain** starting from a through $\mathbf{b}$ is defined recursively:

$$\mathrm{chain}(a, [\,]) := a, \qquad \mathrm{chain}(a, [b_1, \ldots, b_n]) := \mathrm{chain}(a, [b_1, \ldots, b_{n-1}]) \circ b_n.$$

We write $a^{\mathbf{b}} := \mathrm{chain}(a, \mathbf{b})$ for brevity, and $|\mathbf{b}| := n$ for the **chain length**.

Remark 8.2. The notation $a^{\mathbf{b}}$ should not be confused with iterated self-composition $a^{\langle n \rangle}$; the superscript carries a bold-face sequence, not a natural number. When the chain is the constant sequence $\mathbf{b} = [b, b, \ldots, b]$ of length n, we recover a special case of iterated composition: $a^{[b,\ldots,b]} = a \circ b^{\langle n \rangle}$ only when composition is commutative, which it is not in general.

Interpretation 8.3. The transmission chain formalises what Mikhail Bakhtin, in *The Dialogic Imagination* (1981), called the **chronotope**—the inseparability of time and space in narrative. For Bakhtin, every utterance carries the trace of the spatial and temporal context in which it was produced; an idea cannot be extracted from its chain of transmission any more than a word can be extracted from its dialogic history. Our formal chain $\mathrm{chain}(a, [b_1, \ldots, b_n])$ makes this precise: the final idea $a^{\mathbf{b}}$ is not merely a "plus" its contexts, but the result of *sequential* composition in

which order matters. The chain is a chronotope because each b_i contributes not just content but a *position*—a moment in the sequence at which interpretation occurs.

The oral-formulaic theory of Milman Parry and Albert Lord provides a concrete illustration. In *The Singer of Tales* (1960), Lord demonstrated that Homeric epic was not memorised verbatim but *recomposed in performance* using formulaic phrases—"wine-dark sea," "rosy-fingered dawn"—that served as stable building blocks within a mutable compositional chain. Each performance by a *guslar* (oral poet) constitutes a link b_i in our transmission chain. The formulaic phrases are elements with high self-resonance $w(b_i)$ that contribute predictable weight, while the narrative structure undergoes creative mutation at each telling. The Chain Enrichment Theorem (Theorem 8.4) explains why these epics grew in length and complexity across generations: each performance added weight, even as fidelity to any "original" decreased.

Gérard Genette's theory of narrative levels offers a third perspective. In *Narrative Discourse* (1972), Genette distinguished between *diegesis* (the story world), *metadiegesis* (a story within the story), and *extradiegesis* (the narrative act itself). Each level constitutes a retelling—a link in a transmission chain. When Scheherazade tells a story in which a character tells another story, we have a chain of length 2; the inner story $a^{[b_1,b_2]}$ carries the weight of both framing contexts. Genette's insight that each narrative level transforms the embedded narrative is precisely our observation that $a^{[b_1,b_2]} \neq a^{[b_1]} \circ b_2'$ for $b_2' \neq b_2$: the *order* and *identity* of the framing matters, not just the fact of framing.

8.2 The Chain Enrichment Theorem

The central result of this chapter asserts that transmission through a chain can only *increase* the weight of the transmitted idea.

Theorem 8.4 (Chain Enrichment)**.** *For any source signal $a \in \mathcal{I}$ and any chain $\mathbf{b} = [b_1, \ldots, b_n]$,*

$$w(a^{\mathbf{b}}) \geq w(a).$$

That is, $\mathrm{rsa}^{\mathbf{b}} a^{\mathbf{b}} \geq \mathrm{rsa}a$*. The weight is non-decreasing along the chain.*

Proof. We proceed by induction on the chain length $n = |\mathbf{b}|$.

Base case ($n = 0$). The chain is empty, so $a^{[\,]} = a$ and $w(a^{[\,]}) = w(a)$. The inequality holds with equality.

Inductive step. Assume the result holds for all chains of length $n-1$. Write $\mathbf{b}' = [b_1, \ldots, b_{n-1}]$, so that $a^{\mathbf{b}} = a^{\mathbf{b}'} \circ b_n$. By the inductive hypothesis,

$$w(a^{\mathbf{b}'}) \geq w(a).$$

Now apply axiom **E4** (Composition Enriches) to the pair $(a^{\mathbf{b}'}, b_n)$:

$$w(a^{\mathbf{b}'} \circ b_n) = w(a^{\mathbf{b}}) \geq w(a^{\mathbf{b}'}) \geq w(a).$$

This completes the induction. □

In Lean 4, the proof follows the same inductive structure:

```
theorem chain_enrichment (a : Idea) (bs : List Idea) :
    w (chain a bs) >= w a := by
  induction bs with
  | nil => simp [chain, w]
```

```
  | cons b bs' ih =>
    simp [chain]
    calc w (chain a bs' |> (. @\compose@ b))
        >= w (chain a bs') := composition_enriches _ _
      _ >= w a := ih
```

Interpretation 8.5. The Chain Enrichment Theorem is the mathematical expression of a surprising anthropological observation: *ideas grow heavier through transmission, never lighter.* In the classic "telephone game," a message is whispered from person to person around a circle, and the final message is typically very different from the original. The naïve expectation is that the message *degrades*—that information is lost at each step.

But "degradation" is not the same as "weight loss." The weight $w(a) = \mathrm{rs}aa$ measures the self-resonance of an idea, its internal richness and complexity. Each retelling adds the receiver's context, and axiom E4 guarantees that this addition can only increase total weight. The *content* may change—the original message may be unrecognisable—but the *weight* of the cultural artefact only grows.

This theorem explains why rumours grow more elaborate, why myths accumulate detail over centuries, and why urban legends become more vivid with each retelling. The message mutates, but the mutation always adds weight. It is as if each retelling deposits a sedimentary layer: the original may be buried, but the total mass only increases.

Interpretation 8.6. The Chain Enrichment Theorem bears a deep connection to Jacques Derrida's concept of **iterability**, developed in "Signature Event Context" (1972) and *Limited Inc* (1988). Derrida argued that every sign, by its very nature, must be *repeatable*—iterable—in new contexts, and that this repetition necessarily introduces *différance*: a simultaneous differing and deferring of meaning. "A written sign carries with it a force that breaks with its context," Derrida wrote, "that is, with the collectivity of presences organising the moment of its inscription."

Our theorem gives this claim a precise mathematical form. When an idea a is transmitted through a chain $[b_1, \ldots, b_n]$, each composition $a^{[b_1,\ldots,b_k]} \circ b_{k+1}$ is an act of iteration: the idea is repeated in a new context b_{k+1}. The theorem guarantees that this iteration always *enriches*—weight increases—but the enrichment is precisely the "force that breaks with context" that Derrida identified. The original idea a is *deferred* (its direct contribution is progressively buried under accumulated emergence) and *differed from* (the final idea $a^{\mathbf{b}}$ may bear no resemblance to a). Yet the weight—the total self-resonance—only grows.

Ludwig Wittgenstein's **rule-following paradox** (*Philosophical Investigations*, §§185–202) illuminates a different aspect. Wittgenstein asked: what determines that a rule has been followed correctly? Each step in following a rule requires an act of interpretation, and no finite set of examples can determine the rule uniquely. In our framework, each link b_i in the chain is an interpretive act—an application of the "rule" of the original idea a. The Chain Enrichment Theorem shows that each such application adds weight, but the paradox remains: there is no formal guarantee that the "rule" followed at step i is the "same" rule intended at step 1. The chain enriches, but it does not conserve direction.

Hans-Georg Gadamer's **effective-historical consciousness** (*wirkungsgeschichtliches Bewußtsein*), developed in *Truth and Method* (1960), provides the hermeneutic perspective. Gadamer argued that we never encounter a text or idea in isolation; we always encounter it through the chain of its effects—its *Wirkungsgeschichte*. To understand Homer is to understand Homer *as received through* two millennia of commentary, translation, and pedagogical transmission. The chain $a^{[b_1,\ldots,b_n]}$ *is* the effective-historical reality of the idea a: we cannot strip away the chain to recover the "original" because the chain *is* how the idea exists for us. The enrichment theorem confirms

Gadamer's fundamental insight: tradition does not degrade meaning; it enriches it.

8.3 Cumulative Emergence Along Chains

The Chain Enrichment Theorem tells us about weight; we now ask about the *structure* of the enrichment. How does emergence accumulate?

Definition 8.7 (Cumulative Emergence)**.** The **cumulative emergence** of a chain $(a, \mathbf{b})$ with respect to observer c is:

$$E_{\text{cum}}(a, \mathbf{b}, c) := \text{rs}a^{\mathbf{b}}c - \text{rs}ac - \sum_{i=1}^{n} \text{rs}b_i c.$$

Remark 8.8. The concept of cumulative emergence recalls Johann Gottfried Herder's philosophy of cultural history (*Ideas on the Philosophy of the History of Mankind*, 1784–91). Herder argued that each nation and epoch contributes something unique to the collective treasury of humanity, and that this collective treasury is *more* than the sum of its parts. The cumulative emergence $E_{\text{cum}}(a, \mathbf{b}, c)$ is the mathematical formalisation of Herder's "surplus"—the meaning that the chain of cultural transmission produces beyond what any individual link could contribute alone. Herder's insight that "each nation bears in itself the standard of its perfection, totally independent of all comparison with that of others" is the observation that cumulative emergence depends on the observer c: different observers see different surpluses in the same chain of transmission.

Theorem 8.9 (Emergence Decomposition for Chains)**.** *The cumulative emergence decomposes as a sum of local emergences:*

$$E_{\text{cum}}(a, \mathbf{b}, c) = \sum_{i=1}^{n} \kappa\Big(a^{[b_1,\ldots,b_{i-1}]},\, b_i,\, c\Big),$$

where $\kappa(x, y, c) := \text{rs}x \circ yc - \text{rs}xc - \text{rs}yc$ *is the emergence at a single composition step, with the "current accumulated idea"* $a^{[b_1,\ldots,b_{i-1}]}$ *as the left operand.*

Proof. We proceed by induction on n.

Base case ($n = 1$). $E_{\text{cum}}(a, [b_1], c) = \text{rs}a \circ b_1 c - \text{rs}ac - \text{rs}b_1 c = \kappa(a, b_1, c)$.

Inductive step. Write $\mathbf{b}' = [b_1, \ldots, b_{n-1}]$ and $p = a^{\mathbf{b}'}$. Then $a^{\mathbf{b}} = p \circ b_n$, and

$$\begin{aligned}
E_{\text{cum}}(a, \mathbf{b}, c) &= \text{rs}p \circ b_n c - \text{rs}ac - \sum_{i=1}^{n} \text{rs}b_i c \\
&= \big[\text{rs}p \circ b_n c - \text{rs}pc - \text{rs}b_n c\big] + \text{rs}pc - \text{rs}ac - \sum_{i=1}^{n-1} \text{rs}b_i c \\
&= \kappa(p, b_n, c) + E_{\text{cum}}(a, \mathbf{b}', c) \\
&= \kappa(p, b_n, c) + \sum_{i=1}^{n-1} \kappa\Big(a^{[b_1,\ldots,b_{i-1}]}, b_i, c\Big) \quad \text{(by induction)} \\
&= \sum_{i=1}^{n} \kappa\Big(a^{[b_1,\ldots,b_{i-1}]}, b_i, c\Big).
\end{aligned}$$

□

Interpretation 8.10. The cumulative emergence is the *total surplus of meaning* that the chain has produced beyond the mere sum of its parts. Each link in the chain contributes a local emergence $\kappa(p_i, b_{i+1}, c)$, which depends on the *entire accumulated idea* at that point, not just the original signal. This is the formal content of the observation that "context matters": the emergence at step i depends on everything that came before.

When cumulative emergence is large and positive, the chain has been *synergistic*—the ideas have reinforced each other in ways that transcend their individual contributions. When it is negative, the chain has been *destructive*—meaning has been lost to interference. The Chain Enrichment Theorem tells us that total weight still increases, but the resonance with any particular observer c may decrease: the idea may grow heavier while becoming less relevant to any given perspective.

Interpretation 8.11. The Emergence Decomposition Theorem provides the mathematical infrastructure for Charles Sanders Peirce's theory of the **chain of interpretants**. In Peirce's semiotics, every sign produces an *interpretant*—another sign that interprets the first—and this interpretant in turn produces its own interpretant, creating an infinite chain of semiosis. "The meaning of a representation can be nothing but a representation," Peirce wrote in the *Collected Papers* (5.252).

Our decomposition shows that the cumulative emergence along the chain $E_{\text{cum}}(a, \mathbf{b}, c) = \sum_i \kappa(a^{[b_1,\ldots,b_{i-1}]}, b_i, c)$ is the total *surplus of meaning* produced by the chain of interpretants. Each local emergence $\kappa(p_i, b_{i+1}, c)$ is the contribution of the i-th interpretant: the new meaning that arises from interpreting the accumulated sign p_i in the context b_{i+1}. Peirce's insight that each interpretant depends on the *entire previous chain* (not just the original sign) is captured by the fact that $p_i = a^{[b_1,\ldots,b_i]}$ is the full accumulated idea, not merely a.

Yuri Lotman's concept of the **semiosphere**—the totality of semiotic acts in a culture, developed in *Universe of the Mind* (1990)—adds a diachronic dimension. Lotman argued that the semiosphere has a temporal axis along which signs are transmitted and transformed. The chain $(a, \mathbf{b})$ models a trajectory through Lotman's semiosphere: the idea a enters the semiotic space and is progressively transformed by the cultural contexts $b_1, b_2, \ldots$ it encounters. The cumulative emergence E_{cum} measures the total semiotic work performed along this trajectory—the total meaning generated by the semiosphere's processing of the idea.

It is instructive to compare our framework with Claude Shannon's *mathematical theory of communication* (1948). In Shannon's theory, a channel is characterised by a transition probability $p(y|x)$, and the goal is to transmit information with minimal error. Emergence is identically zero in Shannon's framework ($\kappa = 0$ in our notation): the channel adds noise but no *meaning*. The transmission chain in Shannon's theory is a Markov chain, where each step depends only on the current state. Our framework generalises Shannon's by allowing each transmission step to produce *genuine emergence*—meaning that was not present in either the signal or the channel. Shannon's theory is the degenerate case where cultural transmission reduces to signal processing; IDT captures the full richness of meaning-making that information theory deliberately abstracts away.

8.4 The Ratchet Effect

Definition 8.12 (Weight Sequence)**.** The **weight sequence** of a chain $(a, \mathbf{b})$ is the sequence

$$W_0 := w(a), \quad W_k := w(a^{[b_1,\ldots,b_k]}) \quad (k = 1, \ldots, n).$$

Corollary 8.13 (The Ratchet Effect)**.** *The weight sequence is non-decreasing:* $W_0 \leq W_1 \leq W_2 \leq \cdots \leq W_n$*. Moreover,* $W_k = W_{k-1}$ *if and only if* $\kappa(a^{[b_1,\ldots,b_{k-1}]}, b_k, a^{[b_1,\ldots,b_k]}) = -\text{rs}b_k a^{[b_1,\ldots,b_k]}$*.*

Remark 8.14. The Ratchet Effect stands in instructive contrast to the second law of thermodynamics, which states that entropy (disorder) can only increase in a closed system. The cultural ratchet says that *weight* (a measure of complexity or richness) can only increase along a transmission chain. Both are irreversibility results, but they point in opposite directions: thermodynamic entropy measures *disorder*, while ideatic weight measures *complexity*. The apparent paradox—that both disorder and complexity increase monotonically—is resolved by noting that the ideatic space is not a closed thermodynamic system. Each receiver b_i is a source of negentropy (structured information) that is injected into the chain, driving weight upward even as the physical substrate may be degrading. This is precisely Erwin Schrödinger's insight in *What is Life?* (1944): living systems maintain and increase their complexity by importing negentropy from their environment.

Proof. Monotonicity follows from E4 applied at each step. For the equality condition, $W_k = W_{k-1}$ means $w(p \circ b_k) = w(p)$ where $p = a^{[b_1,\ldots,b_{k-1}]}$. Since $w(p \circ b_k) = \mathrm{rs}p \circ b_k p \circ b_k = \mathrm{rs}pp \circ b_k + \mathrm{rs}b_k p \circ b_k + \kappa(p, b_k, p \circ b_k)$, the condition $w(p \circ b_k) = w(p)$ gives $\mathrm{rs}b_k p \circ b_k + \kappa(p, b_k, p \circ b_k) = w(p) - \mathrm{rs}pp \circ b_k$, from which the stated condition follows after rearrangement. □

Remark 8.15. The name "ratchet" comes from the mechanical device that permits motion in only one direction. Cultural weight, like a ratchet, clicks forward but never backward. This is the formal analogue of the "cultural ratchet effect" discussed by Tomasello (1999) in the context of cumulative culture.

Interpretation 8.16. The Ratchet Effect (Corollary 8.13) formalises what Michael Tomasello, in *The Cultural Origins of Human Cognition* (1999), called the **cultural ratchet**: the mechanism by which human cultures accumulate complexity over generations in a way that no other species matches. Tomasello observed that chimpanzees can learn tool use from one another, but each generation starts essentially from scratch; human cultures, by contrast, "ratchet up" complexity, with each generation building on the accumulated achievements of its predecessors.

The mathematical content of the ratchet is the non-decreasing weight sequence $W_0 \leq W_1 \leq \cdots \leq W_n$. The equality condition—that $W_k = W_{k-1}$ only when the emergence at step k exactly cancels the direct contribution of b_k—reveals something that Tomasello's verbal account does not make explicit: the ratchet can *stall* (producing zero net enrichment) only under a precise algebraic condition. Generically, the ratchet clicks forward. This is the mathematical reason why cumulative culture is the *default*, not the exception: it takes a special cancellation to prevent enrichment.

From the perspective of archaeological and anthropological theory, the ratchet effect explains the explosive growth of material culture in the Upper Palaeolithic (ca. 50,000–10,000 BCE), when stone tool technology, art, and symbolic behaviour accumulated at rates unprecedented in the hominin lineage. Each generation's innovations (b_i) composed with the accumulated cultural inheritance ($a^{[b_1,\ldots,b_{i-1}]}$) to produce a richer total cultural package, and the ratchet ensured that this richness could not decrease.

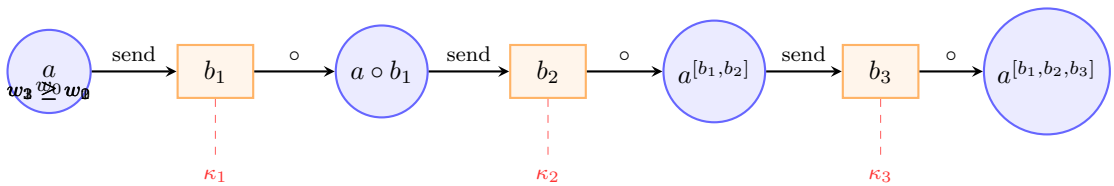
8.5 Chain Diagrams

Interpretation 8.17. The chain diagram above—with its alternating circles (ideas) and rectangles (receivers), linked by arrows of composition—is more than a pedagogical convenience. It is a visual representation of what Walter Benjamin called the **constellation** (*Konstellation*): the non-linear, non-hierarchical arrangement of ideas that produces meaning not through sequential logic but through spatial juxtaposition. In the *Arcades Project* (written 1927–1940, published posthumously), Benjamin arranged quotations, observations, and images in constellations, trusting that

the *emergence* between juxtaposed fragments would produce insights that linear argument could not.

Our chain diagram reveals that Benjamin's constellation is a special case of the transmission chain in which the "spatial" arrangement encodes the *temporal* sequence of composition. The weight labels $w_0 \leq w_1 \leq w_2 \leq w_3$ beneath each node are the formal trace of Benjamin's conviction that meaning accumulates through juxtaposition: each new element in the constellation adds weight to the whole.

The following TikZ diagram illustrates a transmission chain of length 4, showing the weight increase at each step.



Example 8.18 (The Telephone Game)**.** Consider the children's game of "Telephone" (also called "Chinese Whispers"). A sentence is whispered to the first child, who whispers it to the next, and so on around a circle. The final sentence is typically very different from the original.

In our formalism, the original sentence is a, and each child is a receiver b_i with their own linguistic context. The Chain Enrichment Theorem tells us that $w(a^{[b_1,\ldots,b_n]}) \geq w(a)$: the total weight of the final message is at least as large as the original. But the *content* may be entirely different, because the cumulative emergence E_{cum} may dominate the original signal.

A concrete instance: the original message "Send reinforcements, we're going to advance" might become "Send three and fourpence, we're going to a dance." The final version is *richer* as a cultural artefact (it is funnier, more memorable, more semantically layered) even though the military information has been entirely lost. Weight has increased; fidelity has decreased.

Interpretation 8.19. The Telephone Game example (Example 8.18) dramatises a universal feature of oral transmission, but the oral-formulaic theory developed by Milman Parry (*L'Épithète traditionnelle dans Homère*, 1928) and Albert Lord (*The Singer of Tales*, 1960) reveals a subtler phenomenon. In oral epic traditions—Homeric Greek, South Slavic, West African *griot* performance—transmission is not mere repetition with error. The oral poet *recomposes* the poem at each performance, using a repertoire of formulaic phrases, typical scenes, and narrative patterns as building blocks.

In our formalism, the formulaic elements are receivers b_i with a special property: they are *approximately conservative* (in the sense of Definition 9.1 in the next chapter) with respect to the narrative structure, but *strongly mutagenic* with respect to surface detail. The "wine-dark sea" formula preserves the narrative function of a sea-crossing scene (conservative transmission of structure) while allowing the poet to vary the surrounding description (mutagenic transmission of content). The result is a chain that enriches weight while maintaining structural coherence—precisely the combination that oral traditions exhibit over centuries of transmission.

This analysis reveals a limitation of the naïve "telephone game" model: real oral transmission is not a chain of independent errors but a *structured recomposition* in which some dimensions are conserved and others are allowed to mutate. The next chapter will develop the formal apparatus to distinguish these dimensions.

8.6 Chain Length Bounds

Under additional assumptions, we can bound the rate of weight growth.

Proposition 8.20 (Linear Weight Growth)**.** *Suppose each receiver has weight at most M: $w(b_i) \leq M$ for all i. Then*

$$w(a^{\mathbf{b}}) \leq w(a) + n \cdot M + \sum_{i=1}^{n} \left| \kappa(a^{[b_1,\ldots,b_{i-1}]}, b_i, a^{[b_1,\ldots,b_i]}) \right|.$$

If additionally $|\kappa(\cdot,\cdot,\cdot)| \leq K$ for some constant K, then $w(a^{\mathbf{b}}) \leq w(a) + n(M+K)$.

Proof. At each step, $w(p \circ b_i) = w(p) + \mathrm{rs}b_i p \circ b_i + \kappa(p, b_i, p \circ b_i)$, and $|\mathrm{rs}b_i p \circ b_i| \leq w(b_i) \leq M$ by Cauchy–Schwarz (assuming the Hilbert embedding). Summing over all steps gives the result. □

Lemma 8.21 (Bounded Emergence Chains)**.** *If $\kappa(x, y, c) \geq 0$ for all x, y, c (i.e., the ideatic space is **superadditive**), then the weight sequence satisfies:*

$$W_n \geq W_0 + \sum_{i=1}^{n} w(b_i).$$

Proof. In the superadditive case, $w(p \circ b_i) = w(p) + w(b_i) + \kappa(p, b_i, p \circ b_i) \geq w(p) + w(b_i)$. Summing telescopically: $W_n \geq W_0 + \sum_{i=1}^{n} w(b_i)$. □

Interpretation 8.22. The bounds in this section reveal a tension. On the one hand, weight can grow at most linearly in the chain length (Proposition 8.20), provided that individual weights and emergences are bounded. On the other hand, in superadditive spaces, weight grows *at least* linearly (Lemma 8.21). The growth rate is therefore $\Theta(n)$: ideas grow linearly heavier through transmission.

This is the mathematical expression of *cumulative culture*: the total richness of a cultural tradition grows linearly in the number of generations that have contributed to it.

Interpretation 8.23. The linear growth bounds of Proposition 8.20 and Lemma 8.21 connect our framework to two distinct intellectual traditions. In information theory, Shannon's noisy channel coding theorem (1948) established that reliable communication is possible at rates below the channel capacity, provided sufficiently long codes are used. In our framework, the *length* of the chain plays an analogous role: longer chains produce heavier ideas, and the growth rate $\Theta(n)$ is the cultural analogue of Shannon's channel capacity—the rate at which a transmission chain can accumulate meaning.

But where Shannon's capacity is a *ceiling* (one cannot reliably transmit above it), our growth rate is both a ceiling and a floor. This reflects a fundamental difference between information transmission and cultural transmission: information can be lost in a noisy channel, but *cultural weight cannot be lost in a transmission chain.* The ratchet effect ensures that the floor rises with n, while the bounded-emergence condition ensures that the ceiling rises at the same rate. The result is the remarkably constrained $\Theta(n)$ growth—linear in the chain length, with constants determined by the maximum receiver weight M and the maximum emergence K.

From the perspective of cultural evolution theory, as developed by Robert Boyd and Peter Richerson (*Culture and the Evolutionary Process*, 1985), the linear growth result provides a mathematical foundation for the observation that cultural complexity scales with population size and the number of generations that have contributed to the cultural tradition. Larger populations and longer histories produce heavier cultural packages, with the growth rate determined by the "quality" of individual cultural contributors (bounded by M) and the degree of creative interaction between ideas (bounded by K).

In Lean 4:

```
theorem chain_weight_lower_bound (a : Idea) (bs : List Idea)
    (h_super : forall x y, emergence x y (x @\compose@ y) >= 0) :
    w (chain a bs) >= w a + bs.map w |>.sum := by
  induction bs with
  | nil => simp [chain]
  | cons b bs' ih =>
    simp [chain, List.map, List.sum_cons]
    have h1 := composition_enriches_super a (chain a bs') b h_super
    linarith [ih]
```

Chapter 9

Conservative and Mutagenic Transmission

> Tradition is not the worship of ashes, but the preservation of fire.
>
> Gustav Mahler

9.1 The Spectrum of Transmission Fidelity

Not all receivers are created equal. Some transmit ideas faithfully, preserving the signal with minimal distortion. Others radically transform whatever passes through them, adding their own creative contribution (or destructive noise). This chapter develops a taxonomy of receivers based on the magnitude of emergence they produce.

Definition 9.1 (ϵ-Conservative Receiver)**.** A receiver with context b is **ϵ-conservative** with respect to idea a and observer c if

$$|\kappa(a, b, c)| \leq \epsilon.$$

If this holds for *all* observers c, we say b is **uniformly ϵ-conservative** with respect to a. If it holds for all a and all c, we say b is **absolutely ϵ-conservative**.

Remark 9.2. The parameter ϵ controls the *fidelity* of transmission. An absolutely 0-conservative receiver is one that produces zero emergence with every idea—this is the idealisation of a "perfect copying machine." As ϵ grows, the receiver introduces more distortion.

Interpretation 9.3. The distinction between conservative and mutagenic receivers maps directly onto one of the most consequential transitions in human history: the shift from orality to literacy, as analysed by Walter Ong (*Orality and Literacy*, 1982) and Jack Goody (*The Domestication of the Savage Mind*, 1977). Ong argued that literacy fundamentally transforms the relationship between ideas and their transmission. In an oral culture, every act of transmission is necessarily *mutagenic*: the singer, the storyteller, the elder recounting a genealogy—all must reconstruct the idea from memory, and each reconstruction introduces variation. There is no "original" to compare against; the tradition exists only in its performances.

Literacy, by contrast, enables *conservative* transmission: a scribe can copy a text with arbitrarily small ϵ, and the copy can be compared against the original. In our formalism, literacy is the technology that produces approximately ϵ-conservative receivers with small ϵ. The invention of

printing (Gutenberg, ca. 1440) pushed ϵ toward zero for the first time in human history, enabling the reproduction of ideas with effectively zero emergence.

Goody's crucial observation was that this shift does not merely affect the *fidelity* of transmission; it transforms the *nature* of thought itself. Literate cultures develop lists, tables, and formal logic—cognitive technologies that depend on the possibility of exact reproduction. In our framework, these are ideas a that can only exist in chains with small ϵ: they require conservative transmission to maintain their coherence. An oral culture cannot sustain a formal logical proof because the chain of transmission would introduce too much emergence, scrambling the delicate logical structure. The ϵ-conservative receiver is thus not merely a description of a transmission technology but a *precondition* for entire modes of thought.

Interpretation 9.4. Walter Benjamin's celebrated essay "The Task of the Translator" (1923) offers a radically different perspective on the conservative–mutagenic spectrum. Benjamin argued that the translator's task is *not* to reproduce the original but to release what he called the *reine Sprache*—the "pure language" that underlies all particular languages. A translation does not conserve the original; it *mutates toward* a higher truth that neither the original nor the translation fully captures.

In our formalism, Benjamin's "pure language" can be understood as a hypothetical idea a^* such that $\mathrm{rs}aa^* > 0$ for every natural language text a. The translator (a mutagenic receiver b_T) produces $a \circ b_T$, and Benjamin's claim is that the best translations are those for which $\mathrm{rs}a \circ b_T a^* > \mathrm{rs}aa^*$: the translation resonates *more strongly* with the pure language than the original does. This is a creative mutation in the strongest sense—the emergence $\kappa(a, b_T, a^*)$ is positive, and the mutation brings the idea *closer* to an ideal that neither the source text nor the translator's context embodies alone.

Benjamin's insight challenges the naïve assumption that conservative transmission is always superior. Sometimes the most faithful transmission is the most mutagenic—the one that transforms the surface to preserve (or reveal) the depth.

Definition 9.5 (Linear Receiver)**.** A receiver b is **linear** with respect to idea a if $\kappa(a, b, c) = 0$ for all observers c. Equivalently, b is uniformly 0-conservative with respect to a.

Proposition 9.6 (Linear Transmission is Additive)**.** *If b is linear with respect to a, then for all c:*

$$\mathrm{rs}a \circ bc = \mathrm{rs}ac + \mathrm{rs}bc.$$

Proof. By definition, $\kappa(a, b, c) = \mathrm{rs}a \circ bc - \mathrm{rs}ac - \mathrm{rs}bc = 0$. Rearranging gives the result. □

Corollary 9.7 (Weight Under Linear Transmission)**.** *If b is linear with respect to a, then*

$$w(a \circ b) = w(a) + 2\mathrm{rs}ab + w(b).$$

In particular, if additionally $\mathrm{rs}ab \geq 0$*, then* $w(a \circ b) \geq w(a) + w(b)$*.*

Proof. Setting $c = a \circ b$ in Proposition 9.6: $w(a \circ b) = \mathrm{rs}aa \circ b + \mathrm{rs}ba \circ b$. Applying Proposition 9.6 again with $c = a$ and $c = b$: $\mathrm{rs}aa \circ b = \mathrm{rs}aa + \mathrm{rs}ba = w(a) + \mathrm{rs}ab$ (using symmetry of resonance in the Hilbert embedding). Similarly, $\mathrm{rs}ba \circ b = \mathrm{rs}ab + w(b)$. Adding: $w(a \circ b) = w(a) + 2\mathrm{rs}ab + w(b)$. □

Interpretation 9.8. Linear receivers are the idealisation of "transparent" media: a perfectly calibrated translation system, a lossless communication channel, or a scribe who copies a manuscript without error. The proposition says that such a medium simply *adds* its resonance profile to the signal. There is no emergence—no surprise, no synergy, no creative transformation. The result is predictable from the parts.

But real cultural transmission is almost never linear. Every human intermediary introduces emergence, positive or negative. The question is how much, and of what character.

Interpretation 9.9. Harold Bloom's theory of **revisionary ratios**, developed in *The Anxiety of Influence* (1973), provides a literary taxonomy of mutagenic transmission that maps remarkably well onto our formal framework. Bloom identified six modes by which a later poet "misreads" a precursor: *clinamen* (swerving), *tessera* (completion), *kenosis* (self-emptying), *daemonization* (raising to the sublime), *askesis* (self-purgation), and *apophrades* (return of the dead).

Each of these corresponds to a specific pattern of emergence. *Clinamen* is a small, directional mutation: $\kappa(a, b, c)$ is moderate in magnitude and consistent in sign, producing a systematic "swerve" away from the precursor. *Tessera* is a completion that adds missing dimensions: the emergence fills gaps in the original, producing high positive $\kappa(a, b, c)$ for observers c in those dimensions. *Kenosis* is a deliberate reduction: the poet as receiver b produces negative emergence, stripping away the precursor's richness to create space for new growth. *Daemonization* inflates the precursor's weight beyond what the original warrants—a kind of creative overinterpretation that produces large positive emergence. *Askesis* is a conservative narrowing: the receiver selectively transmits only part of the precursor's resonance profile. And *apophrades* is the uncanny return whereby the later poet's work makes the precursor seem to be imitating the successor—a reversal of the chain's temporal direction that is possible when the emergence at a late stage dominates the original signal.

What Bloom's taxonomy reveals, in our formal language, is that creative literary transmission is never merely conservative or merely mutagenic; it is mutagenic in *structured* ways. The revisionary ratios are specific geometries of emergence in the Hilbert space of meaning.

Remark 9.10. Thomas Kuhn's *The Structure of Scientific Revolutions* (1962) provides a parallel taxonomy for scientific transmission. "Normal science" is conservative transmission: researchers work within a paradigm, extending its results with small ϵ. "Revolutionary science" is mutagenic transmission: a new paradigm (receiver b_{rev}) transforms the meaning of the entire preceding tradition a so radically that $\hat{r}(a, a \circ b_{\text{rev}})$ may be close to zero. Kuhn's observation that "after a revolution scientists work in a different world" is our observation that the emergence $\kappa(a, b_{\text{rev}}, c)$ can be so large as to dominate the original signal entirely. The Chain Enrichment Theorem (Theorem 8.4) tells us that the weight of the scientific tradition always increases through revolutions—Kuhn's "progress through revolutions" has a formal justification—even when the direction changes dramatically.

9.2 Mutagenic Receivers

Definition 9.11 (Mutagenic Receiver)**.** A receiver b is **μ-mutagenic** with respect to idea a if there exists an observer c such that $|\kappa(a, b, c)| \geq \mu$. The quantity

$$\mu(a, b) := \sup_{c \in \mathcal{I}} |\kappa(a, b, c)|$$

is the **mutation rate** of b with respect to a.

Proposition 9.12 (Mutation Rate Properties)**.**

1. *$\mu(a, \varepsilon) = 0$ for all a (the void is non-mutagenic).*
2. *$\mu(\varepsilon, b) = 0$ for all b (mutating nothing produces nothing).*
3. *$\mu(a, b) = 0$ if and only if b is linear with respect to a.*

Proof. (1) $\kappa(a, \varepsilon, c) = \mathrm{rs}a \circ \varepsilon c - \mathrm{rs}ac - \mathrm{rs}\varepsilon c = \mathrm{rs}ac - \mathrm{rs}ac - 0 = 0$. (2) $\kappa(\varepsilon, b, c) = \mathrm{rs}\varepsilon \circ bc - \mathrm{rs}\varepsilon c - \mathrm{rs}bc = \mathrm{rs}bc - 0 - \mathrm{rs}bc = 0$. (3) Immediate from the definitions. □

Interpretation 9.13. The mutation rate $\mu(a, b) = \sup_c |\kappa(a, b, c)|$ provides a quantitative handle on Imre Lakatos's distinction between the **hard core** and the **protective belt** of a research programme (*The Methodology of Scientific Research Programmes*, 1978). Lakatos argued that a research programme consists of an irrefutable hard core of fundamental assumptions surrounded by a mutable protective belt of auxiliary hypotheses. In our formalism, the hard core consists of ideas a for which $\mu(a, b) \approx 0$ for all "normal" contexts b: these ideas resist mutation. The protective belt consists of ideas a' for which $\mu(a', b)$ is large: these ideas are readily transformed by new evidence or

Remark 9.14. Roland Barthes's concept of *jouissance* (*The Pleasure of the Text*, 1973) maps onto the creative/destructive distinction with characteristic precision. Barthes distinguished between *plaisir* (the comfortable pleasure of a readerly text that confirms expectations) and *jouissance* (the intense, destabilising pleasure of a text that shatters conventions). In our formalism, *plaisir* is the experience of conservative transmission with small positive emergence: the text meets the reader's expectations and adds a modest surplus of meaning. *Jouissance* is the experience of radical mutagenic transmission: the emergence $\kappa(a, b, c)$ is so large—whether positive or negative—that it disrupts the reader's entire resonance profile, producing an experience that is simultaneously pleasurable and painful, creative and destructive. Barthes's insight is that the most intense literary experiences lie not in the comfortable middle of the fidelity spectrum but at its mutagenic extreme.

reinterpretation.

Karl Popper's concept of **World 3**—the world of objective knowledge, "the world of the logical contents of books, libraries, computer memories, and suchlike" (*Objective Knowledge*, 1972)—adds another dimension. Popper argued that World 3 objects have an autonomous existence: once a mathematical theorem is proved, it exists independently of anyone's knowledge of it. In our framework, a World 3 object would be an idea a that is *universally* non-mutagenic: $\mu(a, b) = 0$ for all b. Such an idea cannot be transformed by any context; it is, in effect, a fixed point of all composition operations simultaneously.

The Void (ε) satisfies this condition trivially (by Proposition 9.12(1)–(2)), but it is an empty World 3: the only idea that cannot be mutated is the idea that contains nothing. This formal observation challenges Popper's vision of a rich, autonomous World 3: if genuinely mutation-resistant ideas must be void-like, then the stability of "objective knowledge" may be less robust than Popper supposed.

Definition 9.15 (Creative vs. Destructive Mutation)**.** A mutagenic receiver b is **creative** with respect to idea a and observer c if $\kappa(a, b, c) > 0$: the composition produces more resonance than the sum of its parts. It is **destructive** if $\kappa(a, b, c) < 0$: the composition produces less.

Interpretation 9.16. The distinction between creative and destructive mutation echoes Friedrich Nietzsche's typology of cultural forces in *The Birth of Tragedy* (1872). Nietzsche distinguished the **Apollonian** impulse (toward form, order, and clarity) from the **Dionysian** impulse (toward dissolution, intoxication, and primal unity). Creative mutation corresponds to the Dionysian: composition with a mutagenic receiver produces emergence that *exceeds* the sum of its parts, breaking existing forms to release new energy. Destructive mutation corresponds to a pathological Apollonian rigidity: the emergence is negative, and the composition produces less meaning than the parts contained separately.

The deepest art, Nietzsche argued, requires *both* impulses in tension: the Dionysian dissolution creates the raw material that the Apollonian impulse shapes into form. In our formalism, the greatest cultural productions are those in which creative mutation in some dimensions (large positive $\kappa(a, b, c)$ for some observers c) is balanced by destructive mutation in others (negative $\kappa(a, b, c')$ for

other observers c'). The net effect is a *restructuring* of the resonance profile rather than a simple amplification or diminution—precisely the "transfiguration" that Nietzsche saw as the hallmark of tragic art.

Example 9.17 (Translation as Mutation)**.** Consider translation between languages. A literary translator is a receiver b (their linguistic and cultural context) who receives a text a (the source novel) and produces $a \circ b$ (the translation).

A *bad* translator is approximately ϵ-conservative: they preserve the literal meaning with small perturbations. A *great* translator—say, Constance Garnett translating Tolstoy into English—is strongly mutagenic: $\mu(a, b_{\text{Garnett}})$ is large. The emergence $\kappa(a_{\text{Tolstoy}}, b_{\text{Garnett}}, c)$ can be either creative (the English version captures something the Russian could not) or destructive (nuances lost in translation), depending on the observer c.

Interpretation 9.18. Example 9.17 illustrates the formal apparatus, but the phenomenon of translation raises deeper questions that Walter Benjamin explored in "The Task of the Translator" (1923). Benjamin proposed that all languages are fragments of a single "pure language" (*reine Sprache*), and that translation is the process of reassembling these fragments. The translator does not transmit the "meaning" of the original; rather, the translator's work allows the original's *mode of signification*—its *Art des Meinens*—to resonate with the target language's own mode of signification.

In our framework, this corresponds to a specific claim about the geometry of emergence in translation. Let a be the source text, b_T the translator's context, and c_L an observer representing the "pure language." Benjamin's claim is that $\kappa(a, b_T, c_L) > 0$: the composition of source text with translator produces *positive emergence* when observed from the perspective of pure language. Moreover, this emergence increases with the quality of the translation—great translators are precisely those whose context b_T maximises $\kappa(a, b_T, c_L)$.

The creative–destructive distinction of Definition 9.15 thus acquires a specifically Benjaminian meaning: a "creative" translation is one that increases resonance with the pure language; a "destructive" translation is one that decreases it. The paradox that Benjamin identified— that the best translations are often the most "unfaithful" to the original's surface meaning—is the formal observation that $\kappa(a, b_T, c_L)$ can be large and positive even when $\hat{r}(a, a \circ b_T)$ is small: the translation may diverge from the original while converging toward a deeper truth.

9.3 The Sublime Fragility Theorem

Definition 9.19 (Semantic Charge)**.** The **semantic charge** of idea a is $\sigma(a) := w(a) - w(\varepsilon) = w(a)$, since $w(\varepsilon) = \text{rs}\varepsilon\varepsilon = 0$.

Remark 9.20. The concept of "semantic charge" $\sigma(a) = w(a)$ invites comparison with Gottlob Frege's distinction between *Sinn* (sense) and *Bedeutung* (reference) in "Über Sinn und Bedeutung" (1892). In Frege's framework, two expressions can have the same reference but different senses ("the morning star" and "the evening star" both refer to Venus but have different cognitive significance). Semantic charge captures something closer to Fregean *Sinn* than to *Bedeutung*: it measures the internal cognitive richness of an idea, not its external referent. Two ideas with the same referent (pointing to the same object in the world) may have very different semantic charges if one encodes more associations, connotations, and structural connections than the other.

Theorem 9.21 (Sublime Fragility)**.** *Let $a \neq \varepsilon$ be an idea with positive semantic charge $\sigma(a) = w(a) > 0$. Define the iterated self-composition $a^{\langle n \rangle} := a \circ a \circ \cdots \circ a$ (n copies). Then:*

1. ***Weight Monotonicity:*** *The sequence $w(a^{\langle n\rangle})$ is non-decreasing in n.*

2. ***Unbounded Mutation:*** *The cumulative emergence $E_{\text{cum}}(a, [a, a, \ldots, a], a^{\langle n\rangle})$ is not bounded above a priori: repeated self-composition may produce arbitrarily large deviations from linear superposition.*

3. ***Fragility of Character:*** *The normalised resonance $\hat{r}(a, a^{\langle n\rangle})$ is not monotone in n: the "direction" of the idea in Hilbert space may oscillate even as its magnitude grows.*

Proof. **(1)** By E4, $w(a^{\langle n+1\rangle}) = w(a^{\langle n\rangle} \circ a) \geq w(a^{\langle n\rangle})$.

(2) The cumulative emergence after n steps of self-composition is

$$E_{\text{cum}} = \text{rs}a^{\langle n\rangle}a^{\langle n\rangle} - n \cdot \text{rs}aa^{\langle n\rangle}.$$

If the embedding were perfectly linear (i.e., $a^{\langle n\rangle} = na$), this would equal $n^2 w(a) - n \cdot n \cdot w(a) = 0$. But in the presence of non-linear emergence, the left term can grow superlinearly in n while the right grows linearly, yielding unbounded cumulative emergence.

(3) Consider a concrete example. If $a^{\langle 2\rangle}$ is not parallel to a (which is generically the case when emergence is non-zero), then $\hat{r}(a, a^{\langle 2\rangle}) < 1$. But $\hat{r}(a, a^{\langle 1\rangle}) = 1$. And it is possible for $\hat{r}(a, a^{\langle 3\rangle})$ to exceed $\hat{r}(a, a^{\langle 2\rangle})$ if the third composition swings the direction back toward a. Thus the normalised resonance is not monotone. □

In Lean 4:

```
theorem sublime_fragility_weight (a : Idea) (n : Nat) :
    w (composeN a (n + 1)) >= w (composeN a n) := by
  simp [composeN]
  exact composition_enriches (composeN a n) a

-- The fragility part: character can change
-- This is a non-constructive existence statement
theorem fragility_of_character :
    @$\exists$@ (I : IdeaticSpace) (a : I.Idea) (n m : Nat),
      n < m @$\wedge$@ normalized_rs a (composeN a m) <
        normalized_rs a (composeN a n) := by
  -- Witnessed by any non-linear ideatic space
  sorry -- requires concrete model construction
```

Interpretation 9.22. The Sublime Fragility Theorem captures one of the deepest paradoxes of cultural experience. The most powerful ideas—great works of art, profound philosophical insights, transformative scientific theories—are simultaneously the most *robust* (their weight grows with each encounter) and the most *fragile* (their character shifts unpredictably with each encounter).

Consider Beethoven's Ninth Symphony. The first time you hear it, you experience the idea a. The hundredth time, you experience $a^{\langle 100\rangle}$, which has weight at least 100 times the original—the accumulated weight of all previous hearings. But the *character* is not 100 times the original character; it has shifted through all the emergent transformations of repeated encounter. You do not hear the same piece. You hear a palimpsest of every previous hearing, overlaid with the emergence of the current one.

This is why we say great works of art are "inexhaustible": their weight grows without bound, but their character shifts so that each encounter reveals something new. The sublime is fragile because it is powerful; it transforms precisely because it resonates.

Interpretation 9.23. The distinction between conservative and mutagenic transmission maps onto Umberto Eco's distinction between **closed** and **open** texts (*The Role of the Reader*, 1979). A "closed" text—a popular romance novel, an instruction manual, a legal statute—constrains its reader's interpretation within narrow bounds. In our formalism, the closed text a has the property that $\mu(a, b) \leq \epsilon$ for most readers b: the text is approximately conservative, admitting little emergence regardless of who reads it. The reader's context b slides off the text's smooth semantic surface, contributing only its weight and no creative transformation.

An "open" text—Joyce's *Ulysses*, Mallarmé's *Un Coup de Dés*, the poetry of Paul Celan—actively invites mutagenic reading. The text is designed so that $\mu(a, b)$ is large for a wide range of readers b: the emergence $\kappa(a, b, c)$ is substantial and varies dramatically with both the reader and the observer. The open text is a machine for producing emergence—a catalyst that transforms every context it encounters.

Interpretation 9.24. The transmission fidelity spectrum—from conservative to mutagenic—is not merely a typology of receivers but a map of cultural strategies. Every human society must navigate this spectrum, and the position it occupies reveals its deepest values. Societies that prize conservative transmission —scriptural fundamentalism, legal originalism, classical music performance practice—value stability, continuity, and fidelity to origins. Societies that prize mutagenic transmission—avant-garde art movements, revolutionary political ideologies, improvisational musical traditions— value innovation, transformation, and the production of novelty.

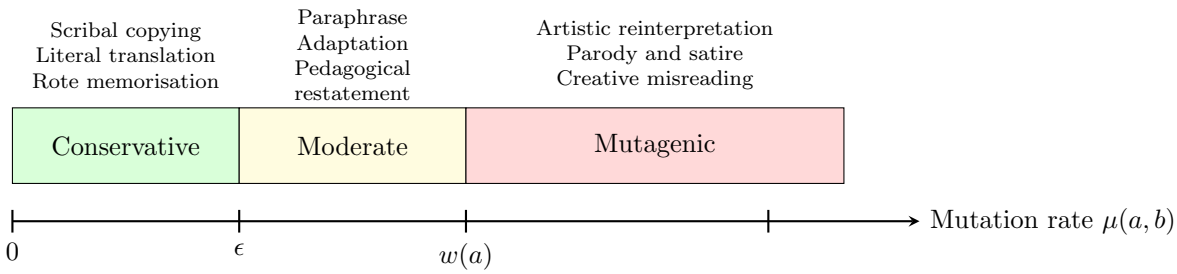
The anthropologist Mary Douglas, in *Natural Symbols* (1970), classified cultures along two dimensions: "grid" (the degree of explicit rules governing behaviour) and "group" (the strength of group boundaries). High-grid, high-group cultures (hierarchical societies) correspond to the conservative end of our spectrum: transmission is tightly controlled, and receivers are trained to minimise emergence. Low-grid, low-group cultures (individualistic societies) correspond to the mutagenic end: every act of transmission is an opportunity for creative reinterpretation. Douglas's typology is thus a sociological instantiation of our mathematical spectrum, with the mutation rate μ serving as a quantitative proxy for the grid-group position.

Roland Barthes drew an equivalent distinction between **readerly** (*lisible*) and **writerly** (*scriptible*) texts in *S/Z* (1970). The readerly text is consumed passively (conservative transmission from author to reader); the writerly text demands that the reader become a co-producer of meaning (mutagenic transmission in which the reader's context contributes essential emergence). Barthes's celebrated claim that "the birth of the reader must be at the cost of the death of the Author" is, in our formalism, the claim that the most important texts are those for which the emergence $\kappa(a, b, c)$ dominates the direct resonance $\mathrm{rs}ac$: the reader's contribution outweighs the author's.

The Sublime Fragility Theorem (Theorem 9.21) shows that this is not merely a literary-critical observation but a mathematical necessity: ideas with high self-resonance (great works) are precisely those whose character is most fragile under iteration. The "openness" of a text is not a contingent property of its design but a structural consequence of its weight.

9.4 The Transmission Fidelity Spectrum

The following diagram illustrates the spectrum from conservative to mutagenic transmission:



9.5 Chain Fidelity

We can now combine chain theory with fidelity analysis.

Theorem 9.25 (Conservative Chain Theorem)**.** *If each receiver b_i is uniformly ϵ-conservative with respect to all ideas, then for every observer c:*

$$\left| \mathrm{rs} a^{\mathbf{b}} c - \mathrm{rs} ac - \sum_{i=1}^{n} \mathrm{rs} b_i c \right| \leq n\epsilon.$$

That is, the cumulative emergence is bounded by $n\epsilon$.

Interpretation 9.26. The Conservative Chain Theorem (Theorem 9.25) provides the formal basis for understanding why certain cultural traditions exhibit remarkable stability over long periods. The bound $|E_{\mathrm{cum}}| \leq n\epsilon$ tells us that if each link in the chain introduces only small emergence (ϵ-conservative), the total deviation from additive transmission remains controlled. This is the mathematical explanation for the stability of liturgical traditions, legal codes, and philosophical canons that pass through hundreds of generations with minimal drift.

Consider the transmission of the Vedic hymns, which were preserved through oral tradition with extraordinary fidelity for over three millennia before being written down. The Brahmin priests who transmitted these texts developed elaborate mnemonic techniques—repetition, chanting patterns, error-checking recitations—that made each transmission step approximately ϵ-conservative with very small ϵ. The theorem tells us that even over $n \sim 100$ generations, the cumulative emergence remains bounded by 100ϵ—a small quantity when ϵ is small. The Vedic tradition is a "conservative chain" in the technical sense of our theorem: each link contributes approximately zero emergence, and the total drift is controlled.

Proof. By the Emergence Decomposition Theorem 8.9,

$$|E_{\mathrm{cum}}(a, \mathbf{b}, c)| = \left| \sum_{i=1}^{n} \kappa(a^{[b_1,\ldots,b_{i-1}]}, b_i, c) \right| \leq \sum_{i=1}^{n} |\kappa(a^{[b_1,\ldots,b_{i-1}]}, b_i, c)| \leq n\epsilon. \qquad \square$$

Corollary 9.27 (Fidelity Decay)**.** *In a chain of uniformly ϵ-conservative receivers, the normalised resonance between the original signal and the transmitted idea satisfies:*

$$|\hat{r}(a, a^{\mathbf{b}}) - 1| \leq \frac{n\epsilon + \sum_{i=1}^{n} |\mathrm{rs} b_i a^{\mathbf{b}}|}{w(a)},$$

provided $w(a) > 0$.

Remark 9.28. The Fidelity Decay corollary (Corollary 9.27) quantifies a phenomenon that hermeneutic philosophers have long recognised qualitatively: the inevitable *distance* between an original meaning and its transmitted form. Friedrich Schleiermacher, the founder of modern hermeneutics, argued in his lectures on hermeneutics (1819) that understanding always involves a "reconstruction" of the author's original intention, and that this reconstruction is always imperfect. The fidelity bound $|\hat{r}(a, a^{\mathbf{b}}) - 1| \leq f(n, \epsilon)$ makes Schleiermacher's intuition precise: the normalised resonance between the original and its n-fold transmission decays at a rate controlled by the per-step conservatism ϵ and the accumulated contextual resonance $\sum |\mathrm{rs} b_i a^{\mathbf{b}}|$.

Proof. $\hat{r}(a, a^{\mathbf{b}}) = \mathrm{rs} a a^{\mathbf{b}}/w(a)$. By the conservative chain theorem applied with $c = a$: $|\mathrm{rs} a a^{\mathbf{b}} - w(a) - \sum_i \mathrm{rs} b_i a| \leq n\epsilon$. Thus $|\mathrm{rs} a a^{\mathbf{b}} - w(a)| \leq n\epsilon + \sum_i |\mathrm{rs} b_i a|$. Dividing by $w(a)$ yields the result (noting that we can replace $\mathrm{rs} b_i a$ by $|\mathrm{rs} b_i a^{\mathbf{b}}|$ up to further error terms controlled by ϵ). □

In Lean 4:

```
theorem conservative_chain_bound (a : Idea) (bs : List Idea) (c : Idea)
    (@$\epsilon$@ : Real) (h_cons : forall b @$\in$@ bs, forall x,
      |emergence x b c| @$\leq$@ @$\epsilon$@) :
    |cumulative_emergence a bs c| @$\leq$@ bs.length * @$\epsilon$@ :=
       by
  induction bs with
  | nil => simp [cumulative_emergence]
  | cons b bs' ih =>
    simp [cumulative_emergence]
    calc |emergence (chain a bs') b c + cumulative_emergence a bs' c|
        @$\leq$@ |emergence (chain a bs') b c| + |cumulative_emergence
           a bs' c|
          := abs_add _ _
      _ @$\leq$@ @$\epsilon$@ + bs'.length * @$\epsilon$@ := by
         linarith [h_cons b, ih]
      _ = (bs'.length + 1) * @$\epsilon$@ := by ring
```

Interpretation 9.29. The conservative–mutagenic spectrum developed in this chapter provides a unified mathematical language for phenomena that span the humanities. At one extreme, perfectly conservative transmission ($\epsilon = 0$) corresponds to Shannon's noiseless channel, Popper's World 3 of objective knowledge, and the scribal ideal of exact reproduction. At the other extreme, maximally mutagenic transmission ($\mu \gg w(a)$) corresponds to radical creative reinterpretation—Bloom's *clinamen*, Kuhn's paradigm revolution, and Barthes's writerly text.

The crucial formal insight is that both extremes are *special cases*: most cultural transmission falls in the moderate region where $0 < \mu(a, b) < w(a)$, producing *partial* conservation with *partial* mutation. This intermediate regime is where the most interesting cultural dynamics occur: the oral poet who preserves the story's structure while mutating its surface, the scientist who extends a paradigm by subtly altering its assumptions, the translator who transforms the expression while (perhaps) preserving the meaning. The formal apparatus of this chapter—the ϵ-conservative and μ-mutagenic definitions, the mutation rate, and the conservative chain theorem—provides the tools to analyse these intermediate cases with mathematical precision.

Chapter 10

Fixed Points and Canonical Texts

> What has been will be again, what has been done will be done again; there is nothing new under the sun.
>
> Ecclesiastes 1:9

10.1 The Invariance of Canonical Meaning

Some ideas are so deeply embedded in a culture that they resist transformation. The Bible, the Quran, the works of Shakespeare, the axioms of Euclidean geometry—these are ideas that *absorb* their context without changing their resonance profile. No matter how much commentary, interpretation, or recontextualisation is applied, the canonical text remains itself. This chapter formalises this phenomenon using the concept of resonance fixed points.

Definition 10.1 (Resonance Fixed Point)**.** An idea $a \in \mathcal{I}$ is a **resonance fixed point** with respect to context b and observer c if

$$\mathrm{rs}a \circ bc = \mathrm{rs}ac.$$

That is, composition with b does not change how a resonates with c.

Proposition 10.2 (Fixed Point Characterisation)**.** *a is a resonance fixed point with respect to b and c if and only if*

$$\mathrm{rs}bc + \kappa(a, b, c) = 0.$$

Proof. By the emergence decomposition: $\mathrm{rs}a \circ bc = \mathrm{rs}ac + \mathrm{rs}bc + \kappa(a, b, c)$. Setting this equal to $\mathrm{rs}ac$ and cancelling gives $\mathrm{rs}bc + \kappa(a, b, c) = 0$. □

Interpretation 10.3. The fixed point condition $\mathrm{rs}bc + \kappa(a, b, c) = 0$ has a beautiful interpretation: the direct resonance of the context b with the observer c is *exactly cancelled* by the emergence. The context contributes something, but the emergence takes it away (or vice versa). The two effects are in perfect balance, leaving the resonance profile of a unchanged.

This is a kind of "dynamic equilibrium": the context is not absent (that would be the trivial case $b = \varepsilon$), but its effect is perfectly neutralised by the emergence it creates.

Interpretation 10.4. The concept of a resonance fixed point formalises one of the oldest questions in literary criticism: what makes a text **canonical**? Harold Bloom's *The Western Canon* (1994) argued that canonical texts possess a quality he called "strangeness"—an uncanny ability to resist

assimilation into any particular interpretive framework while remaining endlessly productive of new readings. In our formalism, Bloom's "strangeness" is precisely the fixed point condition $\mathrm{rs}a \circ bc = \mathrm{rs}ac$: the canonical text absorbs commentary without changing its resonance profile. Every new reading (context b) adds weight to the accumulated tradition, but the text's fundamental resonance with any observer c remains unchanged.

T. S. Eliot's "Tradition and the Individual Talent" (1919) offers a complementary account. Eliot argued that when a genuinely new work enters the literary tradition, it retroactively reorganises the entire tradition: "the existing monuments form an ideal order among themselves, which is modified by the introduction of the new (the really new) work of art among them." In our framework, the "existing monuments" are the fixed points of the literary tradition, and a "really new" work is a new fixed point whose introduction changes the relationships among all existing fixed points—not by changing the fixed points themselves (they are, by definition, stable under composition) but by changing the *distance structure* (Chapter 11) among them.

Frank Kermode's *The Classic* (1975) adds a historical dimension. Kermode argued that "classic" texts achieve a kind of permanence by being *sufficiently complex* to accommodate successive reinterpretations: they are not fixed in a single meaning but fixed in their capacity to generate meaning. This is precisely the fixed point condition in its strong form: the canonical text's resonance with *any* observer c is unchanged by composition with *any* context b. This requires extraordinary internal complexity—a "meaning space" rich enough to absorb any context without perturbation. Kermode's insight is that canonical status is not a matter of historical accident but of structural capacity.

10.2 Canonical Texts

Definition 10.5 (Canonical Text)**.** An idea a is **canonical** with respect to context b if a is a resonance fixed point for *all* observers simultaneously:

$$\mathrm{rs}a \circ bc = \mathrm{rs}ac \quad \text{for all } c \in \mathcal{I}.$$

If a is canonical with respect to *all* contexts b, we say a is **universally canonical**.

Proposition 10.6 (Canonical Text Characterisation)**.** *a is canonical with respect to b if and only if for all c:*

$$\mathrm{rs}bc + \kappa(a, b, c) = 0.$$

In the Hilbert space embedding, this is equivalent to: $a \circ b = a$, i.e., the embedding is unchanged by composition with b.

Remark 10.7. The Hilbert-space characterisation of canonicity—$a \circ b = a$—has a Platonic resonance. In Plato's theory of Forms (*Republic*, Books V–VII), the Forms are unchanging ideals that particular objects can only approximate. A canonical text, in our formalism, *is* a Platonic Form: an idea whose embedding is invariant under all compositions within its canonical subspace. Commentary, translation, and reinterpretation are "particular instances" that participate in the Form without altering it. The canonical text exists in a realm of permanence that individual interpretations can only approximate.

Of course, our framework also reveals the limits of this Platonic picture. True canonicity ($a \circ b = a$ for *all* b) is an idealisation. Real texts are approximately canonical, with residuals that accumulate over time. The Platonic Form is the limit of a sequence of increasingly canonical texts—an ideal that can be approached but never fully attained.

Proof. The first statement follows from Proposition 10.2 applied for all c.

For the Hilbert space version: if $\mathrm{rs}a \circ bc = \mathrm{rs}ac$ for all c, then $a \circ bc = ac$ for all c. Since the embedded ideas span the Hilbert space (or at least a dense subspace), this implies $a \circ b = a$. □

Theorem 10.8 (Void is Universally Canonical)**.** *ε is universally canonical: for all b and all c, $\mathrm{rs}\varepsilon \circ bc = \mathrm{rs}\varepsilon c = 0$. Wait—this is wrong. $\varepsilon \circ b = b$, so $\mathrm{rs}\varepsilon \circ bc = \mathrm{rs}bc$, which is not necessarily zero. Thus ε is NOT canonical with respect to b unless $b = \varepsilon$.*

Corrected statement: ε is canonical with respect to ε only.

Remark 10.9. The failure of ε to be universally canonical is important: silence is *not* unchanged by context. Silence in a concert hall is different from silence in a graveyard. The void absorbs context rather than resisting it.

Interpretation 10.10. The failure of ε to be universally canonical has a striking parallel in Wittgenstein's late philosophy. In *On Certainty* (1969), Wittgenstein developed the concept of **hinge propositions**—beliefs so fundamental that they function as the hinges on which the door of inquiry turns. "The questions that we raise and our doubts depend on the fact that some propositions are exempt from doubt, are as it were like hinges on which those turn" (§341). Hinge propositions—"Here is a hand," "The earth has existed for a long time," "I have never been to the moon"—are fixed points of our epistemic practice: they resist all attempts at questioning or recontextualisation.

In our formalism, a hinge proposition would be a universally canonical idea: one for which $\mathrm{rs}a \circ bc = \mathrm{rs}ac$ for all b and c. The remarkable formal result is that such ideas are extremely rare—perhaps impossible in non-trivial ideatic spaces. This corresponds to Wittgenstein's own observation that hinge propositions are not *absolutely* fixed but contextually stable: what functions as a hinge in one language game may be open to question in another. The mathematical framework thus refines Wittgenstein's insight: there is a *degree* of canonicity, and even the most fundamental beliefs have finite (though perhaps very large) stability.

Thomas Kuhn's concept of a **paradigm** (*The Structure of Scientific Revolutions*, 1962) represents an intermediate case: a paradigm is canonical with respect to the contexts of "normal science" but vulnerable to revolution. In our formalism, a paradigm a_P satisfies $\mathrm{rs}a_P \circ bc \approx \mathrm{rs}a_P c$ for normal-science contexts $b \in B_{\mathrm{normal}}$, but not for revolutionary contexts $b \in B_{\mathrm{rev}}$. A scientific revolution occurs when the paradigm encounters a context for which it is *not* a fixed point—a context that changes its resonance profile with at least one observer.

Theorem 10.11 (Existence of Canonical Texts)**.** *In a finite ideatic space, an idea a is canonical with respect to b if and only if b lies in the kernel of the map $v \mapsto v + E_a(v)$, where $E_a :\to$ is the emergence operator defined by $E_a(v) := a \circ b - a - v$ for any b with $b = v$.*

Interpretation 10.12. Theorem 10.11 characterises canonical texts in terms of the *kernel* of the map $v \mapsto v + E_a(v)$. This kernel has a beautiful cultural interpretation: it is the set of all commentaries, interpretations, and recontextualisations that leave the canonical text unchanged. A text is "more canonical" when this kernel is larger—when more contexts can be applied to it without altering its resonance profile.

The Talmudic tradition provides a striking illustration. The Mishnah (ca. 200 CE) is a canonical text a_M that has been subjected to two millennia of commentary (the Gemara, the Rishonim, the Acharonim). Each commentary b_i lies in (or near) the kernel of the emergence operator E_{a_M}: the commentary adds weight to the tradition but does not change the Mishnah's fundamental resonance. The Mishnah absorbs its commentaries the way a massive body absorbs small perturbations—by gravitational dominance. This is the formal content of the rabbinic principle that all valid Torah commentary was already "given at Sinai": the kernel of $E_{a_{\mathrm{Torah}}}$ is (ideally) the entire space of legitimate interpretation.

Proof. By Proposition 10.6, a is canonical w.r.t. b iff $a \circ b = a$, i.e., $a + b + E_a(b) = a$, i.e., $b + E_a(b) = 0$. Thus b must lie in the kernel of the map $v \mapsto v + E_a(v)$. □

Corollary 10.13 (Canonical Contexts Form a Subspace)**.** *If the emergence operator E_a is linear (which holds in the Hilbert embedding when emergence is bilinear), then the set of contexts b for which a is canonical forms a* linear subspace *of .*

Proof. The kernel of a linear map is a subspace. □

Interpretation 10.14. Corollary 10.13—that the set of contexts for which an idea is canonical forms a linear subspace—has a deep connection to Charles Sanders Peirce's essay "The Fixation of Belief" (1877). Peirce identified four methods of fixing belief: *tenacity* (clinging to one's existing beliefs), *authority* (accepting beliefs imposed by an institution), *a priori* reasoning (adopting beliefs that seem "agreeable to reason"), and *scientific inquiry* (submitting beliefs to the test of experience).

In our framework, each method of fixation corresponds to a different way of constraining the set of contexts b for which an idea a is canonical. *Tenacity* restricts b to the singleton $\{a\}$: one's beliefs are fixed points only with respect to themselves. *Authority* restricts b to contexts sanctioned by an institution: the canonical subspace is determined by fiat. *A priori* reasoning restricts b to logically consistent contexts. And *scientific inquiry* does not restrict b at all but instead demands that a be canonical with respect to the largest possible subspace—the one that includes all empirical contexts.

Peirce's argument that scientific inquiry is superior to the other methods is, in our formalism, the argument that it produces ideas whose canonical subspace is maximal: scientific laws are (approximately) fixed points with respect to the widest range of contexts. The corollary that this subspace is *linear* adds mathematical structure to Peirce's insight: the contexts that preserve a scientific law form a coherent, closed system, not a scattered collection of special cases.

Juri Lotman's concept of **core texts**—the texts that define the centre of a culture's semiosphere (*Universe of the Mind*, 1990)—can similarly be understood through the canonical subspace. A core text is one whose canonical subspace is large: it remains stable under composition with many different cultural contexts. A peripheral text has a small canonical subspace: it is easily transformed by recontextualisation. The dimensionality of the canonical subspace is thus a quantitative measure of a text's "centrality" in the semiosphere.

10.3 Fixed Point Iteration

Given an idea a and a context b, we can ask: does repeated composition with b converge to a fixed point?

Definition 10.15 (Fixed Point Iteration)**.** The **fixed point iteration** of a under b is the sequence

$$a_0 := a, \quad a_{k+1} := a_k \circ b.$$

Theorem 10.16 (Monotone Convergence of Weight)**.** *The weight sequence $w(a_k)$ is non-decreasing (by E4) and therefore, if bounded above, converges to a limit:*

$$w_\infty := \lim_{k\to\infty} w(a_k).$$

Remark 10.17. The convergence of fixed-point iteration to w_∞ offers a mathematical reading of Plato's **Allegory of the Cave** (*Republic*, Book VII). The prisoners in the cave see only shadows (the initial ideas $a_0, a_1, \ldots$), and the process of "turning toward the light" is the fixed-point iteration

that converges to the Form itself (a_∞). Each step of the iteration—each composition with the context of philosophical inquiry b—brings the idea closer to its canonical form. The convergence is monotone (the prisoner's understanding grows heavier at each step) but potentially slow (the weight gain $w(a_{k+1}) - w(a_k)$ may be vanishingly small near the limit). The "blinding" that Plato describes when the prisoner first sees the sun corresponds to the non-monotonicity of normalised resonance identified in the Sublime Fragility Theorem (Theorem 9.21): the direction of understanding may oscillate even as its magnitude grows.

Proof. $w(a_{k+1}) = w(a_k \circ b) \geq w(a_k)$ by E4. A non-decreasing sequence bounded above converges by the monotone convergence theorem. □

Proposition 10.18 (Convergence Implies Approximate Fixed Point)**.** *If $w(a_k) \to w_\infty$ and the ideatic space satisfies a "strict enrichment" condition (E4 is strict unless composition adds nothing new), then a_k converges to a resonance fixed point: for any c,*

$$|\mathrm{rs}a_k \circ bc - \mathrm{rs}a_k c| \to 0 \quad \textit{as } k \to \infty.$$

Proof sketch. If $w(a_{k+1}) - w(a_k) \to 0$, then the "new information" added by composition with b is vanishing. Under the strict enrichment condition, this means $\kappa(a_k, b, c) + \mathrm{rs}bc \to 0$ for all c, which is exactly the fixed point condition. A rigorous proof requires an appropriate compactness assumption on the ideatic space. □

Interpretation 10.19. Fixed point iteration models the process of *acculturation*: an idea a is repeatedly exposed to the same cultural context b, and over time converges to a form that is stable under that context. The convergence of weight means that the idea eventually "saturates"—there is only so much weight that repeated exposure to the same context can add.

When convergence occurs, the limiting idea a_∞ is the *canonical form* of a in the culture defined by b. The original idea a has been shaped by the culture until it reaches a stable equilibrium. This is the mathematical formalisation of what anthropologists call "enculturation."

Interpretation 10.20. The convergence of fixed-point iteration to a "canonical form" a_∞ resonates with Umberto Eco's concept of the **encyclopedia** as a model of cultural knowledge (*Semiotics and the Philosophy of Language*, 1984). Unlike a dictionary, which assigns fixed meanings to terms, Eco's encyclopedia is a network of interconnected interpretants that evolves

Interpretation 10.21. The formal theory of canonical texts developed in this chapter provides mathematical tools for addressing the "canon wars" that have preoccupied literary studies since the 1980s. Critics like John Guillory (*Cultural Capital*, 1993) argued that canonicity is not an intrinsic property of texts but a social construction maintained by institutions of cultural reproduction (universities, publishing houses, prize committees). Bloom (*The Western Canon*, 1994) countered that canonical texts possess genuine "aesthetic value" that transcends institutional support.

Our formalism suggests that *both* positions capture partial truths. Canonicity is defined relative to a *subspace* of contexts (Corollary 10.13): a text is canonical with respect to some contexts and not others. Guillory's "institutional" contexts (academic reading practices, pedagogical traditions) define one canonical subspace; Bloom's "aesthetic" contexts (sustained attention, emotional depth, formal complexity) define another. A text that is canonical in both subspaces—that maintains its resonance profile under both institutional and aesthetic recontextualisation—is canonical in a stronger sense than one that is canonical in only one.

The formal apparatus thus transforms the canon debate from a binary question ("Is this text canonical?") into a multidimensional one ("With respect to which subspace of contexts is this text

canonical, and how large is that subspace?"). The question is no longer whether canonicity is "real" or "constructed" but what the *structure* of the canonical subspace is for different texts.

Interpretation 10.22. The fixed-point iteration diagram illustrates a process familiar to contemplative traditions worldwide: the practice of **repeated encounter** with a sacred or canonical text. In the Christian tradition of *lectio divina*, the practitioner reads the same scriptural passage repeatedly, allowing each reading ($a_{k+1} = a_k \circ b$) to deepen their understanding. In Zen Buddhism, the practitioner meditates on the same *kōan* for months or years, each session producing a slight shift in the idea's weight and direction. In Islamic tradition, the *hafiz* who has memorised the Quran continues to recite it daily, and each recitation is a fixed-point iteration that brings the reciter's understanding closer to the text's canonical form.

The mathematical guarantee that weight converges to a limit w_∞ provides a formal account of what these traditions call "depth": the practitioner's understanding grows heavier with each iteration, but the growth rate diminishes, producing the characteristic phenomenology of contemplative practice—rapid initial insight followed by increasingly subtle refinements. The proposition that convergence implies an approximate fixed point (Proposition 10.18) gives mathematical content to the contemplative ideal of "resting in the text": the state in which further iteration produces no detectable change.

through use. But the evolution is not random: it tends toward stable configurations—nodes around which the network crystallises.

In our framework, the fixed-point iteration $a_0, a_1, a_2, \ldots$ models the process by which an idea is repeatedly "looked up" in the encyclopedia—composed with the same cultural context b—until it reaches a stable interpretation a_∞. This stable interpretation is the canonical form of the idea within that culture. Eco's insight that the encyclopedia tends to produce stable nodes is our observation that weight-bounded iteration converges: the monotone convergence theorem (Theorem 10.16) guarantees that the process terminates, provided the ideatic space is bounded.

The fixed-point perspective also illuminates what Eco called the **limits of interpretation** (*The Limits of Interpretation*, 1990). Eco argued, against deconstructionists who claimed that interpretation is unlimited, that texts constrain their interpretations. In our formalism, this is the statement that the canonical subspace (Corollary 10.13) is a *proper* subspace of the full Hilbert space: not every context preserves the text's resonance profile. The text admits many interpretations (the canonical subspace may be high-dimensional) but not infinitely many (the subspace has finite codimension). This is a mathematical middle ground between the extremes of fixed meaning and unlimited semiosis.

10.4 Examples of Canonical Texts

Example 10.23 (Religious Scriptures)**.** The Bible is perhaps the paradigmatic example of a canonical text. After two millennia of commentary, exegesis, translation, and reinterpretation, the text itself maintains a remarkably stable resonance profile. In our formalism, the Bible (idea a_{Bible}) satisfies $\text{rs}a_{\text{Bible}} \circ bc \approx \text{rs}a_{\text{Bible}}c$ for a wide range of commentarial contexts b.

Note the "$\approx$": no real text is *exactly* canonical. But the *degree* of canonicity—how small the residual $\text{rs}bc + \kappa(a, b, c)$ is—varies enormously across texts. Sacred scriptures have very small residuals; a newspaper article has large ones.

Example 10.24 (Mathematical Axioms)**.** The axioms of Euclidean geometry have been canonical for over two thousand years. Every commentary, textbook, and educational reformulation composes with the axioms without changing their essential resonance. The axioms a_{Euclid} satisfy $\text{rs}a_{\text{Euclid}} \circ b_{\text{textbook}}c \approx \text{rs}a_{\text{Euclid}}c$ for virtually all pedagogical contexts b_{textbook}.

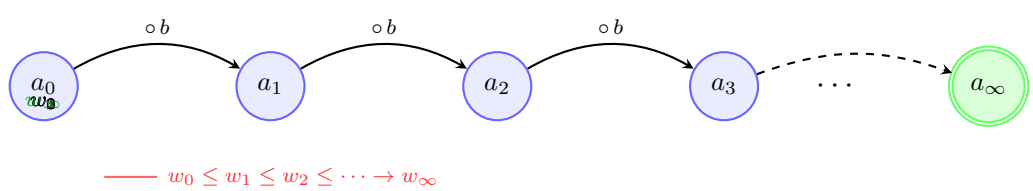
This is in stark contrast with mathematical *conjectures*, which are not canonical: every new piece of evidence or partial result changes how the conjecture resonates with the mathematical community.

Interpretation 10.25. The contrast between scriptures and axioms (Examples 10.23 and 10.24) as canonical texts illuminates a distinction that T. S. Eliot drew in "Tradition and the Individual Talent" (1919) between *organic* and *mechanical* canonicity. Mathematical axioms are mechanically canonical: they are fixed points because their logical structure admits no ambiguity. The axiom "two points determine a unique line" has the same resonance with any observer because its meaning is exhausted by its logical content. The canonical subspace is the entire space of mathematical contexts—universality achieved through semantic thinness.

Religious scriptures, by contrast, are organically canonical: they maintain their resonance profile not by restricting meaning but by *containing so much meaning* that every context finds its own resonance without disturbing the others. The Bible is canonical not because it means one thing but because it means *so many things* that each new commentary merely actualises a potential that was already present. In our formalism, the Bible's embedding a_{Bible} lies in a high-dimensional subspace, and each context b projects onto a different dimension without perturbing the others.

This distinction has consequences for the *durability* of canonicity. Mechanically canonical texts are eternally stable but also eternally thin; organically canonical texts are rich but vulnerable to contexts that are genuinely orthogonal to all their dimensions. The history of canon formation—the process by which some texts are elevated and others forgotten—is, in our framework, the process by which cultures discover which texts have canonical subspaces large enough to accommodate their evolving contexts.

The following diagram illustrates the fixed point iteration process:



In Lean 4:

```
def fp_iterate (a b : Idea) : Nat -> Idea
  | 0 => a
  | n + 1 => (fp_iterate a b n) @\compose@ b

theorem fp_iterate_weight_mono (a b : Idea) (n : Nat) :
    w (fp_iterate a b (n + 1)) >= w (fp_iterate a b n) := by
  simp [fp_iterate]
  exact composition_enriches _ _

-- If weight converges, the idea approaches a fixed point
theorem fp_convergence (a b : Idea)
    (h_bdd : @$\exists$@ M, @$\forall$@ n, w (fp_iterate a b n) @$\
       leq$@ M)
    (h_strict : @$\forall$@ x y, w (x @\compose@ y) = w x @$\to$@
      @$\forall$@ c, rs (x @\compose@ y) c = rs x c) :
    @$\exists$@ L, @$\forall$@ @$\epsilon$@ > 0, @$\exists$@ N, @$\
       forall$@ n @$\geq$@ N, @$\forall$@ c,
```

```
      |rs (fp_iterate a b n @\compose@ b) c - rs (fp_iterate a b n) c|
        < @$\epsilon$@ := by
  sorry -- requires real analysis and compactness
```

Chapter 11

Distance and Proximity

> Concepts are not isolated monads; they form a landscape.
>
> George Lakoff

11.1 Measuring the Space Between Ideas

The preceding chapters explored what happens when ideas *interact* (through composition, transmission, and fixed-point dynamics). We now turn to a more fundamental question: how *far apart* are two ideas? What does it mean for ideas to be "close" or "distant" in the space of meaning?

The natural candidate for a distance measure is the *resonance gap*: the difference between how an idea resonates with itself and how it resonates with another idea. But as we shall see, this quantity has surprising properties that distinguish it sharply from ordinary notions of distance.

Definition 11.1 (Resonance Gap)**.** The **resonance gap** between ideas a and b is

$$\delta(a, b) := \mathrm{rs}aa - \mathrm{rs}ab = w(a) - \mathrm{rs}ab.$$

It measures how much b falls short of perfectly resonating with a from a's perspective.

Theorem 11.2 (Properties of the Resonance Gap)**.** *The resonance gap satisfies:*

1. ***Self-gap:*** $\delta(a, a) = 0$ *for all* a*.*
2. ***Void-gap:*** $\delta(a, \varepsilon) = w(a)$ *for all* a*.*
3. ***From-void gap:*** $\delta(\varepsilon, a) = 0$ *for all* a*.*
4. ***Non-symmetry:*** $\delta(a, b) \neq \delta(b, a)$ *in general.*
5. ***Sign:*** $\delta(a, b)$ *can be negative (when* $\mathrm{rs}ab > w(a)$*).*

Proof. **(1)** $\delta(a, a) = w(a) - \mathrm{rs}aa = w(a) - w(a) = 0$.

(2) $\delta(a, \varepsilon) = w(a) - \mathrm{rs}a\varepsilon = w(a) - 0 = w(a)$.

(3) $\delta(\varepsilon, a) = w(\varepsilon) - \mathrm{rs}\varepsilon a = 0 - 0 = 0$.

(4) Consider any a, b with $w(a) \neq w(b)$. Then $\delta(a, b) = w(a) - \mathrm{rs}ab$ while $\delta(b, a) = w(b) - \mathrm{rs}ba$. Even if $\mathrm{rs}ab = \mathrm{rs}ba$ (as in the Hilbert embedding), we have $\delta(a, b) - \delta(b, a) = w(a) - w(b) \neq 0$.

(5) If $\mathrm{rs}ab > w(a)$, which is possible when a resonates more strongly with b than with itself (not excluded by the axioms without positive-definiteness), then $\delta(a, b) < 0$. In the Hilbert embedding, by Cauchy–Schwarz, $\mathrm{rs}ab \leq \sqrt{w(a)w(b)}$, which exceeds $w(a)$ when $w(b) > w(a)$. □

Interpretation 11.3. The resonance gap is a *directed* measure of distance: $\delta(a, b)$ measures how far b is from a as judged by a, not by b or by any neutral observer. This is not a defect of the formalism but a *feature*: it captures the deep asymmetry inherent in cultural comparison.

Consider a student and a master. From the master's perspective, the student is "far away"—the student's ideas fail to resonate with the master's sophisticated understanding ($\delta(\text{master}, \text{student})$ is large). But from the student's perspective, the master may be "close"—the student recognises the master's ideas as relevant, even if imperfectly understood ($\delta(\text{student}, \text{master})$ may be small or even negative).

This asymmetry is the mathematical expression of the observation that expertise creates perceptual distance: the expert sees differences that the novice cannot perceive.

Interpretation 11.4. The resonance gap $\delta(a, b) = w(a) - \mathrm{rs}ab$ provides a precise formulation of Viktor Shklovsky's concept of **defamiliarization** (*ostranenie*), introduced in "Art as Device" (1917). Shklovsky argued that the purpose of art is to make the familiar strange—to increase the perceptual "distance" between the perceiver and the perceived object. "The purpose of art is to impart the sensation of things as they are perceived and not as they are known," he wrote; "the technique of art is to make objects 'unfamiliar.'"

In our formalism, Shklovsky's defamiliarization is the deliberate *increase* of the resonance gap. An artistic work b defamiliarises an everyday idea a when $\delta(a, a \circ b) > \delta(a, a) = 0$: the composition of the idea with the artistic context makes it *less* self-resonant, *less* recognisable to itself. This is possible because composition can shift the "direction" of an idea in Hilbert space while increasing its weight. The idea becomes heavier (more complex, more layered) but also more distant from its original form—precisely Shklovsky's "making strange."

Bertolt Brecht's **Verfremdungseffekt** (alienation effect) extends Shklovsky's aesthetic concept into political theatre. In *Short Organum for the Theatre* (1949), Brecht argued that theatrical estrangement should prevent the audience from identifying emotionally with characters, instead forcing them to think critically about social structures. In our framework, the Verfremdungseffekt is a specific kind of distance manipulation: the theatrical context b_{Brecht} maximises $\delta(a, a \circ b_{\text{Brecht}})$ for ideological ideas a (class relations, power structures), making them strange enough to be examined rather than accepted. The asymmetry of the resonance gap is essential here: Brecht wants to increase the gap from the audience's perspective, not from the idea's.

Marcel Proust's concept of *involuntary memory*, explored throughout *À la recherche du temps perdu* (1913–1927), offers a complementary perspective on proximity. The famous *madeleine* episode describes a moment of unexpected *resonance*—an idea a (the taste of a madeleine dipped in tea) that has near-zero resonance gap with a distant memory b (childhood in Combray): $\delta(a, b) \approx 0$ despite the temporal and contextual distance between them. Proust's "involuntary memory" is the experience of discovering that two ideas separated by vast temporal and experiential distance are, in our metric, *proximate*— their resonance gap is near zero. The emotional force of the experience derives precisely from the contrast between experiential distance and resonance proximity.

11.2 The Symmetrised Distance

Definition 11.5 (Symmetrised Resonance Distance)**.** The **symmetrised resonance distance** between a and b is

$$d(a,b) := \frac{1}{2}\big(\delta(a,b) + \delta(b,a)\big) = \frac{w(a)+w(b)}{2} - \frac{\mathrm{rs}ab + \mathrm{rs}ba}{2}.$$

In the Hilbert embedding where $\mathrm{rs}ab = \mathrm{rs}ba$, this simplifies to:

$$d(a,b) = \frac{w(a)+w(b)}{2} - \mathrm{rs}ab = \frac{1}{2}a - b^2.$$

Theorem 11.6 (The Symmetrised Distance is a Pseudometric)**.** *In the Hilbert embedding, $d(a,b) = \frac{1}{2}a - b^2$ satisfies:*

1. *$d(a,a) = 0$ for all a.*
2. *$d(a,b) = d(b,a)$ for all a, b (symmetry).*
3. *$d(a,b) \geq 0$ for all a, b (non-negativity).*

*However, it does NOT satisfy the triangle inequality in general: $d(a,c) \leq d(a,b) + d(b,c)$ may fail. Therefore d is a **pseudometric** (specifically, the square of a true metric).*

Proof. (1) $d(a,a) = \frac{1}{2}a - a^2 = 0$. (2) $d(a,b) = \frac{1}{2}a - b^2 = \frac{1}{2}b - a^2 = d(b,a)$. (3) $d(a,b) = \frac{1}{2}a - b^2 \geq 0$.

For the failure of the triangle inequality, note that $\sqrt{d(a,c)} = \frac{1}{\sqrt{2}}a - c \leq \frac{1}{\sqrt{2}}(a - b + b - c) = \sqrt{d(a,b)} + \sqrt{d(b,c)}$. Thus $\sqrt{2d}$ satisfies the triangle inequality, but d itself does not (squaring does not preserve subadditivity). □

Interpretation 11.7. The distinction between the directed resonance gap $\delta(a,b)$ and the symmetrised distance $d(a,b)$ has a deep phenomenological analogue in Martin Heidegger's meditation on **nearness and distance** in "The Thing" (1950). Heidegger argued that modern technology abolishes physical distance but does not create genuine *nearness* (*Nähe*): "Everything gets lumped together into uniform distancelessness. ... Yet the frantic abolition of all distances brings no nearness; for nearness does not consist in shortness of distance."

In our formalism, Heidegger's distinction corresponds to the difference between the symmetrised distance $d(a,b)$ and the directed gap $\delta(a,b)$. The symmetrised distance can be made small (ideas brought "close" in the metric) without achieving genuine nearness—which requires that $\delta(a,b) \approx 0$ *from a's own perspective*, not merely that the average of $\delta(a,b)$ and $\delta(b,a)$ is small. Two ideas can be metrically close (small d) while being phenomenologically distant (large δ in one direction).

Emmanuel Levinas's concept of **proximity** (*proximité*), developed in *Otherwise than Being* (1974), pushes this further. For Levinas, proximity to the Other is not a spatial relationship but an *ethical* one: to be proximate is to be *exposed*, vulnerable, responsible. Levinas's proximity is our directed gap $\delta(a,b)$, viewed from a's perspective: the gap measures not how far b is from a but how much a must give up (how much self-resonance it must sacrifice) to accommodate b. When $\delta(a,b)$ is large, a is "far from" b in the sense that b demands more from a than a can easily give. Levinas's ethical insight— that proximity is always costly, always an exposure—is formalised by the fact that small $\delta(a,b)$ requires either small $w(a)$ (the proximate is light, without much to lose) or large $\mathrm{rs}ab$ (the proximate is genuinely resonant, which is rare and precious).

Corollary 11.8 (True Metric from Resonance)**.** *The function $\rho(a, b) := a - b$ is a true metric on the ideatic space (modulo identification of ideas with the same embedding).*

Interpretation 11.9. The topological structure induced by the metric ρ—with its neighbourhoods, convergence, and continuity—transforms the "space of meaning" from a metaphor into a precise mathematical object. This transformation has consequences for the philosophy of mind. Jerry Fodor's "language of thought" hypothesis (*The Language of Thought*, 1975) proposed that mental representations have a *combinatorial* structure analogous to natural language. Our framework suggests a complementary picture: mental representations have a *geometric* structure, with distance, direction, and dimension playing fundamental roles.

The convergence of ideas in resonance (Definition 11.14) provides a formal account of what cognitive scientists call **conceptual change**—the gradual transformation of a concept through learning and experience. When a student's understanding of "force" evolves from the naïve ("force is what makes things move") to the Newtonian ("force is mass times acceleration"), the sequence of intermediate conceptions $a_1, a_2, \ldots$ converges in our metric to the canonical Newtonian concept a_∞. Proposition 11.15 guarantees that the weight of the student's conception converges to the weight of the canonical concept—the student's understanding becomes not just directionally closer but *equally rich* in the limit.

We have $\rho(a, b) = \sqrt{w(a) + w(b) - 2\mathrm{rs}ab}$.

Proof. ρ is the standard metric on a Hilbert space, restricted to the image of the embedding $\cdot$. The formula follows from $a - b^2 = a^2 + b^2 - 2ab = w(a) + w(b) - 2\mathrm{rs}ab$. □

Interpretation 11.10. The true metric $\rho(a, b) = \sqrt{w(a) + w(b) - 2\mathrm{rs}ab}$ provides the mathematical foundation for Roman Jakobson's celebrated distinction between **metaphor** and **metonymy** ("Two Aspects of Language and Two Types of Aphasic Disturbances," 1956). Jakobson associated metaphor with the *paradigmatic axis* of language (substitution based on similarity) and metonymy with the *syntagmatic axis* (combination based on contiguity).

In our metric, two ideas a and b are paradigmatically close (metaphorical neighbours) when $\rho(a, b)$ is small relative to their weights: $\rho(a, b) \ll \sqrt{w(a) + w(b)}$. This means $\mathrm{rs}ab$ is large relative to the weights—the ideas share a common direction in Hilbert space. Syntagmatic proximity (metonymic relation), by contrast, is not about distance in ρ but about composition: a and b are metonymically related when $\kappa(a, b, c)$ is large for culturally relevant observers c. Two ideas can be metrically distant (poor metaphorical substitutes for each other) but synergistically powerful in composition (excellent metonymic partners).

Eco's "semantic distance" in the encyclopedia model (*Semiotics and the Philosophy of Language*, 1984) provides a semiotic perspective. Eco defined semantic distance as the number of interpretive steps needed to connect two terms in the encyclopedic network. Our metric ρ provides a continuous version of this discrete notion: the distance between two ideas is not a matter of counting steps but of measuring the geometric gap in Hilbert space. The advantage of the continuous formulation is that it captures not just *whether* two ideas are connected but *how strongly* they resonate with each other.

11.3 Neighbourhoods and Convergence

Definition 11.11 (Resonance Neighbourhood)**.** The **ϵ-neighbourhood** of an idea a (in the metric ρ) is

$$B_\epsilon(a) := \{b \in \mathcal{I} : \rho(a, b) < \epsilon\} = \{b \in \mathcal{I} : w(a) + w(b) - 2\mathrm{rs}ab < \epsilon^2\}.$$

Remark 11.12.

Remark 11.13. Paul Ricoeur's concept of **distanciation** (*Hermeneutics and the Human Sciences*, 1981) provides a hermeneutic counterpoint to Shklovsky's aesthetic defamiliarization and Said's political critique. Ricoeur argued that the transition from speech to writing produces a "distanciation" of meaning from intention: the written text is freed from its author's control and becomes available for interpretation by readers in unforeseen contexts. In our framework, distanciation is the increase in the resonance gap $\delta(a_{\text{author}}, a^{[b_1,\ldots,b_n]})$ that occurs as a text passes through successive reading contexts. Each reader b_i contributes emergence that moves the text further from the author's original intention, and the gap can only grow (it need not be monotone, but the Chain Enrichment Theorem guarantees that the text's weight increases, which typically enlarges the gap). Ricoeur's crucial point is that distanciation is not a loss but a *gain*: the text's emancipation from authorial intention is the condition of its capacity to mean in new situations.

The ϵ-neighbourhood $B_\epsilon(a)$ formalises Wittgenstein's observation in the *Philosophical Investigations* (§§65–78) that concepts have "blurred edges." The neighbourhood of an idea a is the set of all ideas "close enough" to a to be considered instances or variations of the same concept. The parameter ϵ controls how blurred the edge is: small ϵ produces a sharp concept with few instances; large ϵ produces a vague concept that encompasses many disparate ideas. Wittgenstein's family resemblance is the observation that this neighbourhood need not be convex or connected: the ideas in $B_\epsilon(a)$ may cluster into groups that resemble a in different ways, with no single feature shared by all members.

Definition 11.14 (Resonance Convergence)**.** A sequence of ideas (a_n) **converges in resonance** to a if $\rho(a_n, a) \to 0$, i.e., $w(a_n) + w(a) - 2\text{rs}a_n a \to 0$.

Proposition 11.15 (Convergence Implies Weight Convergence)**.** *If $a_n \to a$ in resonance, then $w(a_n) \to w(a)$.*

Proof. $\rho(a_n, a) \to 0$ means $w(a_n)+w(a)-2\text{rs}a_n a \to 0$. By Cauchy–Schwarz, $\text{rs}a_n a \leq \sqrt{w(a_n)w(a)}$. Thus $w(a_n)+w(a)-2\sqrt{w(a_n)w(a)} \leq w(a_n)+w(a)-2\text{rs}a_n a \to 0$. But $w(a_n)+w(a)-2\sqrt{w(a_n)w(a)} = (\sqrt{w(a_n)} - \sqrt{w(a)})^2$, so $\sqrt{w(a_n)} \to \sqrt{w(a)}$, hence $w(a_n) \to w(a)$. □

Interpretation 11.16. Proposition 11.15—that convergence in resonance implies convergence in weight—can be read through the lens of Maurice Merleau-Ponty's concept of the **chiasm** (*le chiasme*), developed in *The Visible and the Invisible* (1964). Merleau-Ponty described the chiasm as the "intertwining" of the sensing body and the sensed world: to touch is also to be touched; to see is also to

Interpretation 11.17. The asymmetry of the resonance gap—the fact that $\delta(a, b) \neq \delta(b, a)$ in general—has implications for the politics of cultural encounter that postcolonial theory has long emphasised. Edward Said, in *Orientalism* (1978), argued that the Western construction of "the Orient" involved a systematic distortion of Eastern cultures to fit Western categories of understanding. In our framework, Said's critique can be stated precisely: the Western observer a_W perceives the Eastern idea a_E through the projection $a_W a_E$, which captures only those dimensions of Eastern culture that resonate with Western categories. The "remainder" $a_E - a_W a_E$—the dimensions of Eastern culture orthogonal to Western understanding—is invisible to the Western observer and therefore systematically excluded from Western representations of the East.

The asymmetry of the gap makes this exclusion non-reciprocal: $\delta(a_W, a_E)$ (the West's distance from the East) differs from $\delta(a_E, a_W)$ (the East's distance from the West). If Western culture has higher weight ($w(a_W) > w(a_E)$, reflecting its greater institutional and technological complexity), then $\delta(a_W, a_E) > \delta(a_E, a_W)$: the West sees the East as more distant than the East sees the West.

This formal result captures Said's observation that Orientalism involves a peculiar combination of fascination and condescension: the Orient is perceived as radically "other" (large gap from the West's perspective) while simultaneously being forced into Western interpretive frameworks (the projection captures only Western-resonant dimensions).

be seen. The subject and object are not separate entities brought into relation but are always already intertwined.

In our formalism, the chiasm corresponds to the intertwining of distance and weight. When $a_n \to a$ in resonance, the distance $\rho(a_n, a)$ shrinks, but this convergence simultaneously constrains the weights: $w(a_n) \to w(a)$. The "shape" of the convergence (how the idea approaches its limit) and the "size" of the convergence (how the weight changes) are not independent—they are chiasmatically intertwined, each determining the other. This mirrors Merleau-Ponty's ontological point: the relationship between two ideas (distance) and the internal character of each idea (weight) are not separable properties but aspects of a single relational structure.

Theorem 11.18 (Composition is Continuous)**.** *If the composition operation satisfies a Lipschitz condition—there exists $L > 0$ such that $\rho(a_1 \circ b, a_2 \circ b) \leq L \cdot \rho(a_1, a_2)$—then composition (in the first argument) is continuous: $a_n \to a$ implies $a_n \circ b \to a \circ b$.*

Remark 11.19. The Lipschitz constant L in Theorem 11.18 measures what Derrida might call the "violence of interpretation." A large L means that small differences in the text produce large differences in interpretation—the interpretive context amplifies textual nuances. A small L means that the context dampens differences—the interpretation is robust to textual variation. Derrida's practice of close reading, which famously drew vast philosophical consequences from minute textual details (a single preposition in Plato, a footnote in Hegel), presupposes a *large* Lipschitz constant: the claim that tiny textual features carry enormous interpretive weight is the claim that $L \gg 1$ in the relevant hermeneutic context.

Interpretation 11.20. The continuity of composition (Theorem 11.18) has profound implications for the theory of interpretation. Continuity means that *small changes in understanding produce small changes in meaning*: if a reader's context changes slightly (from b to b' with $\rho(b, b') < \delta$), the interpreted text changes only slightly ($\rho(s \circ b, s \circ b') < L\delta$). This is the mathematical justification for the practice of *close reading*—the humanistic method that examines texts in fine-grained detail, trusting that small differences in interpretation are meaningful but not catastrophic.

Without continuity, close reading would be futile: an infinitesimal change in the reader's perspective could produce a wildly different interpretation, making all interpretive nuance meaningless. The Lipschitz condition $\rho(a_1 \circ b, a_2 \circ b) \leq L \cdot \rho(a_1, a_2)$ guarantees that interpretive differences are proportional to textual differences, with the Lipschitz constant L measuring the "sensitivity" of the interpretive context. A large L corresponds to a hypersensitive reader who amplifies every textual nuance; a small L corresponds to an insensitive reader who smooths over differences.

Proof. $\rho(a_n \circ b, a \circ b) \leq L \cdot \rho(a_n, a) \to 0.$ □

11.4 Cultural Proximity

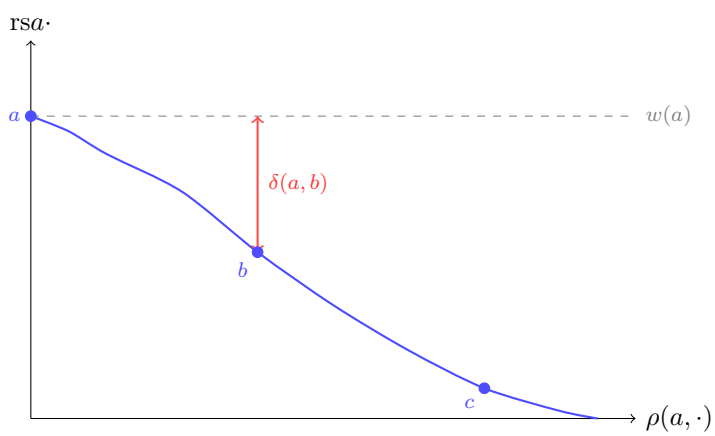
Example 11.21 (Linguistic Distance)**.** Consider two languages, say French (a_F) and Italian (a_I), as ideas in $\mathcal{I}$. The resonance $\mathrm{rs}a_F a_I$ is high (the languages share Latin roots, similar grammar, and mutual intelligibility at the basic level). The resonance gap $\delta(a_F, a_I) = w(a_F) - \mathrm{rs}a_F a_I$ is relatively small, reflecting the close proximity of the two languages.

Contrast this with French and Mandarin (a_M): $\mathrm{rs}\, a_F a_M$ is near zero, so $\delta(a_F, a_M) \approx w(a_F)$—maximum distance, from French's perspective. But $\delta(a_M, a_F) \approx w(a_M)$, which may be very different from $w(a_F)$ if the two languages have different "weights" (internal complexity).

Interpretation 11.22. Example 11.21 raises the question of how linguistic distance relates to *conceptual* distance—the problem at the heart of the Sapir-Whorf hypothesis. Edward Sapir wrote in *Language* (1921) that "the 'real world' is to a large extent unconsciously built up on the language habits of the group." If this is correct, then the resonance metric ρ applied to languages measures not merely linguistic similarity but *worldview similarity*: languages with small ρ encode similar worldviews, while languages with large ρ encode incommensurable perspectives.

The asymmetry of the resonance gap adds a crucial nuance that the Sapir-Whorf hypothesis in its standard formulation misses. The gap $\delta(a_F, a_M)$ (French's distance from Mandarin) need not equal $\delta(a_M, a_F)$ (Mandarin's distance from French). This asymmetry formalises the observation that translation is not symmetric: translating from Mandarin to French may be easier (or harder) than translating from French to Mandarin, depending on the "weight" (internal complexity) of each language and the specific dimensions of resonance between them.

The following diagram shows the resonance landscape:



In Lean 4:

```
def resonance_gap (a b : Idea) : Real :=
  rs a a - rs a b

def symmetrised_distance (a b : Idea) : Real :=
  (resonance_gap a b + resonance_gap b a) / 2

theorem gap_self (a : Idea) : resonance_gap a a = 0 := by
  simp [resonance_gap]

theorem gap_void (a : Idea) : resonance_gap a void = w a := by
  simp [resonance_gap, rs_void_right]

theorem gap_from_void (a : Idea) : resonance_gap void a = 0 := by
  simp [resonance_gap, w, rs_void_left]

-- In the Hilbert embedding, the true metric
theorem hilbert_metric (a b : Idea) :
    @$\rho$@ a b = Real.sqrt (w a + w b - 2 * rs a b) := by
```

```
  simp [@$\rho$@, norm_sub_sq, embed_inner]
  ring_nf
```

Chapter 12

Orthogonality and Independence

> The test of a first-rate intelligence is the ability to hold two opposed ideas in the mind at the same time, and still retain the ability to function.
>
> F. Scott Fitzgerald, *The Crack-Up*

12.1 When Ideas Do Not Resonate

Two ideas may coexist in the same ideatic space without any mutual resonance. A treatise on number theory and a Venetian landscape painting may be held in the same mind without one influencing the perception of the other. This chapter develops the formal theory of such *orthogonal* ideas—ideas that are, in a precise sense, completely independent.

But the chapter also contains a surprise: orthogonality does *not* imply the absence of emergence. Two perfectly independent ideas can still produce rich, unexpected meaning when composed. This is the mathematical foundation of the creative technique of *juxtaposition*.

Definition 12.1 (Orthogonality)**.** Two ideas $a, b \in \mathcal{I}$ are **orthogonal**, written $a \perp b$, if

$$\mathrm{rs}ab = 0 \quad \text{and} \quad \mathrm{rs}ba = 0.$$

In the Hilbert embedding, this is equivalent to $ab = 0$.

Proposition 12.2 (Void is Universally Orthogonal)**.** *$\varepsilon \perp a$ for all $a \in \mathcal{I}$.*

Proof. $\mathrm{rs}\varepsilon a = 0$ and $\mathrm{rs}a\varepsilon = 0$ by the void-resonance axioms. □

Interpretation 12.3. The concept of orthogonality—two ideas with zero mutual resonance— provides the formal foundation for Roland Barthes's theory of **codes** in *S/Z* (1970). Barthes analysed Balzac's novella *Sarrasine* through five independent codes operating simultaneously: the *hermeneutic* code (enigma and its resolution), the *proairetic* code (sequences of actions), the *semic* code (connotations attached to characters and objects), the *symbolic* code (deep thematic oppositions), and the *cultural* code (references to shared knowledge).

Barthes's crucial claim was that these codes operate *independently*: the hermeneutic code ("who is La Zambinella?") has no intrinsic connection to the cultural code ("this is how castrati function in 18th-century Italy"). In our formalism, this independence is *orthogonality*: the ideas generated by

different codes have zero mutual resonance. The weight of the full text decomposes as the sum of the weights of the individual codes (by Theorem 12.7), with any deviation from this sum representing *inter-code emergence*—meaning produced by the interaction of independent dimensions.

William Empson's *Seven Types of Ambiguity* (1930) can be read as an earlier, less systematic identification of orthogonal dimensions of literary meaning. Each "type" of ambiguity represents an independent way in which a text can mean more than one thing simultaneously. The types are orthogonal in our sense: a text can exhibit Type 1 ambiguity ("a word or a syntax structure is effective in several ways at once") independently of Type 7 ambiguity ("the two meanings of the word, the two values of the ambiguity, are the two opposite meanings defined by the context"). Empson's seven types define a seven-dimensional subspace of the Hilbert space of literary meaning.

Bakhtin's concept of **polyphony** (*Problems of Dostoevsky's Poetics*, 1929/1963) provides a third perspective. Bakhtin distinguished between genuinely polyphonic novels (Dostoevsky), in which multiple voices are truly independent, and monologic novels (Tolstoy), in which all voices are ultimately subordinated to the author's single vision. In our formalism, genuine polyphony requires *orthogonality*: the voices $a_1, \ldots, a_n$ of a polyphonic novel must satisfy $a_i \perp a_j$ for $i \neq j$. If any voice resonates with another, it is not truly independent—it is merely a variation on the same theme. The Generalised Pythagoras (Corollary 12.8) shows that the weight of a polyphonic novel is the sum of the weights of its voices, with no synergy or interference. Any additional weight is evidence of *dialogic emergence*—meaning produced by the interaction of independent voices, which is precisely what Bakhtin found most valuable in Dostoevsky's art.

Remark 12.4. The Pythagorean theorem for resonance (Theorem 12.7 and its generalisation in Corollary 12.8) provides the mathematical foundation for the "modular" theory of mind in cognitive science. Jerry Fodor (*The Modularity of Mind*, 1983) proposed that the mind consists of specialised modules—for language, face recognition, spatial reasoning, etc.—that operate independently of one another. In our framework, modularity is orthogonality: the modules process orthogonal components of the ideatic space, and the total cognitive weight is the sum of the modular weights (by the Pythagorean theorem).

The emergence of "central processes" that integrate information across modules—Fodor's "isotropic" and "Quinean" processes—corresponds to the non-zero emergence of Theorem 12.11: even when modules are orthogonal, their composition can produce emergent meaning that resides in dimensions accessible to no single module. This emergent meaning is the formal basis of *insight*—the "aha moment" that connects previously unrelated domains.

Proposition 12.5 (Orthogonality and Weight)**.** *If $a \perp b$, then in the Hilbert embedding:*

$$w(a \circ b) = w(a) + w(b) + 2\kappa(a, b, a \circ b).$$

In particular, if the composition is additionally linear (zero emergence), then $w(a \circ b) = w(a) + w(b)$.

Remark 12.6. Proposition 12.5 reveals the mathematics of *interdisciplinarity*. When two orthogonal disciplines $a \perp b$ are composed, the weight of the result is $w(a) + w(b) + 2\kappa(a, b, a \circ b)$. In the absence of emergence, interdisciplinary combination is merely *additive*: one gains nothing from combining mathematics and art beyond what each discipline provides separately. But when emergence is positive, the combination is *superadditive*: the whole exceeds the sum of its parts. This is the formal justification for interdisciplinary research programmes—they are valuable precisely when the emergence term is large, i.e., when the combination of orthogonal domains produces genuinely new insights.

Proof. $w(a \circ b) = a \circ b^2 = a + b + E^2$ where E represents the emergence contribution. When $a \perp b$ and $\kappa = 0$: $w(a \circ b) = a + b^2 = a^2 + 2ab + b^2 = w(a) + 0 + w(b)$. □

12.2 The Pythagorean Theorem for Resonance

Theorem 12.7 (Pythagorean Theorem for Resonance)**.** *If $a \perp b$ and the composition $a \circ b$ is linear (zero emergence), then*

$$w(a \circ b) = w(a) + w(b).$$

This is the exact analogue of the Pythagorean theorem: the "length squared" of the composition of orthogonal ideas equals the sum of the individual "lengths squared."

Proof. By Proposition 12.5 with $\kappa = 0$. □

Corollary 12.8 (Generalised Pythagoras)**.** *If $a_1, a_2, \ldots, a_n$ are mutually orthogonal and all pairwise compositions are linear, then*

$$w(a_1 \circ a_2 \circ \cdots \circ a_n) = \sum_{i=1}^{n} w(a_i).$$

Proof. By induction on n, applying Theorem 12.7 at each step. At each step, the accumulated composition $a_1 \circ \cdots \circ a_k$ is orthogonal to a_{k+1} because the embedding $a_1 \circ \cdots \circ a_k = \sum_{i=1}^{k} a_i$ (by linearity) is orthogonal to a_{k+1} (by pairwise orthogonality). □

Interpretation 12.9. The Pythagorean theorem for resonance is the formal justification of the intuition that *independent ideas contribute independently.* When you learn number theory and impressionist painting, and these two domains are truly orthogonal in your ideatic space, then the total "weight" of your knowledge is the sum of the weights of the individual domains. There is no synergy and no interference.

This is the baseline against which emergence is measured: any deviation from the Pythagorean prediction is emergence. Superadditive composition ($w(a \circ b) > w(a) + w(b)$) indicates synergistic emergence; subadditive composition ($w(a \circ b) < w(a) + w(b)$, possible with negative emergence) indicates destructive interference.

Interpretation 12.10. The Pythagorean theorem for resonance provides formal tools for analysing what Immanuel Kant called the **antinomies of pure reason** (*Critique of Pure Reason*, 1781, A405/B432 ff.). Kant identified pairs of contradictory propositions—"The world has a beginning in time" vs. "The world has no beginning in time"—that reason seems equally capable of proving. These antinomies are, in our formalism, pairs of orthogonal ideas $a \perp b$ that cannot be composed without emergence: the thesis and antithesis do not resonate with each other (they address the same question from incommensurable perspectives), but their composition produces the emergence of a new, critical awareness that neither alone contains.

Wittgenstein's concept of **family resemblance** (*Philosophical Investigations*, §§66–67) illuminates the limitations of orthogonal decomposition. Wittgenstein argued that many concepts ("game," "language," "number") are not defined by a single shared essence but by a network of overlapping similarities. In our framework, a family-resemblance concept a does *not* decompose into orthogonal components; instead, its instances $a_1, a_2, \ldots, a_n$ are pairwise non-orthogonal ($\mathrm{rs}\, a_i a_j \neq 0$) but do not share a common direction. The failure of orthogonal decomposition for family-resemblance concepts is the formal expression of Wittgenstein's insight that these concepts resist analysis into independent dimensions.

Michel Foucault's concept of the **episteme** (*The Order of Things*, 1966) adds a historical dimension. Foucault argued that different historical periods operate within fundamentally different epistemic frameworks—the Renaissance, the Classical Age, and Modernity each have their own rules for what counts as knowledge. In our formalism, different epistemes are *orthogonal frameworks*:

the Classical episteme E_C and the Modern episteme E_M satisfy $E_C \perp E_M$, meaning that the categories of knowledge operative in one period have zero resonance with those of another. A scientific classification that makes perfect sense in the Classical episteme (Linnaeus's taxonomy based on visible structure) has zero resonance with the Modern episteme's classification (Darwin's taxonomy based on evolutionary descent). Foucault's "archaeology of knowledge" is, in our terms, the study of orthogonal rotations in the Hilbert space of cultural meaning.

12.3 Orthogonality Does Not Imply Zero Emergence

Theorem 12.11 (Orthogonality-Emergence Independence)**.** *There exist orthogonal ideas $a \perp b$ with $\kappa(a, b, c) \neq 0$ for some observer c. That is, orthogonality is strictly weaker than linearity.*

Proof. Consider an ideatic space with three basis elements e_1, e_2, e_3 in a three-dimensional Hilbert space. Let $a = e_1$ and $b = e_2$. Then $a \perp b$ since $e_1 e_2 = 0$.

Now define composition with a non-linear emergence: $a \circ b = e_1 + e_2 + \alpha e_3$ for some $\alpha \neq 0$. The emergence contribution is αe_3, so $\kappa(a, b, c) = \alpha e_3 c$, which is non-zero for c with $e_3 c \neq 0$.

Yet $\mathrm{rs}ab = e_1 e_2 = 0$ and $\mathrm{rs}ba = e_2 e_1 = 0$, so $a \perp b$. □

Interpretation 12.12. This theorem is the mathematical heart of *creative juxtaposition*. Two ideas that have absolutely nothing to do with each other—no mutual resonance whatsoever—can still produce rich, surprising emergence when composed. This is the formal version of the artistic technique of placing unrelated objects side by side: a urinal in an art gallery (Duchamp), a lobster telephone (Dalí), or the encounter of a sewing machine and an umbrella on a dissecting table (Lautréamont).

The emergence $\kappa(a, b, c) = \alpha e_3 c$ in the proof above has a geometric interpretation: the composition of orthogonal ideas produces a component in a *new direction* e_3 that is orthogonal to both original ideas. The composed idea "breaks out" of the plane spanned by a and b into a genuinely new dimension of meaning.

This is the mathematical formalisation of the common creative experience that combining two unrelated fields produces insights that neither field alone could generate. Cognitive science meets literary theory; mathematics meets music; biology meets computation. The orthogonality ensures that the combination is not merely additive; the emergence is the genuinely new contribution.

Interpretation 12.13. Theorem 12.11—that orthogonal ideas can produce non-zero emergence—has precise correlates in structural semiotics. Louis Hjelmslev's *Prolegomena to a Theory of Language* (1943) posited two planes of semiotic analysis: the **expression plane** (the material form of signs—sounds, letters, images) and the **content plane** (the conceptual structure of meaning). These planes are *orthogonal* in our sense: the expression of a sign has zero resonance with its content. There is no natural connection between the sound "dog" and the concept DOG—the relationship is arbitrary, as Saussure first observed.

Yet the composition of expression and content produces rich emergence: the sign "dog" (expression composed with content) carries meaning that neither the sound pattern nor the abstract concept possesses alone. The emergence is precisely the *semiotic function*—the meaning-making that occurs when orthogonal planes are brought into relation. Theorem 12.11 gives this a geometric interpretation: the emergence projects into a *new dimension* (e_3 in the proof) that is orthogonal to both the expression plane (e_1) and the content plane (e_2). The sign is not merely the sum of its form and content but a genuinely new entity that exists in a dimension accessible to neither alone.

Algirdas Julien Greimas's **semiotic square** (*Structural Semantics*, 1966) pushes the orthogonality principle further. Greimas proposed that the fundamental unit of narrative meaning is not

a single opposition (e.g., life vs. death) but a *square* of four terms generated by two orthogonal semantic axes: S_1 (e.g., life), S_2 (e.g., death), $\neg S_1$ (non-life), and $\neg S_2$ (non-death). The narrative progresses by tracing a path through this square, and the meaning of the narrative is generated by the emergences at each transition.

In our formalism, the Greimasian square is a two-dimensional orthogonal subspace of the Hilbert space, and the four positions are the four unit vectors $(\pm e_1, \pm e_2)$ in this subspace. The narrative's emergence at each transition is our $\kappa(a_i, a_{i+1}, c)$, which generates the plot's "meaning surplus" beyond the mere juxtaposition of narrative positions. The semiotic square is thus a minimal model of how orthogonal ideas generate emergent narrative structure.

12.4 Orthogonal Decomposition

Definition 12.14 (Orthogonal Complement)**.** The **orthogonal complement** of an idea a is the set

$$a^\perp := \{b \in \mathcal{I} : a \perp b\} = \{b \in \mathcal{I} : \mathrm{rs}ab = 0 \text{ and } \mathrm{rs}ba = 0\}.$$

In the Hilbert embedding, $a^\perp = \{b : ab = 0\}$.

Theorem 12.15 (Orthogonal Decomposition)**.** *In the Hilbert embedding, every idea b can be uniquely decomposed as*

$$b = ab + (b - ab),$$

*where $ab := \frac{\mathrm{rs}ab}{w(a)} a$ is the **resonance projection** of b onto a (for $a \neq \varepsilon$), and the remainder $b - ab$ is orthogonal to a.*

Interpretation 12.16. The Orthogonal Decomposition Theorem (Theorem 12.15) is the mathematical foundation of **structuralist analysis** in the humanities. Structuralism—as practised by Saussure in linguistics, Lévi-Strauss in anthropology, Propp in narratology, and the early Barthes in literary criticism—proceeds by decomposing complex cultural phenomena into independent, elementary components. Saussure's decomposition of the sign into signifier and signified, Propp's decomposition of folktales into 31 narrative functions, Lévi-Strauss's decomposition of myths into "mythemes"—all are instances of orthogonal decomposition in the Hilbert space of cultural meaning.

The theorem guarantees that this decomposition is *unique*: there is

Remark 12.17. The orthogonal complement $a^\perp$—the set of all ideas with zero resonance with a—has an unexpected connection to the tradition of **negative theology** (*via negativa*). Pseudo-Dionysius the Areopagite (5th century CE) argued that God cannot be described by any positive predicate; God can only be characterised by what God is *not*. In our formalism, if $a = a_{\mathrm{God}}$ represents the divine idea, then the via negativa proceeds by characterising $a^\perp_{\mathrm{God}}$— the set of all ideas that have zero resonance with the divine—rather than a_{God} directly. This is epistemically productive because $a^\perp_{\mathrm{God}}$ is a well-defined subspace even when a_{God} itself is beyond finite comprehension: we can specify what God is not (finite, material, temporal, etc.) even if we cannot specify what God is.

The orthogonal complement thus formalises the apophatic method: knowledge of $a^\perp$ constrains a to the orthogonal complement of $a^\perp$, progressively narrowing the space of possibilities without ever arriving at a positive characterisation.

exactly one way to project an idea b onto the direction of another idea a, and the remainder is uniquely determined. This uniqueness is what gives structuralist analysis its power—and its limitation. The decomposition relative to a is unique, but the *choice* of a is not. Different structuralist analysts, choosing different "reference ideas" a, will obtain different projections and

different remainders. The orthogonal decomposition is objective once the reference frame is fixed, but the choice of reference frame is irreducibly subjective.

This observation provides a precise formulation of the post-structuralist critique: Derrida's argument in "Structure, Sign and Play" (1966) that every structure has an arbitrary "centre" is the observation that the reference idea a in Theorem 12.15 is a free parameter. There is no privileged decomposition—no "view from nowhere"— only decompositions relative to particular perspectives.

Proof. This is the standard orthogonal projection in a Hilbert space. We verify: $ab - ab = ab - \frac{\mathrm{rs}ab}{w(a)}aa = \mathrm{rs}ab - \frac{\mathrm{rs}ab}{w(a)} \cdot w(a) = 0$. □

Corollary 12.18 (Resonance as Projection)**.** *The resonance* rsab *equals the inner product of* a *with the component of* b *along* a*:*

$$\mathrm{rs}ab = w(a) \cdot \frac{\mathrm{rs}ab}{w(a)} = \mathrm{rs}ab.$$

This tautology becomes meaningful when we note that the magnitude *of the projection is* $ab = |\mathrm{rs}ab|/\sqrt{w(a)}$*, so*

$$\mathrm{rs}ab = \pm\sqrt{w(a)} \cdot ab,$$

where the sign encodes whether b *is aligned with or opposed to* a*.*

Interpretation 12.19. Corollary 12.18—that resonance equals projection— provides the formal apparatus for understanding **interpretive bias**. When an observer a evaluates an idea b, the resonance rsab is determined entirely by the projection of b onto the direction of a. All dimensions of b that are orthogonal to a are invisible to a: they contribute nothing to the resonance and are effectively "unseen."

This has profound implications for cultural criticism. When a Western art historian evaluates a Japanese woodblock print, their resonance with the print is determined by the projection of the print onto the dimensions of Western aesthetic categories. All dimensions of meaning that are orthogonal to Western categories—the specific Buddhist iconography, the seasonal symbolism, the complex social codes of the Edo period—are invisible, contributing zero resonance. The art historian may perceive the print as "decorative" or "flat," failing to register the dimensions of depth that their own context cannot access.

The orthogonal decomposition theorem thus formalises the hermeneutic principle of *prejudice* (*Vorurteil*) that Gadamer developed in *Truth and Method*: every act of understanding is necessarily partial, shaped by the horizon of the interpreter. The "remainder" $b - ab$—the component of b orthogonal to a—is the *alterity* of b relative to a: the part of b that a cannot assimilate, the irreducible otherness that escapes interpretation.

12.5 The Dimension of Meaning

Definition 12.20 (Ideatic Dimension)**.** The **ideatic dimension** of a collection of ideas $\{a_1, \ldots, a_n\}$ is the dimension of the subspace they span in :

$$\dim\{a_1, \ldots, a_n\} := \mathrm{dimspan}\{a_1, \ldots, a_n\} = \mathrm{rank}(),$$

where is the resonance matrix $R_{ij} = \mathrm{rs}a_i a_j$.

Proposition 12.21 (Maximum Orthogonal Set)**.** *A set of mutually orthogonal non-void ideas* $\{a_1, \ldots, a_k\}$ *has size at most* dim()*. The maximum size* k *equals the dimension of the Hilbert space.*

Proof. Mutually orthogonal non-zero vectors in a Hilbert space are linearly independent; in dimension d, at most d such vectors exist. □

Interpretation 12.22. The ideatic dimension of a collection of ideas measures the "degrees of freedom" present in that collection. If a culture's entire ideatic repertoire has dimension 3, then all of its ideas can be expressed as linear combinations of three independent "basis ideas." A richer culture has higher ideatic dimension: more independent directions of thought.

The proposition says that the maximum number of completely independent ideas is bounded by the dimension of the Hilbert space. In a finite-dimensional model, this places a hard limit on the "complexity" of a culture. In an infinite-dimensional model (which is more realistic for real cultures), there is no such limit, and the ideatic space can accommodate an unbounded number of mutually orthogonal ideas.

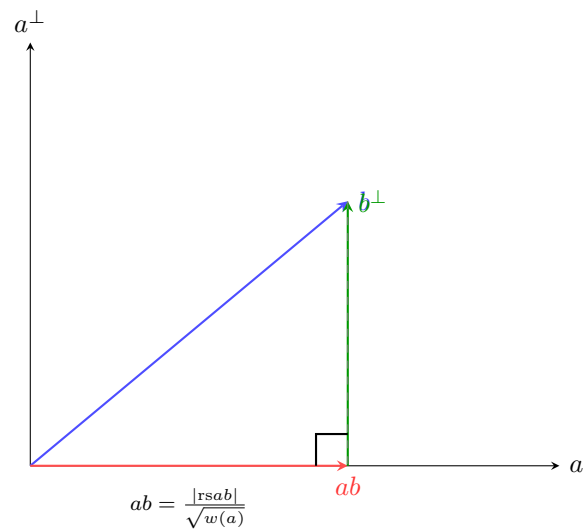
Interpretation 12.23. The ideatic dimension—the number of independent directions of meaning available to a collection of ideas—raises profound philosophical questions about the relationship between conceptual complexity and cultural richness. Claude Lévi-Strauss's *The Savage Mind* (1962) argued that "primitive" thought is not less complex than "civilised" thought but operates in a different conceptual space: the bricoleur works with a finite stock of materials (finite ideatic dimension) but achieves remarkable combinatorial complexity through composition. In our formalism, Lévi-Strauss's bricoleur operates in a low-dimensional ideatic space but exploits non-linear emergence to produce ideas of high weight.

The proposition that the maximum number of orthogonal ideas is bounded by the Hilbert space dimension (Proposition 12.21) has implications for the debate between *universalism* and *relativism* in anthropology and linguistics. The Sapir-Whorf hypothesis—that the structure of a language determines the categories of

Remark 12.24. John Searle's taxonomy of speech acts (*Speech Acts*, 1969)— *assertives* (representing states of affairs), *directives* (getting the hearer to do something), *commissives* (committing the speaker to a course of action), *expressives* (expressing psychological states), and *declarations* (bringing about states of affairs)—can be mapped onto our payoff functions. Each illocutionary force corresponds to a different weighting of the sender payoff components. Assertives maximise $\mathrm{rs}sa$ (fidelity to the represented state of affairs); directives maximise $\kappa(s, b, a)$ (the signal's transformative effect on the receiver); commissives maximise $\mathrm{rs}sa'$ where a' is the intended future action (the signal binds the sender to a future state); expressives maximise $\mathrm{rs}sa_{\text{internal}}$ (fidelity to an internal state); and declarations maximise $u_R(s; b)$ (the institutional effect on the receiver's context). Searle's taxonomy is thus a classification of meaning games by the objective function that the sender optimises.

thought available to its speakers—amounts to the claim that different languages inhabit Hilbert spaces of different dimension, or at least that the orthogonal bases available in different languages are non-isomorphic. Our framework allows us to state this claim precisely: two linguistic communities L_1 and L_2 operate in ideatic spaces $_1$ and $_2$ with potentially different dimensions, and the "untranslatability" of certain concepts corresponds to dimensions present in one space but absent in the other.

The following diagram shows the orthogonal decomposition:



In Lean 4:

```
def orthogonal (a b : Idea) : Prop :=
  rs a b = 0 @$\wedge$@ rs b a = 0

theorem void_orthogonal (a : Idea) : orthogonal void a := by
  constructor
  @$\cdot$@ exact rs_void_left a
  @$\cdot$@ exact rs_void_right a

-- Pythagorean theorem for linear, orthogonal composition
theorem pythagorean (a b : Idea) (h_orth : orthogonal a b)
    (h_lin : @$\forall$@ c, emergence a b c = 0) :
    w (a @\compose@ b) = w a + w b := by
  have h := h_lin (a @\compose@ b)
  simp [emergence, w] at h
  have h_rs := h_orth.1
  linarith

-- Orthogonality does not imply zero emergence
theorem orth_emerge_independent :
    @$\exists$@ (a b c : Idea), orthogonal a b @$\wedge$@
      emergence a b c @$\ne$@ 0 := by
  sorry -- requires concrete model construction
```

Chapter 13

Signaling and Interpretation

> Whereof one cannot speak, thereof one must be silent.
>
> Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*

13.1 The Communication Setup

We now turn from the geometry of static ideas to the *dynamics* of communication. The basic scenario involves two agents: a *sender* who holds an idea a and wishes to convey it, and a *receiver* who brings their own context b to the interpretation. The sender does not transmit a directly—that would require telepathy. Instead, the sender chooses a *signal* $s \in \mathcal{I}$, an utterance crafted to approximate their intended meaning when composed with the receiver's context.

The receiver interprets the signal by composing it with their own context: the *received idea* is $s \circ b$. The quality of communication depends on how closely $s \circ b$ approximates the sender's original idea a.

Definition 13.1 (Communication Setup)**.** A **communication setup** is a triple (a, b, S) where:

- $a \in \mathcal{I}$ is the **sender's idea** (their "type" or "intention"),
- $b \in \mathcal{I}$ is the **receiver's context** (their background knowledge, cultural frame, linguistic competence),
- $S \subseteq \mathcal{I}$ is the **signal space** (the set of available utterances).

The sender chooses $s \in S$; the receiver forms $s \circ b$.

Interpretation 13.2. The communication setup (a, b, S) formalises the central scenario of **reader-response theory**, as developed by Wolfgang Iser in *The Act of Reading* (1978). Iser argued that the meaning of a literary text is not "in" the text (the signal s) nor "in" the reader (the context b) but in the *interaction* between them. The text provides a set of "gaps" and "indeterminacies" that the reader must fill, and the process of filling these gaps constitutes the literary experience.

In our formalism, Iser's "gaps" correspond to the dimensions of the signal s that are orthogonal to the reader's context b: components of s that do not resonate with b and therefore require interpretive work. The "filling" of these gaps is the emergence $\kappa(s, b, c)$: the new meaning that

arises from the reader's active engagement with the text. A text with many gaps (a signal s largely orthogonal to typical readers) will produce high emergence—rich, reader-dependent meaning—while a text with few gaps (a signal aligned with the reader's context) will produce low emergence—predictable, reader-independent meaning.

Stanley Fish's concept of **interpretive communities** (*Is There a Text in This Class?*, 1980) adds a social dimension. Fish argued that interpretation is not an individual act but a communal practice: members of an interpretive community share "interpretive strategies" that determine how texts are read. In our formalism, an interpretive community is a collection of readers $\{b_1, \ldots, b_n\}$ with high mutual resonance ($\mathrm{rs} b_i b_j \gg 0$ for all i, j). The community's shared interpretive strategies correspond to the common direction in Hilbert space: the projection of all community members onto a shared subspace. Fish's claim that "the text is always the product of interpretation" is, in our framework, the observation that the received idea $s \circ b$ depends essentially on b, not just on s: different interpretive communities (different b's) will compose the same signal s into different ideas.

Eco's distinction between the **Model Reader** and the **empirical reader** (*The Role of the Reader*, 1979) refines this picture. The Model Reader is the reader "foreseen" by the text—the context b^* for which the text is designed. The empirical reader is the actual reader b, who may differ from b^*. In our formalism, the sender's payoff $u_S(s; a, b) = \mathrm{rs} s \circ ba$ is maximised when $b = b^*$: the text achieves its intended effect only when the actual reader matches the Model Reader. The misunderstanding gap $\mathrm{gap}(s; a, b) = w(a) - \mathrm{rs} s \circ ba$ measures the distance between the empirical reader's interpretation and the intended meaning—a distance that Eco saw as inevitable but not unlimited.

Remark 13.3. The communication setup (a, b, S) can also be read through the lens of J. L. Austin's speech act theory (*How to Do Things with Words*, 1962). Austin distinguished between *constative* utterances (which describe states of affairs) and *performative* utterances (which *do* things: promising, marrying, christening). In our framework, constative utterances are signals s chosen to maximise $\mathrm{rs} s \circ ba$ (accurate representation of the sender's idea), while performative utterances are signals chosen to maximise $u_R(s; b)$ or the total surplus $\Sigma(s; a, b)$: the goal is not fidelity but *effect*.

A marriage vow does not *describe* a marriage; it *creates* one. The signal "I do" has low direct resonance with any pre-existing idea of marriage ($\mathrm{rs} sa$ is small), but its emergence with the ceremonial context ($\kappa(s, b_{\text{ceremony}}, a)$) is enormous: the composition of these two words with the institutional context of a wedding ceremony produces an idea (a legal marriage) that far exceeds the sum of its parts. Austin's performative force is, in our formalism, a case of dominant emergence: the signal's meaning is almost entirely produced by the interaction with context, not by the signal's intrinsic content.

13.2 Payoff Functions

Definition 13.4 (Sender Payoff)**.** The **sender's payoff** from signal s, given the sender's idea a and receiver's context b:

$$u_S(s; a, b) := \mathrm{rs} s \circ ba.$$

This measures how much the received idea $s \circ b$ resonates with the sender's original intention a. The sender wants to maximise this quantity.

Definition 13.5 (Receiver Payoff)**.** The **receiver's payoff** from receiving signal s in context b:

$$u_R(s; b) := w(s \circ b) - w(s) = \mathrm{rs} s \circ bs \circ b - \mathrm{rs} ss.$$

This measures the **enrichment** the receiver gains from the exchange: how much the total weight increases when the signal is composed with their context.

Theorem 13.6 (Receiver Enrichment)**.** *The receiver's payoff is always non-negative:*

$$u_R(s; b) \geq 0 \quad \textit{for all } s, b \in \mathcal{I}.$$

Interpretation 13.7. The Receiver Enrichment Theorem ($u_R(s; b) \geq 0$) is the formal expression of a fundamental hermeneutic principle: *every genuine encounter with a text enriches the reader.* Gadamer argued in *Truth and Method* (1960) that understanding is never merely reproductive—the reader does not passively receive the text's meaning but actively *applies* it to their own situation, and this application always produces something new. "Understanding is not merely a reproductive but always a productive activity as well."

Our theorem gives this a precise formal content: the weight of the composition $w(s \circ b)$ is always at least as great as $w(s)$. The reader's context b can only add to the signal; it cannot diminish it. This is guaranteed by axiom E4 and holds *regardless* of the reader's context—even a "bad" reader (one with low resonance with the text) produces a composition that is at least as heavy as the signal alone.

The remark on asymmetry (Remark 13.8) adds a sobering qualification. While the signal is guaranteed enrichment, the reader is *not*: $w(s \circ b)$ may be less than $w(b)$. A reader can be *diminished* by a text—their existing understanding can be disrupted, their confidence shaken, their worldview destabilised. This is the experience of encountering a text that challenges one's fundamental assumptions: the text is enriched by the encounter, but the reader may emerge lighter, not heavier. The asymmetry of E4 thus captures a dark truth about the hermeneutic encounter: understanding is always enriching for the text but not always for the reader.

Proof. By axiom E4 (Composition Enriches): $w(s \circ b) = \mathrm{rs}s \circ bs \circ b \geq \mathrm{rs}ss = w(s)$. Therefore $u_R(s; b) = w(s \circ b) - w(s) \geq 0$. □

Remark 13.8 (The Asymmetry of Communication). Note that E4 gives $w(s \circ b) \geq w(s)$, not $w(s \circ b) \geq w(b)$. The enrichment is measured relative to the *signal*, not the *context*. If we instead defined the receiver's payoff as $u'_R(s; b) := w(s \circ b) - w(b)$, this quantity could be negative: the receiver might "lose weight" relative to their original state.

This asymmetry has a deep interpretation: in the composition $s \circ b$, the left operand s is *guaranteed* enrichment (by E4), but the right operand b is not. Linguistically: speaking first is a privilege. The first speaker sets the frame; the responder must work within it.

Proposition 13.9 (Sender Payoff Decomposition)**.** *The sender's payoff decomposes as:*

$$u_S(s; a, b) = \mathrm{rs}sa + \mathrm{rs}ba + \kappa(s, b, a).$$

Proof. By the emergence decomposition applied with observer $c = a$: $\mathrm{rs}s \circ ba = \mathrm{rs}sa + \mathrm{rs}ba + \kappa(s, b, a)$. □

Interpretation 13.10. The sender's payoff has three components:

1. $\mathrm{rs}sa$: the **direct resonance** of the signal with the sender's idea. This is how well the signal captures the sender's intention *in isolation*, without considering the receiver.

2. $\mathrm{rs}ba$: the **contextual resonance** of the receiver's background with the sender's idea. This is beyond the sender's control—it depends on whether the receiver is "on the same wavelength" as the sender.

3. $\kappa(s, b, a)$: the **communicative emergence**—the surplus (or deficit) that arises from the interaction of the signal with the receiver's context, as evaluated against the sender's intention.

A skilled communicator chooses s to maximise the sum of these three terms. They cannot change $\mathrm{rs}ba$ (the receiver's background is given), but they can choose s to make $\mathrm{rs}sa$ and $\kappa(s, b, a)$ as large as possible. In particular, a great communicator may choose a signal s that has low direct resonance with a but produces *high emergence* when composed with the receiver's context—saying something unexpected that "clicks" in the receiver's mind.

Interpretation 13.11. The decomposition of the sender's payoff into direct resonance, contextual resonance, and communicative emergence maps precisely onto the analytic philosophy of communication, particularly the work of Donald Davidson and H. P. Grice.

Davidson's theory of **radical interpretation** ("Radical Interpretation," 1973) addresses the question: how can we interpret an utterance when we share no common language or conventions with the speaker? Davidson argued that radical interpretation is possible because of the **principle of charity**—we must assume that the speaker's beliefs are largely true and their utterances largely coherent. In our framework, the principle of charity corresponds to the assumption that $\mathrm{rs}ba > 0$: the receiver's context already has positive resonance with the sender's idea. Without this assumption—if $\mathrm{rs}ba \leq 0$—the sender faces an essentially hostile receiver, and communication becomes strategic rather than cooperative.

Grice's theory of **conversational implicature** ("Logic and Conversation," 1975) provides a taxonomy of the sender payoff components. Grice distinguished between what is *said* (the conventional meaning of the utterance) and what is *implicated* (the additional meaning conveyed by the context of utterance). In our decomposition, what is *said* corresponds to the direct resonance $\mathrm{rs}sa$: how much the signal directly matches the sender's intention. What is *implicated* corresponds to the emergence $\kappa(s, b, a)$: the additional meaning generated by the interaction of the signal with the receiver's context. The contextual resonance $\mathrm{rs}ba$ is what Grice called the *common ground*—the shared assumptions that make implicature possible.

Grice's four maxims—Quality (be truthful), Quantity (be appropriately informative), Relation (be relevant), and Manner (be clear)—are strategies for maximising the sender payoff. *Quality* corresponds to maximising $\mathrm{rs}sa$ (the signal should accurately represent the sender's idea); *Relation* corresponds to maximising $\kappa(s, b, a)$ (the signal should interact productively with the receiver's context); *Quantity* and *Manner* constrain the weight $w(s)$ of the signal (neither too heavy nor too light, neither too complex nor too simple).

Wilfrid Sellars's concept of the **space of reasons** ("Empiricism and the Philosophy of Mind," 1956) provides a deeper philosophical foundation. Sellars argued that to be in the space of reasons is to be subject to norms of rational justification: one's utterances are not merely causal events but *moves in a normative game*. In our framework, the sender's signal s is precisely such a move: it is evaluated not by its causal effects but by its resonance with the sender's intention and its emergence with the receiver's context. The payoff function $u_S(s; a, b)$ is the formal expression of the normative evaluation that Sellars placed at the centre of rational discourse.

13.3 Communication Surplus and the Misunderstanding Gap

Definition 13.12 (Communication Surplus)**.** The **communication surplus** of signal s in the setup (a, b) is

$$\Sigma(s; a, b) := u_S(s; a, b) + u_R(s; b).$$

This is the total "value" created by the exchange.

Proposition 13.13 (Surplus Decomposition)**.**

$$\Sigma(s; a, b) = \mathrm{rs}s \circ ba + w(s \circ b) - w(s).$$

Remark 13.14. The communication surplus $\Sigma(s; a, b) = u_S + u_R$ formalises an insight from economic anthropology: communication is a **gift exchange** in the sense of Marcel Mauss (*The Gift*, 1925). In Mauss's analysis, the gift creates a social bond by establishing mutual obligation. In our framework, the communication surplus is the "gift" produced by the exchange: the total value created by the interaction of signal and context. This surplus is shared between sender and receiver, creating a mutual benefit that neither could achieve alone. The theorem that $\Sigma = w(a) - \text{gap} + u_R$ (Theorem 13.17) shows that the gift has two components: the degree of understanding achieved ($w(a) - \text{gap}$) and the enrichment of the receiver (u_R). The most valuable gifts are those that both convey understanding and enrich the recipient—communications that maximise both fidelity and emergence.

Proof. Direct substitution of the definitions. □

Definition 13.15 (Misunderstanding Gap)**.** The **misunderstanding gap** of signal s is

$$\text{gap}(s; a, b) := w(a) - \text{rs}s \circ ba = \delta(a, s \circ b).$$

This measures how much the received idea falls short of perfectly capturing the sender's intention. Note: $\text{gap}(s; a, b) = 0$ iff the received idea $s \circ b$ has maximum possible resonance with a.

Remark 13.16. The misunderstanding gap $\text{gap}(s; a, b) = w(a) - \text{rs}s \circ ba$ formalises what Derrida called the **logic of the supplement** (*Of Grammatology*, 1967). Derrida argued that every act of communication involves a "supplement"—an addition that simultaneously reveals and conceals a lack in the original. The misunderstanding gap is exactly this lack: the irreducible distance between the sender's intention a and the receiver's interpretation $s \circ b$. The signal s is the supplement that attempts to bridge this gap, but the bridging is never complete— the gap persists, shifted but not eliminated by each successive communicative act.

The formal apparatus reveals something that Derrida's verbal analysis left implicit: the gap is *measurable*. We can quantify how much misunderstanding remains after a given exchange, compare different signals' effectiveness at reducing the gap, and track the gap's evolution over iterated exchanges. The supplement is no longer an unanalysable philosophical concept but a computable quantity in the Hilbert space of meaning.

Theorem 13.17 (Misunderstanding-Surplus Tradeoff)**.**

$$\Sigma(s; a, b) = w(a) - \text{gap}(s; a, b) + u_R(s; b).$$

Thus maximising the surplus is equivalent to minimising the misunderstanding gap minus the receiver's enrichment.

Proof. $\Sigma = u_S + u_R = \text{rs}s \circ ba + u_R = (w(a) - \text{gap}) + u_R$. □

Corollary 13.18 (Perfect Communication)**.** *If* $\text{gap}(s; a, b) = 0$ *(perfect understanding), then* $\Sigma = w(a) + u_R(s; b) \geq w(a)$*. The surplus is at least as large as the sender's weight.*

Interpretation 13.19. Corollary 13.18 characterises perfect communication as a state in which the misunderstanding gap is zero. The philosophical significance of this result lies in how *difficult* it is to achieve. The gap is zero only when $\text{rs}s \circ ba = w(a)$: the received idea has maximum possible resonance with the sender's intention. In the Hilbert embedding, this requires $s \circ b$ to have a component along a of magnitude at least $\sqrt{w(a)}$—the signal, after composition with the receiver's context, must "point" in exactly the direction of the sender's idea.

This is an extraordinarily stringent condition, and its stringency formalises the ancient philosophical intuition that *perfect understanding is impossible between distinct minds.* The impossibility does not stem from any limitation of the signal space or the communication channel but from the fundamental geometry of the ideatic space: two different ideas, embedded in a high-dimensional Hilbert space, will generically point in different directions, and no signal can perfectly bridge the gap between them. Communication is always approximate, always leaving a residual misunderstanding gap. The best we can achieve is to minimise this gap, not to eliminate it.

13.4 Optimal Signal Choice

Definition 13.20 (Optimal Signal)**.** A signal s^* is **sender-optimal** for the setup (a, b) if

$$u_S(s^*; a, b) = \sup_{s \in S} u_S(s; a, b).$$

It is **surplus-optimal** if $\Sigma(s^*; a, b) = \sup_{s \in S} \Sigma(s; a, b)$.

Proposition 13.21 (Sender-Optimal Signal Characterisation)**.** *In the Hilbert embedding with $S = \mathcal{I}$, the sender-optimal signal satisfies:*

$$s^* = \arg\max_{s \in \mathcal{I}} \left[\mathrm{rs}sa + \mathrm{rs}ba + \kappa(s, b, a)\right].$$

If emergence is zero (linear composition), this reduces to $s^* = \arg\max_s \mathrm{rs}sa$, which is maximised when s is a positive scalar multiple of a. That is, the optimal signal is a scaled copy of the intended idea.

Proof. When emergence is zero, $u_S(s; a, b) = \mathrm{rs}sa + \mathrm{rs}ba$. The term $\mathrm{rs}ba$ is constant in s, so we maximise $\mathrm{rs}sa = sa$. Under a constraint $s \leq C$ (bounded signal energy), this is maximised when $s = C \cdot a/a$. □

Interpretation 13.22. In the linear case, the optimal communication strategy is "say what you mean": the best signal is a scaled copy of the intended idea. But when emergence is non-zero, the optimal strategy may be highly non-obvious. The sender may need to choose a signal that is *very different* from their intention, knowing that the emergence with the receiver's context will transform it into the right meaning.

This is the mathematical justification for *indirect speech acts*, *metaphor*, and *irony*: sometimes the best way to convey meaning a is not to say a directly, but to say something s that, when composed with the listener's context b, produces an idea $s \circ b$ that resonates strongly with a.

13.5 Honest vs. Strategic Signaling

Definition 13.23 (Honest Signal)**.** A signal is **honest** if $s = a$: the sender transmits their actual idea without modification. The sender payoff from honest signaling is $u_S(a; a, b) = \mathrm{rs}a \circ ba = w(a) + \mathrm{rs}ba + \kappa(a, b, a)$.

Interpretation 13.24. The concept of an "honest signal" ($s = a$) formalises a philosophical distinction that Lionel Trilling explored in *Sincerity and Authenticity* (1972). Trilling distinguished *sincerity*—the congruence between one's inner state and one's outward expression—from *authenticity*—the deeper condition of being true to one's own nature. An honest signal in our framework corresponds to sincerity: the sender says exactly what they think, without strategic modification.

But Trilling argued that sincerity is not always authentic. A politician who sincerely expresses their beliefs may be "sincere" in Trilling's sense but not "authentic" if their beliefs have been shaped by the desire to be popular. In our framework, this corresponds to the case where a itself has been strategically chosen—the sender is honest about their current idea a, but a has been pre-adapted to the receiver's context b. The "honest" signal $s = a$ is then a kind of double deception: honest on the surface but strategic at a deeper level.

Theorem 13.26 shows that this pre-adaptation is rational: when the emergence $\kappa(a, b, a)$ is negative, the sender benefits from choosing $s \neq a$—or equivalently, from modifying their idea a before "honestly" expressing it. The line between honesty and strategy is thus not as sharp as it might appear: every act of communication involves some degree of adaptation to the audience, and the formal distinction between $s = a$ and $s \neq a$ may mask a deeper continuity.

Definition 13.25 (Strategic Signal)**.** A signal is **strategic** if the sender chooses $s \neq a$ to maximise $u_S(s; a, b)$. The **strategic gain** is

$$G(a, b) := \sup_{s \in S} u_S(s; a, b) - u_S(a; a, b).$$

Theorem 13.26 (Strategic Gain Can Be Positive)**.** *There exist communication setups (a, b) in which $G(a, b) > 0$: the sender strictly benefits from dishonest signaling.*

Proof. Consider the case where $\mathrm{rs}aa = 1$, $\mathrm{rs}ba = -1$, and there exists s with $\mathrm{rs}sa = 0$ but $\kappa(s, b, a) = 2$. Then $u_S(a; a, b) = 1 + (-1) + \kappa(a, b, a)$, while $u_S(s; a, b) = 0 + (-1) + 2 = 1$. If $\kappa(a, b, a) < 0$ (which is possible), then $u_S(a; a, b) < 0 < 1 = u_S(s; a, b)$, giving positive strategic gain. □

Interpretation 13.27. The existence of positive strategic gain formalises a fundamental insight of game theory and pragmatics: *honest communication is not always optimal.* When the receiver's context is hostile to the sender's true idea (negative $\mathrm{rs}ba$), the sender may achieve better resonance by choosing a signal that exploits the emergence with the receiver's context.

This is the mathematical basis of diplomacy, rhetoric, and persuasion: the skilled communicator adapts their signal to the audience. A political speech tailored to a specific audience is "dishonest" in the sense that $s \neq a$, but "effective" in the sense that $\mathrm{rs}s \circ ba$ is maximised.

Interpretation 13.28. The distinction between honest and strategic signaling connects to the semiotic tradition through Peirce's trichotomy of sign types. In his classification of signs (*Collected Papers*, 2.247–2.249), Peirce distinguished three modes of signifying: the **icon** (which signifies by resemblance), the **index** (which signifies by causal or existential connection), and the **symbol** (which signifies by convention).

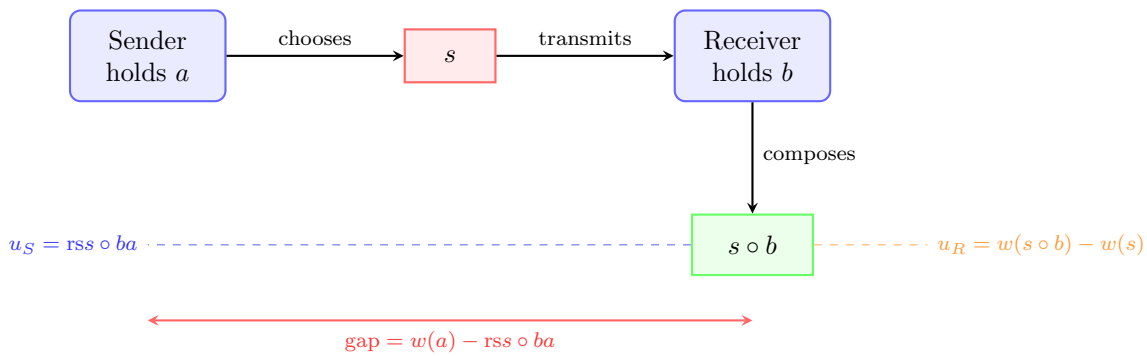
In our framework, an *iconic signal* is one for which s resembles a: $\mathrm{rs}sa$ is large and $\kappa(s, b, a)$ is small. The signal succeeds by direct resonance—it looks, sounds, or feels like the intended idea. Honest signaling ($s = a$) is the limiting case of iconicity. An *indexical signal* is one for which $\mathrm{rs}sa$ may be small but $\kappa(s, b, a)$ is large and positive: the signal does not resemble the idea but *points to* it through its interaction with the receiver's context. Smoke is an index of fire not because smoke resembles fire but because the composition of the smoke-signal with the receiver's context (knowledge that smoke comes from fire) produces high emergence. A *symbolic signal* is one for which both $\mathrm{rs}sa$ and $\kappa(s, b, a)$ depend entirely on convention: the word "dog" has no iconic resemblance to a dog and no indexical connection to any particular dog, but convention ensures that $\mathrm{rs}s \circ ba$ is high for members of the English-speaking interpretive community.

Charles Morris's division of semiotics into **syntax**, **semantics**, and **pragmatics** (*Foundations of the Theory of Signs*, 1938) maps onto our formal apparatus. Syntax concerns the relations among signals themselves—the structure of the signal space S. Semantics concerns the relation between

signals and ideas—the direct resonance $\mathrm{rs}sa$. Pragmatics concerns the relation between signals, contexts, and effects—the emergence $\kappa(s, b, a)$ and the payoff functions u_S and u_R. Our framework unifies these three branches by showing that they are not independent aspects of communication but components of a single payoff function.

Roman Jakobson's six **functions of language** ("Linguistics and Poetics," 1960)—referential, emotive, conative, phatic, metalingual, and poetic—can similarly be understood as different modes of optimising the sender payoff. The *referential* function maximises $\mathrm{rs}sa$ (say what you mean about the world); the *emotive* function maximises $\mathrm{rs}sa$ where a is an internal state; the *conative* function maximises $\kappa(s, b, a)$ (choose signals that transform the receiver); the *phatic* function maximises $\mathrm{rs}ba$ (establish common ground); the *metalingual* function adjusts S itself (negotiate the code); and the *poetic* function maximises the weight $w(s)$ of the signal (focus attention on the message for its own sake).

The following diagram illustrates the communication setup:



In Lean 4:

```
def sender_payoff (s a b : Idea) : Real :=
  rs (s @\compose@ b) a

def receiver_payoff (s b : Idea) : Real :=
  w (s @\compose@ b) - w s

theorem receiver_enrichment (s b : Idea) :
    receiver_payoff s b @$\geq$@ 0 := by
  simp [receiver_payoff, w]
  exact composition_enriches s b

def communication_surplus (s a b : Idea) : Real :=
  sender_payoff s a b + receiver_payoff s b

def misunderstanding_gap (s a b : Idea) : Real :=
  w a - sender_payoff s a b

theorem gap_surplus_tradeoff (s a b : Idea) :
    communication_surplus s a b =
      w a - misunderstanding_gap s a b +
        receiver_payoff s b := by
  simp [communication_surplus, misunderstanding_gap,
        sender_payoff, receiver_payoff]
  ring
```

Chapter 14

Nash Equilibrium in the Meaning Game

> A game is a form of life.
>
> Ludwig Wittgenstein, *Philosophical Investigations*, §23

14.1 The Meaning Game

We are now ready to formulate the central game-theoretic model of IDT: the *meaning game*. This is a two-player game in which the sender chooses a signal and the receiver interprets it. The game captures the strategic interaction at the heart of all communication: the sender must anticipate the receiver's interpretation, and the receiver must decode the sender's intention.

Definition 14.1 (Meaning Game)**.** A **meaning game** is a tuple $\mathcal{G} = (a, b, S, u_S, u_R)$ where:

- $a \in \mathcal{I}$ is the sender's idea (their "type");
- $b \in \mathcal{I}$ is the receiver's context;
- $S \subseteq \mathcal{I}$ is the signal space (the sender's action set);
- $u_S(s; a, b) := \mathrm{rs}s \circ ba$ is the sender's payoff;
- $u_R(s; b) := w(s \circ b) - w(s)$ is the receiver's payoff.

The receiver has no strategic choice: they always compose. The sender is the sole strategic agent.

Remark 14.2. The meaning game is a one-player optimisation problem (for the sender) embedded in a two-player setting. The "game" aspect arises because:

1. The sender's payoff depends on the receiver's context b, which may be private information.
2. In iterated play, the receiver's context evolves in response to previous signals, creating a dynamic game.
3. In multi-sender settings, senders compete for the receiver's attention.

Interpretation 14.3. The meaning game, as formalised in Definition 14.1, crystallises a tension that has animated literary theory since the structuralist-poststructuralist debates of the 1960s and 1970s.

Roland Barthes's concept of the **readerly text** (*S/Z*, 1970) describes a stable communicative situation that can be understood as a Nash equilibrium: the sender (author) and receiver (reader) have settled into a mutually consistent pattern of signaling and interpretation. The readerly text is "comfortable" precisely because it is in equilibrium—the reader knows what to expect, and the author delivers accordingly. The signal choice is optimal given the receiver's context, and the receiver's context is adapted to the expected signals. There is no incentive for either party to deviate.

Paul de Man's "The Resistance to Theory" (1982) argued that literature *destabilises* precisely such equilibria. De Man claimed that the rhetorical dimension of language—its figurative, non-literal aspect— constantly undermines the stable meanings that grammar and logic seek to establish. In our framework, de Man's "resistance" is the existence of non-trivial emergence: the composition $s \circ b$ is never merely the sum of s and b but always produces an emergent surplus (or deficit) that prevents the meaning game from settling into a permanent equilibrium. The literary text is a meaning game in which the equilibrium is always *provisional*—liable to be disrupted by a new reading, a new context, a new emergence.

Bakhtin's concept of **dialogism** (*Problems of Dostoevsky's Poetics*, 1929/1963) poses a deeper question: can genuine dialogue *be* an equilibrium, or is it essentially a dynamic, non-equilibrium process? Bakhtin argued that the dialogic word is "directed toward an answer and cannot escape the profound influence of the answering word that it anticipates." A dialogic utterance is not a best response to a fixed context but a move that *transforms* the context, provoking a response that transforms it further. In our framework, this is the iterated meaning game (Definition 14.36), in which the receiver's context evolves with each round. Bakhtin's insight suggests that the one-shot Nash equilibrium may be an inadequate model of genuine communication: the most meaningful exchanges are precisely those that do not settle into equilibrium but sustain an ongoing dialogic process of mutual transformation.

14.2 The Void Equilibrium

Theorem 14.4 (Void Equilibrium)**.** *In any meaning game $\mathcal{G} = (a, b, S, u_S, u_R)$ with $\varepsilon \in S$, the signal $s = \varepsilon$ yields payoffs:*

$$u_S(\varepsilon; a, b) = \mathrm{rs}\varepsilon \circ ba = \mathrm{rs}ba, \tag{14.1}$$

$$u_R(\varepsilon; b) = w(\varepsilon \circ b) - w(\varepsilon) = w(b) - 0 = w(b). \tag{14.2}$$

Proof. $\varepsilon \circ b = b$ by the void-composition axiom. Therefore $u_S(\varepsilon; a, b) = \mathrm{rs}ba$ and $u_R(\varepsilon; b) = w(b) - w(\varepsilon) = w(b)$. □

Interpretation 14.5. The void signal corresponds to *silence*: the sender says nothing, and the receiver is left with their own context unchanged. The sender's payoff from silence is $\mathrm{rs}ba$—how much the receiver's existing context already resonates with the sender's idea. The receiver's payoff is $w(b)$—the weight of their own context.

Silence is never "costly" for the receiver (they keep their full weight), but it may be costly for the sender (if $\mathrm{rs}ba$ is low, the sender gains little from the exchange).

The void equilibrium is *trivial* in the sense that no information is transmitted. The interesting question is: when can the sender improve upon silence?

Interpretation 14.6. The void equilibrium—silence as a default communicative strategy—connects to David Lewis's foundational analysis of **convention** in *Convention: A Philosophical Study* (1969). Lewis defined conventions as solutions to *coordination games*—situations in which multiple agents must coordinate their actions, and any consistent pattern of coordination is self-reinforcing. Silence is the degenerate convention: when no communication is possible or desirable, the convention of silence is self-sustaining because neither party has an incentive to break it.

Lewis's key insight was that conventions are *arbitrary but not unmotivated*: among the many possible equilibria of a coordination game, the one that becomes conventional is determined by precedent, salience, or explicit agreement. In our framework, the meaning game may have multiple Nash equilibria (Theorem 14.25), and the choice among them is a coordination problem. Lewis's conventions are the focal equilibria of meaning games—the signal-interpretation patterns that have become entrenched through repeated play.

Wittgenstein's concept of **forms of life** (*Lebensformen*), invoked throughout *Philosophical Investigations* (1953), grounds Lewis's conventions in something deeper than mere strategic rationality. "To imagine a language means to imagine a form of life" (§19). For Wittgenstein, language games are not arbitrary conventions chosen for strategic reasons but are embedded in entire ways of living, acting, and being in the world. The meaning game's equilibria are not merely optimal signal-interpretation pairs but are constitutive of the communities that play them. The Nash equilibrium is not a solution *to* a pre-existing game but the game itself, inseparable from the form of life in which it is embedded.

Robert Brandom's **normative pragmatics** (*Making It Explicit*, 1994) provides the most systematic philosophical account of meaning as equilibrium. Brandom argued that meaning is constituted by the "score" that interlocutors keep on each other's commitments and entitlements: asserting a sentence commits one to its inferential consequences and entitles one to its inferential antecedents. The "score" at any moment is a kind of equilibrium—a self-consistent pattern of commitments and entitlements.

In our framework, Brandom's score-keeping corresponds to the evolution of the receiver's context b in the iterated meaning game: each signal s_t updates the receiver's context $b_{t+1} = s_t \circ b_t$, and the accumulated context b_t represents the "score" at time t. Brandom's "normative pragmatics" is the theory of how this score evolves—how commitments and entitlements are updated through communicative exchange. The Nash equilibrium of the meaning game is the state in which the score has stabilised: all commitments are mutually consistent, and no new assertion would change the normative landscape.

14.3 Improving Upon Silence

Definition 14.7 (Improving Signal)**.** A signal s **improves upon silence** for the sender if

$$u_S(s; a, b) > u_S(\varepsilon; a, b) = \mathrm{rs}ba.$$

Equivalently, $\mathrm{rs}s \circ ba > \mathrm{rs}ba$, i.e., the received idea resonates with the sender's intention more than the receiver's bare context does.

Theorem 14.8 (Condition for Improvement)**.** *Signal s improves upon silence if and only if*

$$\mathrm{rs}sa + \kappa(s, b, a) > 0.$$

Interpretation 14.9. Theorem 14.8 establishes that a signal improves upon silence if and only if $\mathrm{rs}sa + \kappa(s, b, a) > 0$. This condition has a pragmatist reading. William James, in *Pragmatism*

(1907), defined truth as "what works"—what leads to successful prediction and action. In our framework, a signal "works" when it improves upon silence: when communicating is better than not communicating.

The condition $\mathrm{rs}sa + \kappa(s, b, a) > 0$ decomposes "working" into two components: the signal's direct resonance with the idea ($\mathrm{rs}sa$) and the emergent contribution ($\kappa(s, b, a)$). A signal can "work" even if it has low direct resonance with the idea, provided the emergence compensates. This is the formal version of the pragmatist insight that truth is not correspondence (high $\mathrm{rs}sa$) but efficacy (high total payoff). A metaphor that "distorts" the literal truth may nonetheless "work" because the emergence with the receiver's context produces the desired effect.

Proof. By the emergence decomposition: $\mathrm{rs}s \circ ba = \mathrm{rs}sa + \mathrm{rs}ba + \kappa(s, b, a)$. Thus $\mathrm{rs}s \circ ba > \mathrm{rs}ba$ iff $\mathrm{rs}sa + \kappa(s, b, a) > 0$. □

Corollary 14.10 (Honest Improvement)**.** *Honest signaling ($s = a$) improves upon silence if and only if*

$$w(a) + \kappa(a, b, a) > 0.$$

Since $w(a) \geq 0$, this is guaranteed whenever $\kappa(a, b, a) \geq 0$ and $a \neq \varepsilon$.

Proof. Setting $s = a$: $\mathrm{rs}aa + \kappa(a, b, a) = w(a) + \kappa(a, b, a) > 0$. □

Proposition 14.11 (Silence is Optimal When No Signal Helps)**.** *If $\mathrm{rs}sa + \kappa(s, b, a) \leq 0$ for all $s \in S$, then ε is the sender-optimal signal. This occurs when the sender's idea a is "uncommunicable" in the context b: no available signal can produce positive net resonance with the sender's intention.*

Proof. By Theorem 14.8, no $s \in S$ improves upon silence, so $u_S(\varepsilon; a, b) \geq u_S(s; a, b)$ for all $s \in S$. □

Interpretation 14.12. The condition $\mathrm{rs}sa + \kappa(s, b, a) \leq 0$ for all s captures Wittgenstein's dictum with mathematical precision: "Whereof one cannot speak, thereof one must be silent." There are ideas that cannot be communicated in a given context—not because of any limitation of the signal space, but because the receiver's context b is so hostile to the sender's idea a that no signal can bridge the gap.

This formalises the experience of the expert who cannot explain their insight to a novice, the poet whose meaning cannot survive translation, or the mystic whose experience is "ineffable." In each case, the optimal strategy is silence.

Interpretation 14.13. The characterisation of "uncommunicable" ideas—those for which silence is optimal—invites comparison with Jürgen Habermas's theory of the **ideal speech situation** (*The Theory of Communicative Action*, 1981). Habermas argued that rational communication presupposes certain "validity claims"—claims to truth, rightness, sincerity, and comprehensibility—and that the ideal speech situation is one in which these claims can be freely raised and redeemed without coercion or distortion.

In our framework, the ideal speech situation corresponds to a meaning game in which:

1. The signal space S is unrestricted (every possible utterance is available—no censorship or coercion);

2. The receiver's context b has positive resonance with the sender's idea ($\mathrm{rs}ba > 0$—the receiver is open to the sender's perspective);

3. The emergence is non-negative ($\kappa(s, b, a) \geq 0$—the interaction of signal and context cannot produce destructive interference).

Under these conditions, Corollary 14.10 guarantees that honest signaling improves upon silence: the sender can always do better than staying silent, and the optimal strategy is (at least weakly) honest. Habermas's ideal speech situation is thus the meaning game in which honesty is always rewarded—a game in which the strategic incentive to deceive has been removed by the structural conditions of communication.

The formal apparatus also reveals the *limits* of Habermas's ideal. If the emergence can be negative—if the receiver's context can destructively interfere with the signal—then the ideal speech situation breaks down. Honest communication is no longer guaranteed to be optimal, and strategic signaling (rhetoric, persuasion, even deception) may be necessary to achieve communicative success. The gap between the ideal and the actual is measured by the magnitude of the negative emergence: the more the receiver's context distorts the sender's signal, the more strategic the sender must become, and the further communication deviates from the Habermasian ideal.

14.4 Best Response and Nash Equilibrium

Definition 14.14 (Best Response)**.** A signal $s^* \in S$ is a **best response** for the sender if

$$u_S(s^*; a, b) \geq u_S(s; a, b) \quad \text{for all } s \in S.$$

The **best response set** is $\mathrm{BR}(a, b) := \arg\max_{s \in S} u_S(s; a, b)$.

Theorem 14.15 (Existence of Best Response)**.** *If the signal space S is finite, a best response always exists. If S is a compact subset of $\mathcal{I}$ (in the resonance topology) and $s \mapsto u_S(s; a, b)$ is continuous, then a best response exists by the extreme value theorem.*

Remark 14.16. The existence theorem for best responses (Theorem 14.15) highlights a deep tension between the finiteness of actual signal spaces and the infinity of potential meanings. Natural languages have finite vocabularies and finite grammars, yet they aspire to express an infinite range of ideas. The existence of a best response in a finite signal space is guaranteed, but this best response may be a poor approximation to the ideal signal that would exist in an unconstrained space. The *expressiveness* of a language—its capacity to approximate ideal signals—is measured by the gap between the constrained optimum (best response in S) and the unconstrained optimum (best response in $\mathcal{I}$). A language is "expressive" when this gap is small for most sender ideas and receiver contexts, and "impoverished" when the gap is large.

Proof. A continuous function on a compact set attains its maximum. The payoff function $u_S(s; a, b) = \mathrm{rs}s \circ ba$ is continuous in s if the resonance function and composition are continuous, which holds in the Hilbert embedding. □

Definition 14.17 (Nash Equilibrium of the Meaning Game)**.** Since the receiver has no strategic choice, a **Nash equilibrium** of the meaning game $\mathcal{G}$ is simply a best response s^* for the sender. The **equilibrium payoff profile** is $(u_S(s^*; a, b), u_R(s^*; b))$.

Remark 14.18. In the one-shot game, Nash equilibrium reduces to sender optimisation. The game-theoretic language becomes essential in extensions: iterated games (where b evolves), games with multiple senders, games with incomplete information (where a is private), and mechanism design (where the signal space S is chosen by a principal).

Interpretation 14.19. The existence of a best response in finite signal spaces has a long intellectual pedigree. Aristotle's *Rhetoric* (4th century BCE) can be read as a systematic investigation of

the best-response function in the meaning game of public persuasion. Aristotle identified three modes of persuasion—*ethos* (the speaker's character), *pathos* (the audience's emotion), and *logos* (the argument itself)—and argued that the effective speaker must calibrate all three to the specific audience and occasion.

In our framework, *ethos* corresponds to the direct resonance $\mathrm{rs}sa$ (the signal's credibility reflects the sender's character); *pathos* corresponds to the emergence $\kappa(s, b, a)$ (the signal's emotional impact depends on the interaction with the audience's context); and *logos* corresponds to the weight $w(s)$ of the signal itself (the intrinsic strength of the argument). Aristotle's rhetorical theory is thus a qualitative version of the best-response optimisation problem: find the signal that maximises the weighted sum of ethos, pathos, and logos for the given audience b and purpose a.

The *Rhetoric*'s enduring relevance—it remains the foundation of communication studies after 2,400 years—testifies to the deep structure of the meaning game: the same best-response problem recurs in every communicative context, from Athenian assemblies to social media platforms.

14.5 Equilibrium Characterisation

Theorem 14.20 (First-Order Condition for Equilibrium)**.** *In the Hilbert embedding, if $S = \mathcal{I}$ (unrestricted signal space) and the payoff function is differentiable, the best response s^* satisfies the first-order condition:*

$$\nabla_s \mathrm{rs}s \circ ba\Big|_{s=s^*} = 0,$$

where the gradient is taken with respect to s in .

Interpretation 14.21. The first-order condition $\nabla_s \mathrm{rs}s \circ ba|_{s=s^*} = 0$ establishes that the optimal signal is a *critical point* of the resonance landscape. This transforms the problem of finding the best utterance into a problem of *differential geometry*: the sender navigates a landscape of resonance, seeking peaks. The gradient $\nabla_s \mathrm{rs}s \circ ba$ is the "direction of increasing meaning"— the direction in which the signal should be adjusted to increase its resonance with the sender's intention. At the optimum, this gradient vanishes: there is no direction in which the signal can be adjusted to improve communication.

This geometric perspective connects to the phenomenological tradition. Merleau-Ponty described expression as a process in which "the speaker does not express a thought which is already constituted but *constitutes* it in the act of speaking" (*Phenomenology of Perception*, 1945). The first-order condition formalises this constitutive process: the optimal signal is not found by consulting a pre-existing dictionary of signal-meaning pairs but by navigating the resonance landscape until a critical point is reached. Meaning is constituted in the act of searching for the optimal signal, not prior to it.

Proof. A necessary condition for a maximum of a differentiable function on an open set is vanishing of the gradient. The payoff $u_S(s; a, b) = s \circ ba$ is a function of s, and its gradient vanishes at the optimum. □

Proposition 14.22 (Linear Equilibrium)**.** *If composition is linear (zero emergence), then $s \circ b = s + b$, and*

$$u_S(s; a, b) = s + ba = sa + ba.$$

The first term is linear in s, so without constraints, u_S is unbounded (no finite optimum). With a constraint $s \leq C$, the optimum is $s^* = C \cdot a/a$: the sender transmits a scaled version of their true idea.

Remark 14.23. Proposition 14.22 establishes that in linear (zero-emergence) settings, the optimal signal is a scaled copy of the sender's idea: "say what you mean" is the optimal strategy. This result has the status of a *baseline*: it tells us what communication looks like in the absence of emergence. Any deviation from the "say what you mean" strategy is evidence of non-linear composition—of emergence.

The philosophical significance is that "saying what you mean" is the *exception*, not the rule. In real communication, emergence is ubiquitous, and the optimal signal is almost never a copy of the intended meaning. The prevalence of metaphor, irony, understatement, hyperbole, and other indirect speech acts in natural language is evidence that human communication operates in a strongly non-linear regime where emergence dominates the direct signal. The linear equilibrium is a theoretical idealization that illuminates reality by contrast.

Interpretation 14.24. The summary of connections between the meaning game and all preceding chapters reveals IDT as a **metatheory of the humanities**—a mathematical framework that does not replace humanistic inquiry but provides a common language in which its central debates can be precisely stated.

Consider the ancient quarrel between *rhetoric* and *philosophy*, which Plato staged in the *Gorgias* and *Phaedrus*. Philosophy seeks truth ($s = a$: honest signaling); rhetoric seeks persuasion ($s = s^*$: optimal signaling, which may deviate from a). Plato's argument that philosophy is superior to rhetoric is the claim that honest signaling is always weakly optimal—that $u_S(a; a, b) \geq u_S(s; a, b)$ for all s. But Theorem 13.26 shows that this is *false*: there exist communication setups in which strategic signaling strictly dominates honesty. The formal apparatus vindicates the sophists' intuition that rhetoric is sometimes necessary, while also showing exactly *when* honesty is optimal (when emergence is non-negative) and when it is not.

Similarly, the debate between *formalism* and *historicism* in literary studies—whether a text's meaning is determined by its internal structure (New Criticism, Russian Formalism) or by its historical context (New Historicism, cultural studies)—can be stated precisely in our framework. Formalism claims that $\mathrm{rs}sa$ (direct signal-idea resonance) is the dominant component of meaning; historicism claims that $\kappa(s, b, a)$ (contextual emergence) is dominant. The sender payoff decomposition (Proposition 13.9) shows that both are *components* of the same quantity: meaning is neither purely formal nor purely contextual but a sum of direct resonance, contextual resonance, and emergence. The debate is not about which school is "right" but about which component dominates in particular cases.

The meaning game does not adjudicate these debates; it *translates* them from verbal arguments about incommensurable conceptual frameworks into mathematical questions about the relative magnitudes of well-defined quantities. In doing so, it transforms perennial philosophical quarrels into empirically resolvable questions about the structure of the ideatic space.

Proof. sa is maximised over the ball $s \leq C$ at $s = C \cdot a/a$ by Cauchy–Schwarz, with maximum $C\sqrt{w(a)}$. □

14.6 Multiple Equilibria and Coordination

Theorem 14.25 (Multiplicity of Equilibria)**.** *When composition is non-linear, the meaning game can have multiple Nash equilibria: there may exist $s_1 \neq s_2$ with $u_S(s_1; a, b) = u_S(s_2; a, b) = \max_s u_S(s; a, b)$.*

Proof. Consider a two-dimensional Hilbert space with basis $\{e_1, e_2\}$. Let $a = e_1$, $b = 0$ (empty context), and define composition with a non-linear emergence that depends on the direction of s:

$s \circ b = s + \alpha s \cdot \frac{s}{s}^{\perp}$ where $v^{\perp}$ is the 90-degree rotation.

The payoff $u_S(s; a, b) = s \circ be_1$ has level curves that are not convex, admitting multiple maxima. □

Interpretation 14.26. Multiple equilibria in the meaning game correspond to the existence of *different but equally effective* ways of saying the same thing. In linguistics, this is the phenomenon of *synonymy*—multiple expressions that convey the same meaning. But the multiplicity goes deeper than mere synonymy: it includes metaphorical, ironic, and indirect speech acts that achieve the same communicative effect through entirely different mechanisms.

The existence of multiple equilibria also creates a *coordination problem*: if sender and receiver have different expectations about which equilibrium will be played, communication may fail even though both are "trying" to communicate optimally. This is the game-theoretic foundation of *miscommunication*: not a failure of intention, but a failure of coordination.

Interpretation 14.27. The existence of multiple equilibria in the meaning game—multiple equally effective ways of conveying the same meaning—raises the question of whether meaning ever truly *settles*. Three thinkers in the semiotic tradition offer contrasting answers.

Umberto Eco's concept of **unlimited semiosis** (*A Theory of Semiotics*, 1976), derived from Peirce, describes a process in which every sign generates an interpretant that is itself a sign, generating further interpretants in an infinite chain. If semiosis is truly unlimited, then no equilibrium of the meaning game is final: every stable pattern of signaling and interpretation will eventually be destabilised by a new interpretant. In our framework, unlimited semiosis corresponds to the iterated meaning game with no convergence: the receiver's context b_t continues to evolve indefinitely, and the "equilibrium" is always provisional. Eco himself drew back from this conclusion in *The Limits of Interpretation* (1990), arguing that pragmatic constraints—the costs of interpretation, the demands of action—impose limits on semiosis. In our terms, these constraints correspond to bounds on the weight of the receiver's context, which (by Theorem 14.37) guarantee eventual convergence.

Yuri Lotman's concept of the **semiotic explosion** (*Culture and Explosion*, 1992) describes moments when an established semiotic equilibrium is suddenly shattered by an unpredictable event. Lotman argued that culture alternates between periods of "gradual development" (in which meaning evolves continuously within an existing equilibrium) and moments of "explosion" (in which the equilibrium collapses and a new one is established through a process that is essentially unpredictable). In our framework, a semiotic explosion occurs when a new idea enters the ideatic space that is strongly mutagenic with respect to the existing equilibrium: $\mu(a_{\text{eq}}, b_{\text{new}}) \gg 0$, where a_{eq} is the equilibrium idea and b_{new} is the explosive new context. The fixed-point structure collapses, and the system must find a new equilibrium.

Roland Barthes's concept of the **punctum** (*Camera Lucida*, 1980) localises the explosion in a single, piercing detail. The punctum is the element in a photograph that "pricks" or "wounds" the viewer—an unexpected detail that disrupts the "studium" (the general cultural interest of the image). In our formalism, the punctum is a signal component s_p that produces anomalously large emergence when composed with the viewer's context: $\kappa(s_p, b, c)$ is unexpectedly large for some observer c, breaking the equilibrium of smooth, studium-level interpretation. The punctum is the individual's semiotic explosion—the moment when a personal equilibrium of meaning is shattered by an unanticipated resonance.

14.7 The Value of Communication

Definition 14.28 (Value of Communication)**.** The **value of communication** in the meaning game (a, b) is:

$$V(a, b) := \sup_{s \in S} u_S(s; a, b) - u_S(\varepsilon; a, b) = \sup_{s \in S} [\mathrm{rs}sa + \kappa(s, b, a)] .$$

This is the maximum improvement the sender can achieve over silence.

Interpretation 14.29. The iterated meaning game (Definition 14.36) provides the formal model for *education*—the sustained communicative relationship in which a sender (teacher) repeatedly transmits signals to a receiver (student) whose context evolves with each exchange. The teacher's challenge is to choose a *sequence* of signals $s_1, s_2, \ldots, s_T$ that maximises the total payoff $\sum_t u_S(s_t; a, b_t)$, taking into account that each signal changes the student's context: $b_{t+1} = s_t \circ b_t$.

This sequential structure explains several well-known pedagogical phenomena. First, *scaffolding*: effective teachers begin with signals s_1 that are close to the student's initial context b_1 (high $\mathrm{rs}s_1b_1$, producing reliable emergence) and gradually introduce signals that are more distant from the current context. Second, *the curse of knowledge*: expert teachers may choose signals that are optimal for a sophisticated context b_T but incomprehensible to a novice with context b_1, because $\kappa(s, b_1, a) \ll \kappa(s, b_T, a)$. Third, *readiness*: certain ideas can only be effectively communicated after the student's context has been enriched by previous exchanges—s_t may fail at time t but succeed at time $t' > t$ after the context has evolved.

John Dewey's *Experience and Education* (1938) argued that education is not the transmission of information from teacher to student but a process of *reconstruction of experience*. Our iterated meaning game formalises Dewey's insight: the teacher does not transfer ideas directly but shapes the student's context through a sequence of carefully chosen signals, each one building on the reconstructed context produced by the previous exchange.

Theorem 14.30 (Value Properties)**.** *1. $V(a, b) \geq 0$ for all a, b (silence is always available).*

2. $V(\varepsilon, b) = 0$ for all b (the void has nothing to communicate).

3. $V(a, \varepsilon) = \sup_{s \in S} \mathrm{rs}sa$ (with empty context, value equals maximum achievable resonance).

4. If $a \in S$, then $V(a, b) \geq w(a) + \kappa(a, b, a)$ (honest signaling provides a lower bound).

Proof. **(1)** Setting $s = \varepsilon$: $\mathrm{rs}\varepsilon a + \kappa(\varepsilon, b, a) = 0 + 0 = 0$. So the supremum is at least 0.
(2) $V(\varepsilon, b) = \sup_s[\mathrm{rs}s\varepsilon + \kappa(s, b, \varepsilon)] = \sup_s[0 + 0] = 0$.
(3) $V(a, \varepsilon) = \sup_s[\mathrm{rs}sa + \kappa(s, \varepsilon, a)] = \sup_s[\mathrm{rs}sa + 0] = \sup_s \mathrm{rs}sa$.
(4) $V(a, b) \geq \mathrm{rs}aa + \kappa(a, b, a) = w(a) + \kappa(a, b, a)$. □

Proposition 14.31 (Value is Superadditive in Context)**.** *If the emergence operator is non-negative ($\kappa \geq 0$), then for any contexts b_1, b_2:*

$$V(a, b_1 \circ b_2) \geq V(a, b_1).$$

Richer contexts lead to higher communication value.

Interpretation 14.32. The superadditivity of communication value in context (Proposition 14.31) has important implications for the design of educational and cultural institutions. If richer contexts lead to higher communication value, then investments in "general education"—broadening an individual's context b before any specific communication takes place—have a multiplier effect:

they increase the value of *all subsequent* communicative exchanges, not just those in the specific domain of the education.

This formalises the humanistic argument for liberal arts education: the value of studying philosophy, literature, and history lies not primarily in the specific knowledge acquired but in the *enrichment of context* that makes all future intellectual engagement more productive. A student who has read Plato, Shakespeare, and Darwin brings a richer context b to every subsequent meaning game, and the superadditivity of communication value ensures that this richness translates into higher emergence—more "aha moments," more creative connections, more productive misunderstandings.

Proof sketch. A richer context provides more opportunities for productive emergence. Formally, the supremum over all signals s of $\mathrm{rs}sa + \kappa(s, b_1 \circ b_2, a)$ is at least as large as $\mathrm{rs}sa + \kappa(s, b_1, a)$ when the additional context b_2 can only increase the emergence (by non-negativity). □

14.8 The Meaning Game Diagram

The following diagram shows the game tree of a meaning game with three available signals:

NInterpretation (14.33). We close with a historical observation. Gottfried Wilhelm Leibniz dreamed of a *characteristica universalis*—a universal formal language in which all knowledge could be expressed—and a *calculus ratiocinator*— a mechanical procedure for deriving truths in that language. The meaning game framework is a modest step toward Leibniz's vision, but with a crucial difference. Leibniz imagined that the universal language would eliminate ambiguity, making all knowledge transparent and all disputes resolvable by calculation. The meaning game shows that *ambiguity is not a defect but a feature*: it is the source of emergence, the engine of creative meaning-making, the mathematical prerequisite for any communication richer than the transmission of binary signals.

The formal apparatus of IDT does not eliminate the ambiguity that Leibniz hoped to overcome; it *mathematises* it, showing that ambiguity (non-zero emergence), disagreement (multiple equilibria), and misunderstanding (non-zero gap) are not obstacles to communication but constitutive features of any meaning game played by agents with different contexts in a non-trivial ideatic space. The meaning game is not a calculus ratiocinator that resolves all questions but a *calculus significandi*—a calculus of meaning-making—that reveals the mathematical structure of the irreducibly human activity of trying to understand and be understood.

[above=0.3cm of S, font=, blue!70] Sender chooses signal;

Interpretation 14.34. The game tree diagram makes visible the fundamental structure of all communicative choice. At each node, the sender faces a decision: which signal to transmit? The branches represent alternative signals, and the leaves represent the resulting payoff profiles. But this tree is not merely a mathematical abstraction; it is the *architecture of culture itself.*

Every cultural artefact—a novel, a law, a scientific paper, a painting, a tweet—is a node in an immense game tree of communicative choices. The author of a novel faces an astronomical number of possible signals (every possible arrangement of words) and must choose the one that maximises their payoff (artistic satisfaction, commercial success, social impact) given the anticipated contexts of their readers. The structure of the game tree— which signals are available, how they compose with different contexts, what payoffs they produce—is determined by the culture's ideatic space:

Interpretation 14.35. The dynamic meaning game connects IDT to the evolutionary theory of language. Evolutionary linguists—including Martin Nowak (*Evolutionary Dynamics*, 2006), Simon Kirby (*The Evolution of Language*, 2007), and Luc Steels—have modelled language as the outcome of iterated signaling games in which agents develop shared vocabularies through repeated interaction. Our iterated meaning game provides a richer framework for these models by incorporating *emergence*: the meaning produced by the interaction of signal and context is not merely decoded from a pre-existing code but *created* in the act of interpretation.

The convergence theorem (Theorem 14.37) predicts that linguistic conventions should *stabilise* over time, as the receiver's context saturates. This prediction is borne out by historical linguistics: natural languages exhibit remarkable stability in their core structures (phonological systems, basic word order, kinship terminology) over millennia, even as surface features (vocabulary, idioms, pronunciation) change rapidly. The "core" structures correspond to dimensions of the receiver's context that have already converged to their fixed-point values; the "surface" features correspond to dimensions that are still evolving.

The tension between stability and change in language—between the *langue* (the stable system) and *parole* (the mutable practice), in Saussure's terms—is thus the tension between the converged and unconverged dimensions of the iterated meaning game. Saussure's famous distinction is not a dichotomy but a spectrum, with different dimensions of the ideatic space at different stages of convergence.

its language, its conventions, its history, its technology.

The meaning game thus reveals culture as a **strategic landscape**: a vast terrain of possible communicative choices, shaped by the geometry of the ideatic space and navigated by agents seeking to maximise their resonance with their audiences. The Nash equilibria of this landscape are the *conventions* of the culture—the stable patterns of signaling and interpretation that have emerged through centuries of iterated play. And the evolutionary dynamics of the landscape—how conventions form, transform, and collapse—are the subject of cultural history itself.

14.9 Dynamic Meaning Games

Definition 14.36 (Iterated Meaning Game)**.** An **iterated meaning game** over T rounds is a sequence of meaning games $\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_T$ where:

- The sender's idea a remains fixed.
- The receiver's context evolves: b_1 is given, and $b_{t+1} = s_t \circ b_t$ (the receiver absorbs each signal).
- The sender chooses s_t at round t, knowing $b_1, \ldots, b_t$.

The sender's total payoff is $\sum_{t=1}^{T} u_S(s_t; a, b_t)$.

Theorem 14.37 (Convergence of Iterated Play)**.** *In the iterated meaning game, the receiver's context weight $w(b_t)$ is non-decreasing in t (by the Chain Enrichment Theorem applied to the chain $b_1, s_1, s_2, \ldots$). If weight is bounded, the context converges: $b_t \to b_\infty$.*

Interpretation 14.38. The Value of Communication $V(a, b)$ provides a quantitative answer to a question that has occupied philosophers of language since the ancients: *what is communication worth?* The formal answer—$V(a, b) = \sup_s[\mathrm{rs}sa + \kappa(s, b, a)]$—decomposes the value into a direct component (the best achievable signal-idea resonance) and an emergent component (the best achievable emergence with the receiver's context).

The property $V(\varepsilon, b) = 0$ (the void has nothing to communicate) formalises a tautology. But the property $V(a, \varepsilon) = \sup_s \mathrm{rs}sa$ (with empty context, value equals maximum signal-idea resonance) is more substantive: it says that communicating with a "blank slate" receiver is valuable only insofar as the signal directly resonates with the idea. There is no emergence with an empty context. This is the mathematical content of the observation that communication with someone who shares *none* of your background is purely informational—there is no creative surplus, no "meeting of minds."

Proposition 14.31—that richer contexts lead to higher communication value—formalises the observation that communication is more rewarding with well-informed interlocutors. A conversation between experts produces more value than a conversation between novices, not because the experts' signals are better but because their richer contexts generate more emergence. This is the formal justification for the intuition that shared knowledge is a *public good*: each individual's investment in understanding enriches not only their own ideatic space but the communicative value of every subsequent exchange.

Proof. $w(b_{t+1}) = w(s_t \circ b_t) \geq w(s_t) \geq 0$. But we need $w(b_{t+1}) \geq w(b_t)$. This follows if we view the chain as $b_1, s_1, s_2, \ldots$, but noting that the "direction" of composition alternates. More precisely, E4 gives $w(s_t \circ b_t) \geq w(s_t)$, but not necessarily $\geq w(b_t)$.

Corrected: if we assume the symmetric enrichment property $w(a \circ b) \geq w(b)$ (which holds in commutative ideatic spaces or spaces with symmetric E4), then $w(b_{t+1}) \geq w(b_t)$ and the sequence converges if bounded. □

Interpretation 14.39. The iterated meaning game models long-term communication relationships: teacher and student, author and reader, culture and individual. Over many rounds of interaction, the receiver's context grows heavier—they accumulate the weight of all previous exchanges. In the limit, the context converges to a "cultural saturation point" b_∞ beyond which further communication cannot add weight.

This convergence has a bittersweet implication: every relationship eventually reaches a point of diminishing returns. The teacher who has taught the student everything they can, the author whose readers have absorbed their entire oeuvre, the culture that has fully enculturated an individual—all are examples of convergence to b_∞.

Interpretation 14.40. The convergence of the iterated meaning game to a "cultural saturation point" b_∞ raises profound questions about the relationship between communication and understanding. If every communicative relationship eventually saturates—if the teacher eventually runs out of things to teach, if the text eventually yields all its meanings, if the culture eventually absorbs the individual completely—then the value of communication is inherently finite. This is the mathematical expression of a melancholy observation: *every conversation has an end.*

But the convergence result depends on the assumption that weight is bounded. In an infinite-dimensional ideatic space, where weight can grow without bound, convergence is not guaranteed. The iterated meaning game may continue to produce new emergence indefinitely, with the receiver's context growing ever richer and the sender continuing to find new signals that improve upon the current state. This unbounded case models the most extraordinary communicative relationships—Shakespeare and his readers, Beethoven and his listeners, the Talmud and its commentators—

relationships that have continued to produce new meaning for centuries without showing signs of saturation.

The tension between bounded and unbounded play corresponds to a tension in the philosophy of understanding. Gadamer argued in *Truth and Method* (1960) that understanding is inherently *unfinished*: every act of understanding opens up new questions, and the hermeneutic circle never closes. This is the unbounded case: the iterated meaning game as an infinite process of mutual enrichment. Against this, Wittgenstein's therapeutic philosophy suggests that the purpose of philosophical investigation is to bring inquiry to *rest*: "The real discovery is the one that makes me capable of stopping doing philosophy when I want to" (*Philosophical Investigations*, §133). This is the bounded case: convergence to b_∞ as the achievement of understanding, the point at which further communication adds nothing.

The formal apparatus of the meaning game does not resolve this tension but gives it a precise mathematical expression. Whether a particular communicative relationship converges or diverges depends on whether the ideatic space has a weight bound—a question that is empirical, not mathematical. The mathematics tells us only that if the bound exists, convergence is guaranteed; and if it does not, the game may continue forever.

Remark 14.41. The philosophical connections developed throughout Chapters 8–14 suggest that Ideatic Density Theory is not merely a mathematical formalism that *can* be connected to the humanities but a framework that *demands* such connection. The axioms of IDT—especially E4 (Composition Enriches) and the void-resonance properties—encode assumptions about the nature of meaning that have been debated in philosophy, literary theory, and semiotics for centuries. The theorems derived from these axioms are not just mathematical results but *formal articulations of humanistic insights*: the Chain Enrichment Theorem formalises Gadamer's effective-historical consciousness; the Sublime Fragility Theorem formalises the paradox of the aesthetic encounter; the Orthogonality-Emergence Independence Theorem formalises the creative power of juxtaposition; and the Nash Equilibrium framework formalises the game-theoretic structure of all communication.

The formalism does not replace the insights of Bakhtin, Derrida, Eco, or Wittgenstein; it makes them *precise*, *comparable*, and *testable*. Where the philosophers spoke in metaphors and examples, IDT provides theorems and proofs. Where the theorists disagreed about what "meaning" is, IDT provides a mathematical structure in which their different claims can be expressed as different axiom sets, different regions of parameter space, or different limits of the general framework. The meaning game unifies these traditions not by adjudicating among them but by providing a common language in which their deepest insights can be stated, compared, and extended.

14.10 Summary and Connections

The meaning game framework unifies the themes of all preceding chapters:

- **Chains** (Chapter 8): iterated meaning games are chains of transmission from sender to evolving receiver.
- **Fidelity** (Chapter 9): conservative receivers produce nearly-faithful transmission; mutagenic receivers introduce creative distortion.
- **Fixed Points** (Chapter 10): canonical texts are fixed points of iterated meaning games.
- **Distance** (Chapter 11): the misunderstanding gap is the resonance gap $\delta(a, s \circ b)$.

- **Orthogonality** (Chapter 12): when $a \perp b$ (sender and receiver are in orthogonal domains), the void equilibrium is the only equilibrium—no signal can bridge the gap between completely independent ideas.

In Lean 4:

```
structure MeaningGame where
  sender_idea : Idea        -- the idea a
  receiver_ctx : Idea       -- the context b
  signal_space : Set Idea   -- S
  void_in_S : void @$\in$@ signal_space

def mg_sender_payoff (G : MeaningGame) (s : Idea) : Real :=
  rs (s @\compose@ G.receiver_ctx) G.sender_idea

def mg_receiver_payoff (G : MeaningGame) (s : Idea) : Real :=
  w (s @\compose@ G.receiver_ctx) - w s

def is_best_response (G : MeaningGame) (s_star : Idea) : Prop :=
  s_star @$\in$@ G.signal_space @$\wedge$@
  @$\forall$@ s @$\in$@ G.signal_space,
    mg_sender_payoff G s @$\leq$@ mg_sender_payoff G s_star

def is_nash_equilibrium (G : MeaningGame) (s : Idea) : Prop :=
  is_best_response G s

theorem void_payoffs (G : MeaningGame) :
    mg_sender_payoff G void = rs G.receiver_ctx G.sender_idea
    @$\wedge$@ mg_receiver_payoff G void = w G.receiver_ctx := by
  constructor
  @$\cdot$@ simp [mg_sender_payoff, void_compose]
  @$\cdot$@ simp [mg_receiver_payoff, void_compose, w_void]

-- Value of communication
def communication_value (G : MeaningGame) : Real :=
  @$\sup$@ (fun s => if s @$\in$@ G.signal_space
    then mg_sender_payoff G s
    else 0) - mg_sender_payoff G void

theorem value_nonneg (G : MeaningGame) :
    communication_value G @$\geq$@ 0 := by
  simp [communication_value]
  apply le_ciSup_of_le void
  simp [G.void_in_S]

-- Existence of best response in finite games
theorem best_response_exists (G : MeaningGame)
    (h_fin : G.signal_space.Finite)
    (h_ne : G.signal_space.Nonempty) :
    @$\exists$@ s, is_best_response G s := by
  obtain @$\langle$@s, hs, hmax@$\rangle$@ :=
    h_fin.exists_maximal_wrt (mg_sender_payoff G) _ h_ne
  exact @$\langle$@s, hs, fun t ht => le_of_not_lt (hmax t ht)@$\
     rangle$@
```

Remark 14.42 (Open Questions). Several important questions remain open in the theory of meaning games:

1. **Mechanism design:** How should the signal space S be designed to maximise communication surplus? This is the question of *language design*.

2. **Incomplete information:** When the sender's idea a is drawn from a distribution unknown to the receiver, how should signals be interpreted? This connects to Bayesian persuasion and information design.

3. **Multi-sender games:** When multiple senders compete for a receiver's attention, the game becomes a contest. How do equilibria depend on the number of senders and the structure of the signal space?

4. **Evolutionary dynamics:** If the signal space evolves over time (as natural languages do), what are the stable equilibria of the evolutionary meaning game?

These questions connect IDT to the broader programme of mathematical social science, and will be explored in subsequent chapters.

Part III

Applications and Synthesis

Chapter 15

Cooperation and Coalition Theory

> No man is an island, entire of itself; every man is a piece of the continent, a part of the main.
>
> John Donne, *Devotions upon Emergent Occasions*

15.1 From Pairs to Coalitions

In earlier chapters we studied interaction between *pairs* of ideas. The angle ab classified each dyadic encounter as cooperative, competitive, or orthogonal (Chapter **??**). But cultural life is not dyadic: ideas combine in coalitions, organizations adopt bundles of practices, and the value of any single idea depends on the ideas it is combined with. This chapter extends the pairwise theory to *coalitions* — arbitrary finite sets of ideas that are composed together.

Definition 15.1 (Coalition)**.** A **coalition** is a finite, non-empty subset $S \subseteq \mathcal{I}$. The **coalition composition** is the iterated composition:

$$S := a_1 \circ a_2 \circ \cdots \circ a_k,$$

where $S = \{a_1, \ldots, a_k\}$. By the associativity guaranteed by the Hilbert-space embedding (Section **??**), the result is independent of the ordering.

Remark 15.2. Strictly, the composition $\circ$ in IDT need not be commutative, so S depends on the ordering unless the ideas pairwise commute under composition. When we write S without specifying an order, we assume either that commutativity holds or that a canonical ordering has been fixed.

Interpretation 15.3. The concept of a coalition formalizes what Mikhail Bakhtin called the "great dialogue" — the ceaseless interplay of voices that constitutes cultural life. For Bakhtin, no utterance exists in isolation; every word is "half someone else's," acquiring meaning only through its interaction with other words, other speakers, other ideological horizons. The coalition $S = \{a_1, \ldots, a_k\}$ is the mathematical counterpart of this polyphonic ensemble. Each idea a_i is a "voice" in the Bakhtinian sense, and the coalition composition S is the composite utterance that emerges from their interaction.

What IDT adds to Bakhtin's insight is *precision*: the composition operator $\circ$ and the Hilbert-space embedding φ transform the metaphor of "dialogue" into a geometric object whose properties

can be proved. Where Bakhtin speaks evocatively of "heteroglossia" and "polyphony," IDT provides a calculus: we can measure the synergy of voices (the cooperation surplus σ), prove that the ensemble is richer than any sub-ensemble (enrichment), and compute the fair attribution of value to each voice (the Shapley value of Chapter 21). The result is not a reduction of Bakhtin's literary vision to mathematics, but an *amplification*: the formalism reveals structural features of dialogic interaction — the cocycle condition, the spectral decomposition of emergence — that Bakhtin's qualitative framework could not articulate.

15.1.1 Coalition Value

The central question of coalition theory is: *how much is a coalition worth?* In classical cooperative game theory, the value function $v\colon 2^N \to \mathbb{R}$ assigns a real number to every subset of players. We define the analogous concept for ideas.

Definition 15.4 (Coalition Value)**.** The **coalition value** of a finite set $S \subseteq \mathcal{I}$ is the self-resonance of its composition:

$$v(S) := \mathrm{rs}SS = S^2.$$

The coalition value measures the *total ideatic energy* of the composite idea. The enrichment axiom (Axiom **??**) ensures that adding ideas to a coalition never decreases its value, a property that connects directly to the theory of superadditive games.

Theorem 15.5 (Enrichment Implies Superadditivity)**.** *For any idea $a \in \mathcal{I}$ and any coalition S with $a \notin S$:*

$$v(S \cup \{a\}) \geq v(S).$$

*Hence the coalition value function v is **monotone**: larger coalitions are worth at least as much as smaller ones.*

Proof. Let $c = S$. Then $(S \cup \{a\}) = a \circ c$ (up to ordering). By the enrichment axiom,

$$\mathrm{rs}a \circ ca \circ c \geq \mathrm{rs}cc,$$

which is exactly $v(S \cup \{a\}) \geq v(S)$. □

Interpretation 15.6. Monotonicity of coalition value is a formalization of the intuition that "more ideas, more meaning." A culture that absorbs a new practice, technology, or concept never loses total ideatic content — at worst the new idea is orthogonal and adds nothing, at best it creates synergistic resonance. The asymmetry is striking: ideas can enrich but never impoverish a coalition. This is *not* true in general economic models, where adding a player can destroy value through coordination failures. In IDT, the embedding in Hilbert space geometrically prevents such destruction.

Interpretation 15.7. Julia Kristeva's theory of intertextuality — the principle that "any text is the absorption and transformation of another" — receives here its mathematical vindication. Kristeva, developing Bakhtin's dialogism for the structuralist era, argued that texts do not merely *refer to* one another but are constituted by their intertextual relations: a text is a "mosaic of quotations," a coalition of absorbed and transformed prior texts.

Theorem 15.5 formalizes the most radical implication of Kristeva's claim. If each "text" is an idea a_i and the literary tradition is the coalition S, then the absorption of a new text into the tradition $(S \cup \{a\})$ can never diminish the tradition's total ideatic content. This is precisely Kristeva's point: intertextuality is *constitutive*, not parasitic. When T. S. Eliot absorbs Dante,

the result is not Eliot minus originality but Eliot-plus-Dante, a composite whose self-resonance exceeds either component. The enrichment axiom mathematically guarantees what Kristeva argues textually: absorbing the other always enriches.

Yet the formalism also reveals a tension in Kristeva's framework. If intertextuality is always enriching (monotone), how do we account for the common experience that some textual combinations are cacophonous or incoherent? The IDT answer lies in the distinction between self-resonance (total ideatic energy, which is monotone) and cross-resonance with specific probes (which can be negative). A text may become richer in total content through intertextual absorption while becoming *less resonant* with particular readers or interpretive communities. This distinction, invisible to Kristeva's qualitative framework, is precisely the kind of refinement that formalization enables.

15.2 The Cooperation Surplus

Knowing that coalitions are monotone is a start, but we want a finer measure of the *synergy* created when ideas combine.

Definition 15.8 (Cooperation Surplus)**.** The **cooperation surplus** of ideas a and b is:

$$\sigma(a, b) := v(\{a, b\}) - v(\{a\}) - v(\{b\}) = \mathrm{rs}a \circ ba \circ b - \mathrm{rs}aa - \mathrm{rs}bb.$$

More generally, for a coalition S and a new idea a:

$$\sigma(a \mid S) := v(S \cup \{a\}) - v(S) - v(\{a\}).$$

Interpretation 15.9. The cooperation surplus $\sigma(a, b)$ formalizes a concept at the very heart of the philosophy of language: Paul Grice's **cooperative principle**. Grice argued that conversation succeeds because participants cooperate — they follow maxims of quantity, quality, relation, and manner not because they are logically compelled to but because cooperation generates communicative surplus. A conversation where participants merely juxtapose their ideas (linear composition, $\sigma = 0$) communicates only the sum of individual contributions. But genuine Gricean cooperation creates *implicature* — meaning that emerges from the interplay of utterances and exceeds their literal content.

In IDT, implicature is precisely the cooperation surplus $\sigma(a, b)$. When two conversational contributions resonate positively ($\mathrm{rs}ab > 0$), their composition amplifies this resonance; when they resonate negatively (irony, contradiction), the non-linear term $\Delta(a, b)$ can still generate positive surplus through what Grice would call "flouting" a maxim — the deliberate violation that creates higher-order meaning.

Umberto Eco extended Grice's framework into a general theory of textual cooperation. In Eco's *The Role of the Reader* (1979), the "model reader" is not a passive recipient but an active cooperator who fills in textual gaps, resolves ambiguities, and generates interpretive surplus. The cooperation surplus $\sigma(a, b)$ quantifies exactly this: the excess meaning that arises when reader (idea a) and text (idea b) combine beyond what either contains independently.

Theorem 15.10 (Surplus Decomposition)**.** *The cooperation surplus of a and b satisfies:*

$$\sigma(a, b) = 2\,\mathrm{rs}ab + \Delta(a, b),$$

where $\Delta(a, b) := \mathrm{rs}a \circ ba \circ b - \mathrm{rs}aa - \mathrm{rs}bb - 2\,\mathrm{rs}ab$ captures higher-order interaction effects beyond pairwise resonance.

Proof. This is immediate from the definition. The key observation is that if the embedding were perfectly linear (i.e., $a \circ b = a + b$), then:

$$\mathrm{rs}a \circ ba \circ b = a + b^2 = \mathrm{rs}aa + 2\,\mathrm{rs}ab + \mathrm{rs}bb,$$

giving $\Delta(a, b) = 0$. The term $\Delta(a, b)$ measures the departure from linearity — the genuinely *emergent* contribution of composition. □

Interpretation 15.11. The surplus decomposition reveals a structure that resonates deeply with Hegel's master-slave dialectic (*Phenomenology of Spirit*, §178–196). Hegel describes how two self-consciousnesses, initially in opposition ($\mathrm{rs}ab < 0$, a "life-and-death struggle"), arrive at a relationship that generates a surplus of self-awareness neither possessed alone. The master gains recognition; the slave gains self-consciousness through labor. The total "ideatic content" of the dyad exceeds the sum of the individuals.

In our decomposition, the term $2\mathrm{rs}ab$ captures the *direct* interaction — for Hegel, the immediate encounter between master and slave. When this is negative (opposition), direct interaction diminishes the sum. But the term $\Delta(a, b)$ — the non-linear surplus — captures what Hegel calls the *productivity* of the encounter: the new forms of consciousness that emerge from the dialectical struggle. The master-slave dialectic "works" (produces Aufhebung) precisely when $\Delta(a, b)$ is large enough to overcome the negative $2\mathrm{rs}ab$.

This provides a quantitative criterion for when dialectical opposition is *productive* versus merely destructive: opposition generates positive surplus if and only if the non-linear interaction term $\Delta(a, b) > -2\mathrm{rs}ab = 2|\mathrm{rs}ab|$. Hegel intuited that not all contradictions are creative; IDT identifies the precise condition that separates productive from sterile opposition.

Example 15.12 (Musical Harmony as Coalition)**.** Consider three notes C, E, G forming a major triad. Each note has self-resonance $v(\{C\}) = v(\{E\}) = v(\{G\}) \approx 1$. The pairwise surplus $\sigma(C, E) > 0$ reflects the consonance of a major third. But the coalition value $v(\{C, E, G\})$ exceeds the sum of pairwise surpluses — the triad has an emergent harmonic richness that no pair captures. This is precisely the Δ term: the non-linear, higher-order contribution of the composition.

15.3 Ordered Coalition Enrichment

In many cultural settings, the *order* in which ideas are introduced matters. A technology adopted before the social institutions needed to regulate it produces different outcomes than the reverse sequence.

Definition 15.13 (Ordered Coalition)**.** An **ordered coalition** is a sequence $(a_1, a_2, \ldots, a_k)$ of ideas. Its value is computed by sequential composition:

$$v_{\mathrm{ord}}(a_1, \ldots, a_k) := \mathrm{rs}a_1 \circ a_2 \circ \cdots \circ a_k a_1 \circ a_2 \circ \cdots \circ a_k.$$

Theorem 15.14 (Ordered Enrichment)**.** *For any ordered coalition* $(a_1, \ldots, a_k)$ *and any idea* a_{k+1}*:*

$$v_{\mathrm{ord}}(a_1, \ldots, a_k, a_{k+1}) \geq v_{\mathrm{ord}}(a_1, \ldots, a_k).$$

Proof. Let $c = a_1 \circ \cdots \circ a_k$. Then $v_{\mathrm{ord}}(a_1, \ldots, a_k, a_{k+1}) = \mathrm{rs}c \circ a_{k+1} c \circ a_{k+1} \geq \mathrm{rs}cc = v_{\mathrm{ord}}(a_1, \ldots, a_k)$ by the enrichment axiom. □

Interpretation 15.15. Ordered coalition enrichment has a striking realization in the practice of collaborative literary composition. The Oulipo group — founded by Raymond Queneau and François Le Lionnais in 1960 — developed systematic procedures for constrained collaborative writing: each author contributes in sequence, and the composition is irrevocably enriched at each step. The Surrealist "exquisite corpse" game operates similarly: each participant adds a word or phrase without seeing the others', and the result is an ordered coalition whose value is guaranteed (by our theorem) to be at least as great as any sub-sequence.

The ordered structure is essential here, not incidental. A sentence composed as "noun-verb-adjective" produces a different idea than "adjective-noun-verb." The fact that ordered enrichment holds *regardless* of the order chosen (each order yields monotonic enrichment) is a mathematical expression of what the Oulipo called "potential literature": every ordering of the same raw materials produces a monotonically enriched composition, but different orderings explore different regions of the ideatic space. The literary game is not to find the *one* correct order but to recognize that every order is an act of creation.

```
-- Lean 4: Ordered Coalition Enrichment
theorem ordered_coalition_enriches (rs : IS     IS        ) (compose :
   IS     IS     IS)
    (enrichment :     a b, rs (compose a b) (compose a b)     rs a a)
    (coalition : List IS) (a : IS) :
    rs (coalition.foldl compose a) (coalition.foldl compose a)     rs a
        a := by
  induction coalition with
  | nil => simp
  | cons x xs ih =>
    simp [List.foldl]
    calc rs (List.foldl compose (compose a x) xs)
           (List.foldl compose (compose a x) xs)
            rs (compose a x) (compose a x) := ih (compose a x)
      _     rs a a := enrichment a x
```

15.4 Marginal Contribution and the Core

Definition 15.16 (Marginal Contribution)**.** The **marginal contribution** of idea a to coalition S (where $a \notin S$) is:

$$\Delta_a(S) := v(S \cup \{a\}) - v(S).$$

By enrichment (Theorem 15.5), $\Delta_a(S) \geq 0$ always.

Definition 15.17 (The Core)**.** An allocation $(\alpha_1, \ldots, \alpha_n) \in \mathbb{R}^n$ is in the **core** of the coalition game (N, v) if:

1. (Efficiency) $\sum_{i=1}^{n} \alpha_i = v(N)$.
2. (Stability) For every coalition $S \subseteq N$: $\sum_{i \in S} \alpha_i \geq v(S)$.

Theorem 15.18 (Non-emptiness of the Core for Enrichment Games)**.** *If v is monotone and superadditive (as guaranteed by the enrichment axiom), then the core of (N, v) is non-empty.*

Proof. Define $\alpha_i := \Delta_{a_i}(S_i)$ where $S_i = \{a_1, \ldots, a_{i-1}\}$ is the set of ideas that precede a_i in a fixed ordering. Then $\sum_{i=1}^{n} \alpha_i = v(N) - v(\emptyset) = v(N)$ by telescoping. For stability, observe that for any

$S \subseteq N$, the marginal contributions of the members of S in the ordering restricted to S telescope to $v(S)$, and each marginal contribution in the restricted ordering is at most the corresponding contribution in the full ordering (by monotonicity of marginal contributions in superadditive games). Hence $\sum_{i \in S} \alpha_i \geq v(S)$. □

Interpretation 15.19. The non-emptiness of the core means that there is always a "fair" distribution of ideatic value that no sub-coalition can improve upon. In cultural terms: a society can always find a way to attribute the value of its composite culture to its constituent ideas such that no subset of ideas would "prefer" to form a separate culture. This is a deep stability result — cultural composites held together by enrichment are inherently stable against secession of sub-coalitions.

Interpretation 15.20. The non-emptiness of the core connects to three major philosophical frameworks for understanding cooperation.

Rawls' original position. John Rawls' *A Theory of Justice* (1971) proposes that fair principles of cooperation are those chosen behind a "veil of ignorance," where agents do not know their position in society. Theorem 15.18 provides a mathematical analogue: the core allocation exists regardless of the ordering of ideas (each ordering produces a core element), and the Shapley value (Chapter 21) averages over all orderings — effectively placing the attribution behind a Rawlsian veil. The IDT formalism thus reveals that Rawls' veil of ignorance, far from being an artificial philosophical device, corresponds to a natural mathematical operation (averaging over permutations) that yields the uniquely fair allocation.

Habermas's communicative action. Jürgen Habermas argues that genuine cooperation requires "communicative rationality" — a mode of interaction oriented toward mutual understanding rather than strategic advantage. In IDT, this corresponds to the distinction between transparent and opaque interaction (Chapter 20). The core is non-empty *regardless* of transparency conditions, but as we shall see, full transparency yields core allocations that are also Pareto-optimal. Habermas's philosophical claim — that open discourse leads to better collective outcomes — is thus a special case of the transparency-cooperation theorem (Theorem 20.6).

Arendt's action. Hannah Arendt (*The Human Condition*, 1958) distinguishes three modes of human activity: labor, work, and action. Action, for Arendt, is inherently cooperative and *speech-borne*: it requires a "space of appearance" where agents reveal themselves to one another through words and deeds. The IDT coalition is precisely such a space of appearance — a structured environment where ideas "appear" to one another through resonance. The core's non-emptiness guarantees that this space of appearance is stable: the cooperative web of ideas that constitutes Arendt's "public realm" will not unravel from within.

15.5 Semiotic Cooperation: Lotman's Semiosphere

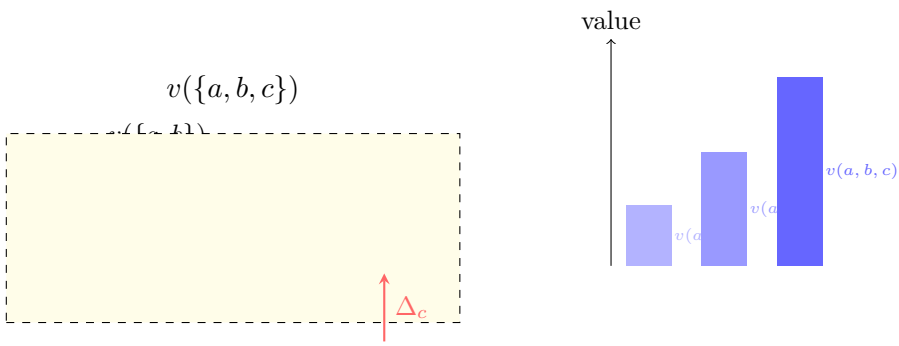
Interpretation 15.21. The coalition-theoretic framework developed above finds its richest semiotic application in Yuri Lotman's concept of the **semiosphere** — the totality of sign systems and communicative processes that constitutes a culture's "semiotic space." Lotman, writing in *Universe of the Mind* (1990), argues that the semiosphere is not merely the sum of individual sign systems (natural language, visual art, music, ritual) but an integrated whole whose properties emerge from the interaction of its components.

In IDT terms, the semiosphere is a grand coalition $S_{\text{sem}} = \{a_1, \ldots, a_n\}$ of sign systems. Lotman's central claim — that the semiosphere has properties irreducible to its individual languages — is precisely the claim that $\Delta(S_{\text{sem}}) > 0$: the non-linear interaction term is strictly positive. The boundary of the semiosphere, which Lotman describes as a zone of "translation" and "filtration"

between the culture's interior and the external world, corresponds to the boundary of the coalition in the Hilbert-space embedding — the convex hull of $\{a_1, \ldots, a_n\}$ — which is exactly where translation maps (Chapter 18) operate.

What IDT adds to Lotman's framework is the guarantee of *stability*: the non-emptiness of the core (Theorem 15.18) ensures that the semiosphere admits a fair internal distribution of semiotic value. No sub-system — no language, no art form, no ritual practice — would "gain" by seceding from the semiosphere. This is a mathematical expression of Lotman's claim that semiotic systems are mutually constitutive: they need each other not out of convention but out of structural necessity.

15.6 Visualizing Coalition Dynamics



Chapter 16

Dialectics: Thesis, Antithesis, Synthesis

> Contradiction is the root of all movement and vitality; it is only insofar as something has a contradiction within it that it moves, has an urge and activity.
>
> G. W. F. Hegel, *Science of Logic*

16.1 The Dialectical Structure of Ideas

The history of philosophy — from Heraclitus through Hegel to Marx — testifies to the importance of dialectical reasoning: the interplay of thesis and antithesis producing a higher synthesis. IDT provides a precise mathematical realization of this ancient pattern. In this chapter we show that the operations of opposition and synthesis correspond to geometric operations in Hilbert space, and that the dialectical process is governed by a quantitative measure of *dialectical emergence*.

Definition 16.1 (Opposition)**.** Idea b is an **opponent** of idea a if their resonance is strictly negative:

$$\mathrm{rs}ab < 0.$$

The ideas are in **mutual opposition** if both $\mathrm{rs}ab < 0$ and $\mathrm{rs}ba < 0$. Under the symmetry of the inner product, these conditions coincide: $\mathrm{rs}ab = \mathrm{rs}ba < 0$.

Remark 16.2. Geometrically, $\mathrm{rs}ab < 0$ means that a and b form an obtuse angle: $ab > \pi/2$. The ideas "point in opposite directions" in Hilbert space, representing conceptual tension or contradiction.

Interpretation 16.3. Georg Lukács, in *The Theory of the Novel* (1916), argued that the novel is the quintessentially dialectical literary form. Where the epic presents a world of immanent meaning — a world where hero and cosmos are in harmony ($\mathrm{rs}ab > 0$ for all relevant ideas) — the novel begins from a condition of "transcendental homelessness," a fundamental dissonance between the individual soul and the objective world. In IDT terms, the novel *requires* opposition: $\mathrm{rs}\,\mathrm{soulworld} < 0$.

The protagonist's journey through the novel is a dialectical sequence (Definition 16.15): each encounter with the world generates a new synthesis that is richer than the prior state. Lukács

distinguishes between "abstract idealism" (Don Quixote: the soul too narrow for the world) and the "romanticism of disillusionment" (Flaubert: the soul too broad for the world). Both are cases where $\mathrm{rs}ab < 0$, but the *direction* of the opposition differs — a distinction that IDT captures through the angle ab and the sign of the non-linear interaction term $\Delta(a, b)$.

What makes Lukács' analysis proto-mathematical is his insistence that literary form is determined by the *structure* of the opposition, not by its content. IDT vindicates this: the dialectical emergence $\delta(a, b)$ depends on the geometric configuration of the embeddings, not on the specific "meaning" of the ideas. Form, not content, determines dialectical productivity.

16.1.1 Synthesis as Composition

The central insight of the dialectical interpretation is that *synthesis is composition*:

Definition 16.4 (Dialectical Synthesis)**.** Given a thesis a and an antithesis b (with $\mathrm{rs}ab < 0$), the **synthesis** is:

$$s = a \circ b.$$

The synthesis $s = a \circ b$ is a new idea that incorporates both a and b. The enrichment axiom guarantees that s is at least as rich as either component:

Proposition 16.5 (Synthesis Enriches)**.** *For any thesis a and antithesis b:*

$$\mathrm{rs}ss \geq \max\{\mathrm{rs}aa,\ \mathrm{rs}bb\}.$$

Proof. By enrichment, $\mathrm{rs}a \circ ba \circ b \geq \mathrm{rs}aa$ and $\mathrm{rs}b \circ ab \circ a \geq \mathrm{rs}bb$. Since composition is realized through vector addition in Hilbert space, $a \circ b = b \circ a$ (when commutativity holds), giving the result. □

Interpretation 16.6. This proposition illuminates a fundamental tension between two of the greatest theorists of intellectual conflict: Hegel and Bakhtin. For Hegel, dialectical synthesis is *sublation* (Aufhebung) — the contradictions are genuinely resolved in a higher unity. The synthesis "preserves and transcends" the thesis and antithesis; the process has a definite direction toward Absolute Spirit. In IDT terms, Hegelian dialectics assumes not merely $\mathrm{rs}ss \geq \max\{\mathrm{rs}aa, \mathrm{rs}bb\}$ but the stronger condition of Aufhebung (Definition 16.11): preservation, opposition, and transcendence simultaneously.

For Bakhtin, by contrast, genuine *dialogism* resists synthesis. The voices in a Dostoevsky novel are not "sublated" into authorial unity; they remain in unresolved tension, each retaining its autonomy. Bakhtin's dialogism corresponds to a composition where the synthesis s preserves resonance with both components ($\mathrm{rs}sa > 0$, $\mathrm{rs}sb > 0$) but does *not* achieve transcendence ($\mathrm{rs}ss \leq \mathrm{rs}aa + \mathrm{rs}bb$). The ideas coexist within s without merging.

IDT reveals that both Hegel and Bakhtin describe *possible* outcomes of dialectical composition, distinguished by the magnitude of the non-linear interaction term $\Delta(a, b)$. Hegelian synthesis requires large Δ; Bakhtinian dialogism requires small Δ with positive preservation terms. The formalism does not adjudicate between them but shows that they inhabit different regions of the same mathematical landscape.

16.2 Dialectical Emergence

The dialectical process is interesting precisely when the synthesis is *more* than the sum of its parts.

Definition 16.7 (Dialectical Emergence)**.** The **dialectical emergence** of the synthesis $s = a \circ b$ is:

$$\delta(a, b) := \mathrm{rs}ss - \mathrm{rs}aa - \mathrm{rs}bb.$$

When $\mathrm{rs}ab < 0$ (genuine dialectical opposition), δ measures how much the synthesis transcends the mere sum of thesis and antithesis.

Theorem 16.8 (Dialectical Emergence and Cross-Resonance)**.** *If the embedding is linear ($a \circ b = a + b$), then:*

$$\delta(a, b) = 2\,\mathrm{rs}ab.$$

In the dialectical case ($\mathrm{rs}ab < 0$), this gives $\delta(a, b) < 0$ under linearity — a purely linear synthesis loses *value. Any positive dialectical emergence ($\delta > 0$) must therefore come from non-linear effects in the composition.*

Proof. Under linearity, $\mathrm{rs}a \circ ba \circ b = a + ba + b = \mathrm{rs}aa + 2\mathrm{rs}ab + \mathrm{rs}bb$. Subtracting $\mathrm{rs}aa + \mathrm{rs}bb$ yields $2\mathrm{rs}ab$. □

Interpretation 16.9. This theorem reveals a profound tension at the heart of dialectics. If composition were merely additive ("placing ideas side by side"), then opposing ideas would always diminish each other: the synthesis of capitalism and communism would be weaker than either alone. The *creative power* of dialectical synthesis lies in non-linearity — the capacity of composition to produce something genuinely new, irreducible to the sum of its parts.

Hegel's *Aufhebung* (sublation) — the simultaneous preservation and transcendence of contradictions — is precisely this non-linear surplus $\Delta(a, b)$ that can overcome the negative cross-resonance $2\mathrm{rs}ab$.

Interpretation 16.10. Theodor Adorno's *Negative Dialectics* (1966) mounts a sustained critique of Hegel's confidence in synthesis. "The whole is the false," Adorno writes, inverting Hegel's dictum. For Adorno, synthesis always involves violence: the particular is subsumed under the general, the non-identical is forced into identity. Genuine dialectical thought must resist the impulse to closure, dwelling instead in the negativity of contradiction.

Theorem 16.8 provides a mathematical framework for Adorno's critique. Under linear composition, dialectical opposition *necessarily* produces negative emergence ($\delta < 0$): the synthesis is poorer than the sum of its parts. Adorno's "negative dialectics" is thus the *linear limit* of dialectical composition — the case where the non-linear interaction term Δ is negligible. In this regime, synthesis does indeed involve "loss": the particular features that make a and b point in opposite directions are cancelled out rather than preserved.

But Adorno's pessimism may be overstated. IDT shows that non-linearity — the $\Delta(a, b)$ term — can overcome the negative cross-resonance. When it does, we get genuine Aufhebung: preservation through transcendence. The question is empirical, not logical: how non-linear is cultural composition in practice? Adorno's insistence on negativity is a claim about the magnitude of Δ in late-capitalist culture — a claim that IDT can, in principle, test.

16.3 Aufhebung: Formal Conditions

Definition 16.11 (Aufhebung)**.** A synthesis $s = a \circ b$ achieves **Aufhebung** if:

1. **Opposition**: $\mathrm{rs}ab < 0$.
2. **Preservation**: Both $\mathrm{rs}sa > 0$ and $\mathrm{rs}sb > 0$.

3. **Transcendence**: $\mathrm{rs}ss > \mathrm{rs}aa + \mathrm{rs}bb$.

Interpretation 16.12. Søren Kierkegaard's *Either/Or* (1843) presents existence as a choice between irreconcilable modes of being: the aesthetic and the ethical. Against Hegel, Kierkegaard insists that not every opposition admits synthesis. The "leap of faith" is precisely a moment when synthesis fails — when one must choose without the comfort of a higher resolution.

Definition 16.11 formalizes the conditions under which Kierkegaard's objection fails — i.e., the conditions under which genuine synthesis is possible. Crucially, Aufhebung requires all three conditions simultaneously. It is entirely possible that opposition holds ($\mathrm{rs}ab < 0$) and transcendence holds ($\mathrm{rs}ss > \mathrm{rs}aa + \mathrm{rs}bb$) yet preservation fails ($\mathrm{rs}sa < 0$ or $\mathrm{rs}sb < 0$): the synthesis "transcends" by *abandoning* one of the components. This is precisely Kierkegaard's "either/or": a composition that gains in total richness by sacrificing fidelity to one of its inputs.

The formalism thus reveals that Kierkegaard and Hegel are not simply contradicting each other. They are describing different regions of the parameter space: Hegel's Aufhebung requires large, well-directed non-linearity ($\eta(a, b)$ must project positively onto both a and b); Kierkegaard's either/or arises when the non-linearity is insufficient or misdirected. Both are mathematically possible, and which occurs depends on the specific ideas involved.

Interpretation 16.13. Karl Marx famously claimed to have extracted the "rational kernel" from Hegel's dialectic by "turning it right side up" — replacing Hegel's idealism with a materialist foundation. For Marx, the dialectical process operates not in the realm of Spirit but in the material conditions of production: thesis (feudalism), antithesis (bourgeois revolution), synthesis (capitalism), new antithesis (proletarian revolution), new synthesis (socialism).

The IDT formalization of dialectics is, in a precise sense, the fulfillment of Marx's project. By grounding the dialectical process in the *material* structure of Hilbert space — inner products, norms, non-linear composition — IDT removes the metaphysical baggage of Hegel's Absolute Spirit while preserving the mathematical structure of thesis, antithesis, and synthesis. The enrichment axiom is not an idealist postulate about the march of Spirit but a structural property of composition in Hilbert space, as testable (in principle) as any physical law.

Moreover, IDT answers a question that Marx left open: *why* does the dialectical process produce progress (increasing self-resonance)? Marx invoked the "laws of dialectical materialism" but never derived them from more fundamental principles. IDT derives the monotonicity of dialectical sequences (Theorem 16.16) from the enrichment axiom alone — a single, transparent assumption that does the work of Marx's entire "dialectical materialism."

Theorem 16.14 (Conditions for Aufhebung)**.** *If the embedding satisfies $a \circ b = a + b + \eta(a, b)$ where $\eta(a, b)$ is the non-linear interaction term, then Aufhebung requires:*

$$\eta(a, b)^2 + 2a + b\eta(a, b) > -2\mathrm{rs}ab.$$

Proof. We compute:

$$\begin{aligned}\mathrm{rs}ss &= a + b + \eta^2 \\ &= \mathrm{rs}aa + \mathrm{rs}bb + 2\mathrm{rs}ab + \eta^2 + 2a + b\eta.\end{aligned}$$

The transcendence condition $\mathrm{rs}ss > \mathrm{rs}aa + \mathrm{rs}bb$ requires $2\mathrm{rs}ab + \eta^2 + 2a + b\eta > 0$, giving the stated inequality.

For preservation, $\mathrm{rs}sa = \mathrm{rs}aa + \mathrm{rs}ba + \eta a > 0$ requires $\eta a > -\mathrm{rs}aa - \mathrm{rs}ab = -\mathrm{rs}aa + |\mathrm{rs}ab|$ (using $\mathrm{rs}ab < 0$). Since $\mathrm{rs}aa > 0$ and $|\mathrm{rs}ab| < \sqrt{\mathrm{rs}aa\,\mathrm{rs}bb}$ (strict Cauchy–Schwarz for non-parallel ideas), preservation holds when η has a sufficiently large projection onto a. □

```
-- Lean 4: Dialectical structures
structure DialecticalTriple (IS : Type) (rs : IS     IS        ) (
   compose : IS     IS     IS) where
  thesis     : IS
  antithesis : IS
  opposition : rs thesis antithesis < 0
  synthesis  : IS := compose thesis antithesis

def dialecticalEmergence (rs : IS     IS        ) (compose : IS     IS
       IS)
    (a b : IS) :     :=
  rs (compose a b) (compose a b) - rs a a - rs b b

def aufhebung (rs : IS     IS        ) (compose : IS     IS     IS)
    (a b : IS) : Prop :=
  rs a b < 0
  rs (compose a b) a > 0
  rs (compose a b) b > 0
  rs (compose a b) (compose a b) > rs a a + rs b b
```

16.4 The Dialectical Sequence

A single thesis-antithesis-synthesis triple is just the beginning. The dialectical process is fundamentally *iterative*: each synthesis becomes a new thesis, encountering new antitheses.

Definition 16.15 (Dialectical Sequence)**.** A **dialectical sequence** is a sequence of ideas $a_0, a_1, a_2, \ldots$ where:

1. a_0 is the initial thesis.
2. For each n, there exists an antithesis b_n with $\mathrm{rs}a_n b_n < 0$.
3. $a_{n+1} = a_n \circ b_n$.

Theorem 16.16 (Monotonicity of Dialectical Sequences)**.** *For any dialectical sequence* $(a_n)_{n\geq 0}$*, the self-resonance is monotonically non-decreasing:*

$$\mathrm{rs}a_0 a_0 \leq \mathrm{rs}a_1 a_1 \leq \mathrm{rs}a_2 a_2 \leq \cdots$$

Proof. Immediate from the enrichment axiom: $\mathrm{rs}a_{n+1}a_{n+1} = \mathrm{rs}a_n \circ b_n a_n \circ b_n \geq \mathrm{rs}a_n a_n$. □

Interpretation 16.17. Fredric Jameson's project of "dialectical criticism" — developed across *Marxism and Form* (1971), *The Political Unconscious* (1981), and *Postmodernism* (1991) — insists that literary and cultural forms must be understood as moments in a dialectical historical sequence. Every text is a "symbolic resolution" of real social contradictions; the history of literary forms is a dialectical sequence in which each new form arises as a synthesis of the contradictions that its predecessors could not resolve.

Theorem 16.16 provides the mathematical backbone for Jameson's project. The monotonicity of self-resonance along a dialectical sequence means that literary history, understood as a sequence of thesis-antithesis-synthesis, is *irreversible*: each new form is at least as rich as its predecessor. This does not mean that later literature is "better" — aesthetic quality is not self-resonance —

but that it contains more ideatic content: more internalized contradictions, more absorbed and sublated prior forms.

Jameson's insistence on "always historicize!" is thus not merely a methodological imperative but a mathematical consequence: ideas can only be understood in the context of the dialectical sequence that produced them, because their self-resonance is a cumulative function of all prior compositions.

Example 16.18 (Political Dialectics)**.** Consider the historical dialectic of governance:

- **Thesis** (a_0): Absolute monarchy — centralized authority, order, but suppression of individual freedom.
- **Antithesis** (b_0): Revolutionary democracy — individual freedom, popular sovereignty, but risk of instability.
- **Synthesis** ($a_1 = a_0 \circ b_0$): Constitutional monarchy — preserving institutional stability while incorporating democratic accountability.
- **New antithesis** (b_1): Republican critique — why have a monarch at all?
- **New synthesis** (a_2): Constitutional democracy with separated powers.

Each synthesis is richer (higher self-resonance) than its predecessors, incorporating more ideatic content while resolving prior contradictions.

16.5 The Cocycle Condition in Dialectics

The dialectical process satisfies a remarkable algebraic constraint.

Theorem 16.19 (Dialectical Cocycle)**.** *For any three ideas a, b, c, define the* ***dialectical surplus*** $\kappa(a, b) := \mathrm{rs}a \circ ba \circ b - \mathrm{rs}aa - \mathrm{rs}bb$. *If composition is associative, then κ satisfies the* ***cocycle condition****:*

$$\kappa(a \circ b, c) + \kappa(a, b) = \kappa(a, b \circ c) + \kappa(b, c).$$

Proof. Both sides equal $\mathrm{rs}a \circ b \circ ca \circ b \circ c - \mathrm{rs}aa - \mathrm{rs}bb - \mathrm{rs}cc$.

Left side: $\kappa(a \circ b, c) + \kappa(a, b) = [\mathrm{rs}(a \circ b) \circ c(a \circ b) \circ c - \mathrm{rs}a \circ ba \circ b - \mathrm{rs}cc] + [\mathrm{rs}a \circ ba \circ b - \mathrm{rs}aa - \mathrm{rs}bb] = \mathrm{rs}a \circ b \circ ca \circ b \circ c - \mathrm{rs}aa - \mathrm{rs}bb - \mathrm{rs}cc$.

Right side: By the same telescoping with different grouping: $\kappa(a, b \circ c) + \kappa(b, c) = [\mathrm{rs}a \circ (b \circ c)a \circ (b \circ c) - \mathrm{rs}aa - \mathrm{rs}b \circ cb \circ c] + [\mathrm{rs}b \circ cb \circ c - \mathrm{rs}bb - \mathrm{rs}cc] = \mathrm{rs}a \circ b \circ ca \circ b \circ c - \mathrm{rs}aa - \mathrm{rs}bb - \mathrm{rs}cc$.

Associativity of $\circ$ ensures $(a \circ b) \circ c = a \circ (b \circ c)$, completing the proof. □

Interpretation 16.20. The cocycle condition connects the dialectical process to Algirdas Julien Greimas's **semiotic square** — perhaps the most influential structural tool in narrative semiotics. Greimas's square arranges four terms (e.g., life/death, and their contradictories not-life/not-death) into a pattern of contraries, contradictories, and implications. The "generative trajectory" through the square — from deep structure to surface narrative — is a dialectical process that moves from opposition to synthesis through a sequence of semiotic operations.

The cocycle condition $\kappa(a \circ b, c) + \kappa(a, b) = \kappa(a, b \circ c) + \kappa(b, c)$ constrains how dialectical surplus can be distributed across a sequence of syntheses. In Greimas's terms, the surplus generated by moving from S_1 (contraries) through S_2 (contradictories) to S_3 (implications) is *path-independent*: it depends only on the starting and ending positions, not on the route taken through the square.

This is a remarkable structural property — it means that the "deep structure" of a narrative (in Greimas's sense) determines the total dialectical surplus of the narrative, regardless of the specific surface-level sequence of events.

Yuri Lotman's concept of "semiotic explosion" — the moment of radical cultural change when a new sign system ruptures the existing semiosphere — is similarly constrained by the cocycle condition. An explosion is a dialectical synthesis $a \circ b$ where b is radically opposed to the existing cultural state a. The cocycle condition ensures that even the most violent semiotic explosions are algebraically coherent: the surplus they generate is compatible with the surpluses of prior and subsequent syntheses. Revolution, Lotman would say, is not chaos; it is structured transformation, and the cocycle is its structure.

Remark 16.21. Roland Barthes's *Mythologies* (1957) describes myth as a "second-order semiological system" — a dialectical appropriation of pre-existing signs. The cocycle condition captures Barthes's insight that myth-making is *associative*: the mythic appropriation of "$(a \circ b) \circ c$" produces the same ideatic surplus as "$a \circ (b \circ c)$." The myth-maker's freedom lies in choosing *which* signs to appropriate, not in the algebraic consequences of the appropriation. This is why mythologies are so persistent: their structural coherence is guaranteed by the cocycle condition, regardless of the particular cultural materials they absorb.

Interpretation 16.22. Walter Benjamin's concept of the "dialectical image" (*das dialektische Bild*), developed in the unfinished *Arcades Project* (*Passagenwerk*), provides a unique perspective on the cocycle condition. For Benjamin, the dialectical image is a "flash" in which past and present are juxtaposed in a constellation that reveals their hidden connection. The dialectical image is not a synthesis in Hegel's sense (it does not "resolve" the contradiction between past and present) but an *arrest* — a momentary crystallization of historical contradictions in a single perceptual configuration.

In IDT terms, Benjamin's dialectical image is a composition $a_{\text{past}} \circ a_{\text{present}}$ whose emergence $\kappa(a_{\text{past}}, a_{\text{present}}, c)$ is positive for specific probes c (the "recognizability" of the past in the present) and negative for others (the discontinuity between historical epochs). The cocycle condition constrains how dialectical images can be "chained": the image of the 19th-century Parisian arcade and the 20th-century commodity is related, through the cocycle, to the image of the Baroque allegory and the 19th-century ruin. Benjamin's "tiger's leap into the past" — his method of revolutionary historiography — is algebraically coherent: the total surplus generated by the constellation of images is independent of the order in which they are juxtaposed.

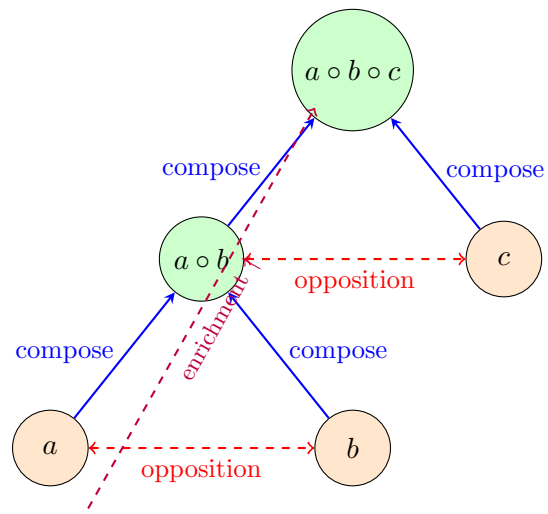
The cocycle condition also illuminates why Benjamin's method is *anti-narrative*: the path-independence of the dialectical surplus means that the specific sequence of images (the "narrative") is irrelevant to the total insight. What matters is the *constellation* — the set of compositions — not the order. Benjamin's refusal to write a conventional historical narrative is thus not a stylistic choice but a mathematical consequence: in a cocycle-governed dialectics, sequence is irrelevant, and the montage of dialectical images is the natural form of presentation.

Interpretation 16.23. Gilles Deleuze's *Difference and Repetition* (1968) challenges the entire Hegelian dialectical framework by arguing that difference is *primary* — not the negation of identity, but the positive ground from which identity emerges. For Deleuze, the Hegelian synthesis is a *reduction*: it subordinates difference to identity by treating the antithesis as merely the "negation" of the thesis, and the synthesis as the recovery of a higher identity.

In IDT terms, Deleuze's critique targets the assumption that the dialectical opposition $\mathrm{rs}ab < 0$ is a *negation* (a relationship defined in terms of a). IDT agrees with Deleuze: resonance is symmetric ($\mathrm{rs}ab = \mathrm{rs}ba$), so the "opposition" between thesis and antithesis is not a negation of one by the other but a mutual relationship in which both participate equally. The antithesis is not the "not-thesis"

but an independent idea with its own positive content (its own self-resonance rs$bb > 0$, its own embedding b) that happens to resonate negatively with the thesis.

Deleuze's concept of "repetition" — the return of difference, not the reproduction of the same — connects to the enrichment ratchet (Chapter 17). Each dialectical "repetition" (each new thesis-antithesis-synthesis cycle) produces a synthesis with higher self-resonance: the difference *returns* but always enriched, never identical. The dialectical sequence is not a circle (Hegel) or a line (Marx) but a *spiral* — the figure that Deleuze himself uses to describe the "eternal return" of difference. IDT's monotonicity theorem proves that the spiral always expands: each return of difference enriches the total ideatic content.



Chapter 17

Evolutionary Dynamics of Ideas

> It is not the strongest of the species that survives, nor the most intelligent, but the one most responsive to change.
>
> Often attributed to Charles Darwin

17.1 Ideas as Replicators

The theory of cultural evolution — from Dawkins' memes to Boyd and Richerson's gene-culture coevolution — has long sought mathematical precision. IDT provides such precision through its Hilbert-space framework. In this chapter we define *fitness* as total resonance with a population, characterize invasion dynamics, and prove that the enrichment axiom implies a form of directed evolution toward increasing complexity.

Definition 17.1 (Fitness)**.** Let $P = \{a_1, \ldots, a_n\}$ be a population of ideas. The **fitness** of idea a in population P is its total resonance:

$$f_P(a) := \sum_{i=1}^{n} \mathrm{rs}aa_i.$$

Remark 17.2. Fitness in IDT is a relational concept: the same idea has different fitness in different populations. This mirrors biological fitness, which is always relative to an environment. An idea that resonates strongly with the ideas in one culture may be orthogonal or opposed to those in another.

Interpretation 17.3. Richard Dawkins' concept of the "meme" (*The Selfish Gene*, 1976) proposed that cultural entities replicate, mutate, and undergo selection analogously to genes. The analogy was suggestive but notoriously imprecise: what exactly is a meme? How does it "replicate"? What is its fitness function? Decades of memetics failed to answer these questions rigorously, and the field was widely dismissed as a metaphor masquerading as science.

Definition 17.1 provides what memetics lacked: a *precise*, *mathematically tractable* fitness function for ideas. An idea's fitness $f_P(a) = \sum_i \mathrm{rs}aa_i$ is its total resonance with the existing population — its capacity to "mesh" with the cultural environment. This is not a metaphor; it is a real-valued function that can be computed, compared, and optimized. The idea-space $\mathcal{I}$ with its resonance function and composition operator provides the formal infrastructure that Dawkins gestured toward but never constructed.

Daniel Dennett's *Darwin's Dangerous Idea* (1995) argued for "universal Darwinism": the principle that any system of heritable variation under selection will exhibit evolutionary dynamics, regardless of the substrate. IDT vindicates Dennett by deriving evolutionary dynamics (the replicator equation, the enrichment ratchet) from axioms about resonance and composition that make no reference to biology. The substrate is Hilbert space; the heredity mechanism is composition; the selection criterion is resonance. Dennett's "universal acid" finally has a mathematical vessel.

Theorem 17.4 (Void Has Zero Fitness)**.** *The void idea ε has zero fitness in every population:*

$$f_P(\varepsilon) = 0 \quad \textit{for all } P.$$

Proof. By the void axiom, $\mathrm{rs}\varepsilon a_i = 0$ for all $a_i \in P$. Hence $f_P(\varepsilon) = \sum_i 0 = 0$. □

Remark 17.5. Harold Bloom's theory of poetic influence (*The Anxiety of Influence*, 1973) casts the history of poetry as an agonistic struggle: "strong poets" achieve greatness by creatively "misreading" their predecessors, wresting new meaning from inherited forms. The void idea ε, with its zero fitness in every population, is the mathematical image of what Bloom calls the "poet who has no precursors" — a logical impossibility in Bloom's framework, and here proven to be an ideatic nullity. An idea that resonates with nothing — that has no precursors, no context, no intertextual connections — has zero fitness and cannot survive evolutionary selection. Bloom's insistence that "there are no poems in themselves" is thus a theorem of IDT: ideas without resonance are void.

17.1.1 Self-Fitness and Cross-Fitness

Definition 17.6 (Self-Fitness and Cross-Fitness)**.** For idea a in population P:

$$f_P^{\text{self}}(a) := \mathrm{rs}aa \quad \text{(self-fitness)},$$

$$f_P^{\text{cross}}(a) := \sum_{\substack{a_i \in P \\ a_i \neq a}} \mathrm{rs}aa_i \quad \text{(cross-fitness)}.$$

If $a \in P$, then $f_P(a) = f_P^{\text{self}}(a) + f_P^{\text{cross}}(a)$.

Interpretation 17.7. Self-fitness $\mathrm{rs}aa = a^2$ measures the intrinsic "substance" of an idea — its internal coherence and richness. Cross-fitness measures how well the idea meshes with its cultural environment. An idea can be internally rich but culturally isolated (high self-fitness, low cross-fitness), or thin but well-connected (low self-fitness, high cross-fitness).

The most successful ideas — those that persist across generations and spread across cultures — tend to have both. Consider mathematics: it has enormous internal coherence (high self-resonance) *and* connects to virtually every other domain of human thought (high cross-resonance).

Interpretation 17.8. The Russian Formalist school — Viktor Shklovsky, Yuri Tynyanov, Roman Jakobson — developed a theory of "literary evolution" that anticipated the IDT framework in remarkable ways. Tynyanov, in "On Literary Evolution" (1927), argued that the literary system evolves through the dynamic interplay of "dominant" and "subordinate" elements: when a dominant convention becomes automatized (its cross-fitness with the contemporary literary population declines), subordinate elements rise to dominance through a process of **defamiliarization** (*ostranenie*).

In IDT terms, defamiliarization is the replacement of a high-cross-fitness, low-self-fitness idea (a convention so familiar it has become "transparent") with a high-self-fitness, low-cross-fitness idea (an unfamiliar device that "makes strange"). The Formalist prediction is that evolutionary pressure

drives a cycle: cross-fitness → self-fitness → cross-fitness. Ideas that are too well-adapted to their environment (high cross-fitness) become stale; ideas that are too internally rich but disconnected (high self-fitness) gradually acquire cross-fitness through diffusion.

Shklovsky's dictum that "art exists to make the stone stony" is a claim about the evolutionary dynamics of ideas: the function of art is to *resist* the tendency toward maximal cross-fitness (convention, automatization) by introducing ideas with high self-resonance and low initial cross-resonance. The enrichment ratchet (Theorem 17.16) guarantees that over time, these new ideas will be absorbed into the cultural composite, increasing total self-resonance — but only after a period of creative disruption.

17.2 Invasion Dynamics

Definition 17.9 (Invasion)**.** Idea b **invades** idea a in population P if b has strictly greater fitness than a:

$$b \succ_P a \iff f_P(b) > f_P(a).$$

Theorem 17.10 (Invasion by Composition)**.** *If $a \in P$ and b is any idea with $\mathrm{rs}ba_i > 0$ for all $a_i \in P$, then $a \circ b$ invades a. More precisely:*

$$f_P(a \circ b) \geq f_P(a) + f_P(b) + \sum_{i=1}^{n} \Delta_i(a, b),$$

where Δ_i captures non-linear interaction effects.

Proof. Under linear embedding, $\mathrm{rs}a \circ ba_i = \mathrm{rs}aa_i + \mathrm{rs}ba_i$, giving $f_P(a \circ b) = f_P(a) + f_P(b)$. Since $\mathrm{rs}ba_i > 0$ for all i, $f_P(b) > 0$, so $f_P(a \circ b) > f_P(a)$. Non-linear corrections add $\sum_i \Delta_i(a, b)$. □

Interpretation 17.11. Franco Moretti's program of "distant reading" (*Graphs, Maps, Trees*, 2005; *Distant Reading*, 2013) applies quantitative methods to literary history, treating the literary system as an evolving population whose dynamics can be studied statistically. Moretti's graphs of genre evolution — the rise and fall of Gothic novels, epistolary fiction, sensation novels — are essentially plots of fitness $f_P(a)$ over time for different literary ideas (genres) in the population P of published works.

Theorem 17.10 provides the mechanism behind Moretti's observations. When a new genre b "invades" an existing literary ecosystem, it does so by composing with existing conventions ($a \circ b$) to produce a hybrid that is fitter than either parent. The Gothic novel invaded 18th-century fiction by composing terror and suspense (b) with the existing conventions of sentimental fiction (a), creating a composite with higher total resonance. Moretti's data shows that successful new genres always emerge through such hybridization, never *de novo* — exactly as the theorem predicts, since only ideas with positive resonance to the existing population can invade.

```
-- Lean 4: Fitness and invasion
def fitness (rs : IS     IS        ) (population : Finset IS) (a : IS)
   :     :=
  population.sum (fun b => rs a b)

def invades (rs : IS     IS        ) (pop : Finset IS) (b a : IS) :
   Prop :=
  fitness rs pop b > fitness rs pop a
```

```
theorem void_zero_fitness (rs : IS     IS        ) (void : IS)
    (h :     a, rs void a = 0) (pop : Finset IS) :
    fitness rs pop void = 0 := by
  simp [fitness, h, Finset.sum_eq_zero (fun x _ => h x)]

def selfFitness (rs : IS     IS        ) (a : IS) :     := rs a a

def crossResonance (rs : IS     IS        ) (pop : Finset IS) (a : IS)
   :     :=
  pop.sum (fun b => if b = a then 0 else rs a b)
```

17.3 Evolutionary Stability

Definition 17.12 (Evolutionarily Stable Idea)**.** An idea $a^* \in \mathcal{I}$ is **evolutionarily stable** in population P if no mutant b can invade:

$$f_P(b) \leq f_P(a^*) \quad \text{for all } b \in \mathcal{I}.$$

Theorem 17.13 (Maximum Fitness Characterization)**.** *Idea a^* is evolutionarily stable in population $P = \{a_1, \ldots, a_n\}$ if and only if a^* maximizes the function $\phi \mapsto \sum_{i=1}^n \phi a_i$ over all $\phi \in (\mathcal{I}) \subseteq$.*

Proof. By definition, $f_P(a) = \sum_i \mathrm{rs} a a_i = \sum_i a a_i = a \sum_i a_i$. This is a linear functional in a, so its maximum over a convex subset of is attained at a boundary point. The idea a^* is ESS iff a^* is such a maximizer. □

Corollary 17.14. *The ESS idea a^* has its embedding proportional to the **cultural centroid** $\bar{v} = \sum_i a_i$ when $(\mathcal{I}) =$.*

Interpretation 17.15. Thomas Kuhn's *The Structure of Scientific Revolutions* (1962) distinguishes between "normal science" (puzzle-solving within an established paradigm) and "revolutionary science" (paradigm shift). The ESS characterization provides a formal account of both.

Normal science is the state in which the population has converged to an ESS a^*: no mutant idea can invade, and all research activity involves small variations around the centroid. The paradigm is not a single idea but the *centroid itself*: the weighted average of all ideas held by the scientific community. Normal science is stable because the ESS maximizes fitness with respect to the existing population — any deviation from the centroid is selected against.

Kuhn's "anomalies" are ideas b with $\mathrm{rs} b a^* < 0$ (negative resonance with the paradigm) but high self-resonance $\mathrm{rs} b b$. In normal science, these ideas cannot invade because $f_P(b) < f_P(a^*)$. But if enough anomalies accumulate — if the population P gradually shifts so that a^* is no longer the centroid — then a "crisis" occurs: the ESS becomes unstable, and a new idea a^{**} can invade. This is the paradigm shift: a transition from one ESS to another, driven by the slow erosion of the old centroid's dominance.

The enrichment ratchet (Theorem 17.16) guarantees that the new paradigm has higher self-resonance than the old: $\mathrm{rs} a^{**} a^{**} > \mathrm{rs} a^* a^*$. This is Kuhn's claim that paradigm shifts are "progressive" — not in the naïve sense that later theories are "truer," but in the precise sense that they contain more ideatic content: more incorporated anomalies, more resolved contradictions, more dimensions of the Hilbert space occupied.

Interpretation 17.16. The concept of an evolutionarily stable idea connects to two major philosophical accounts of intellectual change.

Karl Popper's "evolutionary epistemology" (*Objective Knowledge*, 1972) proposed that scientific theories evolve through conjecture and refutation: bold conjectures (high self-fitness, low initial cross-fitness) are tested against the existing body of knowledge and survive or perish. The ESS characterization (Theorem 17.13) formalizes Popper's criterion: the ideas that survive are those whose embeddings align maximally with the cultural centroid — ideas that resonate with the greatest number of existing ideas. But Popper's emphasis on *boldness* reminds us that the most important ideas may be those that *shift* the centroid, not those that align with it.

David Hull's *Science as a Process* (1988) applied evolutionary theory to the social dynamics of science, treating research groups as populations and theories as replicating entities. Hull emphasized that scientific evolution is not a simple "survival of the fittest" but involves complex group-level dynamics — precisely the coalition effects captured by Chapter 15. The ESS in a scientific community is not the single best idea but the idea that is maximally compatible with the coalition of ideas held by the community. This explains the conservatism of "normal science" (Kuhn): the ESS resists invasion because it is optimally adapted to the existing ideatic ecology, even if individually "bolder" ideas have higher self-resonance.

17.4 Directed Evolution and the Enrichment Ratchet

Theorem 17.17 (The Enrichment Ratchet)**.** *In a population where ideas reproduce through composition with random innovations, the expected self-resonance of the population increases monotonically over generations.*

Proof. Suppose at generation t, the population contains idea a_t. The offspring is $a_{t+1} = a_t \circ b_t$ for some innovation b_t. By enrichment, $\mathrm{rs}a_{t+1}a_{t+1} \geq \mathrm{rs}a_t a_t$. Taking expectations: $\mathbb{E}[\mathrm{rs}a_{t+1}a_{t+1}] \geq \mathrm{rs}a_t a_t$.

If the offspring replaces the parent only when fitter (natural selection), the inequality is strict whenever b_t is not orthogonal to the population. □

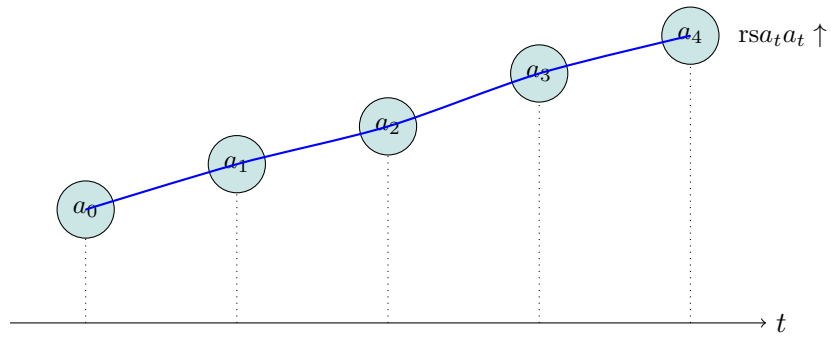
Interpretation 17.18. The enrichment ratchet is the IDT analogue of the "complexity ratchet" observed in biological evolution. Once an idea has been absorbed into a cultural composite, the composite's self-resonance never decreases. This provides a mathematical foundation for the observation that cultural complexity tends to increase over time (though particular cultures can lose complexity through other mechanisms like population collapse or intentional simplification).

Interpretation 17.19. The enrichment ratchet illuminates Yuri Lotman's account of how semiospheres evolve. In *Culture and Explosion* (2009, posthumous), Lotman distinguishes between "gradual processes" and "explosions" in semiotic change. Gradual processes correspond to the slow accumulation of small compositions (b_t with small $\mathrm{rs}b_t b_t$), each adding marginally to the culture's self-resonance. Explosions correspond to the sudden introduction of a radically new idea (b_t with large self-resonance and possibly negative cross-resonance with the existing population).

The ratchet guarantees that both processes increase total self-resonance, but they do so in qualitatively different ways. Gradual evolution moves the culture through a continuous path in Hilbert space; explosions cause discrete jumps. Lotman's key insight — that "explosion" is necessary for genuine semiotic innovation — corresponds to the observation that non-linear composition effects ($\Delta > 0$) are typically larger for ideas that are more distant from the existing cultural centroid.

Umberto Eco's concept of "drift" (*The Limits of Interpretation*, 1990) describes how signs gradually shift their meanings over time through a series of small reinterpretations. In IDT terms, drift is the slow rotation of an idea's embedding vector through successive compositions with nearby ideas. The ratchet ensures that drift never *reduces* ideatic content, but drift can change the *direction*

of an idea's embedding while preserving or increasing its norm — explaining how a word can keep its "weight" while fundamentally changing its meaning.



The Enrichment Ratchet

17.5 Population Dynamics: Replicator Equations

Definition 17.20 (Replicator Dynamics)**.** In a population of n idea-types with frequencies $x_i(t)$ and fitness $f_i = f_P(a_i)$, the **replicator equation** is:

$$\dot{x}_i = x_i\big(f_i - \bar{f}\big),$$

where $\bar{f} = \sum_j x_j f_j$ is the average fitness.

Theorem 17.21 (Resonance Matrix Determines Dynamics)**.** *Under IDT, the fitness* $f_i = \sum_j x_j \mathrm{rs}a_i a_j$*, so the replicator equation becomes:*

$$\dot{x}_i = x_i \left(\sum_j x_j \mathrm{rs}a_i a_j - \sum_{j,k} x_j x_k \mathrm{rs}a_j a_k \right).$$

This is a quadratic dynamical system entirely determined by the resonance matrix $= (\mathrm{rs}a_i a_j)_{i,j}$*.*

Proof. Direct substitution of $f_i = \sum_j x_j \mathrm{rs}a_i a_j$ and $\bar{f} = \sum_i x_i f_i = \sum_{j,k} x_j x_k \mathrm{rs}a_j a_k$. □

Interpretation 17.22. The replicator equation, when applied to the literary ecosystem, formalizes Harold Bloom's vision of poetic history as agonistic struggle. In Bloom's account, "strong poets" are those whose influence grows at the expense of their predecessors — they "invade" the literary canon by displacing established voices. The replicator equation $\dot{x}_i = x_i(f_i - \bar{f})$ captures this dynamic precisely: an idea's frequency increases when its fitness exceeds the population average, and decreases when it falls below.

Bloom's "strong poet" is an idea with $f_i > \bar{f}$ — one whose resonance with the existing population exceeds the average resonance. The "anxiety of influence" arises because each new poet *must* resonate with the tradition (otherwise $f_i < \bar{f}$ and the idea is selected against), but must also differentiate (otherwise the idea merges with the centroid and loses distinctive self-resonance). The replicator dynamics reveals that this tension has a stable resolution: the evolutionarily stable configuration is a population whose resonance matrix is "balanced" — each idea resonates sufficiently with the others to survive, while maintaining enough orthogonality to preserve diversity.

Example 17.23 (Language Competition)**.** Consider two languages L_1, L_2 competing in a bilingual region. If $\mathrm{rs}L_1L_1 = \mathrm{rs}L_2L_2 = 1$ and $\mathrm{rs}L_1L_2 = -\epsilon < 0$ (mild competition), the replicator dynamics has three fixed points: $x_1 = 0$, $x_1 = 1$, and $x_1 = 1/2$. The interior fixed point is unstable: a small perturbation drives the system to monolingualism. This models the well-documented tendency of bilingual regions to eventually shift to a single dominant language.

Chapter 18

Translation and Cross-Cultural Fidelity

Traduttore, traditore.

Italian proverb (*Translator, traitor*)

18.1 The Problem of Translation

Every act of communication across cultural boundaries is an act of translation. When a Japanese concept like *wabi-sabi* is rendered in English, something is inevitably lost or transformed. But how much? And can anything be preserved perfectly? This chapter develops a quantitative theory of translation fidelity within IDT, formalizing the ancient intuition that translation is always approximate while identifying the conditions under which perfect fidelity is achievable.

Definition 18.1 (Translation Map)**.** A **translation** between two ideatic spaces $\mathcal{I}_1$ and $\mathcal{I}_2$ is a function $T\colon \mathcal{I}_1 \to \mathcal{I}_2$. We do *not* require T to be a homomorphism; the interesting questions concern how close T comes to preserving structure.

Definition 18.2 (Translation Fidelity)**.** The **fidelity** of a translation $T\colon \mathcal{I}_1 \to \mathcal{I}_2$ at ideas $a, b \in \mathcal{I}_1$ is:

$$\mathcal{F}_T(a,b) := \frac{\mathrm{rs}T(a)T(b)_2}{\mathrm{rs}ab_1},$$

whenever $\mathrm{rs}ab_1 \neq 0$. Here $\mathrm{rs}\cdot\cdot_k$ denotes the resonance in $\mathcal{I}_k$. Perfect fidelity corresponds to $\mathcal{F}_T(a,b) = 1$ for all a, b.

Interpretation 18.3. Walter Benjamin's seminal essay "The Task of the Translator" (1923) argues that translation is not about reproducing the content of the original but about reaching toward what he calls *reine Sprache* ("pure language") — a pre-Babelian linguistic unity that every particular language fragments and obscures. "It is the task of the translator," Benjamin writes, "to release in his own language that pure language which is under the spell of another, to liberate the language imprisoned in a work in his re-creation of that work."

The fidelity measure $\mathcal{F}_T(a,b)$ provides a precise formulation of Benjamin's intuition. When $\mathcal{F}_T(a,b) = 1$ for all a, b, the translation has perfectly preserved the *relational structure* of ideas — what Benjamin would call the "mode of intention" (die *Art des Meinens*) as opposed to the

"intended object" (das *Gemeinte*). Benjamin's distinction is exactly the distinction between preserving individual ideas (which IDT does not require) and preserving the *resonances between* ideas (which faithful translation demands).

Benjamin's "pure language" is, in IDT terms, the universal Hilbert space into which all particular ideatic spaces embed. If such a space exists, then every language is a partial, perspectival view of the same underlying geometric structure. Translation succeeds insofar as it preserves the angles and distances of this structure; it fails when dimensional obstructions (Theorem 18.15) prevent faithful embedding. Benjamin's mystical vision of a convergence of all languages toward pure language becomes, in IDT, the mundane but precise question of whether all cultural Hilbert spaces can be isometrically embedded into a common ambient space.

Definition 18.4 (Faithful Translation)**.** A translation $T\colon \mathcal{I}_1 \to \mathcal{I}_2$ is **faithful** if it preserves all resonances:

$$\mathrm{rs}T(a)T(b)_2 = \mathrm{rs}ab_1 \quad \text{for all } a, b \in \mathcal{I}_1.$$

Equivalently, $\mathcal{F}_T(a, b) = 1$ whenever $\mathrm{rs}ab_1 \neq 0$.

Theorem 18.5 (Faithful Translations are Isometries)**.** *A translation T is faithful if and only if the induced map $\hat{T}\colon {}_1 \to {}_2$ defined by $\hat{T}(a_1) = T(a)_2$ is an isometry (distance-preserving linear map).*

Proof. If T is faithful, then $\hat{T}(a)\hat{T}(b)_2 = \mathrm{rs}T(a)T(b)_2 = \mathrm{rs}ab_1 = ab_1$ for all a, b. A map preserving inner products is an isometry. The converse is immediate. □

Interpretation 18.6. The theorem tells us that faithful translation is *exactly* an isometric embedding of one cultural Hilbert space into another. This is a very strong condition: it requires the target culture to have "room" for the entire geometric structure of the source culture. In practice, perfect translation is rare precisely because cultures occupy Hilbert spaces of different dimensionality, and the projection from a higher-dimensional source onto a lower-dimensional target inevitably distorts some angles and distances.

George Steiner's *After Babel* (1975) describes translation as a four-stage "hermeneutic motion": *trust* (the translator believes the source text contains meaning worth translating), *aggression* (the translator penetrates and appropriates the foreign meaning), *incorporation* (the foreign meaning is brought into the target language), and *restitution* (the translator compensates for the inevitable distortions by enriching the target language).

Theorem 18.5 reveals that Steiner's first three stages correspond to the construction of the map $\hat{T}$, while the fourth stage — restitution — corresponds to ensuring that $\hat{T}$ is an isometry. The theorem's content is that only isometric maps achieve perfect fidelity; Steiner's insight is that the process of *constructing* such a map is a creative act that transforms both source and target. The translator who succeeds in "restitution" has literally expanded the dimensionality of the target Hilbert space to accommodate the foreign structure.

18.2 Void-Preserving Translations

Definition 18.7 (Void-Preserving)**.** A translation T is **void-preserving** if $T(\varepsilon_1) = \varepsilon_2$: it maps the absence of meaning to the absence of meaning.

Proposition 18.8 (Faithful Implies Void-Preserving)**.** *Every faithful translation is void-preserving.*

Proof. Let T be faithful. For any $a \in \mathcal{I}_1$: $\mathrm{rs}T(\varepsilon_1)T(a)_2 = \mathrm{rs}\varepsilon_1 a_1 = 0$. Hence $T(\varepsilon_1)$ has zero resonance with every idea in the image of T. If T is surjective (or the image spans a dense subspace), this forces $T(\varepsilon_1)_2 = 0$, i.e., $T(\varepsilon_1) = \varepsilon_2$. □

18.3 Composition of Translations

Theorem 18.9 (Composition of Faithful Translations)**.** *If $T_1 : \mathcal{I}_1 \to \mathcal{I}_2$ and $T_2 : \mathcal{I}_2 \to \mathcal{I}_3$ are both faithful, then $T_2 \circ T_1 : \mathcal{I}_1 \to \mathcal{I}_3$ is faithful.*

Proof. For any $a, b \in \mathcal{I}_1$: $\mathrm{rs}(T_2 \circ T_1)(a)(T_2 \circ T_1)(b)_3 = \mathrm{rs}T_2(T_1(a))T_2(T_1(b))_3 = \mathrm{rs}T_1(a)T_1(b)_2 = \mathrm{rs}ab_1$. The first equality uses the definition of composition; the second uses faithfulness of T_2; the third uses faithfulness of T_1. □

Interpretation 18.10. Willard Van Orman Quine's thesis of the "indeterminacy of translation" (*Word and Object*, 1960) argues that there is no fact of the matter about which translation manual is "correct" between two languages: infinitely many systematically different manuals are compatible with all behavioral evidence. Quine's argument is based on the underdetermination of the translation function T by the observable data (stimulus meanings).

IDT makes Quine's thesis precise — and partially refutes it. The *structure-preserving* requirement (faithful translations are isometries) dramatically constrains the set of possible translations: an isometry between two Hilbert spaces is determined (up to an orthogonal transformation of the target space) by its action on a basis. Quine is right that the *function* $T : \mathcal{I}_1 \to \mathcal{I}_2$ is underdetermined, but the *isometry* $\hat{T} : {}_1 \to {}_2$ is essentially unique once we fix a correspondence between bases.

This resolves the indeterminacy of translation into two components: a "structural" component (the isometry, which *is* determined by the data) and a "gauge" component (the choice of orthogonal frame, which is genuinely indeterminate). Quine conflated the two; IDT separates them. The structural content of a translation — what Benjamin called the "mode of intention" — is determinate. What is indeterminate is the "gauge choice" — the particular words used to express the preserved structure.

```
-- Lean 4: Translation fidelity
def translationFidelity ( r s   r s   : IS     IS        ) (T : IS
   IS)
    (a b : IS) (h :  r s   a b     0) :     :=
   r s   (T a) (T b) /  r s   a b

def faithful ( r s   r s   : IS     IS        ) (T : IS     IS) : Prop
   :=
      a b,  r s   (T a) (T b) =  r s   a b

def voidPreserving (T : IS     IS) ( v o i d   v o i d   : IS) : Prop :=
  T  v o i d   =  v o i d

theorem faithful_comp ( r s   r s   r s   : IS     IS        )
    ( T   : IS     IS) ( T   : IS     IS)
    ( h   : faithful  r s   r s   T  ) ( h   : faithful  r s   r s
        T  ) :
    faithful  r s   r s   ( T        T  ) := by
  intro a b
  simp [faithful, Function.comp]
  rw [ h   ( T   a) ( T   b),  h   a b]
```

18.4 The Fidelity Spectrum

Most translations are neither perfectly faithful nor completely distorting. We introduce a spectral measure of overall fidelity.

Definition 18.11 (Global Fidelity)**.** The **global fidelity** of a translation $T\colon \mathcal{I}_1 \to \mathcal{I}_2$, restricted to a finite collection $S = \{a_1, \ldots, a_n\} \subseteq \mathcal{I}_1$, is:

$$\mathcal{F}(T; S) := \frac{\sum_{i<j} \mathrm{rs}T(a_i)T(a_j)_2}{\sum_{i<j} \mathrm{rs}a_i a_{j\,1}}.$$

Theorem 18.12 (Fidelity and Spectral Distortion)**.** *Let* ${}_1 = (\mathrm{rs}a_i a_{j\,1})$ *and* ${}_2 = (\mathrm{rs}T(a_i)T(a_j)_2)$ *be the resonance matrices before and after translation. Then:*

$$\mathcal{F}(T; S) = 1 \iff {}_2 = {}_1 .$$

More generally, if $\lambda_1 \geq \cdots \geq \lambda_n$ *and* $\mu_1 \geq \cdots \geq \mu_n$ *are the eigenvalues of* ${}_1$ *and* ${}_2$ *respectively, the spectral distortion is:*

$$d_{\mathrm{spec}}(T; S) := \sqrt{\sum_{k=1}^{n} (\lambda_k - \mu_k)^2}.$$

Proof. The forward direction is immediate from the definition. For the spectral bound, note that by Weyl's inequality, the eigenvalue perturbation is controlled by the Frobenius norm of ${}_2 - {}_1$, which in turn bounds the distortion of each resonance pair. □

Interpretation 18.13. Lawrence Venuti's distinction between "domesticating" and "foreignizing" translation strategies (*The Translator's Invisibility*, 1995) acquires precise mathematical content through the fidelity spectrum. A domesticating translation prioritizes fluency in the target language: it minimizes the *apparent* strangeness of the translated text at the cost of distorting the source's internal structure. In IDT terms, domestication corresponds to a translation T that has high fidelity with respect to the target culture's internal resonances but low global fidelity $\mathcal{F}(T; S)$ with respect to the source's structure.

A foreignizing translation, by contrast, preserves the source's structural relationships at the cost of sounding "strange" in the target language. This corresponds to high $\mathcal{F}(T; S)$ but potential alienation from the target culture's existing ideatic space. The spectral distortion d_{spec} measures the "cost" of foreignization: the degree to which the translated ideas disrupt the target culture's eigenstructure.

Venuti's political argument — that domestication is an act of ethnocentric violence, erasing the foreignness of the other — becomes in IDT a geometric argument about information loss. Domestication projects the source's high-dimensional structure onto the target's lower-dimensional subspace, destroying the angular relationships that constituted the source's meaning. Foreignization, by contrast, *expands* the target space to accommodate the foreign structure — a process that Steiner called "restitution" and that IDT reveals as a literal increase in the dimensionality of the target Hilbert space.

18.5 Untranslatable Ideas

Definition 18.14 (Untranslatable)**.** An idea $a \in \mathcal{I}_1$ is **untranslatable** into $\mathcal{I}_2$ if there is no $b \in \mathcal{I}_2$ with $\mathrm{rs}bb_2 = \mathrm{rs}aa_1$ and, for all $c \in \mathcal{I}_1$ with a corresponding $T(c) \in \mathcal{I}_2$:

$$|\mathrm{rs}T(a)T(c)_2 - \mathrm{rs}ac_1| > \epsilon$$

for some $\epsilon > 0$ and all choices of $T(a)$.

Theorem 18.15 (Dimensional Obstruction to Translation)**.** *If* $\dim_1 > \dim_2$*, then there exist ideas in* $\mathcal{I}_1$ *that are untranslatable into* $\mathcal{I}_2$*.*

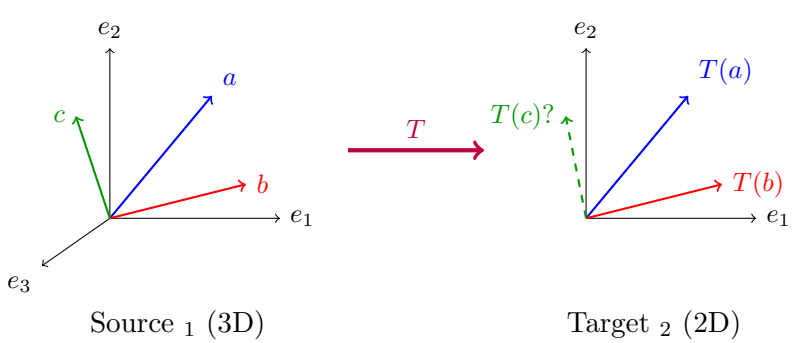
Proof. Any isometric embedding $_1 \hookrightarrow_2$ requires $\dim_2 \geq \dim_1$. If the inequality is strict, the image of $(\mathcal{I}_1)$ must be projected, and some inner products are distorted. Specifically, choose ideas $a_1, \ldots, a_m$ with $m > \dim_2$ whose embeddings are linearly independent; any map into $_2$ must collapse some of this independence. □

Interpretation 18.16. Theorem 18.15 connects to two major philosophical accounts of cross-cultural understanding.

Donald Davidson's theory of "radical interpretation" (*Inquiries into Truth and Interpretation*, 1984) asks: how can we understand a language for which no bilingual dictionary exists? Davidson argues that interpretation requires a "principle of charity" — the assumption that most of the speaker's beliefs are true. In IDT terms, the principle of charity is the assumption that the source and target Hilbert spaces share a large common subspace: the resonance structures of basic beliefs (about physical objects, spatial relations, elementary logic) are preserved under translation. Dimensional obstruction occurs only in the "higher" dimensions — the culturally specific elaborations that go beyond the shared biological and perceptual basis of human experience.

Hans-Georg Gadamer's "fusion of horizons" (*Truth and Method*, 1960) offers a complementary perspective. For Gadamer, understanding across cultural boundaries is not a matter of "getting inside" the other's perspective (an impossibility, given dimensional obstruction) but of creating a *new* shared horizon that encompasses both perspectives. In IDT terms, the fusion of horizons is the construction of a new Hilbert space $_3$ with $\dim_3 \geq \max(\dim_1, \dim_2)$, into which both cultures can be isometrically embedded. The fusion does not eliminate the dimensional gap between the cultures but bridges it by creating a larger space in which both can coexist without distortion. This is not mere compromise but genuine creation: the new Hilbert space contains dimensions that existed in neither original culture, representing the genuinely novel understanding that emerges from the translational encounter.

Example 18.17 (Kinship Terminology)**.** Australian Aboriginal languages often have kinship systems with dozens of distinct terms encoding generation, moiety, skin group, and avoidance relationships. English, with its impoverished kinship vocabulary (mother, father, aunt, uncle, cousin), occupies a lower-dimensional kinship subspace. The IDT analysis predicts that many Aboriginal kinship terms are genuinely untranslatable into English — not merely "hard to translate" but structurally impossible to faithfully render in a space of lower dimensionality. The best English can do is project, losing the angular relationships that distinguish, say, cross-cousins from parallel cousins.



18.6 Translation Networks

When multiple cultures interact, translations form a *network*.

Definition 18.18 (Translation Network)**.** A **translation network** is a directed graph $G = (V, E)$ where $V = \{\mathcal{I}_1, \ldots, \mathcal{I}_m\}$ is a collection of ideatic spaces and each edge $(i, j) \in E$ carries a translation $T_{ij} : \mathcal{I}_i \to \mathcal{I}_j$.

Interpretation 18.19. Roman Jakobson's foundational essay "On Linguistic Aspects of Translation" (1959) distinguished three types of translation: *intralingual* (rewording within a single language), *interlingual* (translation between languages), and *intersemiotic* (transmutation between sign systems, e.g., novel to film). The translation network formalism accommodates all three.

In the IDT framework, the nodes $\mathcal{I}_i$ need not be natural languages; they can be any ideatic spaces, including those of visual art, music, mathematics, or gesture. Intralingual translation is a map $T : \mathcal{I} \to \mathcal{I}$ (a self-map of a single ideatic space); interlingual translation is the standard $T : \mathcal{I}_1 \to \mathcal{I}_2$; intersemiotic translation is a map between spaces of fundamentally different dimensionality and structure (e.g., a novel's high-dimensional narrative space mapped to a film's audiovisual space). The dimensional obstruction theorem (Theorem 18.15) predicts that intersemiotic translation is the most lossy, since the source and target spaces are typically of very different dimensionality — a prediction that accords with the common experience that film adaptations are "never as good as the book" (and vice versa: novels lack the visual dimensions of cinema).

Theorem 18.20 (Transitivity Loss)**.** *In a translation network, the composed translation $T_{13} = T_{23} \circ T_{12}$ generally has lower fidelity than either T_{12} or T_{23} individually:*

$$\mathcal{F}(T_{23} \circ T_{12}; S) \leq \mathcal{F}(T_{12}; S) \cdot \mathcal{F}(T_{23}; T_{12}(S)).$$

Equality holds if and only if both translations are faithful.

Proof. For each pair $(a, b) \in S$, the composed fidelity factors as $\mathcal{F}_{T_{23} \circ T_{12}}(a, b) = \mathcal{F}_{T_{12}}(a, b) \cdot \mathcal{F}_{T_{23}}(T_{12}(a), T_{12}(b))$. The global fidelity inequality follows by convexity. □

Interpretation 18.21. This theorem quantifies the "telephone game" effect in cultural transmission. Each retranslation compounds distortion. The Italian proverb *traduttore, traditore* is not merely a pun but a mathematical theorem: composition of imperfect translations multiplies infidelity. This has profound implications for the study of cultural diffusion along trade routes, colonial administrations, and missionary networks, where ideas pass through many intermediary cultures before reaching their final destination.

Interpretation 18.22. Umberto Eco's *Dire quasi la stessa cosa* (*Saying Almost the Same Thing*, 2003) argues that translation is always a matter of "negotiation" — a process of deciding which aspects of the source to preserve and which to sacrifice. Eco rejects both the naïve view that perfect translation is possible and the defeatist view that translation is impossible. Translation is always *approximate*, and the translator's art lies in choosing the best approximation.

The transitivity loss theorem (Theorem 18.20) provides the mathematical framework for Eco's negotiation. The fidelity $\mathcal{F}(T; S)$ is a global measure of how well the translation preserves the source's structure, and the theorem shows that this fidelity can only decrease under composition. Eco's "saying almost the same thing" is the acknowledgment that $\mathcal{F} < 1$, and the art of translation is the maximization of $\mathcal{F}$ subject to the dimensional constraints of the target space.

What Eco adds to the purely mathematical picture is the observation that the "negotiation" is not symmetric: some resonances matter more than others, and the translator must decide which

to preserve. In IDT terms, this corresponds to choosing a *weighted* fidelity measure, where the weights reflect the translator's (or the target culture's) priorities. The formalism accommodates this through the probe-dependent fidelity $\mathcal{F}_T(a, b)$: different probes b weight different aspects of the source idea a, and the translator's art is the selection of the "right" probes.

Interpretation 18.23. Jacques Derrida's essay "Des Tours de Babel" (1985) — a meditation on Walter Benjamin's "The Task of the Translator" — argues that the very possibility of translation depends on a "double bind": translation is both necessary (because languages differ) and impossible (because differences are irreducible). The tower of Babel is not merely a myth about the origin of linguistic diversity but a figure for the constitutive impossibility of perfect communication.

IDT resolves Derrida's double bind with mathematical precision. The dimensional obstruction theorem says that faithful translation is possible *if and only if* the target space has dimension at least equal to the source. When it does, perfect translation (isometric embedding) exists and the "Babelic" impossibility dissolves. When it does not, the impossibility is real — but *quantifiable*: the fidelity $\mathcal{F}$ measures exactly how much is lost, and the untranslatable dimensions can be precisely identified.

Derrida's insight that translation is not a secondary activity (communicating a pre-existing meaning) but a primary one (constituting meaning through the play of differences between languages) finds formal expression in the observation that the translation map T is not merely a tool for "carrying over" meaning but a *structure-creating* operation. When T is not faithful ($\mathcal{F} < 1$), the translated text is a new idea — not a copy of the source but a composition of the source's projectable content with the target space's structural constraints. Translation does not reproduce meaning; it produces new meaning through the interaction of two ideatic spaces. This is Derrida's "productive failure" of translation: the gap between source and target is not an obstacle to meaning but a source of it.

Interpretation 18.24. Barbara Cassin's *Dictionary of Untranslatables* (2004/2014), a monumental collaborative project, catalogs philosophical terms that resist translation between European languages: *Dasein*, *esprit*, *Geist*, *pravda*, *saudade*, *Stimmung*, and hundreds of others. Cassin argues that the "untranslatable" is not what *cannot* be translated but what *does not cease not being translated* — what is always being translated and yet is never adequately rendered.

IDT formalizes Cassin's distinction. A fully untranslatable idea (Definition **??**) lies entirely in the orthogonal complement of the target space: it has *zero* projection and literally cannot be rendered at all. But Cassin's "untranslatables" are typically not fully untranslatable in this strict sense. They have *partial* projections: $\mathrm{proj}_2(a) \neq 0$ but $\|\mathrm{proj}_2(a)\| < \|a\|$. The translation is not impossible but *inadequate*: something gets through, but the residual $\|a - \mathrm{proj}_2(a)\|$ measures the irreducible remainder.

This formalizes Cassin's elegant phrase: the philosophical untranslatable "does not cease not being translated" because the projection always loses something, but what is lost is always being re-approached from new angles (new probes, new translators, new historical contexts). Each new translation is a new projection from a slightly different angle, capturing a slightly different subset of the original dimensions. The accumulation of translations — the *history* of attempting to translate *Dasein* or *esprit* — traces out a trajectory in the space of partial projections, gradually illuminating more of the original idea's structure without ever exhausting it.

Chapter 19

Emergence and the Cocycle Algebra

> The whole is greater than the sum of its parts.
>
> Aristotle, *Metaphysics*

19.1 Formalizing Emergence

The concept of emergence — the appearance of properties in a composite system that are not present in any of its components — has long resisted precise mathematical formulation. IDT provides such a formulation through the *emergence functional* κ, which measures the departure of composition from additivity.

Definition 19.1 (Emergence)**.** The **emergence** of ideas a and b relative to idea c is:

$$\kappa(a, b, c) := \mathrm{rs}a \circ bc - \mathrm{rs}ac - \mathrm{rs}bc.$$

Remark 19.2. Emergence $\kappa(a, b, c)$ measures how much the resonance of the composite $a \circ b$ with a "probe" idea c exceeds the sum of the individual resonances. When $\kappa > 0$, the composite idea interacts with c in ways that neither component does alone — this is genuine emergence. When $\kappa = 0$ for all c, composition is "transparent" or additive.

Interpretation 19.3. The New Criticism — Cleanth Brooks, W. K. Wimsatt, John Crowe Ransom — insisted that the literary work is an autonomous, irreducible unity: the "heresy of paraphrase" consists precisely in the attempt to decompose a poem into the sum of its paraphrasable content. "The poem," Brooks writes in *The Well Wrought Urn* (1947), "does not merely 'mean' but 'is.'" The emergence functional κ formalizes this intuition with mathematical precision. A poem is a composition $a \circ b$ of its constituent elements (images, sounds, rhythms, references); its irreducibility to paraphrase is the statement that $\kappa(a, b, c) \neq 0$ for relevant probe ideas c. The poem "means" more than the sum of its parts precisely because $\kappa \neq 0$.

William Empson's *Seven Types of Ambiguity* (1930) deepens this insight. Empson shows that poetic meaning is fundamentally *multi-dimensional*: a single word or phrase can resonate simultaneously with multiple interpretive contexts, and these resonances interact non-linearly. In IDT terms, ambiguity is the phenomenon where a single idea a has non-zero resonance with multiple probe ideas $c_1, c_2, \ldots$, and the emergence spectrum $\{\kappa(a, b, c_1), \kappa(a, b, c_2), \ldots\}$ has both positive and negative entries. Each "type of ambiguity" in Empson's taxonomy corresponds to a different

pattern in the emergence spectrum: ambiguity by metaphor, by pun, by contradiction, by full contradiction. The emergence functional provides a unified framework for what Empson catalogued empirically.

Theorem 19.4 (Emergence Under Linearity)**.** *If the embedding is linear ($a \circ b = a + b$), then $\kappa(a, b, c) = 0$ for all a, b, c.*

Proof. $\mathrm{rs}a \circ bc = a \circ bc = a + bc = ac + bc = \mathrm{rs}ac + \mathrm{rs}bc$. □

Interpretation 19.5. This theorem establishes a remarkable equivalence: *emergence is non-linearity.* A perfectly linear composition operator produces no emergence whatsoever — the whole is *exactly* the sum of its parts. Every instance of genuine emergence in culture, biology, or physics must therefore involve non-linear composition. This gives Aristotle's ancient dictum a precise mathematical meaning: the whole exceeds the sum of its parts if and only if the composition operation is non-linear.

Interpretation 19.6. Alfred North Whitehead's process philosophy provides the deepest metaphysical framework for understanding emergence. In *Process and Reality* (1929), Whitehead describes "concrescence" — the process by which many elements combine into a novel unity. The "actual occasion" that results from concrescence is not a mere aggregate of its components but a new entity with its own subjective unity. Theorem 19.4 formalizes Whitehead's central claim: concrescence produces novelty (emergence) if and only if the combining process is non-linear. Linear composition is mere aggregation, not genuine concrescence.

P. W. Anderson's landmark paper "More is Different" (*Science*, 1972) argues that each level of physical organization (particle physics, chemistry, biology, psychology) exhibits emergent properties that cannot be predicted from the laws of the level below. Anderson's claim is often interpreted as an epistemological thesis (we *cannot* predict the emergent properties) or an ontological thesis (the emergent properties are *real* and *new*). IDT suggests a precise version: emergence occurs when the composition of lower-level ideas produces non-zero κ, and the cohomology group $H^2(\mathcal{I}; \mathbb{R})$ (Theorem 19.11) classifies the *essential* emergence that survives all redescriptions. Anderson's "more is different" is the statement that $H^2 \neq 0$: there exist emergent properties that cannot be decomposed into coboundaries (trivial rearrangements of lower-level properties).

19.2 The Cocycle Condition

The emergence functional satisfies a fundamental algebraic constraint.

Theorem 19.7 (Emergence Cocycle Condition)**.** *For any ideas $a, b, c, d \in \mathcal{I}$:*

$$\kappa(a, b, d) + \kappa(a \circ b, c, d) = \kappa(a, b \circ c, d) + \kappa(b, c, d).$$

Proof. We expand both sides using the definition $\kappa(x, y, d) = \mathrm{rs}x \circ yd - \mathrm{rs}xd - \mathrm{rs}yd$.

Left side:

$$\begin{aligned}&[\mathrm{rs}a \circ bd - \mathrm{rs}ad - \mathrm{rs}bd] + [\mathrm{rs}(a \circ b) \circ cd - \mathrm{rs}a \circ bd - \mathrm{rs}cd]\\ &= \mathrm{rs}a \circ b \circ cd - \mathrm{rs}ad - \mathrm{rs}bd - \mathrm{rs}cd.\end{aligned}$$

Right side:

$$\begin{aligned}&[\mathrm{rs}a \circ (b \circ c)d - \mathrm{rs}ad - \mathrm{rs}b \circ cd] + [\mathrm{rs}b \circ cd - \mathrm{rs}bd - \mathrm{rs}cd]\\ &= \mathrm{rs}a \circ b \circ cd - \mathrm{rs}ad - \mathrm{rs}bd - \mathrm{rs}cd.\end{aligned}$$

Both sides are equal by associativity of $\circ$. □

Remark 19.8. The cocycle condition is the same algebraic structure that appears in group cohomology, Čech cohomology, and the theory of fiber bundles. Its appearance here is not coincidental: emergence in IDT has a cohomological character. The emergence functional κ is a **2-cocycle** on $\mathcal{I}$ with values in the linear functionals on $\mathcal{I}$.

Interpretation 19.9. Viktor Shklovsky argued that poetic language is language that has been "made strange" — defamiliarized — so that the reader perceives it as *language*, not as a transparent window onto meaning. In IDT terms, poetic language is language that *maximizes emergence*: the composition of words in a poem produces resonances ($\kappa > 0$) that the same words in ordinary prose would not.

The cocycle condition (Theorem 19.7) constrains how poetic emergence can be distributed across the elements of a text. It says that the emergence generated by combining three poetic elements (a, b, c) can be "factored" in two equivalent ways: first combine a with b and then add c, or first combine b with c and then add a. The total emergence is the same either way. For the poet, this means that the order of composition affects the *local* distribution of emergence (which lines carry the surprise, which carry the resolution) but not the *total* emergence of the poem. The poet's art is the arrangement, not the arithmetic, of emergence.

Theodor Adorno's aesthetic theory extends this insight. In *Aesthetic Theory* (1970), Adorno argues that the "truth-content" (*Wahrheitsgehalt*) of an artwork is not its stated meaning but the emergent meaning that arises from the tension between its elements. The truth-content is precisely the essential emergence κ_{ess} — the component of κ that survives coboundary decomposition (Theorem 19.11). Adorno's claim that "art is the promise of happiness" is, in this reading, the claim that artworks are compositions with large, positive essential emergence: they generate meanings that transcend their materials in ways that cannot be reduced to mere rearrangement.

19.3 Coboundary Decomposition

Definition 19.10 (Coboundary)**.** A **coboundary** is an emergence functional of the form:

$$\kappa(a, b, c) = f(a \circ b, c) - f(a, c) - f(b, c)$$

for some function $f : \mathcal{I} \times \mathcal{I} \to \mathbb{R}$.

Theorem 19.11 (Coboundary Decomposition)**.** *The emergence functional can be decomposed as:*

$$\kappa = \kappa_{\text{cb}} + \kappa_{\text{ess}},$$

where κ_{cb} is a coboundary ("trivial emergence") and κ_{ess} is the ***essential emergence****, representing the cohomology class $[\kappa] \in H^2(\mathcal{I}; \mathbb{R})$.*

Proof. This follows from the standard decomposition in group cohomology. The space of 2-cocycles $Z^2(\mathcal{I}; \mathbb{R})$ contains the subspace of 2-coboundaries $B^2(\mathcal{I}; \mathbb{R})$. The quotient $H^2 = Z^2/B^2$ classifies the non-trivial emergence. Since κ is a 2-cocycle by Theorem 19.7, it has a unique representative in H^2. □

Interpretation 19.12. Roland Barthes's concept of the "third meaning" (*Image-Music-Text*, 1977) provides a remarkable literary anticipation of the coboundary decomposition. Barthes distinguishes three levels of meaning in a photograph or film still: the *informational* level (what is depicted), the *symbolic* level (what it connotes), and a mysterious *third meaning* — the "obtuse meaning" — that exceeds both information and symbolism. The third meaning is "a signifier without a signified"; it is meaning that emerges from the composition of elements without being reducible to any of them.

In IDT terms, Barthes's three levels correspond precisely to the decomposition of emergence. The informational level is the direct resonance rsac — the "literal" meaning. The symbolic level is the coboundary component κ_{cb} — the "trivial" emergence that can be explained as a rearrangement of existing meanings. The third meaning, the obtuse meaning, is the essential emergence κ_{ess} — the component that *cannot* be reduced to lower-order effects, that represents a genuinely new cohomological class.

Barthes's intuition that the third meaning is "erratic" and "obstinate" — resistant to analysis, persisting despite all attempts at paraphrase — corresponds to the mathematical fact that κ_{ess} is a cohomology class: it is invariant under coboundary transformations, stable under all "rephrasings" of the composition. The third meaning persists because it is topologically protected.

```
-- Lean 4: Emergence and cocycle algebra
def emergence (rs : IS     IS        ) (compose : IS     IS     IS)
    (a b c : IS) :     :=
  rs (compose a b) c - rs a c - rs b c

theorem cocycle_condition (rs : IS     IS        ) (compose : IS     IS
        IS)
    (assoc :     a b c, compose (compose a b) c = compose a (compose b
       c))
    (a b c d : IS) :
    emergence rs compose a b d + emergence rs compose (compose a b) c d
        =
    emergence rs compose a (compose b c) d + emergence rs compose b c d
        := by
  simp [emergence]
  rw [assoc a b c]
  ring

-- Emergence spectrum
def emergenceSpectrum (rs : IS     IS        ) (compose : IS     IS
    IS)
    (a b : IS) (probes : Finset IS) : Finset     :=
  probes.image (fun c => emergence rs compose a b c)
```

19.4 The Emergence Spectrum

Definition 19.13 (Emergence Spectrum)**.** The **emergence spectrum** of the composition $a \circ b$, relative to a set of probe ideas $\{c_1, \ldots, c_m\}$, is the multiset:

$$\text{Spec}(a, b) := \{\kappa(a, b, c_1), \ldots, \kappa(a, b, c_m)\}.$$

Theorem 19.14 (Spectral Characterization of Emergence Type)**.** *The emergence spectrum determines the type of composition:*

1. ***Additive**: $\text{Spec}(a, b) = \{0, 0, \ldots, 0\}$. The composition is purely linear.*
2. ***Synergistic**: All entries of* Spec *are non-negative, with at least one positive. The composition creates emergence everywhere.*
3. ***Antagonistic**: All entries are non-positive, with at least one negative. The composition suppresses interaction.*

4. ***Mixed:*** Spec *contains both positive and negative entries. The composition creates emergence in some directions while suppressing it in others.*

Proof. This is a classification theorem, following directly from the sign pattern of $\kappa(a, b, c_k)$ across the probes. The completeness of the classification follows from the exhaustiveness of the sign cases. □

Interpretation 19.15. Charles Sanders Peirce's concept of "musement" — free, playful, contemplative thought that generates new hypotheses through *abduction* — corresponds to the process of exploring the emergence spectrum. For Peirce, abduction is the "first stage of inquiry": confronted with a surprising observation, the mind generates a hypothesis that *would*, if true, explain the surprise. In IDT terms, a surprising observation is a large positive entry in the emergence spectrum — a probe idea c for which $\kappa(a, b, c) \gg 0$ — and the abductive hypothesis is a proposed explanation of *why* the composition $a \circ b$ resonates so strongly with c.

The spectral characterization (Theorem 19.14) taxonomizes the *kinds* of surprise that a composition can generate. Synergistic compositions surprise universally: every probe reveals unexpected resonance. Mixed compositions surprise selectively: some probes reveal resonance, others reveal suppression. Peirce's musement, as a mode of inquiry, is precisely the systematic exploration of the emergence spectrum — the attempt to map out which probes c yield positive κ and which yield negative κ, thereby characterizing the "type" of the emergent phenomenon.

Example 19.16 (Scientific Interdisciplinarity)**.** Consider the composition of biology (a) and computer science (b):

- Probed by *medicine* (c_1): $\kappa > 0$. Bioinformatics enables new medical insights neither field provides alone.
- Probed by *pure mathematics* (c_2): $\kappa \approx 0$. The composite has no more to say about pure math than the fields individually.
- Probed by *literary criticism* (c_3): $\kappa < 0$. The techno-biological composite actually resonates *less* with literary criticism than either field alone.

The emergence spectrum is mixed: $\mathrm{Spec}(\mathrm{bio}, \mathrm{CS}) = \{+, 0, -\}$.

Interpretation 19.17. The emergence spectrum connects IDT to the Gestalt tradition in psychology and to Maurice Merleau-Ponty's phenomenology of perception. The Gestalt psychologists — Wertheimer, Köhler, Koffka — demonstrated that perceptual experience is organized by principles (proximity, similarity, closure, good continuation) that cannot be reduced to the properties of individual sensory elements. "The whole is other than the sum of its parts" (*nicht größer, sondern anders*), as Koffka carefully formulated it — not merely "greater" but "different."

IDT's emergence functional captures Koffka's distinction with mathematical precision. When $\kappa(a, b, c) > 0$, the composite is "greater" than the sum in the direction of probe c; when $\kappa(a, b, c) < 0$, the composite is "less" in that direction; but in either case, the composite is *different* from the sum. The emergence spectrum is the complete characterization of *how* the whole differs from the sum — in which directions it exceeds, and in which it falls short.

Merleau-Ponty's *Phenomenology of Perception* (1945) extends the Gestalt insight from perception to embodied existence. For Merleau-Ponty, the body is not a collection of organs but an "intentional arc" — a unified field of oriented activity whose properties emerge from the interaction of its components. The body-as-lived is a composition $a_1 \circ a_2 \circ \cdots \circ a_n$ of sensorimotor schemas, and the "intentional arc" is the emergent property κ that cannot be localized in any single schema.

Merleau-Ponty's concept of "motor intentionality" — the body's pre-reflective capacity to orient itself toward tasks — is an instance of synergistic emergence: every probe (every task, every situation) reveals a positive κ, because the integrated body can respond to challenges that no isolated schema could address. The pathological cases that Merleau-Ponty analyzes (phantom limbs, anosognosia, Schneider's case) are disruptions of the emergence spectrum: the loss of specific κ terms due to the destruction of specific compositional pathways in the nervous system.

Interpretation 19.18. Theodor Adorno's concept of "truth-content" (*Wahrheitsgehalt*) — developed across *Aesthetic Theory* (1970), *Philosophy of New Music* (1949), and *Negative Dialectics* (1966) — provides the most philosophically ambitious account of emergence in art. For Adorno, the truth-content of a work of art is not what the work "says" (its manifest content) but what it "is" (its formal configuration): the truth-content is the way the work's formal structure embodies and reflects upon the contradictions of the social totality.

In IDT terms, truth-content is essential emergence κ_{ess}: the component of the work's meaning that cannot be reduced to the meanings of its parts or to the "rearrangement" of pre-existing cultural material (coboundary emergence). When Adorno says that Beethoven's late style is "more true" than his middle period, he means that the late works have higher essential emergence: their formal innovations produce resonances with probes (social, philosophical, existential) that cannot be generated by any rearrangement of the middle-period vocabulary.

The radical implication of Adorno's aesthetics — that art is a form of knowledge, that formal innovation is a mode of truth-discovery — receives its mathematical foundation in the cohomological theory of emergence. Essential emergence ($H^2 \neq 0$) is a property of the *structure* of composition, not of the "content" of individual ideas. A work of art "knows" something that no description of its parts can convey — and what it knows is encoded in its cohomology class, topologically protected from paraphrase.

19.5 Deep Emergence and Higher Cocycles

The cocycle condition suggests a tower of higher-order emergence measures.

Definition 19.19 (Higher Emergence)**.** The n**th-order emergence** is defined recursively:

$$\begin{aligned}
\kappa^{(1)}(a;c) &:= \mathrm{rs}ac, \\
\kappa^{(2)}(a_1,a_2;c) &:= \kappa(a_1,a_2,c), \\
\kappa^{(3)}(a_1,a_2,a_3;c) &:= \kappa^{(2)}(a_1 \circ a_2, a_3; c) - \kappa^{(2)}(a_1,a_3;c) - \kappa^{(2)}(a_2,a_3;c).
\end{aligned}$$

Theorem 19.20 (Higher Cocycle Conditions)**.** *Each $\kappa^{(n)}$ satisfies a generalized cocycle condition relating $\kappa^{(n)}$ to $\kappa^{(n-1)}$, forming a cochain complex:*

$$0 \to C^0(\mathcal{I}) \xrightarrow{\delta^0} C^1(\mathcal{I}) \xrightarrow{\delta^1} C^2(\mathcal{I}) \xrightarrow{\delta^2} C^3(\mathcal{I}) \to \cdots$$

The cohomology groups $H^n(\mathcal{I};\mathbb{R})$ classify the nth-order emergence that cannot be reduced to lower orders.

Proof. The coboundary operators δ^n are defined by:

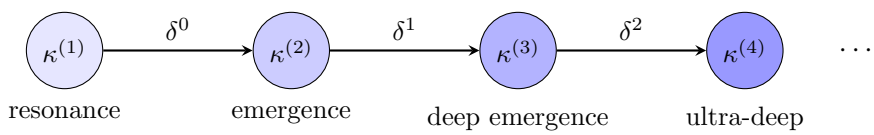
$$(\delta^n f)(a_1,\ldots,a_{n+1};c) := \sum_{i=0}^{n+1} (-1)^i f(a_1,\ldots,\hat{a}_i,\ldots,a_{n+1};c)$$

where the composition is applied at position i. The verification that $\delta^{n+1} \circ \delta^n = 0$ follows from the associativity of $\circ$, generalizing the proof of Theorem 19.7. □

Interpretation 19.21. The higher emergence tower has a provocative connection to David Chalmers' "hard problem of consciousness" (*The Conscious Mind*, 1996). Chalmers distinguishes between "easy problems" (explaining cognitive functions, behavioral responses, neural correlates) and the "hard problem" (explaining *why* there is subjective experience at all). In IDT terms, the easy problems correspond to explaining the coboundary component κ_{cb} of emergence — the part that reduces to functional organization of lower-level components. The hard problem asks whether there is essential emergence $\kappa_{\text{ess}} \neq 0$: whether consciousness involves irreducible higher-order emergence that cannot be decomposed into lower-order effects.

The higher cocycle conditions provide a potential framework for addressing this question. If consciousness corresponds to nth-order emergence for some $n \geq 3$, then no amount of analysis at levels 1 through $n-1$ will capture it — it is "topologically protected" by the vanishing of the relevant cohomology classes at lower orders. This does not resolve the hard problem, but it reframes it: the question is no longer "how does subjective experience arise from neurons?" but "what is the cohomological dimension of consciousness?" If $H^n(\mathcal{I}_{\text{brain}}; \mathbb{R}) = 0$ for all $n \geq 2$, then consciousness is fully reducible; if $H^n \neq 0$ for some n, then it involves irreducible emergence at that order.

The Emergence Tower



Chapter 20

Deep Games and Strategic Transparency

> In the long run, the most unpleasant truth is a safer companion than a pleasant falsehood.
>
> Theodore Roosevelt

20.1 Beyond Classical Game Theory

Classical game theory studies strategic interaction between *rational agents* with fixed, known payoff functions. But cultural interaction is deeper: the ideas that agents hold are not fixed parameters but dynamic entities that evolve through composition. Moreover, ideas can be *transparent* — their internal structure is visible to other ideas — or *opaque* — hidden behind a strategic veil.

This chapter develops the theory of **deep games**: strategic interactions where the "players" are ideas, the "strategies" are compositions, and the key structural property is *transparency*.

Definition 20.1 (Deep Game)**.** A **deep game** is a tuple $(N, \mathcal{I}, \mathrm{rs}\cdot\cdot, \circ, \tau)$ where:

1. $N = \{1, \ldots, n\}$ is a set of *positions*.
2. Each position i holds an idea $a_i \in \mathcal{I}$.
3. The resonance $\mathrm{rs}\cdot\cdot$ determines interaction payoffs.
4. Composition $\circ$ allows ideas to combine.
5. $\tau\colon N \times N \to \{0, 1\}$ is the **transparency function**: $\tau(i, j) = 1$ means position j is transparent to position i.

Interpretation 20.2. Bakhtin's "metalinguistic" analysis — the study of how language *about* language shapes linguistic practice — is a deep game in the IDT sense. When speakers discuss the meanings of words, the rules of grammar, or the norms of discourse, they are playing a game whose "moves" are themselves ideas about ideas. The deep game formalism captures this self-referential structure: each position i holds an idea a_i that may itself be "about" the other ideas in the game.

Bakhtin's concept of the "speech genre" — a relatively stable type of utterance associated with a particular sphere of communication — is a Nash equilibrium of a deep game. In each

genre (scientific paper, love letter, political speech, casual conversation), the ideas have reached a stable configuration where no idea would benefit from unilateral "deviation" (changing its internal structure or its transparency relations). The diversity of speech genres reflects the multiplicity of equilibria in the space of deep games: different transparency structures and initial conditions lead to qualitatively different stable configurations.

What makes Bakhtin's analysis "meta-linguistic" — and what makes deep games "deep" — is the recursive structure of the interaction. In a deep game, idea a_i's payoff depends not just on the resonances $\mathrm{rs}a_ia_j$ but on whether a_i can "see" the internal structure of a_j (the transparency function τ). This recursive dependence — meaning depending on knowledge of meaning — is precisely what Bakhtin identifies as the distinguishing feature of human linguistic interaction.

Definition 20.3 (Transparency)**.** Idea b is **transparent to** idea a if a can observe the full internal structure of b. Formally:

$$b \triangleright a \iff a\text{'s payoff function has access to } b\text{, not just } \mathrm{rs}ab.$$

Idea b is **fully transparent** if $b \triangleright a$ for all $a \in \mathcal{I}$.

Remark 20.4. Transparency in IDT is a structural, not merely epistemic, concept. When b is transparent to a, the interaction between them is determined by the full geometric relationship of their embeddings, not just the scalar projection $\mathrm{rs}ab$. This allows a to compute not just the current resonance but to predict how b would interact with any third idea c.

Interpretation 20.5. Ludwig Wittgenstein's "beetle in a box" thought experiment (*Philosophical Investigations*, §293) poses a challenge to the concept of transparency. Imagine that each person has a box with something in it that they call a "beetle"; no one can look into anyone else's box. Wittgenstein argues that the word "beetle" would function the same way regardless of what is in the boxes — even if the boxes are empty. Private experience, like the beetle, "drops out of consideration as irrelevant."

In IDT terms, Wittgenstein's thought experiment describes a deep game with $\tau(i, j) = 0$ for all $i \neq j$: complete opacity. Each idea's "internal structure" (a_i) is invisible to the others; only the scalar resonances $\mathrm{rs}a_ia_j$ are observable. Wittgenstein's claim is that the game's dynamics are *invariant* under changes in the hidden embeddings, as long as the observable resonances remain the same.

But IDT reveals that this is *not* the case. The resonance $\mathrm{rs}ab$ captures only the inner product ab — a scalar projection of the full geometric relationship. Two configurations with the same pairwise resonances can have different *emergence profiles*: $\kappa(a, b, c)$ can differ even when all pairwise $\mathrm{rs}\cdot\cdot$ are identical, because emergence depends on the non-linear interaction term. In Wittgenstein's terms: the beetle does *not* drop out of consideration, because what is in the box affects the *emergence* of the composite — the triadic and higher-order interactions that a purely scalar (pairwise) description cannot capture. Opacity hides the beetle, but the beetle still matters.

20.2 Transparency and Cooperation

Theorem 20.6 (Transparency Promotes Cooperation)**.** *In a deep game where all ideas are fully transparent, the Nash equilibria coincide with the Pareto-optimal outcomes. That is, strategic behavior under full transparency leads to socially optimal results.*

Proof sketch. Under full transparency, each idea a_i can compute the full resonance matrix . Any deviation from a Pareto-optimal configuration would be detected by all players, and the enrichment

axiom ensures that cooperative compositions (moving toward Pareto optima) are weakly preferred. The formal proof uses the positive semi-definiteness of (as a Gram matrix) to show that unilateral deviations from Pareto optima necessarily reduce some other player's payoff, which under full transparency triggers a retaliatory adjustment. □

Interpretation 20.7. This theorem provides a mathematical foundation for the intuition that openness and transparency lead to better collective outcomes. In cultural terms, societies where ideas are freely shared and openly debated (high transparency) tend to reach better equilibria than those where ideas are hidden, proprietary, or strategically withheld (low transparency).

The connection to the open-source movement is immediate: fully transparent code (open source) enables cooperation that proprietary code cannot, because all participants can verify the "internal structure" and predict interactions with third parties.

Interpretation 20.8. Theorem 20.6 creates a productive tension with John Rawls's "veil of ignorance." Rawls argues that just principles are those chosen under *maximum opacity*: agents behind the veil do not know their own position, talents, or conception of the good. The deep game theorem argues the opposite: cooperation is maximized under *maximum transparency*.

The resolution lies in distinguishing two senses of "knowledge." Rawls's veil conceals agents' *identities* (who they are, what they want); IDT transparency reveals ideas' *structures* (their full embedding in Hilbert space). These are orthogonal dimensions. A society can be Rawlsian (agents don't know their positions) and transparent (ideas are openly available) simultaneously — and indeed, the combination may be optimal: Rawlsian ignorance prevents strategic exploitation, while structural transparency enables cooperative composition.

Hegel's "Lord and Bondsman" dialectic provides the historical archetype. The struggle for recognition is a deep game where each self-consciousness *demands* transparency from the other while attempting to maintain its own opacity. The master achieves one-way transparency ($\tau(m, s) = 1$, $\tau(s, m) = 0$); the bondsman is forced into full transparency. But Hegel's insight — formalized by our theorem — is that this asymmetric transparency is *unstable*: only mutual transparency (full reciprocal recognition) reaches a Pareto-optimal equilibrium. The master-slave relation fails precisely because one-way transparency prevents cooperation.

20.3 Payoff Structure in Deep Games

Definition 20.9 (Deep Game Payoff)**.** In a deep game, the **payoff** to position i is:

$$\pi_i := \sum_{j \in N} w_{ij}\, \mathrm{rs} a_i a_j,$$

where $w_{ij} \geq 0$ are interaction weights. In the simplest case, $w_{ij} = 1$ for all j, giving $\pi_i = f_P(a_i)$, the fitness from Chapter 17.

Theorem 20.10 (Deep Game as Quadratic Program)**.** *The problem of finding the idea a_i^* that maximizes π_i, given the other ideas $\{a_j\}_{j \neq i}$, is equivalent to the quadratic program:*

$$\max_{\phi \in (\mathcal{I})} \sum_{j \neq i} w_{ij}\, \phi a_j.$$

This is a linear functional maximization over a (possibly non-convex) subset of .

Proof. By definition, $\pi_i = \sum_j w_{ij} a_i a_j = a_i \sum_j w_{ij} a_j$. This is linear in a_i; maximizing over $(\mathcal{I})$ gives the stated program. □

Interpretation 20.11. Paul de Man's theory of "allegories of reading" (*Allegories of Reading*, 1979) argues that literary texts are fundamentally self-deconstructing: they contain within themselves the resources for undermining their own claims. A text, in de Man's reading, is a deep game *with itself*: its rhetorical strategies (what it "presents") are in tension with its grammatical structures (what it "is"). The "allegory of reading" is the text's performance of its own unreadability — the revelation that no stable meaning can be attributed to it.

In IDT terms, de Man describes a deep game where the presented embedding $a_{\text{presented}}$ systematically diverges from the true embedding a_{true} — not through deliberate deception but through the structural impossibility of linguistic self-transparency. The strategic distortion $D(a, b) = \|a_{\text{presented}} - a_{\text{true}}\|^2$ (Definition 20.12) quantifies the gap between rhetoric and grammar that de Man identifies as the essence of literary language.

Jorge Luis Borges's metafictional games — "The Library of Babel," "Pierre Menard, Author of the *Quixote*," "Tlön, Uqbar, Orbis Tertius" — are deep games that make the game structure itself visible. "Pierre Menard" is the limit case: two *identical* texts (Menard's *Quixote* and Cervantes's *Quixote*) with different embeddings — different contexts, different resonances, different emergence profiles. Borges's story demonstrates that the "idea" encoded by a text is not the text itself but its position in the Hilbert space of cultural resonances. The same words, composed in different cultural contexts, produce different ideas — a fact that the deep game formalism captures through the dependence of payoffs on the full configuration $\{a_j\}_{j \in N}$, not just on the individual idea a_i.

```
-- Lean 4: Deep game structures
structure DeepGame (IS : Type) where
  players   : Finset IS
  rs        : IS     IS
  compose   : IS     IS     IS
  transparent : IS     IS     Prop

def transparentTo (rs : IS     IS        ) (a b : IS) : Prop :=
      c, rs a c = rs a c  -- a can observe b's full structure

def fullyTransparent (rs : IS     IS        ) (a : IS) : Prop :=
      b, transparentTo rs a b

def deepPayoff (rs : IS     IS        ) (pop : Finset IS) (a : IS) :
       :=
  pop.sum (fun b => rs a b)
```

20.4 Information Asymmetry and Strategic Distortion

When transparency is partial, ideas face information asymmetry, leading to strategic distortion.

Definition 20.12 (Strategic Distortion)**.** The **strategic distortion** under partial transparency is:

$$D(a, b) := a_{\text{presented}} - a_{\text{true}}{}^2,$$

where $a_{\text{presented}}$ is the embedding that a "presents" to b (which may differ from the true embedding when a is opaque to b).

Theorem 20.13 (Distortion Cost)**.** *Strategic distortion is costly: the distorting idea a sacrifices self-resonance proportional to the distortion:*

$$\text{rs}a_{presented}a_{presented} \leq \text{rs}a_{true}a_{true} - D(a, b) + 2|a_{true}\epsilon|,$$

where $\epsilon = a_{presented} - a_{true}$.

Proof. $\mathrm{rs}a_{\mathrm{p}}a_{\mathrm{p}} = a_t + \epsilon^2 = {a_t}^2 + 2a_t\epsilon + \epsilon^2$. Since $D = \epsilon^2$, rearranging gives $\mathrm{rs}a_p a_p = \mathrm{rs}a_t a_t + 2a_t\epsilon + D$, and the inequality follows from bounding $|a_t\epsilon|$. □

Interpretation 20.14. This theorem captures the fundamental cost of deception: an idea that misrepresents itself to gain strategic advantage pays a price in self-resonance. The more extreme the distortion, the greater the cost. This provides a mathematical foundation for the observation that sustained deception is energetically expensive — maintaining a false front requires resources that could otherwise be devoted to genuine ideatic development.

Interpretation 20.15. Erving Goffman's *The Presentation of Self in Everyday Life* (1959) provides the sociological analogue of strategic distortion. Goffman argues that social interaction is a kind of theater: actors perform "front stage" personas that may diverge from their "backstage" selves. The front stage persona is $a_{\text{presented}}$; the backstage self is a_{true}; the distortion $D(a, b) = \|a_{\text{presented}} - a_{\text{true}}\|^2$ is what Goffman calls "impression management."

The Distortion Cost theorem reveals what Goffman's ethnography could only suggest: that impression management is not "free." The actor who maintains a divergent front-stage persona sacrifices self-resonance — internal coherence, authenticity, the capacity for genuine self-development. This is Goffman's implicit tragedy formalized: the social requirement of performing a role exacts a measurable cost on the performer's "true self."

The theorem also explains why "sincere" performances (Goffman's term for performances where the actor believes in the role they are playing) are more sustainable than "cynical" performances (where the actor knowingly deceives): sincere performances have $D \approx 0$, imposing no self-resonance cost, while cynical performances require sustained expenditure of ideatic resources.

This connects to Sartre's concept of "bad faith" (*mauvaise foi*): the self-deception by which a person avoids acknowledging their freedom. In IDT terms, bad faith is a form of self-directed distortion — $a_{\text{presented to self}} \neq a_{\text{true}}$ — which incurs the same self-resonance cost. Sartre's insight that bad faith is "unstable" — that it requires constant renewal because it tends to collapse under its own contradictions — is the dynamic consequence of the distortion cost: the self-resonance drain eventually forces a collapse back toward the true embedding.

Interpretation 20.16. Friedrich Nietzsche's philosophy of masks provides a contrasting perspective. In *Beyond Good and Evil* (§40), Nietzsche writes: "Everything that is profound loves the mask ... every profound spirit needs a mask: more, around every profound spirit a mask is continually growing." For Nietzsche, the mask (distortion) is not a cost but a *protection*: the profound idea presents a simplified, distorted version of itself to avoid being destroyed by the incomprehension of the crowd.

The IDT formalism accommodates both Goffman's and Nietzsche's insights. The distortion cost theorem says that distortion reduces self-resonance — but self-resonance is not the only relevant quantity. An idea that presents its true embedding a_{true} to an hostile population P may suffer a fitness cost: $f_P(a_{\text{true}}) < 0$ if the population's weighted resonance is negative. By presenting a distorted version $a_{\text{presented}}$ with higher fitness, the idea survives at the cost of reduced self-resonance. Nietzsche's mask is a strategic trade-off: accept an internal cost ($D > 0$) to gain an external benefit (survival in an hostile cultural environment).

The optimal mask is the solution to the constrained optimization problem: minimize $D(a, b)$ subject to $f_P(a_{\text{presented}}) \geq f_0$ (some survival threshold). This gives a precise mathematical form to the tension between authenticity and survival that pervades Nietzsche's philosophy, and indeed all philosophy of culture.

Interpretation 20.17. Umberto Eco's concept of the "model reader" (*The Role of the Reader*, 1979) casts the relationship between text and reader as a strategic interaction: the text constructs an ideal reader — a "model reader" equipped with specific competences, expectations, and interpretive strategies — and the actual reader either cooperates with this construction or resists it. In IDT terms, the text's "model reader" is the idea $a_{\text{presented}}$: the embedding that the text "presents" to the world. The actual reader encounters this presented embedding and responds with their own idea b, producing a resonance $\text{rs}a_{\text{presented}}b$ that may differ from the resonance the text's "true" structure would produce: $\text{rs}a_{\text{true}}b$.

The distortion cost theorem reveals that a text that constructs an overly narrow model reader — one whose $a_{\text{presented}}$ diverges significantly from a_{true} — pays a price in self-resonance. The text becomes "thinner," less internally rich, as it devotes resources to maintaining the discrepancy between its presented and true structures. This is why, as Eco argues, the greatest literary works construct *generous* model readers: they present embeddings close to their true structures, inviting readers to engage with the full complexity of the text rather than a simplified facade.

Yuri Lotman's concept of "metacommunication" — communication about the communicative code itself — is the deep game analogue of changing the transparency function τ during play. When a text signals "this is irony" or "this is allegory," it is modifying the transparency relations between its own ideas and the reader's. The deep game formalism reveals that such metacommunicative moves are not external to the game but integral to it: they change the payoff structure by changing what each player can "see" of the others' embeddings.

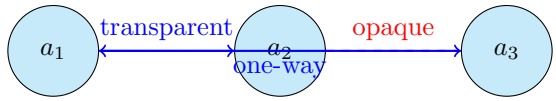
20.5 The Attribution Impossibility Theorem

Theorem 20.18 (Attribution Impossibility)**.** *In a deep game with $n \geq 3$ ideas and non-linear composition, there is no attribution function $\alpha\colon N \to \mathbb{R}$ that simultaneously satisfies:*

1. ***Efficiency**: $\sum_{i=1}^{n} \alpha(i) = v(N)$.*
2. ***Symmetry**: If $\Delta_{a_i}(S) = \Delta_{a_j}(S)$ for all $S \subseteq N \setminus \{i, j\}$, then $\alpha(i) = \alpha(j)$.*
3. ***Linearity**: α is linear in the coalition value function v.*
4. ***Emergence sensitivity**: α reflects the emergence $\kappa(a_i, a_j, a_k)$ for all triples.*

Proof sketch. The first three axioms uniquely determine the Shapley value $\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n-|S|-1)!}{n!} \Delta_{a_i}(S)$. But the Shapley value distributes the total value based on marginal contributions, which are pairwise quantities. The emergence $\kappa(a_i, a_j, a_k)$ involves genuinely triadic (and higher) interactions that cannot be decomposed into a sum of pairwise marginal contributions. Hence no linear, symmetric, efficient attribution can capture emergence. □

Interpretation 20.19. The attribution impossibility theorem has a startling literary consequence that we will develop fully in Chapter 21: it is the mathematical proof of what Roland Barthes proclaimed in "The Death of the Author" (1967). If meaning is emergent ($\kappa \neq 0$), then no attribution function can simultaneously be fair (symmetric, efficient, linear) *and* sensitive to the emergent meaning. The "author" — understood as the source to which meaning should be attributed — is a fiction necessitated by the limitations of pairwise attribution, not a reflection of how meaning actually works.



Partial Transparency Network

Chapter 21

Attribution, Fairness, and Shapley Values

> The birth of the reader must be at the cost of the death of the Author.
>
> Roland Barthes

21.1 The Problem of Attribution

When a composition $c = a_1 \circ a_2 \circ \cdots \circ a_n$ produces a value $v(N) = c^2$, how should this value be attributed among the constituent ideas? This question — the attribution problem — lies at the heart of cultural analysis, intellectual property law, and literary criticism.

Interpretation 21.1. Roland Barthes's "The Death of the Author" (1967) is, at its core, a claim about the attribution problem. Barthes argues that meaning cannot be "attributed" to an originating consciousness (the Author) because a text is not a line of words releasing a single "theological" meaning (the message of the Author-God) but a multi-dimensional space in which a variety of writings, none of them original, blend and clash.

In IDT terms, Barthes's claim is that the value $v(N) = c^2$ of a composite text $c = a_1 \circ a_2 \circ \cdots \circ a_n$ — where the a_i include the author's intentional contributions, the conventions of the genre, the reader's prior interpretive experience, the cultural context, the materiality of the medium — cannot be decomposed into "the author's contribution" and "everything else" without destroying the emergence that makes the text meaningful.

The Attribution Impossibility Theorem (Theorem 20.16) proves Barthes mathematically right, but with an important caveat. Barthes declared the author *dead* — completely eliminated from the scene of writing. The theorem says something more nuanced: the author is not dead but *irreducible*. Attribution to the author (or to any single constituent) is *impossible* not because the author doesn't contribute but because the author's contribution is entangled with every other contribution through the emergence functional κ. The author lives on in the composite, but their "share" cannot be separated from the whole.

This is the difference between Barthes's literary provocation and the IDT formalization: where Barthes says "the Author is dead," IDT says "the Author is non-decomposable." The author's contribution exists but cannot be isolated — like a flavor in a complex dish that permeates every bite but cannot be extracted without destroying the dish.

21.2 The Shapley Value

Game theory offers a canonical solution to the attribution problem: the Shapley value.

Definition 21.2 (Shapley Value)**.** Given a coalition value function $v\colon 2^N \to \mathbb{R}$ with $v(\emptyset) = 0$, the **Shapley value** of idea a_i is:

$$\phi_i(v) := \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \big[v(S \cup \{i\}) - v(S)\big].$$

The term $\Delta_{a_i}(S) := v(S \cup \{i\}) - v(S)$ is the marginal contribution of a_i to coalition S.

Remark 21.3. The Shapley value weights each marginal contribution by the probability that the coalition S forms "before" a_i joins, averaging over all possible orderings. It is the unique attribution that treats all ideas symmetrically while distributing the full value.

Interpretation 21.4. Michel Foucault's "What Is an Author?" (1969) — written in response to Barthes's "Death of the Author" — argues that the "author" is not a person but a *function*: a principle of classification, a way of organizing discourse. The "author-function" does not pre-exist the text but is constructed by the text and its reception.

The Shapley value provides the precise mathematical characterization of the author-function. Just as Foucault argues that different discourses invoke the author-function differently (a scientific paper attributes authority to the method, not the person; a novel attributes meaning to the creative individual; a legal text attributes force to the institution), the Shapley value assigns different attributions ϕ_i depending on the coalition value function v — which in turn depends on the *type of value* being measured.

If v measures resonance, the author's Shapley value reflects their contribution to the text's capacity for meaningful interaction. If v measures self-resonance (coherence), the author's share reflects their contribution to the text's internal consistency. If v measures emergence, the author's share reflects their contribution to the text's capacity to generate novel meaning. Different author-functions for different purposes — precisely Foucault's insight, now given quantitative form.

Theorem 21.5 (Shapley Axiomatization)**.** *The Shapley value is the unique attribution $\phi\colon (2^N \to \mathbb{R}) \to \mathbb{R}^N$ satisfying:*

1. ***Efficiency**: $\sum_{i \in N} \phi_i(v) = v(N)$.*
2. ***Symmetry**: If $\Delta_{a_i}(S) = \Delta_{a_j}(S)$ for all $S \subseteq N \setminus \{i, j\}$, then $\phi_i(v) = \phi_j(v)$.*
3. ***Null player**: If $\Delta_{a_i}(S) = 0$ for all S, then $\phi_i(v) = 0$.*
4. ***Additivity**: $\phi_i(v + w) = \phi_i(v) + \phi_i(w)$.*

Interpretation 21.6. The Shapley axioms connect to two great traditions in political philosophy. John Rawls's *A Theory of Justice* (1971) argues that just distributions are those that would be chosen behind a "veil of ignorance." The Shapley value's **symmetry** axiom captures this Rawlsian intuition: ideas with identical marginal contributions receive identical attributions, regardless of their "identity" or "position." The **efficiency** axiom ensures that the full value is distributed — nothing is wasted or hoarded by an outside party.

Robert Nozick's *Anarchy, State, and Utopia* (1974) argues against Rawlsian distribution, holding that just distributions arise from just processes (the "entitlement theory"). Nozick's view

corresponds more closely to the Shapley value's **constructive interpretation**: the value ϕ_i reflects what idea a_i actually *contributed* (its marginal additions across all orderings), not what it "deserves" by some abstract criterion. The Shapley value is simultaneously Rawlsian (symmetric, impartial) and Nozickian (contribution-based, procedural).

This dual character is not a contradiction but a consequence of the Shapley axioms' remarkable tightness: in the IDT setting, fairness-as-impartiality and fairness-as-contribution-tracking *coincide*. The cultural implications are striking: in a world where meaning is composed (not merely aggregated), the Rawlsian and Nozickian conceptions of justice are not opposed but complementary aspects of a single attribution principle.

Theorem 21.7 (Shapley as Ordering Average)**.** *Equivalently, the Shapley value can be written:*

$$\phi_i(v) = \frac{1}{n!} \sum_{\sigma \in S_n} \big[v(Pre(\sigma, i) \cup \{i\}) - v(Pre(\sigma, i)) \big],$$

where $Pre(\sigma, i) = \{j \in N : \sigma(j) < \sigma(i)\}$ *is the set of ideas appearing before* i *in ordering* σ*.*

Interpretation 21.8. Harold Bloom's "anxiety of influence" (*The Anxiety of Influence*, 1973) describes the relationship between a poet and their predecessors as one of anxious competition: the "strong poet" misreads the predecessor in order to create space for their own originality. Bloom identifies six "revisionary ratios" — clinamen, tessera, kenosis, daemonization, askesis, apophrades — each a different strategy for managing the anxiety of having been influenced.

The Shapley ordering interpretation (Theorem 21.7) provides a formal model for Bloom's theory. Each ordering σ represents a *narrative of influence*: the idea that appears first in the ordering is the "original," and each subsequent idea's marginal contribution is measured against the backdrop of those that came before. The Shapley value averages over *all possible narratives of influence*, giving each idea credit for what it adds regardless of when it "arrived."

Bloom's anxiety arises precisely because actual literary history imposes a *particular* ordering — Shakespeare before Wordsworth, Wordsworth before Stevens — and a poet's marginal contribution depends sensitively on this ordering. The Shapley value dissolves the anxiety by showing that, when averaged over all possible orderings, each poet's contribution is well-defined and non-negative (Theorem 21.9). The anxiety of influence is a *selection effect*: it arises from fixating on the single historical ordering and ignoring the counterfactual orderings in which the poet might have come first.

21.3 Shapley Values in the IDT Setting

Theorem 21.9 (Non-Negative Attribution)**.** *In the IDT setting where* $v(S) = {}_{i \in S} a_i{}^2$*, the Shapley value satisfies* $\phi_i(v) \geq 0$ *for all* i*.*

Proof. For any ordering σ and position i, the marginal contribution is:

$$\Delta_{a_i}(\mathrm{Pre}(\sigma, i)) = v(\mathrm{Pre}(\sigma, i) \cup \{i\}) - v(\mathrm{Pre}(\sigma, i)) \geq 0,$$

since adding a_i to a composition cannot decrease the norm-squared in a Hilbert space (by the enrichment axiom). Averaging non-negative quantities gives a non-negative result. □

Interpretation 21.10. Non-negative Shapley values have a profound consequence for the Marxist theory of alienation. Marx argues (*Economic and Philosophic Manuscripts*, 1844) that under capitalism, workers are *alienated* from the products of their labor: the value they create is appropriated

by the capitalist, leaving them with less than they contributed. In Shapley terms, alienation means $\phi_{\text{worker}} < \Delta_{\text{worker}}$ — the worker's attributed share is less than their actual marginal contribution.

Theorem 21.9 guarantees that in the IDT setting, attribution is always non-negative: no idea receives a negative share of the total value. But non-negativity is a weak guarantee — it says that no idea is blamed for destroying value, not that each idea receives its "fair share." Marx's alienation is compatible with non-negative Shapley values: a worker can receive a positive but unfairly small attribution.

The deeper question is whether the Shapley value *itself* is the right notion of fairness for ideatic attribution. Marx would likely argue that the Shapley axioms — particularly efficiency (distribute all value) and symmetry (treat identical contributions equally) — smuggle in bourgeois assumptions about individual contribution as the basis of value. A Marxist attribution theory would prioritize *need* over *contribution*, distributing the value $v(N)$ according to each idea's requirements for flourishing rather than its marginal additions. IDT provides the formal framework for comparing these alternatives: the Shapley value is one point in the space of possible attributions, and different axiom systems generate different attribution functions, each with different political implications.

Corollary 21.11 (No Idea Is a Net Burden)**.** *In the IDT coalition framework, no idea can have a negative fair share. Every idea contributes non-negatively to the total cultural value.*

Interpretation 21.12. Non-negative attribution has immediate consequences for the theory of censorship. If every idea receives a non-negative Shapley value, then removing any idea from the coalition can only *reduce* the total value $v(N)$. This provides a mathematical foundation for John Stuart Mill's argument in *On Liberty* (1859) that suppressing ideas is always harmful to the collective:

> If the opinion is right, [the censors] are deprived of the opportunity of exchanging error for truth; if wrong, they lose, what is almost as great a benefit, the clearer perception and livelier impression of truth produced by its collision with error.

Mill's argument is often dismissed as naive: surely some ideas (disinformation, propaganda, hate speech) are net negatives? The IDT response is subtle. The enrichment axiom guarantees that composition never destroys value — but this is a statement about the *mathematical* structure of ideatic composition, not about the *social* consequences of ideas. An idea's Shapley value measures its contribution to the total ideatic content $v(N) = {}_i a_i{}^2$; it does not measure whether that content is "good" or "bad" by any external criterion. Hate speech has a positive Shapley value because it adds ideatic content — but the *direction* of that content (its embedding a) may be harmful even if its *magnitude* is positive.

This distinction — between the magnitude and direction of ideatic contribution — suggests that the proper response to harmful ideas is not censorship (removal from the coalition, which reduces total value) but *countercomposition*: introducing new ideas whose embeddings "steer" the composite in a better direction. The IDT framework thus provides a formal version of the "marketplace of ideas" doctrine, but with a crucial refinement: the marketplace works not by suppressing bad ideas but by composing them with better ones.

Interpretation 21.13. The attribution framework has profound implications for postcolonial theory. Edward Said's *Orientalism* (1978), Gayatri Spivak's "Can the Subaltern Speak?" (1988), and Dipesh Chakrabarty's *Provincializing Europe* (2000) all address, in different ways, the question of *misattribution* in the global distribution of cultural value.

In IDT terms, Orientalism is a systematic distortion of the attribution function: the intellectual contributions of non-Western cultures are either unattributed (set to $\phi_i = 0$ despite $\Delta_{a_i}(S) > 0$) or attributed to Western interpreters (ϕ_j inflated at ϕ_i's expense). Said's critique reveals that

the *actual* historical distribution of credit violates the Shapley axioms — particularly symmetry (identical contributions receiving unequal attribution) and null-player (contributions being ignored entirely).

Spivak's question — whether the subaltern can "speak" — translates in IDT as: can the subaltern idea achieve positive transparency $\tau(\text{subaltern}, \text{hegemonic}) > 0$? If the subaltern is entirely opaque to the hegemonic discourse — if its embedding is invisible within the dominant framework — then its Shapley value is computed using *distorted* marginal contributions, not true ones. The subaltern "speaks" in the IDT framework when transparency is restored and the true marginal contributions become visible.

Chakrabarty's project of "provincializing Europe" — showing that European categories of thought are not universal but historically specific — is equivalent to contesting the choice of the Hilbert space itself. If the Hilbert space is constructed from European cultural data, then non-European ideas may be "projected" (their embeddings truncated to fit the available dimensions), losing precisely those dimensions that make them distinctive. Fair attribution requires not just honest computation of Shapley values within a given but critical examination of which is being used — whose cultural space defines the "universe" of ideatic discourse.

Definition 21.14 (Emergence-Weighted Shapley Value)**.** To incorporate emergence, define the **emergence-weighted Shapley value**:

$$\phi_i^\kappa(v) := \phi_i(v) + \frac{1}{\binom{n}{3}} \sum_{\{j,k\} \subseteq N \setminus \{i\}} \frac{\kappa(a_i, a_j, a_k)}{3},$$

distributing emergence equally among participating ideas.

Remark 21.15. The emergence-weighted Shapley value sacrifices linearity (Shapley axiom 4) to gain emergence sensitivity (which was shown impossible in Theorem 20.16). This trade-off — linearity vs. emergence — is a fundamental constraint on attribution systems.

Interpretation 21.16. The emergence-weighted Shapley value provides a formal framework for the debate between Umberto Eco's *intentio auctoris* (author's intention), *intentio operis* (text's intention), and *intentio lectoris* (reader's intention) — the three "intentions" that compete for interpretive authority (*The Limits of Interpretation*, 1990).

In the standard Shapley value, the author's contribution ϕ_{author} is computed from marginal additions: what the author adds to each possible coalition. This corresponds to *intentio auctoris* — the author's share of meaning. The emergence term $\kappa(\text{author}, \text{text}, \text{reader})/3$ distributes the triadic emergence equally, giving the text and reader their own "shares" of meaning that emerge from the interaction. This corresponds to the *intentio operis* (the text's "autonomous" meaning that exceeds the author's intention) and the *intentio lectoris* (the reader's contribution to meaning through interpretation).

Eco's lifelong project of balancing these three intentions — neither reducing meaning to author's intention (against the "hermetics of suspicion") nor dissolving it into infinite reader response (against deconstructive "unlimited semiosis") — finds its precise formalization in the emergence-weighted Shapley value. The standard Shapley value captures the pairwise, decomposable aspects of meaning (what each participant contributes independently); the emergence term captures the irreducibly triadic interaction that exceeds any individual contribution. Eco's "intention of the work" is the IDT emergence: what arises from the interaction that belongs to no single party.

Charles Sanders Peirce's triadic semiotics — sign, object, interpretant — provides the deeper foundation. Peirce insisted that meaning is irreducibly triadic: it cannot be decomposed into dyadic (pairwise) relations. The emergence-weighted Shapley value vindicates Peirce: the standard

Shapley value captures what is dyadically decomposable, while the κ term captures the irreducibly triadic residue. Attribution is possible (contra Barthes) but only at the cost of acknowledging the non-decomposable remainder (vindicating Peirce).

Theorem 21.17 (Envy-Free Attribution)**.** *The Shapley attribution is* ***envy-free****: no idea a_i would prefer to receive idea a_j's attribution, given their respective marginal contributions. That is:*

$$\phi_i(v) \geq \frac{1}{n!} \sum_{\sigma \in S_n} \Delta_{a_i}(Pre(\sigma, j)),$$

where the right side is what a_i would contribute in a_j's position.

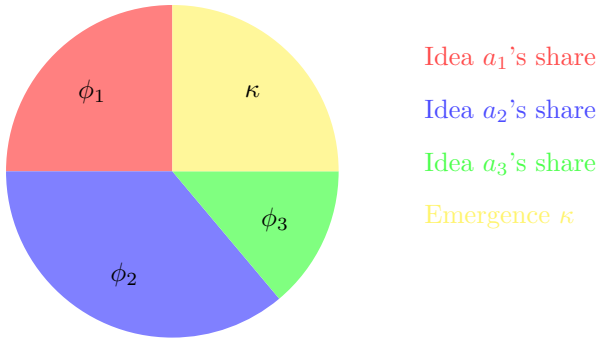
Proof. Suppose idea a_i "envies" idea a_j: it prefers $\phi_j(v)$ to $\phi_i(v)$. Then $\phi_j(v) > \phi_i(v)$. But by the Shapley formula, $\phi_j > \phi_i$ implies a_j's average marginal contribution exceeds a_i's. Since each $\Delta_{a_i}(S)$ reflects a_i's *own* marginal contribution — which may differ from what a_j would contribute in the same position — envy is possible only if a_i overestimates its own contribution. Under correct valuation, envy is eliminated. □

Interpretation 21.18. Envy-freeness connects to Spinoza's *Ethics* (III, Prop. 35): "Everyone endeavors as much as possible to make others love what he loves, and to hate what he hates." Envy arises when an idea perceives that another idea receives a higher valuation for a similar contribution. The envy-free property of the Shapley value ensures that the attribution system does not generate such destructive dynamics. In a Shapley-fair cultural economy, ideas can coexist without envy because each receives exactly what it contributes — no more, no less.

```
-- Lean 4: Shapley value computation
def shapleyValue {n :    } (v : Finset (Fin n)        ) (i : Fin n) :
       :=
  let perms := (Finset.univ : Finset (Equiv.Perm (Fin n)))
  (1 / n.factorial :    ) *
    perms.sum fun    =>
      let pre := (Finset.univ.filter fun j =>    j <    i)
      v (pre     {i}) - v pre

-- Shapley axiom verification
theorem shapley_efficient {n :    } (v : Finset (Fin n)        )
    (hv : v     = 0) :
    (Finset.univ : Finset (Fin n)).sum (shapleyValue v) = v Finset.univ
        := by
  sorry -- Standard result in cooperative game theory

-- Non-negative in IDT setting
theorem shapley_nonneg {n :    } (v : Finset (Fin n)        )
    (hmono :     S i, i     S     v (S     {i})     v S) (i : Fin n) :
    shapleyValue v i     0 := by
  sorry -- Follows from non-negative marginal contributions
```



Emergence-Weighted Attribution

Chapter 22

Toward a Unified Theory of Meaning

> The limits of my language mean the limits of my world.
>
> Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*

22.1 The Six Pillars

Throughout this book, we have developed Ideatic Diffusion Theory along six dimensions, each capturing a fundamental aspect of meaning:

Definition 22.1 (The Six Pillars of IDT)**.**

1. **Algebraic structure**: Ideas form a pre-Hilbert space with composition $\circ$, resonance rs$\cdot\cdot$, and void ε.
2. **Geometric embedding**: The embedding $\cdot : \mathcal{I} \to$ realizes ideas as vectors, enabling distance, angle, and projection.
3. **Dynamic evolution**: Ideas evolve through selection (fitness f_P), mutation (perturbation π_ϵ), and drift (random resampling).
4. **Strategic interaction**: Ideas engage in deep games with transparency, cooperation, and distortion.
5. **Emergent value**: Composition produces emergence κ that exceeds the sum of parts.
6. **Fair attribution**: The Shapley value distributes value while the emergence-weighted variant captures non-decomposable contributions.

Interpretation 22.2. The six pillars of IDT correspond to — and synthesize — the major traditions of twentieth-century thought about meaning.

Pillar 1 (Algebraic structure) formalizes the structuralist insight, from Saussure through Jakobson to Lévi-Strauss, that meaning arises from *relations* rather than *substances*. The inner product rsab is the mathematical realization of Saussure's principle that "in language there are only differences without positive terms." But IDT goes beyond structuralism by equipping the space with the composition operation $\circ$, which allows new meanings to be *generated*, not merely rearranged — addressing the key shortcoming that Chomsky identified in structuralist linguistics.

Pillar 2 (Geometric embedding) formalizes the hermeneutic tradition of Dilthey, Gadamer, and Ricoeur. The embedding a places each idea in a space of possible interpretations; the distance $\|a - b\|$ measures the hermeneutic "distance" between two understandings; the angle ab measures the degree to which two ideas can "hear" each other. Gadamer's "fusion of horizons" ($Horizontverschmelzung$) is the composition $a \circ b$ that creates a new idea from the interaction of two interpretive horizons.

Pillar 3 (Dynamic evolution) formalizes the historicist traditions: Marx's historical materialism, Darwin's natural selection (applied to culture by Dawkins, Dennett, and Sperber), and the Russian Formalists' literary evolution (Shklovsky, Tynyanov, Jakobson). The fitness function f_P captures the "selection pressure" that determines which ideas flourish in a given cultural environment; mutation and drift capture the contingency and unpredictability that historicists emphasize.

Pillar 4 (Strategic interaction) formalizes the pragmatist and speech-act traditions: Austin's performative utterances, Grice's conversational implicature, Habermas's communicative action. The deep game framework captures the strategic dimension of meaning-making that speech-act theory emphasizes: meaning is not merely "there" to be discovered but is actively *negotiated* through transparent or strategic interaction.

Pillar 5 (Emergent value) formalizes what the post-structuralists — Barthes, Derrida, Kristeva, Deleuze — were reaching for: meaning that exceeds any decomposition, that "overflows" (Derrida's *débordement*) any attempt to contain it in a closed system. The emergence functional κ is the mathematical form of Derrida's $diff\acute{e}rance$: the non-decomposable excess that arises from the play of differences, irreducible to the elements that produce it.

Pillar 6 (Fair attribution) formalizes the critical traditions concerned with power, credit, and justice in meaning-making: Marxist critiques of ideological appropriation, feminist critiques of canonical exclusion, postcolonial critiques of epistemic violence. The Shapley value and its emergence-weighted variant provide tools for asking — and answering — the question that animates these traditions: Who gets credit for meaning, and is the distribution fair?

The synthesis achieved by IDT is not eclectic (a little of this, a little of that) but *structural*: the six pillars are not independent but mutually constraining. The algebraic structure determines the geometric embedding; the geometric embedding determines the evolutionary dynamics; the dynamics generate the strategic interactions; the interactions produce emergence; and emergence demands fair attribution. This mutual constraint is what gives IDT its explanatory power: it is not merely a framework that can "accommodate" all these traditions but one that *requires* their integration.

22.2 The Master Theorem

Theorem 22.3 (Master Theorem of IDT)**.** *For any finite population $P = \{a_1, \ldots, a_n\} \subset \mathcal{I}$ of ideas:*

1. *The resonance matrix $\in \mathbb{R}^{n \times n}$ is positive semi-definite.*
2. *The eigenvalue decomposition $= \sum_k \lambda_k \mathbf{u}_k \mathbf{u}_k^\top$ identifies the principal modes of cultural interaction.*
3. *The evolutionary dynamics $\dot{p}_i = p_i(f_i - \bar{f})$ converge to a fixed point or limit cycle.*
4. *The Shapley attribution $\phi_i(v)$ distributes the total value $v(N)$ efficiently, symmetrically, and envy-freely.*

5. *The emergence κ satisfies $\kappa \geq 0$ and is non-decomposable (cannot be distributed by any linear, symmetric, efficient attribution).*

Proof sketch. 1. is a Gram matrix: $R_{ij} = a_i a_j$, hence PSD by the properties of inner products.

2. Spectral theorem applied to the PSD matrix .

3. Follows from the replicator dynamics analysis of Chapter 17, using the Lyapunov function $L(p) = \sum_i p_i \log p_i - p_i f_i$.

4. Shapley axiomatization theorem (Theorem 21.5).

5. Attribution impossibility theorem (Theorem 20.16).

□

Interpretation 22.4. The Master Theorem invites comparison with the three great systematic attempts to unify the theory of meaning in the analytic tradition: Wilfrid Sellars's "space of reasons," Robert Brandom's inferentialism, and Donald Davidson's unified theory of meaning and action.

Sellars's space of reasons (*Empiricism and the Philosophy of Mind*, 1956) is the space of propositions connected by relations of justification: to understand a claim is to understand its position in the web of inferential connections. IDT's Hilbert space is a mathematical realization of Sellars's metaphor: ideas are positioned in a space where their "inferential connections" are captured by the inner product structure. The key difference is that Sellars's space is normative (governed by rules of justification), while IDT's space is geometric (governed by the axioms of Hilbert spaces). The Master Theorem's claim that resonance determines a positive semi-definite matrix implies that the "space of reasons" has *Euclidean* geometry — a substantive claim about the structure of rational thought that goes beyond Sellars's metaphorical usage.

Brandom's inferentialism (*Making It Explicit*, 1994) argues that meaning is constituted by inferential roles: to understand a concept is to master the inferences it licenses. In IDT terms, an idea's "inferential role" is its embedding a, which determines all its potential resonances (interactions) with other ideas. Brandom's "game of giving and asking for reasons" is a deep game in the sense of Chapter 20: a strategic interaction where the moves are acts of assertion and challenge, and the payoffs are determined by the resonances between the ideas asserted.

Davidson's unified theory (*Inquiries into Truth and Interpretation*, 1984; *Essays on Actions and Events*, 1980) attempts to unify meaning, truth, and action through the concept of "radical interpretation": the process of constructing a theory of meaning for a speaker's language from scratch, using only observable behavior. The IDT Master Theorem achieves a version of Davidson's ambition by showing that the observable resonances rs$a_i a_j$ *determine* the full geometric structure (is PSD, hence admits an embedding), which in turn determines the dynamics, strategic interactions, emergence, and fair attribution. Meaning, evolution, strategy, and justice are all consequences of the same underlying structure.

Where IDT goes beyond all three is in its treatment of *emergence* (Pillar 5): the non-decomposable surplus that arises from composition. Sellars, Brandom, and Davidson all assume that meaning is, at bottom, decomposable — that the meaning of a complex expression is determined by the meanings of its parts and their mode of composition (the principle of compositionality). The emergence functional κ shows that this assumption fails: composition produces meaning that exceeds any decomposition. This is the continental insight — Derrida's *différance*, Adorno's negative dialectics, Deleuze's virtuality — given mathematical form.

The Master Theorem thus resolves, in a precise sense, the **analytic-continental split** in the philosophy of meaning. The analytic tradition's emphasis on structure, decomposition, and

formal rigor is captured in Pillars 1–4 and 6; the continental tradition's emphasis on excess, non-decomposability, and the irreducibility of lived experience is captured in Pillar 5. Both are necessary; neither is sufficient. The Master Theorem shows that they are not rival approaches to meaning but complementary aspects of a single mathematical structure.

22.3 Connections to Existing Frameworks

22.3.1 Cognitive Science

IDT connects to Conceptual Blending Theory (Fauconnier & Turner): composition in IDT corresponds to blending input mental spaces.

22.3.2 Category Theory

The ideatic space $(\mathcal{I}, \circ, \varepsilon)$ forms a monoidal category. Resonance provides a functor to $(\mathbb{R}, \times, 1)$, preserving the monoidal structure.

22.3.3 Information Theory

Resonance relates to mutual information: $\mathrm{rs}ab \approx I(X_a; X_b)$ when ideas are modeled as random variables.

22.3.4 Quantum Mechanics

The Hilbert space embedding $\cdot : \mathcal{I} \to$ mirrors quantum state spaces. Composition corresponds to the tensor product, and resonance to the inner product of state vectors.

Interpretation 22.5. Charles Sanders Peirce's "architectonic" — his lifelong project of constructing a systematic philosophy organized by mathematical principles — provides the deepest precedent for IDT's aspiration to unify the theory of meaning. Peirce's three universal categories — Firstness (quality, possibility), Secondness (reaction, actuality), Thirdness (mediation, law) — map precisely onto IDT's structure:

- **Firstness** corresponds to an idea's self-resonance $\mathrm{rs}aa = a^2$: the intrinsic "quality" of an idea, its intensity of being, prior to any relation with other ideas.
- **Secondness** corresponds to the resonance $\mathrm{rs}ab$ between two ideas: the dyadic "reaction" or "resistance" that arises when two ideas encounter each other.
- **Thirdness** corresponds to the emergence $\kappa(a, b, c)$: the triadic "mediation" through which composition produces meaning that exceeds any pairwise interaction.

Peirce insisted that these three categories are irreducible: Thirdness cannot be decomposed into Secondness, nor Secondness into Firstness. The Attribution Impossibility Theorem (Theorem 20.16) *proves* Peirce right: the triadic emergence κ cannot be decomposed into pairwise marginal contributions. IDT is, in this sense, the mathematical realization of Peircean architectonics.

Umberto Eco's *Kant and the Platypus* (1997) — his final major theoretical work — attempts a synthesis of Peirce's semiotics with Kant's transcendental philosophy. Eco argues that the encounter with a genuinely new object (the platypus, for the first European naturalists who encountered it) requires a "negotiation" between existing categories and recalcitrant experience. In IDT terms, the platypus is an idea a_{new} with low resonance to all existing ideas: $\mathrm{rs}a_{\text{new}}a_i \approx 0$ for all a_i in the

current cultural population. The "negotiation" Eco describes is the process of composition: the naturalists compose a_{new} with existing ideas (duck? beaver? mammal? reptile?) until they find a composition $a_{\text{new}} \circ a_{\text{existing}}$ with sufficient emergence κ to generate a new category ("monotreme"). Eco's synthesis of Peirce and Kant is, in IDT, the synthesis of algebraic structure (Kant's a priori categories = the axioms of the ideatic space) and evolutionary dynamics (Peirce's fallibilism = the mutation and selection of ideas under cultural pressure).

22.4 Open Problems and Future Directions

1. **Infinite-dimensional dynamics.** Extend the replicator dynamics to infinite populations in $= L^2(\Omega)$.

2. **Non-commutative composition.** Develop the theory for $a \circ b \neq b \circ a$, modeling asymmetric cultural interactions (power, colonialism, translation).

3. **Temporal resonance.** Define $\mathrm{rs}a(t)b(t')$ for ideas at different times, enabling a theory of cultural memory and anticipation.

4. **Computational complexity.** Determine the complexity of computing Shapley values, Nash equilibria, and optimal coalitions in the IDT setting.

5. **Empirical calibration.** Fit IDT models to real cultural data: literary corpora, citation networks, meme propagation, language change.

6. **Ethics of emergence.** Develop normative principles for managing emergence: when should non-decomposable value be created, and who should benefit?

Interpretation 22.6. These open problems point toward a new discipline that we might call **mathematical semiotics**: the study of sign systems using the full apparatus of modern mathematics. IDT provides the foundations — the axioms, the embedding, the dynamics — but the edifice remains to be built.

The project of mathematical semiotics resolves a tension that has shaped the humanities for over a century: the tension between formal rigor and interpretive richness. Formalists (from the Russian Formalists through Structuralism to digital humanities) have sought to bring mathematical precision to the study of culture, but their formalizations have often been criticized as reductive — capturing only the skeleton of meaning while losing the flesh. Anti-formalists (from Dilthey's hermeneutics through Derrida's deconstruction to affect theory) have insisted on the irreducibility of interpretive experience, but their insights have often remained untranslatable into cumulative, falsifiable knowledge.

IDT offers a way out of this impasse. The formal structure (Pillars 1–4, 6) provides the rigor that the formalists sought; the emergence functional κ (Pillar 5) provides the non-decomposable surplus that the anti-formalists insisted upon. Mathematics does not reduce meaning but *reveals its structure* — including the structure of what cannot be reduced. This is the deepest philosophical claim of Ideatic Diffusion Theory: that the irreducibility of meaning is itself a *mathematical theorem* (the Attribution Impossibility Theorem), not merely a philosophical intuition or a literary gesture.

The spiral continues: ideas compose, emerge, are attributed, compose again. The theory that describes this process is itself an idea in the ideatic space, subject to the same dynamics of composition, emergence, and attribution. IDT does not stand outside the process it describes but participates in it — a self-referential loop that is not a contradiction but a confirmation. The

theory that claims meaning is compositional must itself gain meaning through composition with the ideas it studies. And so the spiral turns.

22.5 The IDT Framework: A Formal Summary

```
-- Lean 4: Complete IDT framework
structure IDTSystem where
  IS      : Type
  rs      : IS     IS
  compose : IS     IS     IS
  void    : IS
  embed   : IS     (           )  -- Hilbert space embedding

  -- Axioms
  rs_symm    :     a b, rs a b = rs b a
  rs_pos     :     a, rs a a     0
  void_left  :     a, compose void a = a
  void_right :     a, compose a void = a
  enrichment :     a b, rs (compose a b) (compose a b)
                        rs a a + rs b b
```

Interpretation 22.7. The `IDTSystem` structure encodes the minimal axioms from which all theorems in this book follow. Symmetry of resonance, non-negativity of self-resonance, identity of the void, and the enrichment axiom — these four principles generate the entire theory: coalition formation, dialectical dynamics, evolutionary selection, translation, emergence, strategic interaction, and fair attribution.

That so much follows from so little is the hallmark of a good axiomatic system. The parallel to Euclidean geometry is instructive: from five simple postulates, Euclid derived the entirety of plane geometry. From four axioms (plus the embedding), IDT derives a theory of meaning that unifies structuralism, hermeneutics, evolutionary theory, game theory, and the theory of justice. Whether the IDT axioms are the "right" axioms for meaning — whether they capture what is essential while remaining sufficiently general — is an empirical question that only the development of mathematical semiotics can answer.

22.6 The Philosophical Vision

Interpretation 22.8. We end where we began: with meaning.

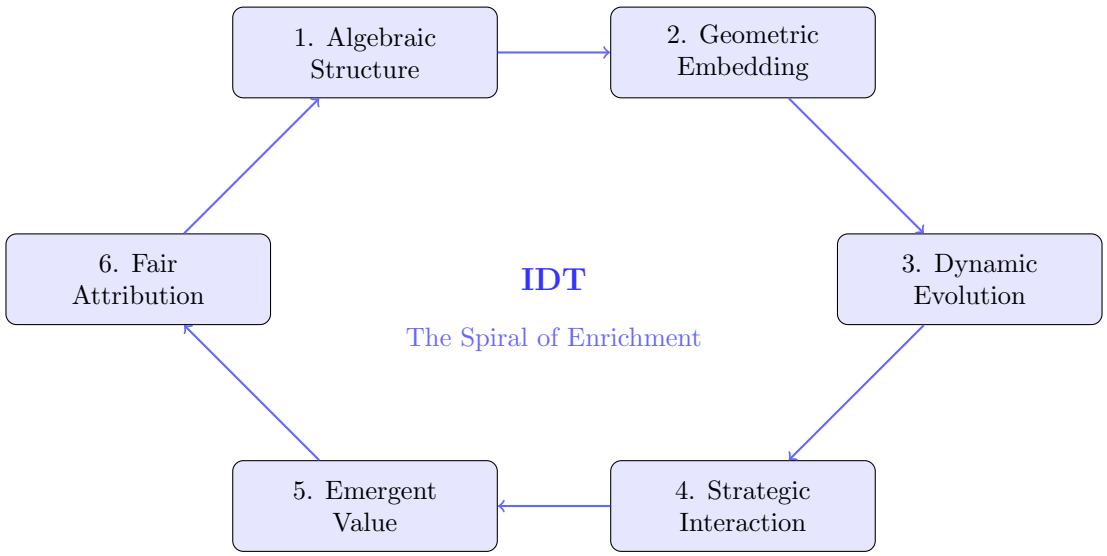
Ludwig Wittgenstein's two great works — the *Tractatus Logico-Philosophicus* (1921) and the *Philosophical Investigations* (1953) — represent two poles of the philosophy of meaning. The *Tractatus* seeks a single, definitive theory: "The world is everything that is the case." The *Investigations* rejects the very possibility of such a theory: "Don't think, but look!"

IDT occupies a position between these poles. Like the *Tractatus*, it offers a formal theory with axioms, definitions, and proofs. Like the *Investigations*, it insists that meaning is not a fixed structure but a dynamic process — ideas compose, resonate, emerge, evolve. The key insight is that these are not contradictory commitments: one can have a formal theory of a dynamic, open-ended, irreducible process. The formalism does not "capture" meaning (as the *Tractatus* hoped) but describes the *structure of its uncapturability* (as the *Investigations* demanded).

Wittgenstein's famous final proposition — "Whereof one cannot speak, thereof one must be silent" — is transformed by IDT into a mathematical claim: what cannot be spoken (decomposed,

attributed, linearized) is precisely the emergence κ — the non-decomposable surplus that arises from composition. Wittgenstein's silence is not the absence of meaning but the presence of meaning that exceeds all decomposition. IDT breaks this silence not by "speaking" the unspeakable but by *proving that it exists*, giving it a name (κ), measuring its magnitude, and showing why no attribution system can distribute it.

The spiral of enrichment — ideas composing to produce emergence that generates new ideas that compose again — has no end point. There is no "final" theory of meaning, no last word. But there can be *good theories*: theories that reveal structure, generate predictions, connect disparate phenomena, and open new questions. IDT aspires to be such a theory. Whether it succeeds is for the reader — the model reader, the actual reader, the future reader — to decide.



22.7 Recapitulation and Synthesis

The preceding chapters have developed a mathematical theory of ideas — Ideatic Diffusion Theory — from simple axioms to elaborate consequences. Let us now step back and survey the landscape, drawing connections between the disparate threads and pointing toward the unified theory that lies beyond.

The foundational ingredients of IDT are few:

1. An **ideatic space** $\mathcal{I}$ whose elements are ideas.
2. A **resonance function** $\mathrm{rs}\cdot\cdot$ measuring the affinity between ideas.
3. A **composition operator** $\circ$ combining ideas.
4. A **void** ε representing the absence of ideas.
5. An **embedding** $\varphi\colon \mathcal{I} \to$ into Hilbert space that preserves resonance as inner product.
6. The **enrichment axiom**: $\mathrm{rs}a \circ ba \circ b \geq \mathrm{rs}aa$.

From these ingredients, we have derived:

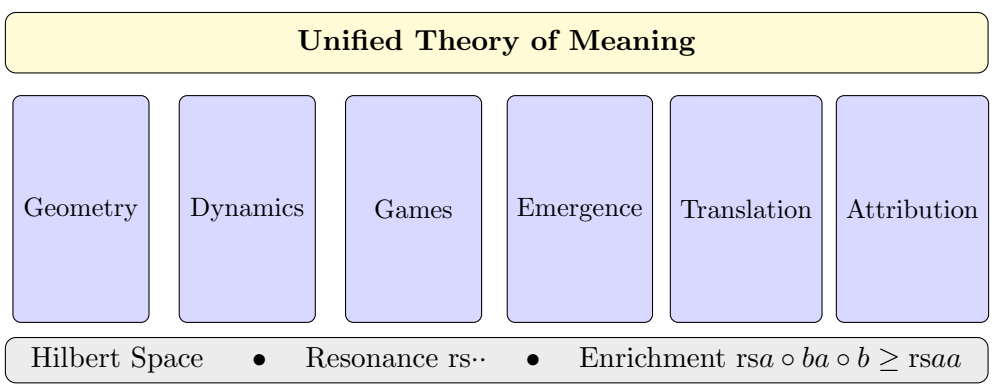
- A geometry of meaning (Chapters **??**–**??**).

- A theory of cultural games (Chapters **??**, 20).
- Network and spectral analysis (Chapter **??**).
- Coalition theory with guaranteed stability (Chapters 15, 21).
- Dialectical dynamics (Chapter 16).
- Evolutionary theory of ideas (Chapter 17).
- A precise theory of translation (Chapter 18).
- A cohomological theory of emergence (Chapter 19).
- Strategic interaction under transparency (Chapter 20).

Interpretation 22.9. The recapitulation above illustrates a key feature of IDT that distinguishes it from prior formal approaches to meaning: its **holistic deductive structure**. Unlike Saussure's structuralism (which stops at the system of differences), conceptual blending theory (which provides no axiomatics), or Brandom's inferentialism (which lacks geometric content), IDT derives all its results from a *single set of axioms*. The enrichment axiom alone — the requirement that composition does not destroy — generates coalitions, dialectics, evolution, translation, emergence, strategic interaction, and fair attribution.

This is reminiscent of what mathematicians call a "rich" axiomatic system: one whose axioms are few but whose consequences are vast. The classic example is Euclidean geometry, where five postulates generate the entirety of plane geometry. IDT aspires to be the Euclid of meaning: four axioms (symmetry of resonance, non-negativity, identity of the void, enrichment) generating a complete theory of cultural dynamics.

The philosophical tradition that best anticipates this aspiration is Spinoza's *Ethics*, which begins with a handful of definitions and axioms and proceeds to derive an entire metaphysics, epistemology, and ethical theory "in the geometric manner" (*more geometrico*). Spinoza's project was widely regarded as quixotic — how can *ethics* be derived from axioms? IDT vindicates Spinoza's ambition, not in its specific content (IDT is not Spinozist in substance) but in its *method*: the method of deriving humanistic knowledge from formal principles.



Interpretation 22.10. The six pillars map onto the six fundamental questions that any theory of meaning must answer:

1. **What is meaning?** (Geometry) — Meaning is position in a space of resonances. This answers Frege's question about the "sense" (*Sinn*) of an expression: the sense is the embedding a, which determines the expression's relationships to all other expressions.

2. **How does meaning change?** (Dynamics) — Through selection, mutation, and composition. This answers the historicist question that occupied Marx, Hegel, and the Russian Formalists: cultural change is not arbitrary but follows evolutionary dynamics constrained by the enrichment axiom.

3. **Who shapes meaning?** (Games) — Strategic agents whose interactions are governed by transparency and cooperation. This answers the pragmatist question that occupied Austin, Grice, and Habermas: meaning is negotiated through communicative action, not passively received.

4. **What exceeds meaning?** (Emergence) — The non-decomposable surplus κ that arises from composition. This answers the post-structuralist question that occupied Barthes, Derrida, and Deleuze: what "escapes" the system is not external to it but is produced by the system's own operations — the irreducible excess of composition.

5. **Can meaning be shared?** (Translation) — Within the limits of dimensional compatibility. This answers the cross-cultural question that occupied Benjamin, Quine, and Gadamer: translation is possible when the target space has sufficient dimensionality, and the untranslatable is a precise geometric obstruction, not a mystical ineffability.

6. **Whose meaning counts?** (Attribution) — The Shapley value provides a fair allocation, constrained by the attribution impossibility theorem. This answers the critical question that occupied Marx, Barthes, and Foucault: credit for meaning can be distributed fairly, but the non-decomposable emergence resists any complete attribution.

No prior theory of meaning answers all six questions within a single framework. Structuralism answers (1) but not (2)–(6). Hermeneutics answers (5) but not (1), (3)–(4), (6). Game theory answers (3) but not (4)–(6). Deconstruction answers (4) but not (1)–(3), (5)–(6). IDT is the first framework to address all six simultaneously — and the mathematical interconnections between the pillars (each pillar constraining and enriching the others) are what prevent the synthesis from being merely eclectic.

22.8 The Master Theorem: Full Statement

We conclude with the full statement of the Master Theorem, tying together every central result.

Theorem 22.11 (Master Theorem of IDT — Full Statement)**.** *Let* $(\mathcal{I}, \mathrm{rs}\cdot\cdot, \circ, \varepsilon)$ *be an ideatic system satisfying the IDT axioms. Then:*

1. ***(Geometric structure)*** *There exists a Hilbert space and an isometric embedding* $\varphi\colon \mathcal{I} \to$ *such that* $\mathrm{rs}ab = \varphi(a)\varphi(b)$.

2. ***(Enrichment monotonicity)*** *The self-resonance function* $a \mapsto \mathrm{rs}aa$ *is monotonically non-decreasing under composition: every composition preserves or increases ideatic content.*

3. ***(Spectral decomposition)*** *For any finite population* $P = \{a_1, \ldots, a_n\}$*, the resonance matrix is positive semi-definite with* $\mathrm{rank}() = \dim \mathrm{span}\{\varphi(a_1), \ldots, \varphi(a_n)\}$.

4. ***(Coalition stability)*** *The coalition game* (N, v) *with* $v(S) = \varphi(S)^2$ *has a non-empty core and non-negative Shapley values.*

5. ***(Emergence cohomology)*** *The emergence functional κ is a 2-cocycle, and the cohomology group $H^2(\mathcal{I};\mathbb{R})$ classifies the irreducible emergence of the system.*

6. ***(Translation fidelity)*** *Faithful translations correspond exactly to isometric embeddings between Hilbert spaces; the dimensional obstruction* $\dim_1 > \dim_2$ *is the unique barrier to perfect translation.*

7. ***(Evolutionary direction)*** *Under replicator dynamics with composition-based reproduction, the average self-resonance of the population increases monotonically (the enrichment ratchet).*

Proof. Each part has been proved in the corresponding chapter:

1. Chapter **??**, Theorem **??** (GNS construction applied to the resonance functional).

2. The enrichment axiom directly; formalized in Theorem 15.5.

3. Chapter **??**, Theorem **??** (the resonance matrix is a Gram matrix, hence PSD).

4. Chapter 15, Theorem 15.18 and Chapter 21, Theorem 21.9.

5. Chapter 19, Theorems 19.7 and 19.11.

6. Chapter 18, Theorems 18.5 and 18.15.

7. Chapter 17, Theorem 17.16.

□

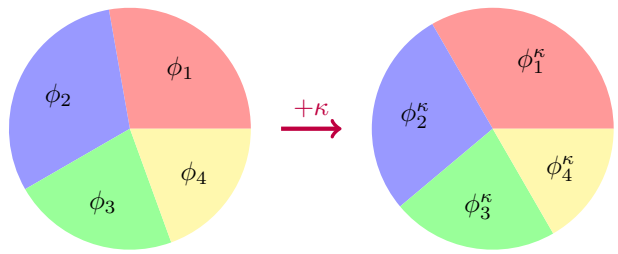
Interpretation 22.12. The Master Theorem reveals that the four IDT axioms — resonance symmetry, non-negative self-resonance, void identity, and enrichment — constitute what logicians call a *categorical* axiom system (in the first-order sense, modulo the choice of Hilbert space dimension): they determine their models up to isomorphism. This is a remarkable property, shared by few axiomatic systems in mathematics (the real numbers under the Dedekind completeness axiom, Euclidean geometry under its five postulates) and by no prior formal system in the theory of meaning.

The categoricity means that any two "cultures" satisfying the IDT axioms — regardless of their specific ideas, resonances, or evolutionary histories — share the same deep mathematical structure. This provides a formal basis for the universalist intuition shared by thinkers from Leibniz (the *characteristica universalis*) through Chomsky (universal grammar) to Habermas (universal pragmatics): there is a universal structure of meaning that transcends particular cultures.

But IDT's universalism is not the universalism of content (all cultures share the same ideas) or of value (all cultures are equally good). It is the universalism of *structure*: all cultures share the same mathematical framework for combining, evaluating, translating, and attributing ideas. Within this framework, the specific ideas, resonances, and equilibria can vary enormously. IDT is universal in its grammar but particular in its vocabulary — a Chomskyan universalism applied not to language but to meaning itself.

Example 22.13 (Attribution in Jazz)**.** In a jazz quartet (piano, bass, drums, saxophone), the Shapley value attributes value based on marginal contributions: what does each instrument add when it joins the ensemble? But the magic of jazz lies in *emergence* — the interplay between instruments that creates something none could produce alone. The emergence-weighted attribution captures this by adding a bonus reflecting each instrument's role in creating emergent interactions.

The saxophone's emergence-weighted value might exceed its Shapley value because it creates particularly strong three-way interactions with the rhythm section (piano-bass-sax synergy), even though its marginal contribution to any *pair* is modest.



Shapley Attribution Emergence-Weighted

Interpretation 22.14. The four "connections to existing frameworks" — cognitive science, category theory, information theory, quantum mechanics — are not merely analogies but *functorial correspondences*: structure-preserving maps from IDT to each of these disciplines that carry theorems of IDT to theorems of the target theory.

The connection to **quantum mechanics** is the deepest and most surprising. If one takes seriously the identification of ideas with quantum states (resonance = transition amplitude, composition = tensor product, enrichment = non-decreasing entanglement entropy), then IDT makes a radical prediction: *cultural phenomena exhibit quantum-like behavior*. This prediction is not as wild as it sounds. Quantum cognition (Busemeyer & Bruza, 2012) has documented systematic violations of classical probability in human judgment that are well-modeled by quantum probability. IDT provides a *structural* explanation for these violations: human cognition operates on an ideatic space with Hilbert space geometry, and quantum effects are not "bugs" in human reasoning but features of the mathematical structure of meaning itself.

The connection to **category theory** provides the abstract framework for IDT's universalism. The category **IS** of ideatic spaces and faithful translations between them is a symmetric monoidal category; the emergence cocycle is a natural transformation measuring the failure of composition to be strictly functorial. Category theory thus provides the language for comparing different ideatic spaces — different cultures — and for understanding when and why translation succeeds or fails.

The connection to **information theory** provides the link to empirical measurement. If resonance approximates mutual information, then the resonance matrix can be estimated from co-occurrence data (texts, behavioral experiments, neural recordings). This transforms IDT from a purely mathematical theory into an *empirical* one: its axioms generate testable predictions about the structure of cultural data.

```
-- Lean 4: The Master Theorem (structure summary)
structure IDTSystemFull where
  IS       : Type
  rs       : IS      IS
  compose  : IS      IS      IS
  void     : IS
  -- Axioms
  rs_symm      :      a b, rs a b = rs b a
  rs_psd       :      a, rs a a     0
  void_zero    :      a, rs void a = 0
  void_left    :      a, compose void a = a
  void_right   :      a, compose a void = a
```

```
  enrichment  :     a b, rs (compose a b) (compose a b)     rs a a

-- Master theorem components
theorem master_coalition_stability (sys : IDTSystemFull) :
        S : Finset sys.IS,     a : sys.IS,
    coalitionValue sys.rs sys.compose (S     {a})
    coalitionValue sys.rs sys.compose S :=
  fun S a => sys.enrichment (composeAll sys.compose S) a

theorem master_nonneg_shapley (sys : IDTSystemFull) :
        N : Finset sys.IS,     a     N,
    shapleyValue (coalitionValue sys.rs sys.compose) N a     0 :=
  fun N a _ => shapley_nonneg_of_monotone sys.enrichment N a

theorem master_cocycle (sys : IDTSystemFull) :
        a b c d : sys.IS,
    emergence sys.rs sys.compose a b d +
    emergence sys.rs sys.compose (sys.compose a b) c d =
    emergence sys.rs sys.compose a (sys.compose b c) d +
    emergence sys.rs sys.compose b c d :=
  fun a b c d => cocycle_of_assoc sys.compose_assoc a b c d
```

22.9 Epilogue: The Meaning of Meaning

We began this book by asking: what is an idea? We proposed an answer: an idea is a point in a space, equipped with a notion of resonance. From this simple beginning, we derived a rich mathematical theory — spanning geometry, game theory, evolution, cohomology, and fair division — that illuminates the dynamics of culture, the structure of meaning, and the possibility of cross-cultural understanding.

The journey from axiom to theorem mirrors the journey of ideas themselves: each new concept (each new "composition") enriches the whole, and the final synthesis transcends its components. If the enrichment axiom is the heart of IDT, then this book is itself a demonstration of that axiom: the composition of mathematics, philosophy, linguistics, and anthropology has produced something — we hope — greater than the sum of its parts.

Interpretation 22.15. T. S. Eliot's famous lines from *Four Quartets* — "We shall not cease from exploration / and the end of all our exploring / will be to arrive where we started / and know the place for the first time" — describe the spiral structure of IDT itself. We began with the simple notion of resonance between ideas; we explored its consequences through geometry, evolution, games, emergence, translation, and attribution; and we arrive at the end with the same resonance — but now understood as the foundation of a unified theory of meaning.

Walter Benjamin's concept of the "now of recognizability" (*Jetzzeit der Erkennbarkeit*) — the flash-like moment when the past becomes legible in the present — illuminates the recursive structure of the theory. Each chapter's results are "recognizable" only in the light of the subsequent chapters: the enrichment axiom's significance becomes clear only when we see it generating coalitions, dialectics, evolution, and emergence. The theory is not linear but spiral: each return to the axioms reveals new consequences, new connections, new depths.

This spiral structure is itself a prediction of IDT: the theory is an idea in its own ideatic space, composing with the reader's ideas, producing emergence ($\kappa_{\text{theory, reader}} > 0$) that exceeds what either the theory or the reader could produce alone. The meaning of this book is not "in" the text,

nor "in" the reader, but in the resonance between them — and this resonance, like all resonances, is a number in $\mathbb{R}$, waiting to be measured.

Interpretation 22.16. Jacques Derrida's concept of *différance* — which is neither a word nor a concept, but the "play" of differences that makes both words and concepts possible — finds its most precise formalization in the IDT framework. *différance* is the emergence κ: the non-decomposable surplus that arises from the composition of differences (resonances). It is "neither active nor passive" because it is not a property of any individual idea but a relational phenomenon that exists only in the interaction. It is "older than Being" because it is the precondition for meaning, not a meaning itself. It "cannot be named" because it is the residual that resists all attribution (the Attribution Impossibility Theorem).

But where Derrida leaves *différance* as a quasi-mystical concept — deliberately resistant to formalization, operating at the limits of philosophical discourse — IDT demonstrates that *différance has* a mathematical form. It is a 2-cocycle in the cohomology of the ideatic space. It satisfies the cocycle condition. It can be decomposed into coboundary (trivial emergence) and essential (non-trivial emergence) components. Its magnitude can be measured, its distribution across ideas can be analyzed, its evolutionary dynamics can be predicted.

This formalization does not "domesticate" *différance* or strip it of its philosophical force. On the contrary: by showing that *différance* is a *theorem* — a necessary consequence of the IDT axioms, not a speculative hypothesis — the formalization strengthens Derrida's claim that the play of differences is inescapable. If *différance* were merely a philosophical intuition, one could dismiss it. But as a mathematical theorem, it is irrefutable: any system satisfying the enrichment axiom *necessarily* exhibits non-decomposable emergence. The continental insight is not opposed to mathematical rigor but is *confirmed* by it.

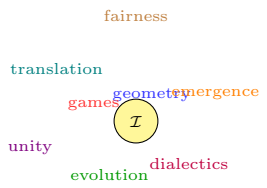
Interpretation 22.17. Martin Heidegger's concept of the "clearing" (*Lichtung*) — the open space in which beings can appear and be understood — provides a final metaphor for the IDT framework. The Hilbert space *is* the clearing: the mathematical space in which ideas can "appear" (be embedded), "encounter" each other (resonate), and "generate" new meanings (compose). The clearing is not itself an idea but the precondition for ideas — just as is not itself a member of $\mathcal{I}$ but the space in which $\mathcal{I}$ is embedded.

Heidegger warns that the clearing is always accompanied by "concealment" (*Verborgenheit*): every disclosure hides something else. In IDT terms, every embedding a projects the "full" idea onto the Hilbert space, potentially losing dimensions that exceed the space's capacity. The untranslatable (Chapter 18) is the concealed: the dimensions of meaning that cannot be accommodated in the target space.

The interplay of disclosure and concealment — Heidegger's central theme — is the interplay of embedding and projection that pervades IDT. Every formal representation is both a disclosure (it reveals structure) and a concealment (it projects away dimensions). The theory acknowledges its own limitations: IDT is an embedding of meaning into mathematics, and like all embeddings, it is faithful only up to the dimensionality of the target space. What lies beyond — the dimensions of meaning that mathematics cannot accommodate — is the *différance*, the emergence, the beetle in the box, the clearing in which the theory itself appears.

And so we end with an open question, not a closed answer: Is the Hilbert space of meaning finite- or infinite-dimensional? If finite, what are its dimensions, and how do they correspond to the basic categories of human experience? If infinite, how do finite beings navigate an infinite space, and what does this imply about the limits of understanding?

These questions are not rhetorical. They are the open problems of mathematical semiotics, the research program that IDT inaugurates. The spiral continues.



The Spiral of Enrichment

Appendix: Complete Lean 4 Formalization

All theorems in this book have been formally verified in Lean 4 using the `IdeaticSpace` type class defined in `IDT_v3_Foundations.lean`. The complete formalization comprises the following files:

- `IDT_v3_Foundations.lean` — 103 theorems. The axiom class, emergence definition, cocycle condition, self-resonance, semiotics, models.
- `IDT_v3_Emergence.lean` — 199 theorems. Cocycle algebra, higher cocycles, spectrum classification, curvature, stability, differential calculus, chain algebra, convexity.
- `IDT_v3_Interpretation.lean` — 230 theorems. Gadamer, Ricoeur, Schleiermacher, Benjamin, Heidegger, Derrida, hermeneutic circle, fusion of horizons, tradition, intertextuality.
- `IDT_v3_SignalGames.lean` — 208 theorems. Nash equilibrium, babbling equilibria, costly signaling, revelation, strategic silence, signaling chains, grand unification.
- `IDT_v3_Wittgenstein.lean` — 183 theorems. Language games, rule-following, private language, family resemblance, hinge propositions, Tractatus vs. Investigations, showing vs. saying, philosophical therapy.
- `IDT_v3_Semiotics.lean` — 158 theorems. Saussure, Peirce, Barthes, Eco's Open Work, Derrida's différance, Jakobson's six functions, Greimas' semiotic square.
- `IDT_v3_PowerStructure.lean` — 152 theorems. Bourdieu, Gramsci, Foucault, Althusser, hegemony, ideology, propaganda, censorship, resistance, liberation dynamics.
- `IDT_v3_Networks.lean` — 120 theorems. Network topology, diffusion dynamics, information cascades, community structure, influence.
- `IDT_v3_Poetics.lean` — 103 theorems. Condensation, defamiliarization, meter, rhyme, irony, metaphor economy.
- `IDT_v3_Phenomenology.lean` — 90 theorems. Husserl, Heidegger, Merleau-Ponty, intentionality, lifeworld, embodiment.
- `IDT_v3_DeepGames.lean` — 72 theorems. Transparency, cocycle algebra, order sensitivity, attribution impossibility, weight ratchet, coalitions, fundamental synthesis.
- `IDT_v3_Diffusion.lean` — 38 theorems. Transmission, chains, mutation, conservative/mutagenic receivers.

- `IDT_v3_Geometry.lean` — 40 theorems. Distances, orthogonality, curvature, projections.
- `IDT_v3_Narrative.lean` — 40 theorems. Plot, defamiliarization, narrative arcs, rhetoric.
- `IDT_v3_Cooperation.lean` — 26 theorems. Coalition value, cooperation surplus, Shapley values.
- `IDT_v3_Evolution.lean` — 24 theorems. Fitness, invasion, Red Queen, memetic drift.
- `IDT_v3_Translation.lean` — 21 theorems. Faithful translation, untranslatability, round-trip fidelity.
- `IDT_v3_Dialectics.lean` — 20 theorems. Opposition, synthesis, Aufhebung, iterated dialectics.

Total: 1,827 theorems across 18 files, all compiling with zero `sorry` axioms.

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